Programming Guide 03/2004 Edition

sinumerik

SINUMERIK 840D/840Di/810D
Cycles

SIEMENS

SIEMENS

SINUMERIK 840D/840Di/810D

Cycles

Programming Guide

Valid for

Control	*Software version*
SINUMERIK 840D powerline	7
SINUMERIK 840DE powerline (export version)	7
SINUMERIK 840Di	3
SINUMERIK 840DiE	3
SINUMERIK 810D powerline	7
SINUMERIK 810DE powerline (export version)	7

03.04 Edition

General 1

Drilling Cycles and Drilling Patterns 2

Milling Cycles 3

Turning Cycles 4

Error Messages and Error Handling 5

Appendix A

SINUMERIK® Documentation

Printing history

Brief details of this edition and previous editions are listed below.

The status of each edition is indicated by the code in the "Remarks" columns.

Status code in the "Remarks" column:

A .... New documentation.

B .... Unrevised reprint with new Order No.

C .... Revised edition with new status.

If factual changes have been made on the page since the last edition, this is indicated by a new edition coding in the header on that page.

Edition	Order No.	Remarks
02.95	6FC5298-2AB40-0BP0	O
04.95	6FC5298-2AB40-0BP1	C
03.96	6FC5298-3AB40-0BP0	C
08.97	6FC5298-4AB40-0BP0	C
12.97	6FC5298-4AB40-0BP1	C
12.98	6FC5298-5AB40-0BP0	C
08.99	6FC5298-5AB40-0BP1	C
04.00	6FC5298-5AB40-0BP2	C
10.00	6FC5298-6AB40-0BP0	C
09.01	6FC5298-6AB40-0BP1	C
11.02	6FC5298-6AB40-0BP2	C
03.04	6FC5298-7AB40-0BP0	C

This manual is included in the documentation on CD-ROM (**DOCONCD**)

Edition	Order No.	Remarks
03.04	6FC5298-7CA00-0BG0	C

Trademarks

SIMATIC®, SIMATIC HMI®, SIMATIC NET®, SIROTEC®, SINUMERIK® and SIMODRIVE® are Siemens trademarks. Other product names used in this documentation may be trademarks which, if used by third parties, could infringe the rights of their owners.

More information is available on the Internet at:
http://www.ad.siemens.com/mc

This document has been created with WinWord V 9.0
and Designer V 7.0.
The reproduction, transmission or use of this document or its contents is not permitted without express written authority. Offenders will be liable for damages. All rights, including rights created by patent grant or registration or a utility model or design, are reserved.

© Siemens AG, 1995-2004. All rights reserved

Other functions not described in this documentation might be executable in the control. This does not, however, represent an obligation to supply such functions with a new control or when servicing.

We have checked the contents of this manual for agreement with the hardware and software described. Nonetheless, differences might exist and therefore we cannot guarantee that they are completely identical. The information contained in this document is, however, reviewed regularly and any necessary changes will be included in the next edition. We welcome all recommendations and suggestions.

Subject to change without prior notice

Preface

Structure of the Documentation

The SINUMERIK documentation is organized in 3 parts:

- General Documentation
- User Documentation
- Manufacturer/Service Documentation

Audience

This Manual is intended for machine-tool users. This publication provides detailed information that the user requires for operating the SINUMERIK 840D powerline and 810D powerline controls.

Standard scope

This Programming Guide describes the functionality afforded by standard functions. Extensions or changes made by the machine tool manufacturer are documented by the machine tool manufacturer.

More detailed information about other publications concerning SINUMERIK 810D and 840D and publications that apply to all SINUMERIK controls (e.g. Universal Interface, Measuring Cycles...) can be obtained from your local Siemens branch office.

Other functions not described in this documentation might be executable in the control. This does not, however, represent an obligation to supply such functions with a new control or when servicing.

Validity

This Programming Guide applies to the following controls:

SINUMERIK 840D powerline	SW7
SINUMERIK 840DE powerline (export version)	SW7
SINUMERIK 840Di	SW3
SINUMERIK 840DiE (Export Version)	SW3
SINUMERIK 810D powerline	SW7
SINUMERIK 810DE powerline (export version)	SW7

with operator panels OP 010, OP 010C, OP 010S, OP 12 or OP 15 (PCU 20 or PCU 50)

© Siemens AG 2004 All rights reserved.

SINUMERIK 840D powerline

From 09.2001

- SINUMERIK 840D powerline and
- SINUMERIK 840DE powerline

will be available with improved performance. For a list of the available **powerline** modules, please refer to Section 1.1 of the Hardware Description /PHD/.

SINUMERIK 810D powerline

From 12.2001

- SINUMERIK 810D powerline and
- SINUMERIK 810DE powerline

will be available with improved performance. For a list of the available powerline modules, please refer to Section 1.1 of the Hardware Description /PHC/.

Structure of descriptions

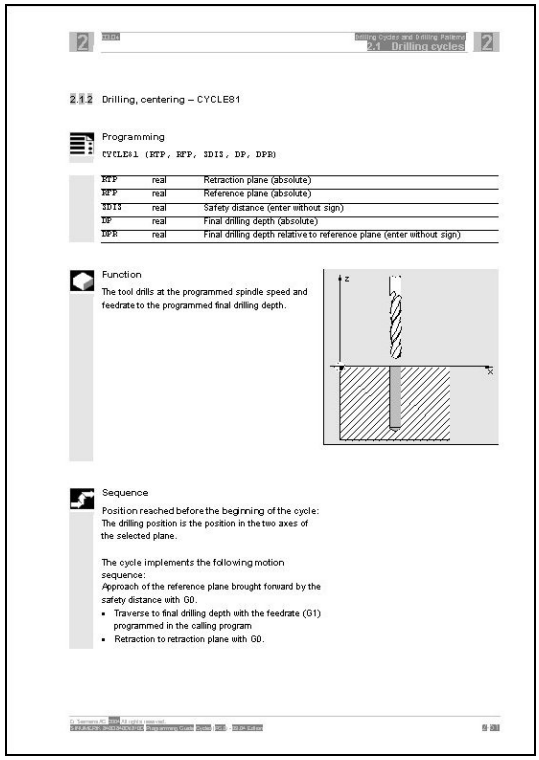
All cycles and programming options have been described according to the same internal structure as far as this is meaningful and practicable. The various levels of information have been organized such that you can selectively access the information you need for the task in hand.

1. A quick overview

If you need to look up a cycle that is rarely used or the meaning of a parameter, you will see at a glance how the function is programmed together with an explanation of the cycles and parameters.

This information is always displayed at the top of the page.

Note:
Due to lack of space, it is not possible to specify all modes of display offered by the programming language for individual cycles and parameters. For this reason, the method of programming shown is the one most commonly used in workshops.



2 2.1 Drilling cycles 2

2.1.2 Drilling, centering – CYCLE81

Programming

CYCLE81 (RTP, RFP, SDIS, DP, DPR)

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Final drilling depth (absolute)
DPR	real	Final drilling depth relative to reference plane (enter without sign)

Function

The tool drills at the programmed spindle speed and feedrate to the programmed final drilling depth.

Sequence

Position reached before the beginning of the cycle:
The drilling position is the position in the two axes of the selected plane.

The cycle implements the following motion sequence:
Approach of the reference plane brought forward by the safety distance with G0.
- Traverse to final drilling depth with the feedrate (G1) programmed in the calling program
- Retraction to retraction plane with G0.

© Siemens AG 2004 All rights reserved.

2. Detailed explanations

You will find detailed answers to the following questions in the theory section:

What is the cycle used for?

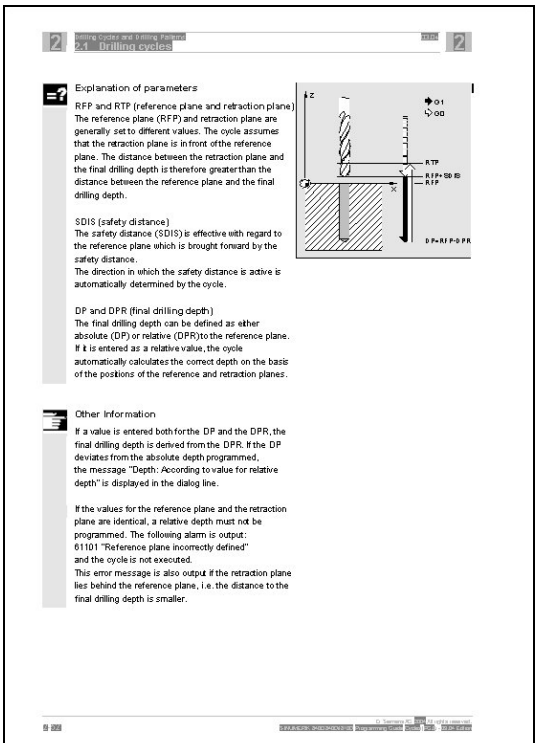
What does the cycle do?

How is it programmed and executed?

What do the parameters do?

What else do I need to know?

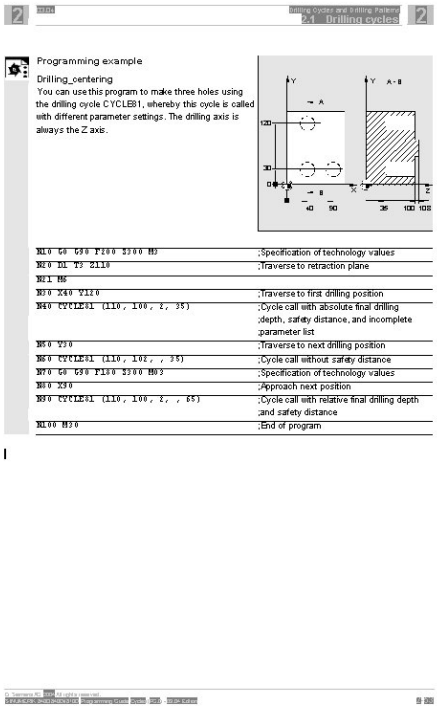
The theoretical sections are primarily intended as learning material for the NC entry-level user. You should work through the manual at least once to get an idea of the functional scope and capability of your SINUMERIK control.



3. From theory to practice

The programming example shows you how to include the cycles in an operating sequence.

An application example of almost all the cycles is provided after the theoretical section.



© Siemens AG 2004 All rights reserved.

Explanation of symbols



Sequence



Explanation



Function



Parameters



Programming example



Programming



Other Information



Cross-references to other documentation or sections



Danger notes and sources of error



Additional notes or background information

© Siemens AG 2004 All rights reserved.

Warnings
The following warnings with varying degrees of severity appear in this document.



Danger

Indicates an imminently hazardous situation which, if not avoided, **will** result in death or serious injury or in substantial property damage.



Warning

Indicates a potentially hazardous situation which, if not avoided, **could** result in death or serious injury or in substantial property damage.



Caution

Used with the safety alert symbol indicates a potentially hazardous situation which, if not avoided, **may** result in minor or moderate injury or in property damage.

Caution

Used without safety alert symbol indicates a potentially hazardous situation which, if not avoided, **may** result in property damage.

Notice

Used without the safety alert symbol indicates a potential situation which, if not avoided, **may** result in an undesirable result or state.

Principle
Your SIEMENS 810D and 840D have been designed and constructed to the latest standards of technology and recognized safety rules, standards and regulations.

Supplementary devices
The applications of SIEMENS controls can be expanded for specific purposes through the addition of special add-on devices, equipment and expansions supplied by SIEMENS.

© Siemens AG 2004 All rights reserved.

Personnel

Only **properly trained, authorized, reliable personnel** must be allowed to use this equipment. No-one without the necessary training must be allowed to operate the control, even temporarily.

The **areas of responsibility** assigned to personnel involved in setting up, operating and maintaining the equipment must be clearly **specified** and their compliance **verified**.

Procedure

Before the control is started up, personnel responsible for its operation must have read and understood the Operator's Guides. The company used this equipment is also obliged to carry out **continuous monitoring** of the overall technical condition of the equipment (with a view to identifying externally visible defects and damage as well as changes in the operating behavior of the control).

Servicing

Repairs to equipment may only be carried out by **personnel specially trained and qualified** for the application in question in accordance with the provisions specified in the maintenance and servicing guides. All relevant safety regulations must be followed.

Hotline

If you have any questions, please get in touch with our hotline:
A&D Technical Support Phone.: +49 (0) 180 5050 – 222
Fax: +49 (0) 180 5050 – 223
E-mail: adsupport@siemens.com

Please send any queries about the documentation (suggestions or corrections) to the following fax number or email address:
Fax: +49 (0) 0131 98 – 2176
E-mail: motioncontrol.docu@erlf.siemens.de
Fax form: See the reply form at the end of the document.

Internet address http://www.siemens.com/mc

© Siemens AG 2004 All rights reserved.

Note

The following are deemed as **improper usage** and **exclude the manufacturer from all liability**:

Any usage or application incompatible with or beyond the scope of the items specified above.

Cases where the control is operated **in a technically imperfect condition**, without due provision for safety considerations and/or hazards, or in contravention with any of the instructions in the Instruction Manual.

Cases where faults which could affect safety are not remedied **before** starting up the control.

Any **modification, bypassing** or **decommissioning** of equipment on the control whose intended purpose is to ensure proper functioning, unrestricted use of equipment and/or active and passive safety.

Unforeseen danger can arise with reference to:

- life and limb of personnel,
- the control, machine or other assets of the owner and the user.

© Siemens AG 2004 All rights reserved.

Notes

© Siemens AG 2004 All rights reserved.

Contents

General 1-17

1.1 General notes ... 1-18

1.2 Overview of cycles ... 1-18
1.2.1 Drilling cycles, drill pattern cycles, milling cycles and turning cycles ... 1-19
1.2.2 Auxiliary cycle routines ... 1-20

1.3 Programming the cycles ... 1-21
1.3.1 Call and return conditions ... 1-21
1.3.2 Messages during execution of a cycle ... 1-22
1.3.3 Cycle call and parameter list ... 1-23
1.3.4 Cycle simulation ... 1-26

1.4 Cycle support in the program editor ... 1-27
1.4.1 Menus, cycle selection ... 1-27
1.4.2 New input screen form functions ... 1-28

1.5 Cycle support for user cycles ... 1-35
1.5.1 Overview of the files required ... 1-35
1.5.2 Accessing cycle support ... 1-35
1.5.3 Configuring cycle support ... 1-36
1.5.4 Bitmap size and screen resolution ... 1-37
1.5.5 Storing bitmaps in data management for HMI Advanced ... 1-38
1.5.6 Handling bitmaps for HMI Embedded ... 1-38

1.6 Starting up the cycles ... 1-40
1.6.1 Machine data ... 1-40
1.6.2 Definition files for cycles GUD7.DEF and SMAC.DEF ... 1-41
1.6.3 New form of delivery for cycles in HMI Advanced ... 1-43
1.6.4 Upgrading the cycles SW 6.4 and later in HMI Advanced SW 6.3 and later ... 1-43

1.7 Additional cycle functions ... 1-44

Drilling Cycles and Drilling Patterns 2-47

2.1 Drilling cycles ... 2-48
2.1.1 Preconditions ... 2-50
2.1.2 Drilling, centering – CYCLE81 ... 2-51
2.1.3 Drilling, counterboring – CYCLE82 ... 2-54
2.1.4 Deep-hole drilling – CYCLE83 ... 2-56
2.1.5 Rigid tapping – CYCLE84 ... 2-63
2.1.6 Tapping with compensating chuck – CYCLE840 ... 2-70
2.1.7 Boring 1 – CYCLE85 ... 2-78
2.1.8 Boring 2 – CYCLE86 ... 2-81
2.1.9 Boring 3 – CYCLE87 ... 2-85
2.1.10 Boring 4 – CYCLE88 ... 2-87
2.1.11 Boring 5 – CYCLE89 ... 2-89

2.2 Modal call of drilling cycles ... 2-91

© Siemens AG 2004 All rights reserved.

2.3 Drilling pattern cycles 2-94
2.3.1 Preconditions 2-94
2.3.2 Row of holes – HOLES1 2-95
2.3.3 Circle of holes – HOLES2 2-99
2.3.4 Point grid – CYCLE801 2-102

Milling Cycles 3-105

3.1 General notes 3-106
3.2 Preconditions 3-107
3.3 Thread milling - CYCLE90 3-109
3.4 Elongated holes on a circle - LONGHOLE 3-116
3.5 Slots on a circle - SLOT1 3-121
3.6 Circumferential slot - SLOT2 3-129
3.7 Milling rectangular pockets - POCKET1 3-135
3.8 Milling circular pockets - POCKET2 3-139
3.9 Milling rectangular pockets - POCKET3 3-143
3.10 Milling circular pockets - POCKET4 3-152
3.11 Face milling - CYCLE71 3-157
3.12 Path milling - CYCLE72 3-163
3.13 Milling rectangular spigots - CYCLE76 3-173
3.14 Milling circular spigots - CYCLE77 3-178
3.15 Pocket milling with islands - CYCLE73, CYCLE74, CYCLE75 3-182
3.15.1 Transfer pocket edge contour - CYCLE74 3-183
3.15.2 Transfer island contour - CYCLE75 3-185
3.15.3 Contour programming 3-186
3.15.4 Pocket milling with islands - CYCLE73 3-188
3.16 Swiveling - CYCLE800 3-210
3.16.1 Operation, parameter assignment, input screen form 3-213
3.16.2 Operating instructions 3-217
3.16.3 Parameters 3-218
3.16.4 Starting up CYCLE800 3-222
3.16.5 User cycle TOOLCARR.spf 3-239
3.16.6 Error messages 3-245
3.17 High speed settings - CYCLE832 (SW 6.3 and later) 3-246
3.17.1 Call CYCLE 832 in HMI menu tree 3-249
3.17.2 Parameters 3-252
3.17.3 Customizing the technology 3-253
3.17.4 Customizing additional program parameters CYC_832T 3-255
3.17.5 Interfaces 3-257
3.17.6 Error messages 3-258

© Siemens AG 2004 All rights reserved.

3.18 Engraving cycle CYCLE60 (SW 6.4 and later) ... 3-259

Turning cycles 4-269

4.1 General notes ... 4-270
4.2 Preconditions ... 4-271
4.3 Grooving cycle – CYCLE93 ... 4-274
4.4 Undercutting cycle – CYCLE94 ... 4-283
4.5 Stock removal cycle – CYCLE95 ... 4-287
4.6 Thread undercut – CYCLE96 ... 4-300
4.7 Thread cutting – CYCLE97 ... 4-304
4.8 Thread chaining – CYCLE98 ... 4-311
4.9 Thread recutting ... 4-317
4.10 Extended stock removal cycle - CYCLE950 ... 4-319

Error messages and error handling 5-341

5.1 General notes ... 5-342
5.2 Error handling in cycles ... 5-342
5.3 Overview of cycle alarms ... 5-343
5.4 Cycle messages ... 5-349

Appendix A-351

A Abbreviations ... A-352
B Definition of Terms ... A-361
C References ... A-381
D Identifiers ... A-393
E Index ... I-397
F Commands, identifiers ... I-399

© Siemens AG 2004 All rights reserved.

Notes

© Siemens AG 2004 All rights reserved.

General

1.1 General notes ... 1-18

1.2 Overview of cycles ... 1-18
1.2.1 Drilling cycles, drill pattern cycles, milling cycles and turning cycles ... 1-19
1.2.2 Auxiliary cycle routines ... 1-20

1.3 Programming the cycles ... 1-21
1.3.1 Call and return conditions ... 1-21
1.3.2 Messages during execution of a cycle ... 1-22
1.3.3 Cycle call and parameter list ... 1-23
1.3.4 Cycle simulation ... 1-26

1.4 Cycle support in the program editor ... 1-27
1.4.1 Menus, cycle selection ... 1-27
1.4.2 New input screen form functions ... 1-28

1.5 Cycle support for user cycles ... 1-35
1.5.1 Overview of the files required ... 1-35
1.5.2 Accessing cycle support ... 1-35
1.5.3 Configuring cycle support ... 1-36
1.5.4 Bitmap size and screen resolution ... 1-37
1.5.5 Storing bitmaps in data management for HMI Advanced ... 1-38
1.5.6 Handling bitmaps for HMI Embedded ... 1-38

1.6 Starting up the cycles ... 1-40
1.6.1 Machine data ... 1-40
1.6.2 Definition files for cycles GUD7.DEF and SMAC.DEF ... 1-41
1.6.3 New form of delivery for cycles in HMI Advanced ... 1-43
1.6.4 Upgrading the cycles SW 6.4 and later in HMI Advanced SW 6.3 and later ... 1-43

1.7 Additional cycle functions ... 1-44

© Siemens AG 2004 All rights reserved.

1.1 General notes

The first section provides you with an overview of the available cycles. The following sections describe the general conditions that apply to all cycles regarding

- Programming the cycles and
- Operator guidance for calling the cycles.

are programmed.

1.2 Overview of cycles

Cycles are generally applicable technology subroutines with which you can implement specific machining processes such as tapping a thread or milling a pocket. These cycles are adapted to individual tasks by defining parameters.

The system provides you with the technologies

- Drilling
- Milling
- Turning

in various standard cycles.

© Siemens AG 2004 All rights reserved.

1.2.1 Drilling cycles, drill pattern cycles, milling cycles and turning cycles

You can run the following cycles with the SINUMERIK 810D, 840D, and 840Di controls:

Drilling cycles

CYCLE81	Drilling, centering
CYCLE82	Drilling, counterboring
CYCLE83	Deep-hole drilling
CYCLE84	Rigid tapping
CYCLE840	Tapping with compensating chuck
CYCLE85	Boring 1
CYCLE86	Boring 2
CYCLE87	Boring 3
CYCLE88	Boring 4
CYCLE89	Boring 5

Drill pattern cycles

HOLES1	Machining a row of holes
HOLES2	Machining a circle of holes
CYCLE801	Dot matrix

Milling Cycles

LONGHOLE	Milling pattern of elongated holes on a circle
SLOT1	Groove milling pattern arranged on a circle
SLOT2	Circumferential groove milling pattern
POCKET1	Rectangular pocket milling (with face cutter)
POCKET2	Circular pocket milling (with face cutter)
CYCLE90	Thread milling
POCKET3	Rectangular pocket milling (with any milling tool)
POCKET4	Circular pocket milling (with any milling tool)
CYCLE71	Face milling
CYCLE72	Contour milling
CYCLE73	Pocket milling with islands
CYCLE74	Transfer of pocket edge contour
CYCLE75	Transfer of isolated contour
CYCLE76	Mill a rectangular spigot
CYCLE77	Mill a circular spigot

© Siemens AG 2004 All rights reserved.

Turning cycles

CYCLE93	Recess
CYCLE94	Undercut (shape E and F according to DIN)
CYCLE95	Stock removal with relief cut
CYCLE96	Thread undercut (shapes A, B, C and D according to DIN)
CYCLE97	Thread cutting
CYCLE98	Chaining of threads
CYCLE950	Extended stock removal

1.2.2 Auxiliary cycle routines

The following auxiliary routines are part of the cycles package

- PITCH and
- MESSAGE.

These must always be loaded in the control.

© Siemens AG 2004 All rights reserved.

1.3 Programming the cycles

A standard cycle is defined as a subroutine with a name and a parameter list. The conditions described in "SINUMERIK Programming Guide Part 1: Fundamentals" apply when calling a cycle.

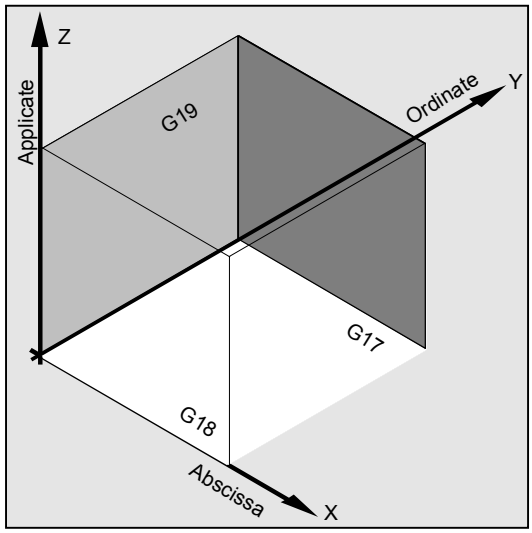
The cycles are supplied on diskette or, in the case of HMI Advanced, with the corresponding software release. They are loaded into the parts program memory of the control via the RS-232 interface (see Operator's Guide).

1.3.1 Call and return conditions

The G functions active before the cycle is called and the programmable frame remain active beyond the cycle.
Define the machining plane (G17, G18, G19) before calling the cycle. A cycle operates in the current plane with

- Abscissa (1st geometry axis)
- Ordinate (2nd geometry axis)
- Applicate (3rd geometry axis of the plane in space).

In drilling cycles, the hole is machined in the axis that corresponds to the applicate of the current plane. The depth infeed is performed in this axis with milling applications.



Plane and axis assignments

Command	Plane	Perpendicular infeed axis
G17	X/Y	Cen.
G18	Z/X	Y
G19	Y/Z	X

© Siemens AG 2004 All rights reserved.

1.3.2 Messages during execution of a cycle

For some cycles, messages that refer to the state of machining are displayed on the control screen during execution.
These messages do not interrupt program processing and continue to be displayed on the screen until the next message appears.
The message texts and their meanings are listed together with the cycle to which they refer.



You will find a summary of all the relevant messages in Appendix A of this Programming Guide.

Block display during execution of a cycle
The cycle call is displayed in the current block display for the duration of the cycle.

© Siemens AG 2004 All rights reserved.

1.3.3 Cycle call and parameter list

The standard cycles use user-defined variables. You can transfer the defining parameters for the cycles via the parameter list when the cycle is called.



Cycle calls must always be programmed in a separate block.

Basic instructions regarding assignment of standard cycle parameters

The Programming Guide describes the parameter list of every cycle together with the

- sequence and
- type.

The sequence of the defining parameters must be observed.

Each defining parameter of a cycle is of a specific data type. The parameter type in use must be specified when the cycle is called. The parameter list is used to pass

- variables or
- constants.

If variables are transferred to the parameter list, they must first be defined as such and assigned values in the calling program. Cycles can be called

- with an incomplete parameter list or
- by leaving out parameters.

aufgerufen werden.

If you want to exclude the last transfer parameters that have to be written in a call, you can prematurely terminate the parameter list with ")". To leave out parameters in the middle of the sequence, use a comma, "..., ,..." as a placeholder.

© Siemens AG 2004 All rights reserved.

No plausibility checks are made of parameter values with a discrete or limited value range unless an error response has been specifically described for a cycle.

If the cycle is called with a parameter containing more items than are defined as parameters in the cycle, the general NC alarm 12340 "Too many parameters" is output and the cycle is not executed.

Cycle call

The various methods for writing a cycle call are shown in the following example, CYCLE100, which requires the following input parameters.

Example

FORM	Definition of the machined shape Values: E and F
MID	Infeed depth (enter without sign)
FFR	Feed
VARI	Machining type Values: 0, 1 or 2
FAL	Final machining allowance

The cycle is called with the command
Cycle100 (FORM, MID, FFR, VARI, FAL).

1. Parameter list with constant values
Rather than input individual parameters, you can directly enter the concrete values to be used in the cycle.

Example

```
CYCLE100 ("E", 5, 0.1, 1, 0)                ;Cycle call
```

2. Parameter list with variables as transfer parameters
You can transfer the parameters as arithmetic variables that you define and assign values before you call the cycle.

© Siemens AG 2004 All rights reserved.

Example

`DEF CHAR FORM="E"`	;Definition of a parameter, ;value assignment
`DEF REAL MID=5, FFR, FAL` `DEF INT VARI=1`	;Definition of parameters with or without ;value assignments
`N10 FFR=0.1 FAL=0`	;Value assignments
`N20 CYCLE100 (FORM, MID, FFR, ->` `-> VARI, FAL)`	;Cycle call

3. Use of predefined variables as transfer parameters

For defining cycles with parameters you may use variables such as R parameters (R variables).

Example

`DEF CHAR FORM="E"`	;Definition of a parameter, ;value assignment
`N10 R1=5 R2=0.1 R3=1 R4=0`	;Value assignments
`N20 CYCLE100 (FORM, R1, ->` `-> R2, R3, R4)`	;Cycle call

As R parameters are predefined as real, it is important to ensure that the type of the target parameter in the cycle is compatible with the type real.



More detailed information about data types and type conversion and compatibility is given in the Programming Guide. If the types are incompatible, the following alarm is output:
12330 "Parameter type … incorrect".

4. Incomplete parameter list and omission of parameters

If a defining parameter is not required for a cycle call or it is to be assigned the value zero, it can be omitted from the parameter list. A comma, "..., ,... " must be written in its place to ensure correct assignment of the following parameters or the parameter list must be concluded prematurely with ")".

© Siemens AG 2004 All rights reserved.

Example

```
CYCLE100 ("F", 3, 0.3, , 1)          ;Cycle call,
                                     ;omit 4th parameter (i.e. zero setting)
CYCLE100 ("F", 3, 0.3)               ;Cycle call
                                     ;the value zero is assigned to the last two
                                     ;parameters (i.e. they have been left out)
```

5. Expressions in the parameter list

Expressions, the result of which is assigned to the corresponding parameter in the cycle are also permitted in the parameter list.

Example

```
DEF REAL MID=7, FFR=200              ;Definition of the parameters,
                                     ;value assignments
CYCLE100 ("E", MID*0.5, FFR+100,1)   ;Cycle call
                                     ;Infeed depth 3.5, feedrate 300
```

1.3.4 Cycle simulation

Programs with cycle calls can be tested initially by the simulation function.



Function

In HMI Embedded configurations, the program runs normally in the NC and the traversing motion is traced on screen during simulation.

In HMI Advanced configurations, the program is simulated on the HMI only. This makes it possible to run cycles without tool data or without prior selection of a tool offset in the MMC.
The finished contour is then traversed in the case of cycles that have to include tool offset data in the calculation of their traversing motion (e.g. milling pockets and grooves, turning with recess) and a message indicates that simulation without tool is active. This function can be used, for example, to check the position of the pocket.

© Siemens AG 2004 All rights reserved.

1.4 Cycle support in the program editor

The program editor provides extended cycle support for Siemens and user cycles.

Function

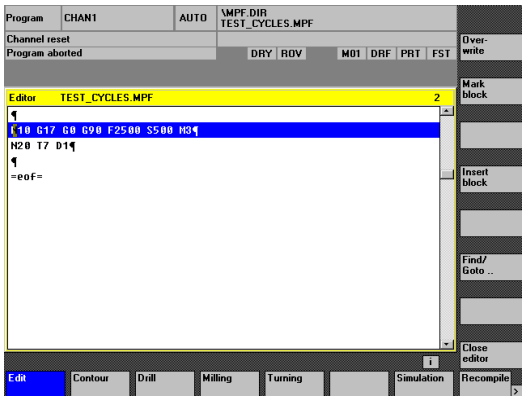
The cycle support offers the following functions:

- Cycle selection via soft keys
- Input screen forms for parameter assignment with help displays
- Online help for each parameter (HMI Advanced only)
- Support of contour input.

Retranslatable code is generated from the individual screen forms.

1.4.1 Menus, cycle selection

The cycle selection is carried out technology-oriented via soft keys:



Contour	Geometry input via the geometry processor or contour definition screen forms.
Drilling	Input screen forms for drilling cycles and drilling patterns.
Milling	Input screen form for milling cycles.
Turning	Input screen forms for turning cycles.

After confirming the screen form input by clicking OK, the technology selection bar is still visible.
Similar cycles are supplied from shared screen forms. Within one screen form, the user may switch between cycles via soft key, e.g. with tapping

© Siemens AG 2004 All rights reserved.

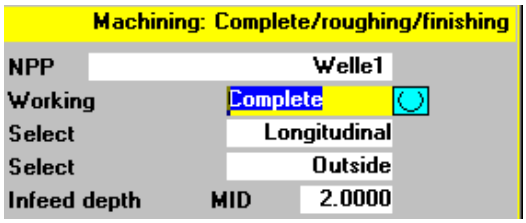
or undercut.

The editor cycle support also contains screen forms that insert a multi-line DIN code in the program instead of a cycle call, e.g. contour definition screen forms and the input of any drilling position.

1.4.2 New input screen form functions



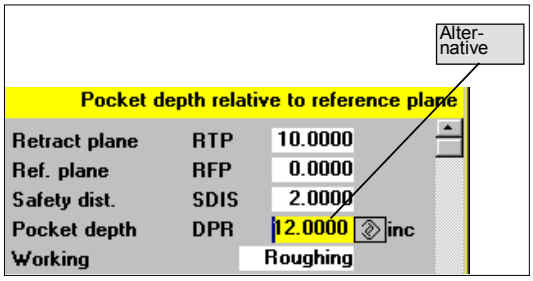
Function



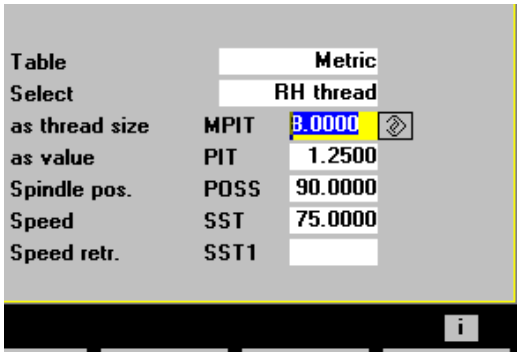
- Many cycles allow you to influence the processing type via the VARI parameter. It contains several settings composing one code. Each individual setting has its own input field on the screen forms of the new cycle support. You can switch between input fields with the toggle key.
- The input screen forms are changed dynamically. Only the input fields required for the selected processing type are displayed. Unnecessary input fields are not shown. In the example, this is the case with the parameter for the dressing feedrate.
- One input may therefore automatically assign several depending parameters. This is the case with threading which presently supports metric thread tables. With the threading cycle CYCLE97, for example, entering 12 in the thread size input field (MPIT parameter) automatically assigns 1.75 to the thread pitch input field and 1.137 to the thread depth input field (TDEP parameter). This function is not active if the metric thread table has not been selected.
- If a screen form is displayed a second time, the last entered values are assigned to all fields.
  When cycles are called up several times in a row in the same program (e.g. pocket milling when roughing and dressing), only few parameters then have to be changed.

© Siemens AG 2004 All rights reserved.

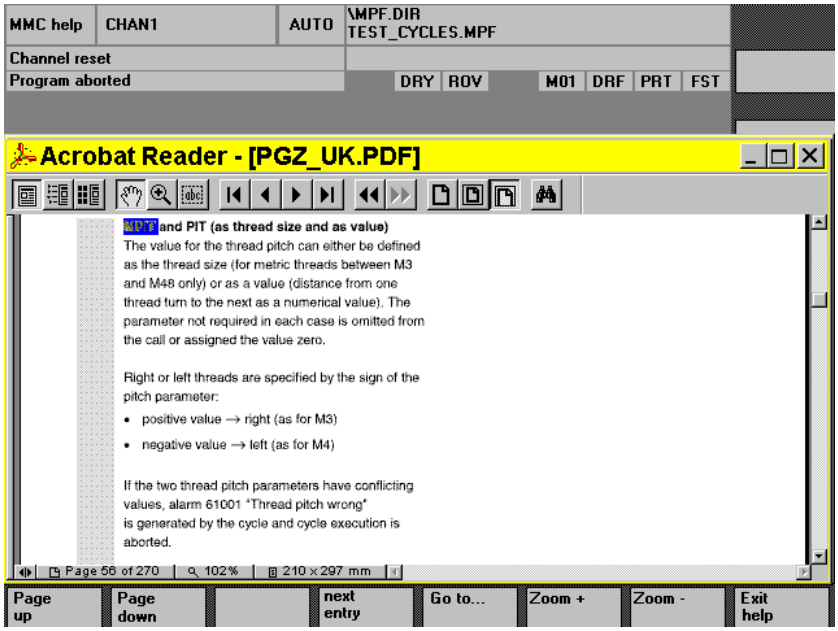
- In screen forms of drilling and milling cycles, certain parameters may be input as absolute or incremental values. The abbreviation ABS for absolute and INC for incremental values is displayed behind the input field. To switch between them, use the "Alternative" soft key. This setting will remain with the next call of these screen forms.



- HMI Advanced allows you to view additional information on each cycle parameter in the online help. Simply place the cursor on a parameter and the help icon [i] will appear bottom right on the screen. Now you can activate the help function.



By pressing the info key the parameter explanation is displayed from the Cycle Programming Guide.



© Siemens AG 2004 All rights reserved.

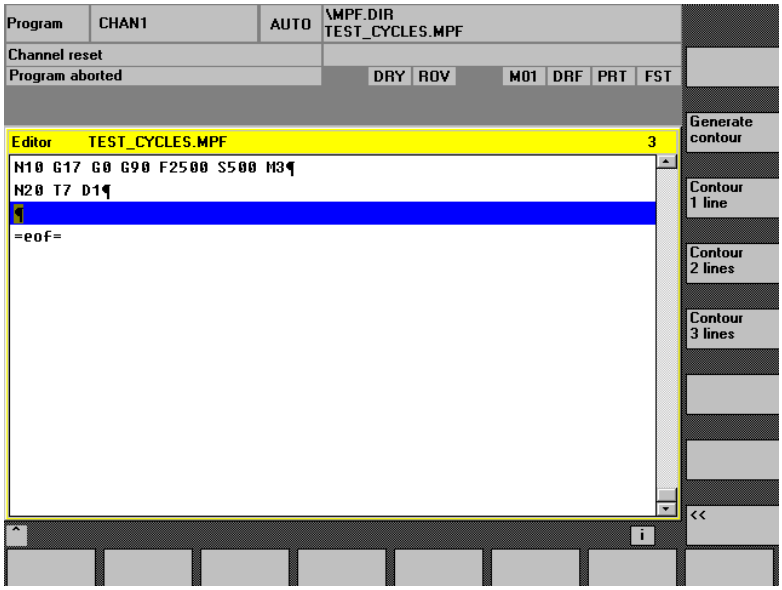
Operator commands in the help display

Page up	Paging backward in the documentation.
Page down	Paging forward in the documentation.
next entry	Enables the jump to another piece of text included in the help display.
Go to	Enables the jump to a selected piece of text.
Zoom +	Zoom the text in the help window.
Zoom -	Reduce the text in help window.
Exit help	Return to the cycle screen form.

Contour input support

Free contour programming

Generate contour Starts free contour programming for entering assembled contour sections (refer to References: /OG/, Chapter 6).



Blueprint programming

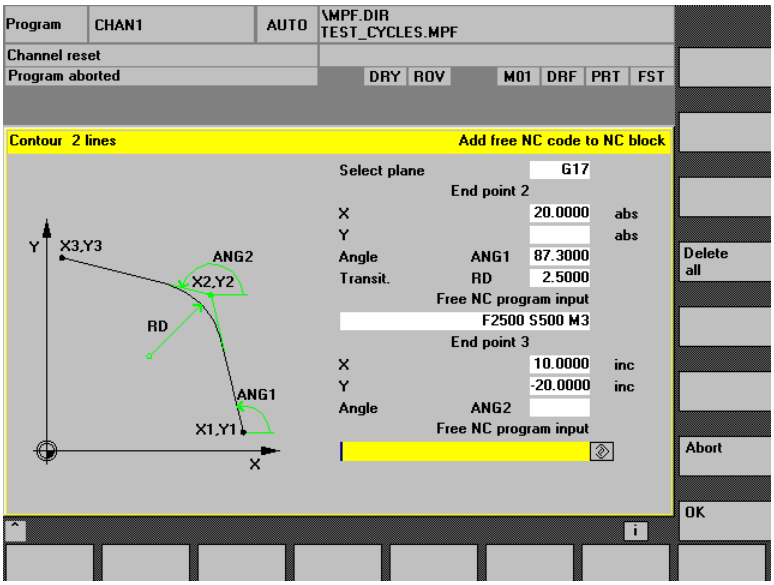
Contour 1 line
Contour 2 lines
Contur 3 lines These soft keys support the blueprints that are possible.

It consists of one or several straight lines with contour transition elements in-between (radii, chamfers). Each contour element may be preassigned by means of end points or point and angle and supplemented by a free DIN code.

© Siemens AG 2004 All rights reserved.

Example

The following DIN code is created from the following input screen form for a 2-straight-line contour definition creates the following DIN code:



X=AC(20) ANG=87.3 RND=2.5 F2000 S500 M3
X=IC(10) Y=IC(-20); end point incremental

Drilling support

The drilling support includes a selection of drilling cycles and drilling patterns.

Drilling centering

Deep hole drilling

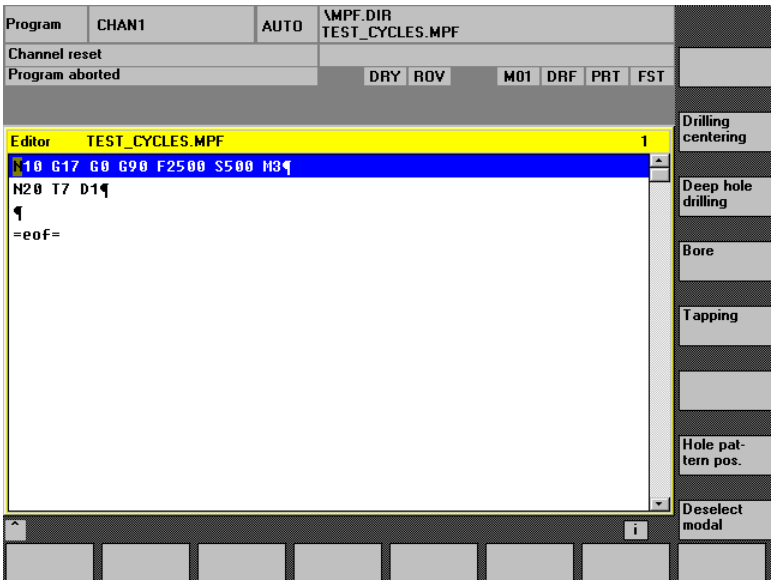
Bore

Tapping

Selection of drilling patterns

Hole pattern pos.

Deselect modal



Cycles CYCLE81, CYCLE87 and CYCLE89 cannot be parameterized with this support.
The function of CYCLE81 is covered by CYCLE82 (softkey "Drilling center."), as is the function of CYCLE89. Die Funktion des CYCLE87 ist abgedeckt durch die Funktion von CYCLE88 (Softkeys „Bohren Zentrier." → „Bohren mit Stop").

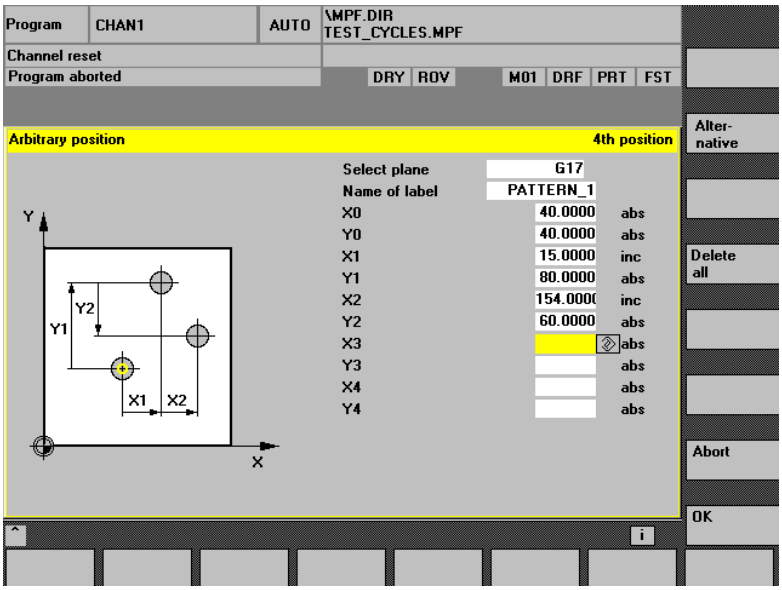
© Siemens AG 2004 All rights reserved.

Drilling patterns may be repeated if, for example, drilling and tapping are to be executed in succession. A name is simply assigned to the drilling pattern and entered later in screen form "Repeat position".

Programming example generated by cycle support

N100 G17 G0 G90 Z20 F2000 S500 M3	;Main block
N110 T7 M6	;Change drilling machine
N120 G0 G90 X50 Y50	;Initial drilling position
N130 MCALL CYCLE82(10,0,2,0,30,5)	;Modal drilling cycle call
N140 Circle of holes 1:	;Label – Name of drilling pattern
N150 HOLES2(50,50,37,20,20,9)	;Call drilling pattern cycle
N160 ENDLABEL:	
N170 MCALL	;Deselect modal call
N180 T8 M6	;Change tap
N190 S400 M3	
N200 MCALL CYCLE84(10,0,2,0,30,,3,5,0.8,180,300,500)	;Modal call of tapping cycle
N210 REPEAT Circle of holes 1	;Repeat drilling pattern
N220 MCALL	;Deselect modal call

Moreover, any drilling position may be entered as repeatable drilling pattern by means of screen forms.



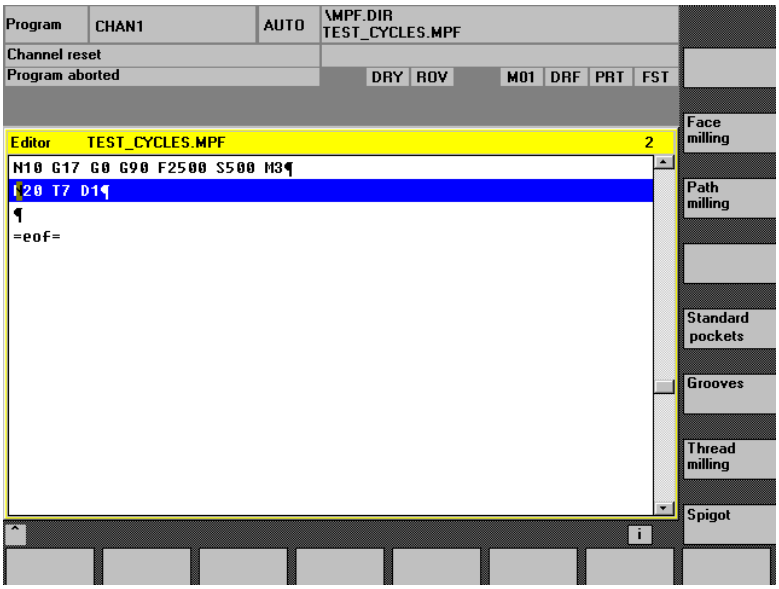
Up to five positions may be programmed in the plane; all values are either absolute or incremental (use the "Alternat." soft key to toggle between them). The "Delete all" soft key creates an empty screen form.

Milling support

The milling support includes the following selection possibilities:

© Siemens AG 2004 All rights reserved.

Face milling	Thread milling
Path milling	Swiveling cycles
Standard pockets	
Grooves	
Spigot	
>>	<<



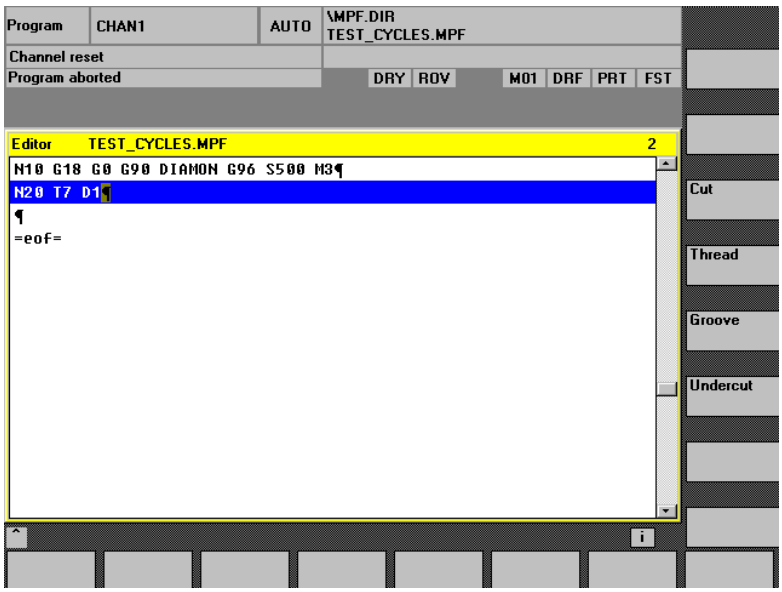
The "Standard pockets", "Slots" and "Spigots" soft keys each branch into submenus offering a selection of pocket, groove and spigot milling cycles.

Pocket milling cycles POCKET1 and POCKET2 cannot be parameterized with this support.

Turning support

The turning support includes the following selection possibilities:

Cut
Thread
Groove
Undercut



The undercut cycles for the E and F forms (CYCLE94) as well as for the thread undercuts of the A to D forms (CYCLE96) are all grouped under the "Undercut" soft key.

The "Thread" soft key contains a submenu for selecting between single thread cutting or thread chaining.

© Siemens AG 2004 All rights reserved.

Retranslation

The retranslation of program codes serves to change an existing program with the help of the cycle support. Place the cursor on the line to be changed and press the "Retranslation" soft key.

Thus, the corresponding input screen form which created the program piece is reopened and values may be modified.

Directly entering modifications in the created DIN code may prevent retranslation. Therefore, consistent use of the cycle support is required and modifications are to be carried out with the help of retranslation.

Configuring support for user cycles

References:

/IAM/,Installation & Start-Up Guide HMI

BE1 "Expand the Operator Interface"
IM2 "Installation and Start-up of HMI Embedded"
IM4 "Installation and Start-up of HMI Advanced"

© Siemens AG 2004 All rights reserved.

1.5 Cycle support for user cycles

1.5.1 Overview of the files required

The following files form the basis for cycle support:

Assignment	File	Application	File type
Cycle selection	aeditor.com	Standard and user cycles	Text file
	common.com (HMI Embedded only)	Standard and user cycles	Text file
Input screen form for parameter setting	*.com	Standard or user cycles	Text file
Help screens	*.bmp	Standard or user cycles	Bitmap
Help in the Editor (HMI Advanced only)	pgz_<language>.pdf and pgz_<language>.txt	standard cycles only	pdf-file

Any names can be chosen for the cycle support configuration files (*.com).

1.5.2 Accessing cycle support

Function

The horizontal soft key HS6 in program editor is designated the Entry soft key for user cycles. Its function must be configured in file aeditor.com. Assign a text to the soft key and configure a function in the press block for soft key operation.

Example:

```
//S(Start)
...
HS5=($80270,,se1)
PRESS(HS5)
LS("Turning",,1)
END_PRESS
HS6=("Usercycle","se1")          ; HS6 is configured with the "Usercycle" text
PRESS(HS6)
LS("SK_Cycles1","cycproj1")      ; operating a soft key loads the soft key bar from
                                 ; file cycproj1.com
END_PRESS
```

© Siemens AG 2004 All rights reserved.

A detailed description of the configuration is given in:

References: /IAM/, Installation & Start-Up Guide HMI

BE1 "Expand the Operator Interface"

With HMI Embedded, make the entry in file common.com to activate this soft key as follows:

```
%_N_COMMON_COM
;$PATH=/_N_CUS_DIR
...
[MMC_DOS]
...
SC315=AEDITOR.COM
SC316=AEDITOR.COM
```

1.5.3 Configuring cycle support

Function

The soft key bars and input screen forms of cycle support can be configured in any files and stored as type *.com in the HMI of the control.

A detailed description of the configuration is given in:

References: /IAM/, Installation & Start-Up Guide HMI

BE1 "Expand the Operator Interface"

In HMI Advanced, the *.com files are stored in data management in the directories:

- dh\cst.dir
- dh\cma.dir or
- dh\cus.dir

and the usual search sequence is followed:
cus.dir, cma.dir, cst.dir. Die Files werden nicht in die NCU geladen.
For HMI Embedded, the *.com files can be loaded into the NCU (read in via "Services" through RS-232-C). But as they occupy NC memory there, it is better to integrate them in the HMI. They must be packed and incorporated into the application software of the HMI version. The tool to pack the files is included with the standard cycle software under \hmi_emb\tools.

© Siemens AG 2004 All rights reserved.

Step sequence for creation

- Copy file arj.exe from directory \hmi_emb\tools to an empty directory on a PC.
- Copy the separate configuration files *.com to this directory.
- Pack each individual com file with the command
  arj a <name target file> <name source file>.
  The target files must have the extension **co_**.
  Example: pack configured file cycproj1.com to:
  arj a cycproj1.co_ cycproj1.com.
- Copy the *.co_ files to the relevant directory of the HMI application software and create a version.

References: /BEM/, HMI Embedded Operator's Guide
/IAM/, Installation & Start-Up Guide HMI

IM2 "Installation and Start-up of HMI Embedded"

1.5.4 Bitmap size and screen resolution

With SW 6.2, there are three different screen resolutions in the HMI.
For each of the resolutions, there is a maximum bitmap size for the cycle screen forms (see the following table), to be observed when creating your own bitmaps.

Screen resolution	Bitmap size
640 * 480	224* 224 pixels
800 * 600	280* 280 pixels
1024 * 768	352* 352 pixels

Bitmaps are created and stored as 16-color bitmaps.

© Siemens AG 2004 All rights reserved.

1.5.5 Storing bitmaps in data management for HMI Advanced

New paths have been set up In data management (as from HMI 6.2) for the different screen resolutions, so that the bitmaps can be stored in parallel in various sizes.

Standard cycles:
- dh\cst.dir\hlp.dir\640.dir
- dh\cst.dir\hlp.dir\800.dir
- dh\cst.dir\hlp.dir\1024.dir

Manufacturer cycles:
- dh\cma.dir\hlp.dir\640.dir
- dh\cma.dir\hlp.dir\800.dir
- dh\cma.dir\hlp.dir\1024.dir

User cycles:
- dh\cus.dir\hlp.dir\640.dir
- dh\cus.dir\hlp.dir\800.dir
- dh\cus.dir\hlp.dir\1024.dir

The search begins in the appropriate directory for the current resolution (e.g. in dh\...\hlp.dir\640.dir for 640 * 480) and then moves to dh\...\hlp.dir. Otherwise the search sequence cus.dir, cma.dir, cst.dir applies.

1.5.6 Handling bitmaps for HMI Embedded



Function

With HMI Embedded, the bitmaps are incorporated in the HMI software. They are grouped together in a package, **cst.arj**. Bitmaps can always be integrated in *.bmp format. However, to save more space and display faster, use a *.bin binary format. To do this, you will need the tools supplied with the standard cycle software in the \hmi_emb\tools :
- arj.exe, bmp2bin.exe, and
- sys_conv.col
- arj_idx.exe (SW 6.3 and later)

and the script files:
- mcst_640.bat,
- mcst_800.bat or
- mcst1024.bat.

© Siemens AG 2004 All rights reserved.

File cst.arj contains all the standard and user cycle bitmaps. So you will have to link together the standard cycle bitmaps and your own bitmaps.

Step sequence for creation

- Copy all the files from directory \hmi_emb\tools to an empty directory on a PC.
- Create a subdirectory \bmp_file there.
- Copy your own bitmaps *.bmp to this subdirectory \bmp_file.
- Depending on the resolution for which a cst.arj is being created, start mcst_640.bat / mcst_800.bat or mcst1024.bat.
- The cst.arj that you create will then be in the same directory as the generation tools.
- In SW 6.3 and later, a further file, cst.idx, is also created in this directory. This is integrated into the HMI software along with cst.arj.

cst.arj is then integrated in the HMI software as described in Subsection 1.4.6.

References: /BEM/, HMI Embedded Operator's Guide
/IAM/, Installation & Start-Up Guide HMI

IM2 "Installation and Start-up of HMI Embedded"

© Siemens AG 2004 All rights reserved.

1.6 Starting up the cycles

1.6.1 Machine data

The following machine data are used for the cycles.
The minimum values for these machine data are given in the table below.

Relevant machine data

MD number	MD name	Minimum value
18118	MM_NUM_GUD_MODULES	7
18130	MM_NUM_GUD_NAMES_CHAN	20
18150	MM_GUD_VALUES_MEM	2 * Number of channels
18170	MM_NUM_MAX_FUNC_NAMES	40
18180	MM_NUM_MAX_FUNC_PARAM	500
28020	MM_NUM_LUD_NAMES_TOTAL	200
28040	MM_NUM_LUD_VALUES_MEM	25



*This information applies **to Siemens standard cycles only**. The relevant values have to be **added** for user cycles. When using ShopMill or ShopTurn, comply with the information relevant to these products.*

The following machine data settings are also required:

MD number	MD name	Value
20240	CUTCOM_MAXNUM_CHECK_BLOCK	4

The machine data files are delivered with these defaults by the machine manufacturer.
Remember: any changes you make to these machine data will not apply until you perform a power ON.



Axis-specific machine data MD 30200: NUM_ENCS must also be noted with respect to cycle CYCLE840 (tapping with compensating chuck).

© Siemens AG 2004 All rights reserved.

1.6.2 Definition files for cycles GUD7.DEF and SMAC.DEF

Standard cycles need Global User Data definitions (GUDs) and macro definitions. These are stored in definition files GUD7.DEF and SMAC.DEF, supplied with the standard cycles.

To make it easier for start-up engineers to compile GUDs and macros in a module without editing the original SIEMENS files, from software version 6.3 onwards files

- GUD7.DEF
- MAC.DEF

are supplied complete with the "standard cycles". These two files contain no definitions and are simply shortcuts to predefined, product-specific definition files. The call-up mechanisms now included in these cycles allow automatic call-up and compilation of all product-specific GUD and macro definitions.
Each package will now only have its own definitions. New cycle files GUD7_xxx.DEF and SMAC_xxx.DEF will be introduced, that will be in data management in the definition directory DEF.DIR.

The new files for the standard cycles are:

- GUD7_SC.DEF and
- SMAC_SC.DEF.

For other cycle packages the following file codes have been assigned by SIEMENS:
(xxx stands for "GUD7" or "SMAC")

- xxx_JS JobShop cycles in general
- xxx_MC Measuring cycles
- xxx_MJ Measuring in JOG
- xxx_MT ManualTurn
- xxx_SM ShopMill
- xxx_ST ShopTurn
- xxx_ISO ISO compatibility
- xxx_C950 Extended stock removal
- xxx_C73 Pocket with islands

Other codes not listed here may be used by the system at a later date.

© Siemens AG 2004 All rights reserved.

Note

According to the definition of "defining user data" (see Programming Guide Advanced Chapter 3.4 "Defining user data") modules GUD7 and SMAC.DEF are NOT available to the machine manufacturer/user. MGUD, UGUD, GUD4,8,9 and MMAC, UMAC should preferably be used for user applications.
However, to enable users to integrate existing in-house definitions into these modules in this system, the following codes have been kept free.

xxx_CMA Manufacturer
xxx_CUS User

Startup, upgrades for standard cycles:

- if a GUD7.DEF is already active in the control, use "Services", "Data out", "NC active data" to select the user data of GUD7 and back up the current values in an archive or onto a diskette;
- read in files GUD7_SC.DEF and SMAC_SC.DEF from the diskette and load them to the NCU;
- read in and activate GUD7.DEF and SMAC.DEF;
- turn on the NCU;
- read the archive of saved values back in;

Loading an additional cycle package:

- unload GUD7.DEF and SMAC.DEF (having backed up the values if necessary),
- read in cycles GUD7_xxx.DEF and SMAC_xxx.DEF of the package and load them to the NCU;
- reactivate GUD7.DEF and SMAC.DEF.

Note

When loading or unloading individual definition files, if a call-up file is already loaded it must be unloaded and then reloaded. Otherwise the NC retains the previous GUD/macro configuration.

Handling in the HMI Advanced simulation:

After upgrading the cycle version in the NCU, machine data adjustment followed by an NC reset of the simulation is necessary as soon as you start the simulation, in order to activate the modified definition files.

© Siemens AG 2004 All rights reserved.

1.6.3 New form of delivery for cycles in HMI Advanced



As from HMI Advanced 6.3, the delivery form of standard cycles in the HMI will change. The cycle files will no longer be stored directly in the relevant data management directories, but will be available as archive files under:
→ Archives/Cycle archives.

This will enable the previously existing cycle version in data management to be retained unchanged when the HMI is upgraded.

To upgrade, these archive files must be imported via "Data in". If these archive files are read in, after the upgrade there are no longer different versions of the cycles in the NCU and on the hard disk. The loaded cycles are overwritten in the NCU, not loaded on the hard drive. The new cycle files are always stored on the hard disk.



References: for current infos see:
- the file "siemensd.txt" from the delivery software (standard cycles) or
- in the case of HMI Advanced F:\dh\cst.dir\HLP.dir\siemensd.txt

1.6.4 Upgrading the cycles SW 6.4 and later in HMI Advanced SW 6.3 and later



In HMI_Advanced SW 6.3 and later, the cycles are stored as archives under "Archive", "Cycle archives". Upgrading is performed by importing an archive contained in the software supplied from directory HMI_Adv. This unpacks the cycle archives and overwrites the existing archives of the previous cycle version.
The archives from the cycle archive of the HMI now have to imported for the required technology (turning, milling, …) and language via "Data in".



References: for current infos see:
- the file "siemensd.txt" from the delivery software (standard cycles) or
- in the case of HMI Advanced F:\dh\cst.dir\HLP.dir\siemensd.txt

© Siemens AG 2004 All rights reserved.

1.7 Additional cycle functions

Function

To provide an overview and for diagnosis of the cycle versions and their definition files, it will be possible as from SW 6.3 to display and use version screens. These can be found in the HMI under "Diagnosis" => "Service displays" => "Version" => "Cycles" or "Definitions". A LOG file can be created in ASCII format and read in "Services" => "Diagnosis" => "LOG files".

This function can also only be run with HMI software versions SW 6.3 and higher.

Different overviews are possible with cycle version display:

- An overview of all available cycles
- An overview of the individual directories of data management for user cycles (CUS.DIR), manufacturer cycles (CMA.DIR) and Siemens cycles (CST.DIR)
- A package overview of all the cycle packages available in the control
- Details on each package and cycle file

References: /BAD/, Operator's Guide HMI Advanced
/BEM/, HMI Embedded Operator's Guide
Service display chapter

The version display includes all the *.SPF cycle files and all the *.COM files of cycle support.

No additional files are required for version display via directories or of all cycles.
To display overviews of individual cycle packages, each cycle package must contain a package list of all associated files.

Package lists

A new file type is introduced for package lists
***.cyp** (for cycle package),
in plain text, cycle package list.
The user can create package lists for his own cycle packages. They must look like this:

© Siemens AG 2004 All rights reserved.

Structure of a package list:

1. Line:	version entry (after the vocabulary word **;VERSION:**) and package name (after the vocabulary word **;PACKAGE:**)
from second line:	list of the files associated with the cycle package with name and type
last line:	M30

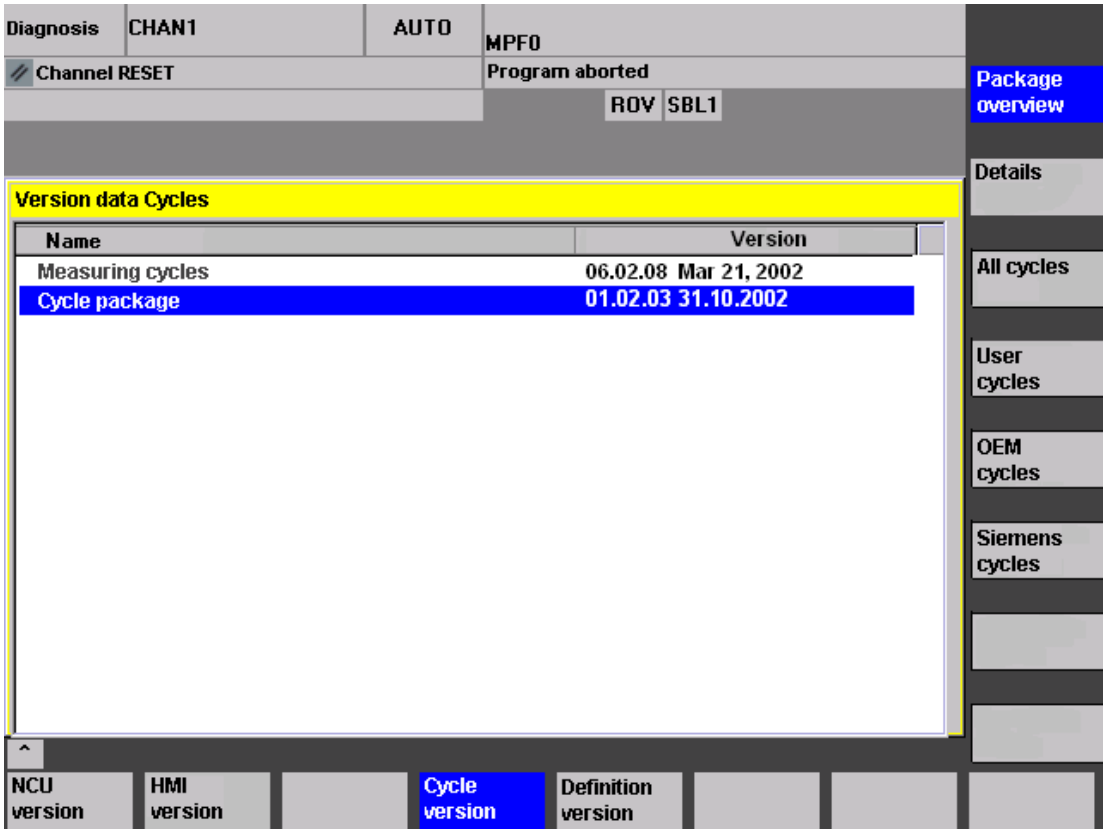
Example:

```
%_N_CYC_USER1_CYP
;$PATH=/_N_CUS_DIR
;VERSION: 01.02.03 31.10.2002 ;PACKAGE: $85200
ZYKL1.SPF
ZYKL2.SPF
ZYKL3.COM
M30
```

Input in the text file uc.com:

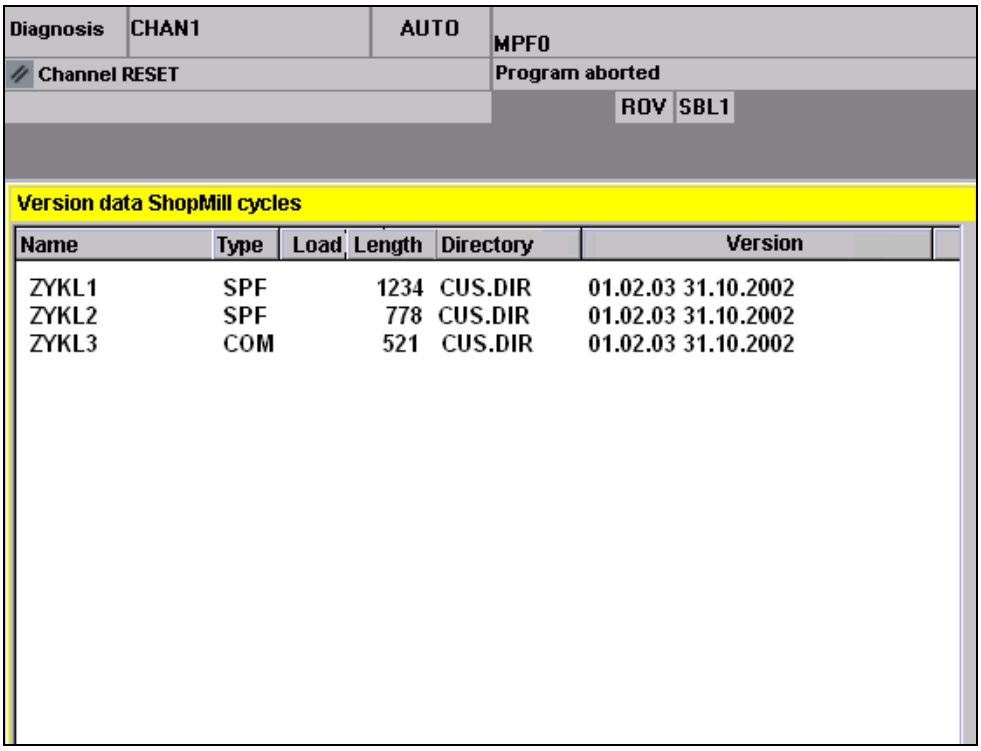
85200 0 0 "Cycle package 1"

The following is displayed in the package overview:



© Siemens AG 2004 All rights reserved.

The following is displayed in the file overview:



Other Information

The cycle package name after the vocabulary word PACKAGE can also be written as a string in " ", but then it is language-specific.

Version entries in cycles

Just as with package lists, the entry after the vocabulary word ";VERSION:" is interpreted as the version ID. The version entry has to be in the first ten lines of the cycle, the search does not go any further.

Example:

```
%_N_ZYKL1_SPF
;$PATH=/_N_CUS_DIR
;VERSION: 01.02.03 31.10.2002
;Comment
PROC ZYKL1(REAL PAR1)
...
```

■

© Siemens AG 2004 All rights reserved.

Drilling Cycles and Drilling Patterns

2.1 Drilling cycles 2-48
2.1.1 Preconditions 2-50
2.1.2 Drilling, centering – CYCLE81 2-51
2.1.3 Drilling, counterboring – CYCLE82 2-54
2.1.4 Deep-hole drilling – CYCLE83 2-56
2.1.5 Rigid tapping – CYCLE84 2-63
2.1.6 Tapping with compensating chuck – CYCLE840 2-70
2.1.7 Boring 1 – CYCLE85 2-78
2.1.8 Boring 2 – CYCLE86 2-81
2.1.9 Boring 3 – CYCLE87 2-85
2.1.10 Boring 4 – CYCLE88 2-87
2.1.11 Boring 5 – CYCLE89 2-89
2.2 Modal call of drilling cycles 2-91
2.3 Drilling pattern cycles 2-94
2.3.1 Preconditions 2-94
2.3.2 Row of holes – HOLES1 2-95
2.3.3 Circle of holes – HOLES2 2-99
2.3.4 Point grid – CYCLE801 2-102

© Siemens AG 2004 All rights reserved.

2.1 Drilling cycles

The following sections describe how
- drilling cycles and
- Drill pattern cycles

are programmed.
These Sections are intended to guide you in selecting cycles and assigning them with parameters. After a detailed description of the function of each cycle and its parameters, you will also find a programming example at the end of each section to familiarize you with using cycles.

The Sections are structured as follows:
- Programming
- Parameters
- Function
- Sequence
- Explanation of parameters
- Other Information
- Programming example

"Programming" and "Parameters" explain the use of cycles sufficiently for the experienced user, whereas beginners can find all the information they need for programming cycles under "Function", "Sequence of operations", "Explanation of parameters", "Additional notes" and the "Programming example".

© Siemens AG 2004 All rights reserved.

Drilling cycles are motion sequences defined according to DIN 66025 for drilling, boring, tapping, etc.
They are called in the form of a subroutine with a defined name and a parameter list.

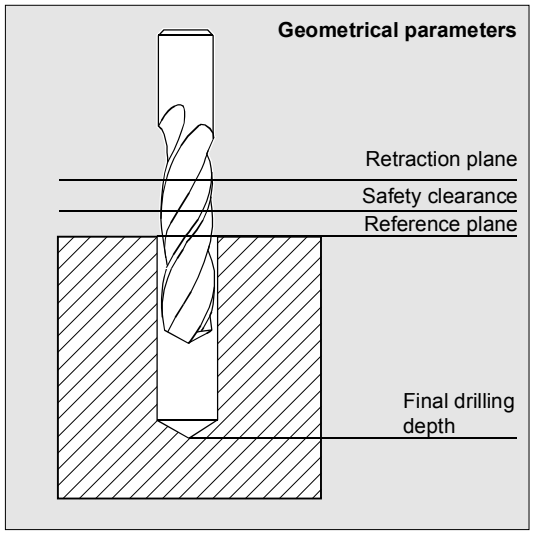
Five cycles are available for boring. They all follow a different technological procedure and are therefore parameterized differently:

Boring cycle		Special parameterization features
Boring 1 -	CYCLE85	Different feedrates for boring and retraction
Boring 2 -	CYCLE86	Oriented spindle stop, definition of retraction path, retraction in rapid traverse, definition of spindle direction of rotation
Boring 3 -	CYCLE87	Spindle stop M5 and program stop M0 at drilling depth, continued machining after NC Start, retraction in rapid traverse, definition of spindle direction of rotation
Boring 4 -	CYCLE88	As for CYCLE87 plus dwell time at drilling depth
Boring 5 -	CYCLE89	Boring and retraction at the same feedrate

Drilling cycles can be modal, i.e. they are executed at the end of each block that contains motion commands. Other cycles written by the user can also be called modally (see Section 2.2).

There are two types of parameter:
- Geometrical parameters and
- Machining parameters.

Geometrical parameters are identical for all drilling cycles, drilling pattern cycles and milling cycles. They define the reference and retraction planes, the safety distance and the absolute and relative final drilling depths. Geometrical parameters are written once in the first drilling cycle CYCLE81.
The machining parameters have a different meaning and effect in each cycle. They are therefore written in each cycle.



© Siemens AG 2004 All rights reserved.

2.1.1 Preconditions

Call and return conditions

Drilling cycles are programmed independently of the actual axis names. The drilling position must be approached in the higher-level program before the cycle is called.

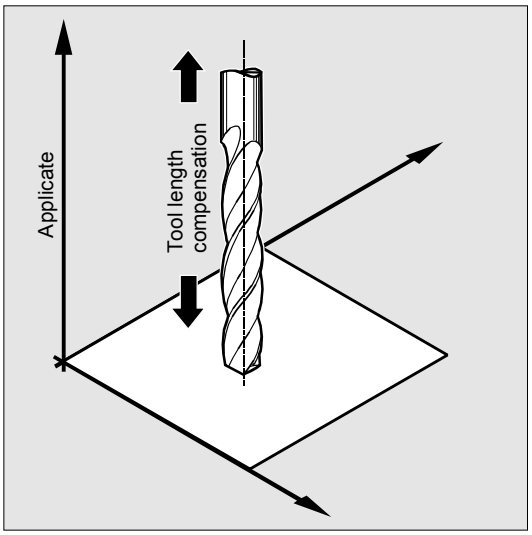
The required values for the feedrate, spindle speed and spindle direction of rotation must be programmed in the parts program if there are no assignment parameters for these values in the drilling cycle.

The G function and current frame active before the cycle was called remain active beyond the cycle.

Level definition

Drilling cycles generally assume that the current workpiece coordinate system in which the machining operation is to be performed has been defined by selecting a plane (G17, G18 or G19) and activating a programmable frame. The drilling axis is always the applicate of this coordinate system.

A tool length compensation must be selected before the cycle is called. Its effect is always perpendicular to the selected plane and remains active even after the end of the cycle (see also Programming Guide).



Spindle handling

The drilling cycles are written in such a way that the spindle commands always refer to the master spindle control. If you want to use a drilling cycle on a machine with several spindles, you must first define the spindle that is to be used for the operation as the master spindle (see also Programming Guide).

Dwell time programming

The parameters for the dwell times in the drilling cycles are always assigned to the F word and must therefore be assigned with values in seconds. Any deviations from this procedure must be expressly stated.

© Siemens AG 2004 All rights reserved.

2.1.2 Drilling, centering – CYCLE81

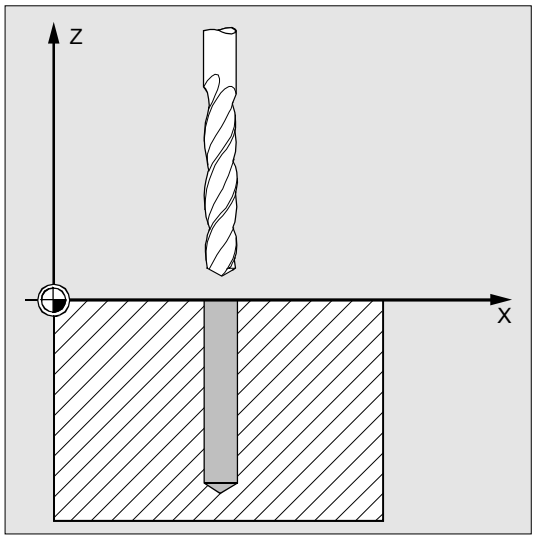
Programming

```
CYCLE81 (RTP, RFP, SDIS, DP, DPR)
```

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Final drilling depth (absolute)
DPR	real	Final drilling depth relative to reference plane (enter without sign)

Function

The tool drills at the programmed spindle speed and feedrate to the programmed final drilling depth.



Sequence

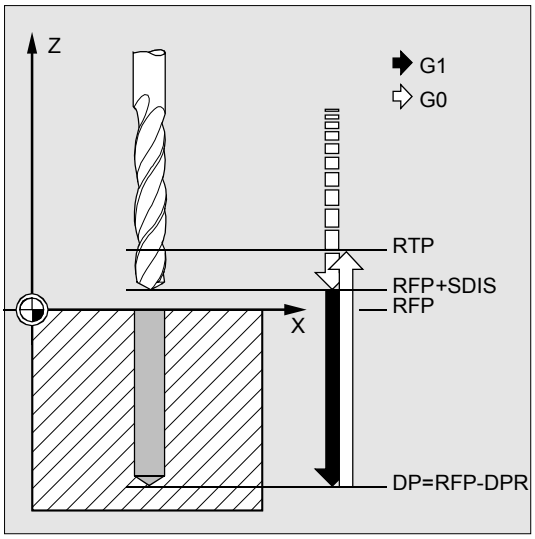
Position reached before the beginning of the cycle:
The drilling position is the position in the two axes of the selected plane.

The cycle implements the following motion sequence:
Approach of the reference plane brought forward by the safety distance with G0.

- Traverse to final drilling depth with the feedrate (G1) programmed in the calling program
- Retraction to retraction plane with G0.

© Siemens AG 2004 All rights reserved.

Explanation of parameters



RFP and RTP (reference plane and retraction plane)
The reference plane (RFP) and retraction plane are generally set to different values. The cycle assumes that the retraction plane is in front of the reference plane. The distance between the retraction plane and the final drilling depth is therefore greater than the distance between the reference plane and the final drilling depth.

SDIS (safety distance)
The safety distance (SDIS) is effective with regard to the reference plane which is brought forward by the safety distance.
The direction in which the safety distance is active is automatically determined by the cycle.

DP and DPR (final drilling depth)
The final drilling depth can be defined as either absolute (DP) or relative (DPR) to the reference plane. If it is entered as a relative value, the cycle automatically calculates the correct depth on the basis of the positions of the reference and retraction planes.

Other Information

If a value is entered both for the DP and the DPR, the final drilling depth is derived from the DPR. If the DP deviates from the absolute depth programmed, the message "Depth: According to value for relative depth" is displayed in the dialog line.

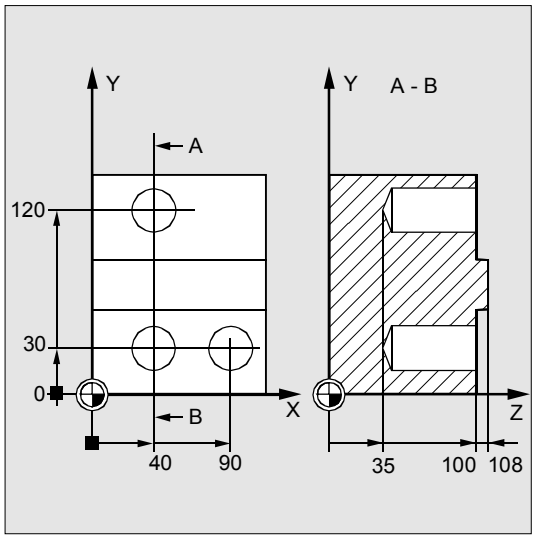
If the values for the reference plane and the retraction plane are identical, a relative depth must not be programmed. The following alarm is output:
61101 "Reference plane incorrectly defined"
and the cycle is not executed.
This error message is also output if the retraction plane lies behind the reference plane, i.e. the distance to the final drilling depth is smaller.

© Siemens AG 2004 All rights reserved.

Programming example

Drilling_centering

You can use this program to make three holes using the drilling cycle CYCLE81, whereby this cycle is called with different parameter settings. The drilling axis is always the Z axis.



N10 G0 G90 F200 S300 M3	;Specification of technology values
N20 D1 T3 Z110	;Traverse to retraction plane
N21 M6	
N30 X40 Y120	;Traverse to first drilling position
N40 CYCLE81 (110, 100, 2, 35)	;Cycle call with absolute final drilling ;depth, safety distance, and incomplete ;parameter list
N50 Y30	;Traverse to next drilling position
N60 CYCLE81 (110, 102, , 35)	;Cycle call without safety distance
N70 G0 G90 F180 S300 M03	;Specification of technology values
N80 X90	;Approach next position
N90 CYCLE81 (110, 100, 2, , 65)	;Cycle call with relative final drilling depth ;and safety distance
N100 M30	;End of program

© Siemens AG 2004 All rights reserved.

2.1.3 Drilling, counterboring – CYCLE82

Programming

```
CYCLE82 (RTP, RFP, SDIS, DP, DPR, DTB)
```

Parameters

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Final drilling depth (absolute)
DPR	real	Final drilling depth relative to reference plane (enter without sign)
DTB	real	Dwell time at final drilling depth (chip breaking)

Function

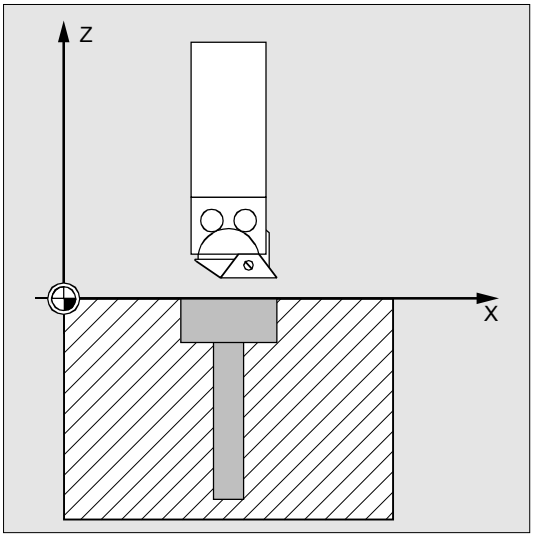
The tool drills at the programmed spindle speed and feedrate to the programmed final drilling depth. A dwell time can be allowed to elapse when the final drilling depth has been reached.

Sequence

Position reached before the beginning of the cycle:
The drilling position is the position in the two axes of the selected plane.

The cycle implements the following motion sequence:

- Approach of the reference plane brought forward by the safety distance with G0.
- Traverse to final drilling depth with the feedrate (G1) programmed in the calling program
- Dwell time at final drilling depth
- Retraction to retraction plane with G0.

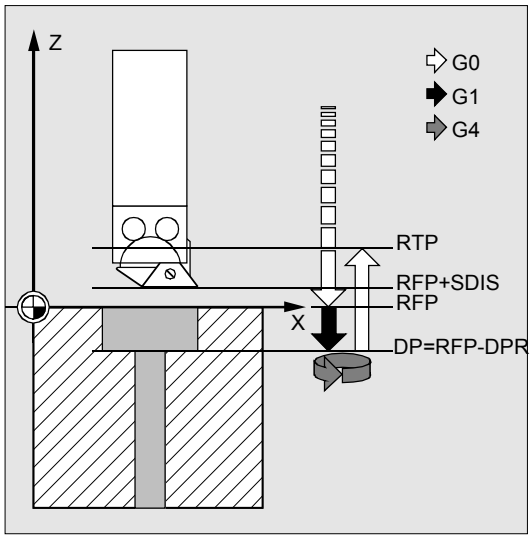


© Siemens AG 2004 All rights reserved.

Explanation of parameters

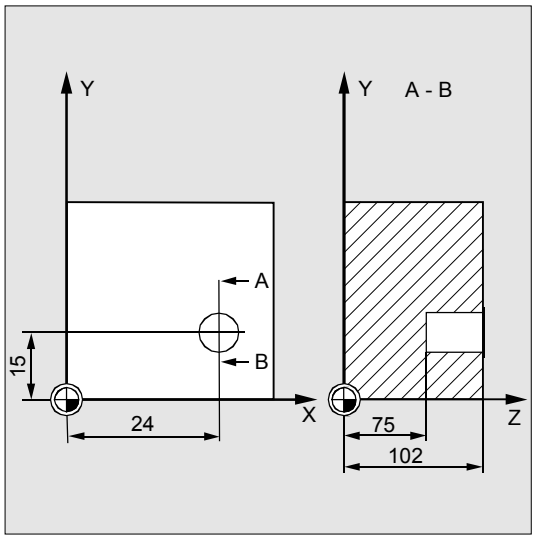
For parameters RTP, RFP, SDIS, DP, DPR see Subsection 2.1.2. (Drilling, Centering – CYCLE81)

DTB (dwell time)
Parameter DTB is the dwell time at the final drilling depth (chip breaking) in seconds.



Programming example

Boring_counterboring
This program machines a single hole to a depth of 27mm at position X24, Y15 in the XY plane with cycle CYCLE82.
The dwell time programmed is 2 sec, the safety distance in the drilling axis Z is 4mm.



`N10 G0 G90 F200 S300 M3`	;Specification of technology values
`N20 D1 T3 Z110`	;Traverse to retraction plane
`N21 M6`	
`N30 X24 Y15`	;Traverse to drilling position
`N40 CYCLE82 (110, 102, 4, 75, , 2)`	;Cycle call with absolute final drilling ;depth and safety distance
`N50 M30`	;End of program

© Siemens AG 2004 All rights reserved.

2.1.4 Deep-hole drilling – CYCLE83



Programming

```
CYCLE83 (RTP, RFP, SDIS, DP, DPR, FDEP, FDPR, DAM, DTB, DTS, FRF, VARI,
_AXN, _MDEP, _VRT, _DTD, _DIS1)
```



Parameters

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Final drilling depth (absolute)
DPR	real	Final drilling depth relative to reference plane (enter without sign)
FDEP	real	First drilling depth (absolute)
FDPR	real	First drilling depth relative to reference plane (enter without sign)
DAM	real	Degression: (enter without sign) Values: > 0 degression as value < 0 degression factor = 0 no degression
DTB	real	Dwell time at drilling depth (chip breaking) Values: > 0 in seconds < 0 in revolutions
DTS	real	Dwell time at starting point and for swarf removal Values: > 0 in seconds < 0 in revolutions
FRF	real	Feedrate factor for first drilling depth (enter without sign) Value range: 0.001...1
VARI	int	Type of machining: Values: 0 chip breaking 1 swarf removal
_AXN	int	Tool axis: Values: 1 = 1. Geometry axis 2 = 2. Geometry axis or else 3rd geometry axis
_MDEP	real	Minimum drilling depth
_VRT	real	Variable retraction distance for chip breaking (VARI=0): Values: > 0 is retraction distance 0 = setting is 1mm
_DTD	real	Dwell time at final drilling depth Values: > 0 in seconds < 0 in revolutions = 0 value as for DTB
_DIS1	real	Programmable limit distance on re-insertion in hole (VARI=1 for swarf removal) Values: > 0 programmable value applies = 0 automatic calculation

© Siemens AG 2004 All rights reserved.

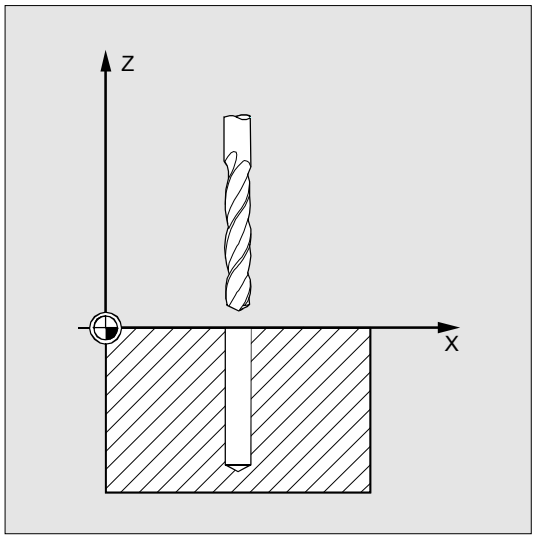
Function

The tool drills at the programmed spindle speed and feedrate to the programmed final drilling depth.
Deep hole drilling is performed with a depth infeed of a maximum definable depth executed several times, increasing gradually until the final drilling depth is reached.
The drill can either be retracted to the reference plane + safety distance after every infeed depth for swarf removal or retracted by the programmed retraction distance for chip breaking.

Sequence

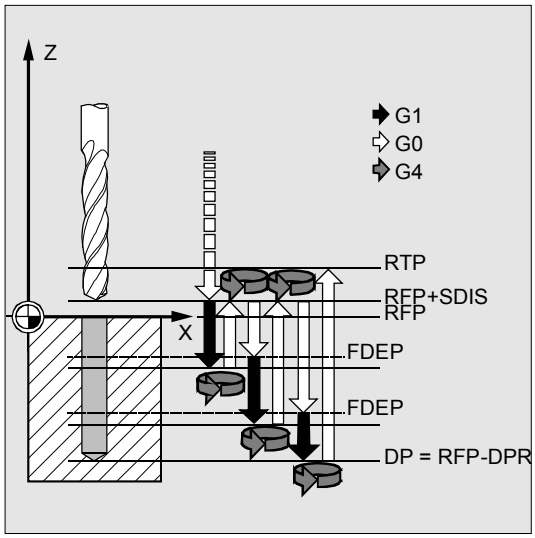
Position reached before the beginning of the cycle:
The drilling position is the position in the two axes of the selected plane.

The cycle implements the following motion sequence:



Deep-hole drilling with swarf removal (VARI=1):

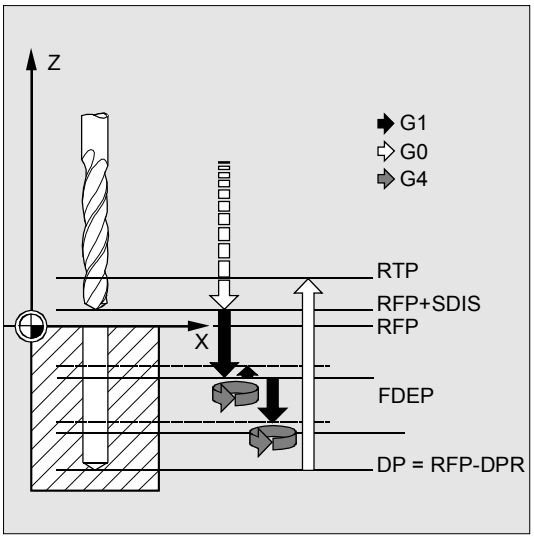
- Approach of the reference plane brought forward by the safety distance with G0.
- Traverse to the first drilling depth with G1, the feedrate for which is derived from the feedrate defined with the program call which is subject to parameter FRF (feedrate factor)
- Dwell time at final drilling depth (parameter DTB)
- Retraction to the reference plane brought forward by the safety distance with G0 for swarf removal
- Dwell time at starting point (parameter DTS)
- Approach last drilling depth reached, reduced by the calculated (by cycle) or programmable limit distance with G0
- Traverse to next drilling depth with G1 (sequence of motions is continued until the final drilling depth is reached)
- Retraction to retraction plane with G0.



© Siemens AG 2004 All rights reserved.

Deep-hole drilling with chip breaking (VARI=0):

- Approach of the reference plane brought forward by the safety distance with G0.
- Traverse to the first drilling depth with G1, the feedrate for which is derived from the feedrate defined with the program call which is subject to parameter FRF (feedrate factor)
- Dwell time at final drilling depth (parameter DTB)
- Variable retraction (parameter _VRT) from the current drilling depth with G1 and the feedrate programmed in the calling program (for chip breaking)
- Traverse to next drilling depth with G1 and the programmed feedrate (sequence of motions is continued until the final drilling depth is reached)
- Retraction to retraction plane with G0.



Explanation of parameters

See Subsection 2.1.2 (Drilling, Centering – CYCLE81)

FDEP and FDPR (first drilling depth absolute or relative)
The first drilling depth is programmed by either one of these two parameters. The parameter FDPR has the same effect in the cycle as parameter DPR. If the values for the reference and retraction plane are identical, the first drilling depth can be defined as a relative value.

DAM (degression)
With deep holes, drilled in several steps, it is useful to work with degressive values for the individual drilling strokes (degression). This allows the swarf to get away and there is no tool breakage.
In the parameter, either program an incremental degression value in order to reduce the first drilling depth step by step or a % value to act as a degression factor.
DAM=0 no degression
DAM>0 degression as value
The current depth is derived in the cycle as follows:

- In the first step, the depth parameterized with the first drilling depth FDEP or FDPR is traversed as long as it does not exceed the total drilling depth.

© Siemens AG 2004 All rights reserved.

- From the second drilling depth on, the drilling stroke is obtained by subtracting the amount of degression from the stroke of the last drilling depth, provided that the latter is greater than the programmed amount of degression.
- The next drilling strokes correspond to the amount of degression, as long as the remaining depth is greater than twice the amount of degression.
- The last two drilling strokes are divided equally and traversed and are therefore always greater than half of the amount of degression.
- If the value for the first drilling depth is opposed to the total depth, error message 61107 "First drilling depth incorrectly defined" is generated and the cycle not executed.

Example:
Programming the values RTP=0, SDIS=0, DP=-40, FDEP=-12 and DAM=3 results in the following drilling strokes:

-12	is the first drilling depth
-21	the incremental difference 9 results from the first drilling depth 12 minus the amount of degression 3
-27	previous drilling depth minus the amount of degression 3
-30, -33, -36	amount of degression
-38, -40	remaining depth divided into two cuts

© Siemens AG 2004 All rights reserved.

DAM<0 (-0.001 to -1) degression factor

The current depth is derived in the cycle as follows:

- The first step is traversing the depth parameterized with the first drilling depth FDED or FDPR provided it does not exceed the total drilling depth.
- The next drilling strokes are calculated from the last drilling stroke multiplied by the degression factor for as long as the stroke remains larger or equal to the minimum drilling depth.
- The last two drilling strokes are divided equally and traversed and are therefore always greater than half of the minimum drilling depth.
- If the value for the first drilling depth is opposed to the total depth, error message 61107 "First drilling depth incorrectly defined" is generated and the cycle not executed.

Example:
Programming the values RTP=0, SDIS=0, DP=-40, FDEP=-10, DAM=-0.8 and MDEP=5 results in the following drilling strokes:

-10	is the first drilling depth
-18	the incremental difference 8 is 0.8 *of the first drilling depth
-24.4, -29.52	the previous drilling depth * degression factor in each case
-34.52	minimum drilling depth MDEP takes effect
-37.26, -40	remaining depth divided into two cuts

DTB (dwell time)

The dwell time at final drilling depth (chip breaking) is programmed in DTB in seconds or revolutions of the main spindle.

➢ 0 in seconds
< 0 in revolutions

DTS (dwell time)

The dwell time at the starting point is only performed if VARI=1 (swarf removal).

➢ Value > 0 in seconds
Value < 0 in revolutions

FRF (feedrate factor)

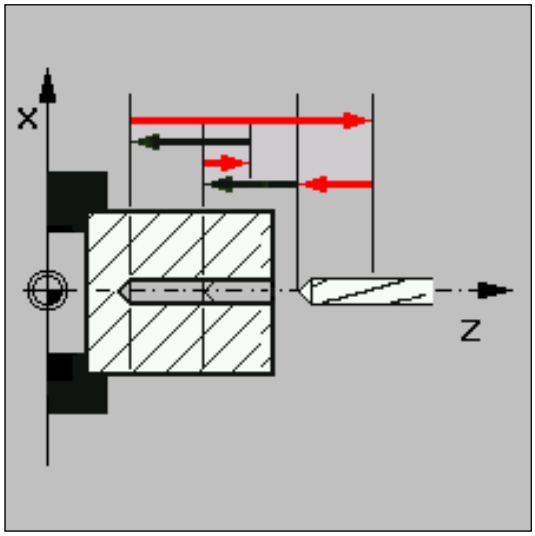
With this parameter you can enter a reduction factor for the active feedrate which only applies to the approach to the first drilling depth in the cycle.
If the FRF is programmed too large, no alarm is output.
The factor is limited to 1 within the cycle.

© Siemens AG 2004 All rights reserved.

VARI (machining type)
If parameter VARI=0 is set, the drill retracts by the programmed retraction distance after reaching each drilling depth for chip breaking. When VARI=1 (for swarf removal), the drill traverses in each case to the reference plane moved forward by the safety distance.

_AXN (tool axis)
By programming the drilling axis via _AXN, it is possible to omit the switchover from plane G18 to G17 when the deep hole drilling cycle is used on turning machines.
Key:
_AXN=1 1. axis of the current plane
_AXN=2 2. 2nd axis of the current plane
_AXN=3 3. axis of the current plane.
For example, to machine a center drilling (in Z) in plane G18, you program:

```
G18
_AXN=1
```



_MDEP (minimum drilling depth)
You can define a minimum drilling depth for drill stroke calculations based on degression factor. If the calculated drilling stroke becomes shorter than the minimum drilling depth, the remaining depth is machined in strokes equaling the length of the minimum drilling depth.

_VRT (variable retraction value for chip breaking with VARI=0)
You can program the retraction path for chip breaking.
Value > 0 retraction value
Value = 0 retraction value 1mm

_DTD (dwell time at final drilling depth)
The dwell time at final drilling depth can be entered in seconds or revolutions.
Value > 0 in seconds
Value < 0 in revolutions
Value = 0 dwell time as programmed in DTB

_DIS1 (programmable limit distance when VARI=1)
The limit distance after re-insertion in the hole can be programmed.

© Siemens AG 2004 All rights reserved.

Value > 0 position at programmed value
Value = 0 automatic calculation

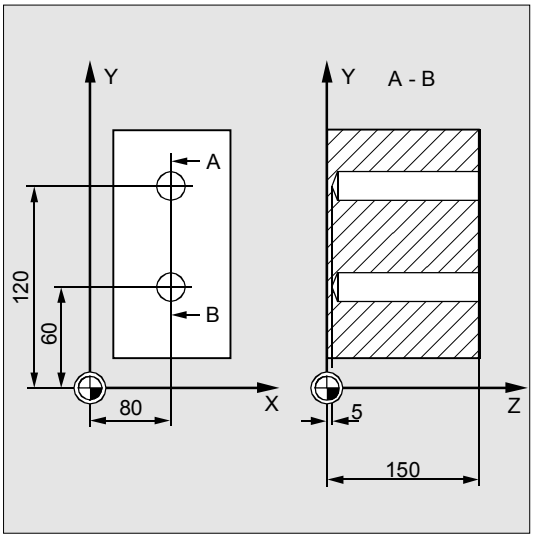
The limit distance is calculated within the cycle as follows:

- up to a drilling depth of 30 mm, the value is set to 0.6 mm
- for larger drilling depths, the limit distance is calculated as (RFP + SDIS – current depth) / 50. If this calculated value >7, the limit distance is limited to a maximum of 7mm

Programming example

Deep-hole drilling

This program executes the cycle CYCLE83 at positions X80 Y120 and X80 Y60 in the XY plane. The first hole is drilled with a dwell time zero and machining type chip breaking.
The final drilling depth and the first drilling depth are entered as absolute values. In the second cycle call, a dwell time of
1 sec is programmed. Machining type swarf removal is selected, the final drilling depth is relative to the reference plane. The drilling axis in both cases is the Z axis.
The drilling stroke is calculated on the basis of a degression factor and must not become shorter than the minimum drilling depth of 8 mm.




```
DEF REAL RTP=155, RFP=150, SDIS=1, DP=5,  ;Definition of parameters
DPR=145, FDEP=100, FDPR=50, DAM=20,
DTB=1, FRF=1, VARI=0, _VRT=0.8, _MDEP=10,
_DIS1=0.4
N10 G0 G17 G90 F50 S500 M4                ;Specification of technology values
N20 D1 T42 Z155                           ;Traverse to retraction plane
N30 X80 Y120                              ;Traverse to first drilling position
N40 CYCLE83 (RTP, RFP, SDIS, DP, ,->      ;Cycle call, depth parameter with
-> FDEP, , DAM, , , FRF, VARI, , , _VRT)  ;absolute values
N50 X80 Y60                               ;Traverse to next drilling position
N55 DAM=-0.6 FRF=0.5 VARI=1               ;Value assignment
N60 CYCLE83 (RTP, RFP, SDIS, , DPR, , ->  ;Cycle call with relative data for final drilling
-> FDPR, DAM, DTB, , FRF, VARI, , _MDEP,  ;depth and 1st final drilling depth the safety
-> , , _DIS1)                             ;distance is 1mm the feedrate is 0.5
N70 M30                                   ;End of program
```

Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

2.1.5 Rigid tapping – CYCLE84



Programming

```
CYCLE84 (RTP, RFP, SDIS, DP, DPR, DTB, SDAC, MPIT, PIT, POSS, SST, SST1,
_AXN, _PTAB, _TECHNO, _VARI, _DAM, _VRT)
```



Parameters

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Final drilling depth (absolute)
DPR	real	Final drilling depth relative to reference plane (enter without sign)
DTB	real	Dwell time at thread depth (chip breaking)
SDAC	int	Direction of rotation after end of cycle Values: 3, 4 or 5
MPIT	real	Pitch as thread size (with sign) Value range: 3 (for M3) ... 48 (for M48), the sign determines the direction of rotation in the thread
PIT	real	Pitch as value (with sign) Value range: 0.001 ... 2000.000 mm), the sign determines the direction of rotation in the thread: If PTAB=0 or 1: in mm (as before) If _PTAB=2 in thread starts per inch
POSS	real	Spindle position for oriented spindle stop in the cycle (in degrees)
SST	real	Speed for tapping
SST1	real	Speed for retraction
_AXN	int	Tool axis Values: 1 = 1. Geometry axis 2 = 2. Geometry axis or else 3rd geometry axis
_PTAB	int	Evaluation of thread pitch PIT Values: 0...according to programmed system of units inch/metric 1...pitch in mm 2...pitch in thread starts per inch 3...pitch in inches/revolution

© Siemens AG 2004 All rights reserved.

_TECHNO	int	Technological settings UNITS DIGIT: Exact stop behavior Values: 0...and as programmed before cycle call 1...(G601) 2...(G602) 3...(G603) TENS DIGIT: Feedforward control Values: 0...and as programmed before cycle call 1...with feed-forward control (FFWON) 2...without feed-forward control (FFWOF) HUNDREDS DIGIT: Acceleration Values: 0...and as programmed before cycle call 1...soft acceleration of axes (SOFT) 2...brisk acceleration of axes (BRISK) 3...reduced acceleration of axes (DRIVE) THOUSANDS DIGIT: Values: 0...activate spindle mode again (with MCALL) 1...stay in position-control mode (with MCALL)
_VARI	int	Type of machining: Values: 0...tapping in one pass 1...deep hole drilling with chip breaking 2...deep hole drilling with stock removal
_DAM	real	Incremental drilling depth Value range: 0 <= max. value
_VRT	real	Variable retraction distance for chip breakage Value range: 0 <= max. value

Function

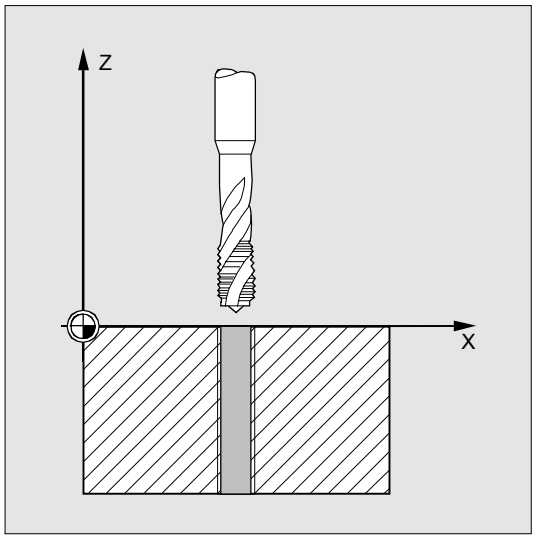
The tool drills at the programmed spindle speed and feedrate to the programmed thread depth.
With cycle CYCLE84 you can perform rigid tapping operations.
The cycle is also capable of performing tapping operations in several stages (deep hole drilling).

Cycle CYCLE84 can be used if the spindle to be used for the boring operation is technically able to go into position-controlled spindle operation.

There is a separate cycle CYCLE840 for tapping with compensating chuck (see Subsection 2.1.6).

© Siemens AG 2004 All rights reserved.

Sequence

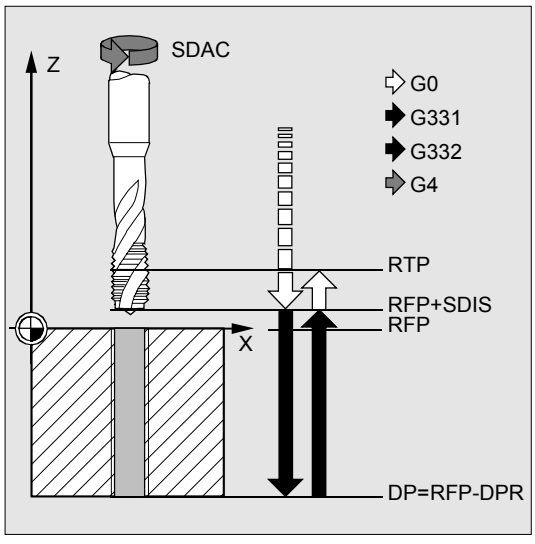


Position reached before the beginning of the cycle:
The drilling position is the position in the two axes of the selected plane.

The cycle implements the following motion sequence:

- Approach of the reference plane brought forward by the safety distance with G0.
- Oriented spindle stop with SPOS (value in parameter POSS) and conversion of spindle to axis mode
- Tapping to final drilling depth with G331 and speed SST
- Dwell time at thread depth (parameter DTB)
- Retraction to the reference plane brought forward by the safety distance with G332, spindle speed SST1 and reversal of direction of rotation
- Retraction to the retraction plane with G0, spindle mode is reintroduced by reprogramming the spindle speed active before the cycle was called and the direction of rotation programmed under SDAC.

Explanation of parameters



See Subsection 2.1.2 (Drilling, Centering – CYCLE81)

DTB (dwell time)
You program the dwell time in seconds. It is recommended that the dwell time is omitted for the tapping of blind holes.

SDAC (direction of rotation after end of cycle)
Under SDAC you program the direction of rotation after completion of the cycle.
For tapping, the direction is changed automatically by the cycle.

© Siemens AG 2004 All rights reserved.

MPIT and PIT (as thread size and as value)
The value for the thread pitch can either be defined as the thread size (for metric threads between M3 and M48 only) or as a value (distance from one thread turn to the next as a numerical value). The parameter not required in each case is omitted from the call or assigned the value zero.

Right or left threads are specified by the sign of the pitch parameter:

- Positive value → right (like M3)
- Negative value → left (like M4)

If the two pitch parameters have contradictory values, the cycle generates alarm 61001 "Incorrect pitch" and cycle execution is aborted.

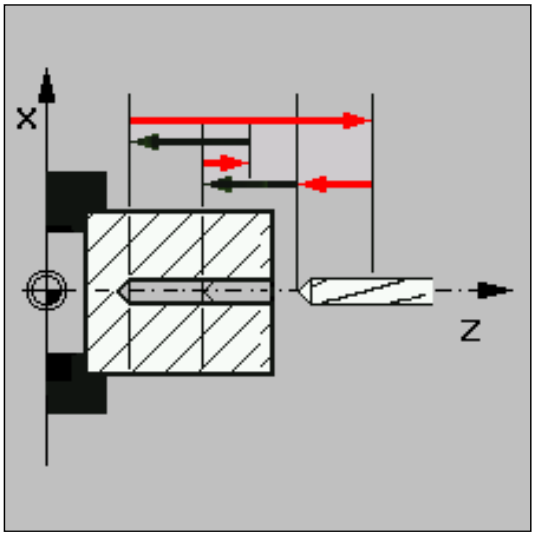
POSS (spindle position)
Before tapping starts in the cycle, oriented spindle stop is performed with command SPOS and the spindle is put in position control.
You program the spindle position for this spindle stop under POSS.

SST (speed)
Parameter SST contains the spindle speed for the tapping block with G331.

SST1 (retraction speed)
Under SST1 you program the speed for the retraction out of the thread hole in the hole with G332.
If this parameter is assigned the value zero, the retraction movement is performed with the speed programmed under SST.

_AXN (tool axis)
Programming the drilling axis via *AXN enables you to omit the switchover from plane G18 to G17 when the deep-hole tapping cycle is used on turning machines*.
Key:
AXN=1 1. axis of the current plane
AXN=2 2. axis of the current plane
AXN=3 3. axis of the current plane.
For example, to machine a center drilling (in Z) in plane G18, you program:

```
G18
_AXN=1
```



© Siemens AG 2004 All rights reserved.

_PTAB (evaluation of pitch PIT)

Parameter _PTAB determines the unit of measurement of the thread pitch.

- 0=according to programmed system of units inch/metric
- 1=pitch in mm
- 2=pitch in thread starts per inch
- 3=pitch in inches/revolution.

This parameter is needed in connection with the option for selecting different thread tables in the cycles support.

_TECHNO (technological settings

Parameter _TECHNO can be set to select technological characteristics for tapping operations.

Possible settings are:
Units digit (exact stop behavior):

- 0=as programmed before cycle call
- 1=(G601)
- 2=(G602)
- 3=(G603)

Tens digit (feedforward control):

- 0=as programmed before cycle call
- 1=with feedforward control (FFWON)
- 2=without feedforward control (FFWOF)

Hundreds digit (acceleration):

- 0=as programmed before cycle call
- 1=soft acceleration of axes (SOFT)
- 2=brisk acceleration of axes (BRISK)
- 3=reduced acceleration of axes (DRIVE)

Thousands digit:

- 0=activate spindle mode again (with MCALL)
- 1=stay in position-control mode (with MCALL)



With input field "Technology" "Yes", both the machine manufacturer and the operator/programmer can customize the technology for tapping.

Adaptation by the machine manufacturer (SW 6.4 and later)

- Preconditions: The manufacturer password is set, input field "Technology" = "Yes"
- When the interactive cycle screen form CYCLE84 is opened, the parameters are preassigned with the value of GUD7 variable _SC_MASK[0]. When the parameters are changed, the values are written directly to this GUD7 variable.
- This enables the machine manufacturer to customize the default settings to the characteristics of his machine.

© Siemens AG 2004 All rights reserved.

Adaptation by the operator/programmer
(SW 6.4 and later)

- Preconditions: The manufacturer password is deleted, input field "Technology" = "Yes"
- The modified data is used to generate CYCLE84. The settings of the machine manufacturer will take effect the next time a tapping cycle is called.

Input field "Technology" and the following input fields are not shown if the keyswitch is in position 0 or 1.

Deep-hole tapping _VARI, _DAM, _VRT

Parameter _VARI can be set to select between simple tapping (_VARI = 0) and deep-hole tapping (_VARI ¹ 0). In conjunction with deep hole tapping, it is possible to choose between chip breakage (retraction by variable distance from current drilling depth, parameter _VRT, _VARI = 1) and stock removal (withdrawal from reference plane _VARI = 2). These functions work analogously to the normal deep hole drilling cycle CYCLE83.

The incremental drilling depth for one pass is specified via parameter _DAM. The cycle internally calculates the intermediate depth as follows:

- The programmed incremental depth is drilled in each pass until the remaining distance to the final drilling depth is < than 2 * _DAM
- The remaining depth is halved and drilled in 2 passes, thereby ensuring that the minimum drilling depth is not less than _DAM/2.



Other Information

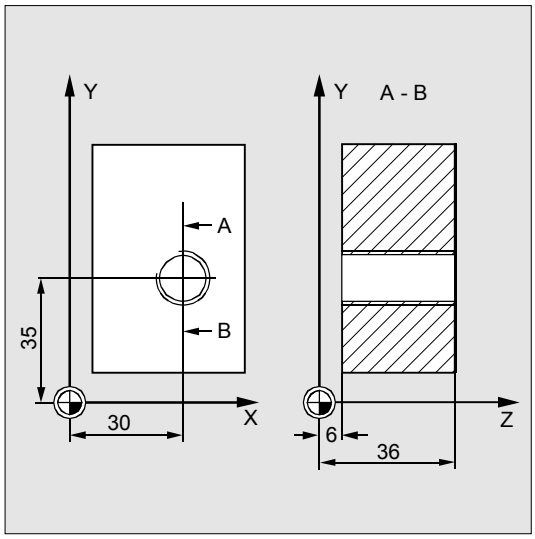
The direction of rotation is always reversed automatically for tapping in cycle.

© Siemens AG 2004 All rights reserved.

Programming example

Rigid tapping

A thread is tapped without a compensating chuck at position X30 and Y35 in the XY plane, the tapping axis is the Z axis. No dwell time is programmed. The depth is programmed as a relative value. The parameters for the direction of rotation and the pitch must be assigned values. A metric thread M5 is tapped.



`N10 G0 G90 T4 D1`	;Specification of technology values
`N20 G17 X30 Y35 Z40`	;Traverse to drilling position
`N30 CYCLE84 (40, 36, 2, , 30, , 3, 5, -> ->, 90, 200, 500)`	;Cycle call, parameter PIT has been omitted, ;no value is entered for the absolute depth ;or the dwell time. Spindle stop at 90 ;degrees, speed for tapping is 200, speed ;for retraction is 500
`N40 M30`	End of program

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

2.1.6 Tapping with compensating chuck – CYCLE840

Programming

```
CYCLE840 (RTP, RFP, SDIS, DP, DPR, DTB, SDR, SDAC, ENC, MPIT, PIT- AXN,
_PTAB, _TECHNO)
```

Parameters

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Final drilling depth (absolute)
DPR	real	Final drilling depth relative to reference plane (enter without sign)
DTB	real	Dwell time at thread depth: always operative now if >0 is programmed Value range: 0<=DTB
SDR	int	Direction of rotation for retraction Values: 0 (automatic reversal of direction of rotation) 3 or 4 (for M3 or M4)
SDAC	int	Direction of rotation after end of cycle Values: 3, 4 or 5 (for M3, M4 or M5)
ENC	int	Tapping with/without encoder Values: 0 = with encoder, without dwell time 1 = without encoder, first programs the cycle, then the feed 11 = without encoder, calculates the feed in the cycle 20 = with encoder, with dwell time
MPIT	real	Thread pitch as thread size Value range: 3 (for M3) ... 48 (for M48)
PIT	real	Thread pitch as value Value range: 0.001 ... 2000.000 mm If PTAB=0 or 1: in mm If _PTAB=2 in thread starts per inch
_AXN	int	Tool axis: Values: 1 = 1. Geometry axis 2 = 2. Geometry axis or else 3rd geometry axis
_PTAB	int	Evaluation of thread pitch PIT Values: 0...according to programmed system of units inch/metric 1...pitch in mm 2...pitch in thread starts per inch 3...pitch in inches/revolution

© Siemens AG 2004 All rights reserved.

_TECHNO	int	Technological settings UNITS DIGIT: Exact stop behavior Values: 0...and as programmed before cycle call 1...(G601) 2...(G602) 3...(G603) TENS DIGIT: Feedforward control Values: 0...and as programmed before cycle call 1...with feed-forward control (FFWON) 2...without feed-forward control (FFWOF) HUNDREDS DIGIT: Brake application point Values: 0...without calculation 1...with calculation

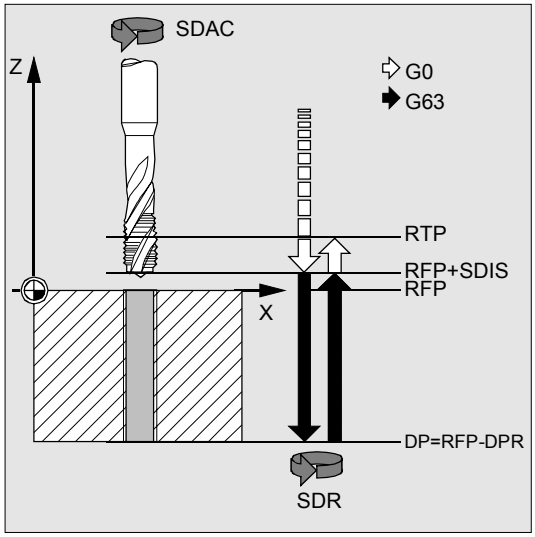
Function

The tool drills at the programmed spindle speed and feedrate to the programmed thread depth.
This cycle can be used to perform tapping with compensating chuck

- without encoder and
- with encoder.

Sequence

Tapping with compensating chuck without encoder (ENC=1)



Position reached before the beginning of the cycle:
The drilling position is the position in the two axes of the selected plane.

The cycle implements the following motion sequence:

- Approach of the reference plane brought forward by the safety distance with G0.
- Tapping to the final drilling depth with G63
- Retraction to the reference plane brought forward by the safety distance with G63
- Retraction to retraction plane with G0.

The spindle override must be set to 100%.

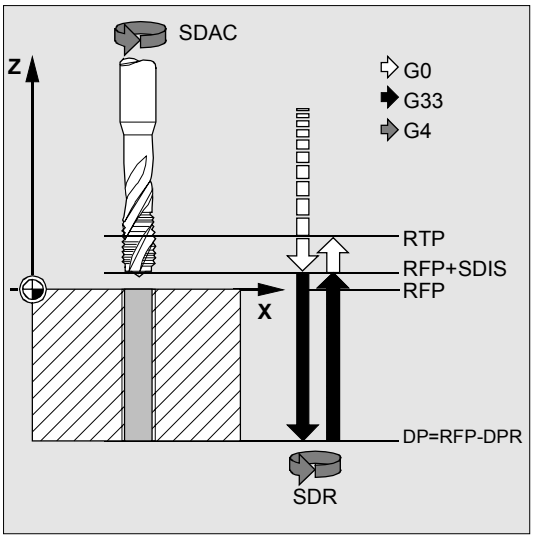
© Siemens AG 2004 All rights reserved.

Tapping with compensating chuck without encoder with encoder (ENC=0)

Position reached before the beginning of the cycle:
The drilling position is the position in the two axes of the selected plane.

The cycle implements the following motion sequence:

- Approach of the reference plane brought forward by the safety distance with G0.
- Tapping to the final drilling depth with G33
- Dwell time at thread depth (parameter DTB)
- Retraction to the reference plane brought forward by the safety distance with G33
- Retraction to retraction plane with G0.



Explanation of parameters

See Subsection 2.1.2 (Drilling, Centering – CYCLE81)

DTB (dwell time)
You program the dwell time in seconds. It works in parameter ENC depending on which technological variant you selected.

SDR (direction of rotation for retraction)
The direction of rotation for retraction when tapping is programmed in parameter SDR. For tapping with an encoder, the direction of spindle travel can be reversed automatically, for which SDR=0 is set.

SDAC (direction of rotation)
As the cycle can also be called modally (see Section 2.2), it requires a direction of rotation for tapping further threads. This is programmed in parameter SDAC and corresponds to the direction of rotation programmed before the first call in the higher-level program. If SDR=0, the value assigned to SDAC is of no significance in the cycle and can be omitted from the parameterization.

© Siemens AG 2004 All rights reserved.

ENC (tapping)

To perform tapping without encoder although an encoder exists, parameter ENC must be set to 1.

However, if no encoder exists and the parameter is assigned the value 0, it is ignored in the cycle.

- **Tapping without encoder with input of the lead:**
  You can calculate the correlation between feedrate and speed via the programmed lead within the cycle for tapping without encoder (G63 thread). The speed must be programmed before the cycle is called.
  You can specify the lead either via MPIT (metric thread unit) or via PIT (thread lead as value) just as for tapping without encoder.

  The feed is then calculated in the cycle from the lead and the speed. After the cycle has ended, the last programmed feed applies again.
  Programming:
  ENC=11, programming the lead in MPIT or PIT
- **Tapping with encoder with dwell time:**
  As an alternative, you can program a dwell time in the parameter DTB for tapping with encoder (G33 thread). It applies after tapping and before retraction to the retraction plane RTP, and is required for machines with unfavorable spindle dynamics.
  Programming:
  ENC=20, entering the dwell time in parameter DTB

MPIT and PIT (as thread size and as value)

The parameter for the spindle pitch only has a meaning if tapping is performed with encoder. The cycle calculates the feedrate from the spindle speed and the pitch.
The value for the thread pitch can either be defined as the thread size (for metric threads between M3 and M48 only) or as a value (distance from one thread turn to the next as a numerical value). The parameter not required in each case is omitted from the call or assigned the value zero.
If the two thread pitch parameters have conflicting values, alarm 61001 "Thread pitch wrong"
is generated by the cycle and cycle execution is aborted.

© Siemens AG 2004 All rights reserved.

_AXN (tool axis)

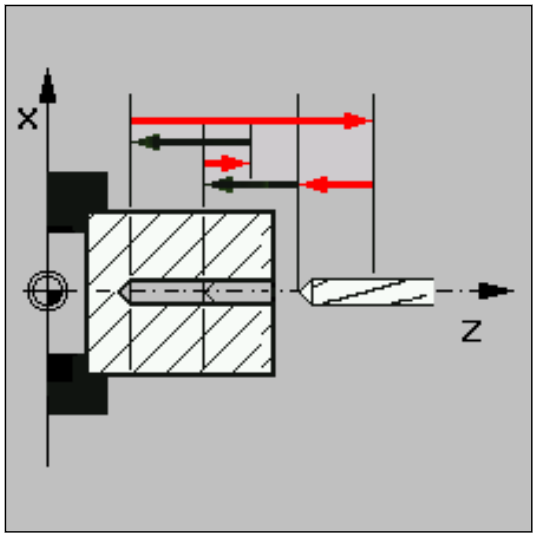
By programming the drilling axis via _AXN, it is possible to omit the switchover from plane G18 to G17 when the deep hole tapping cycle is used on turning machines.

Key:

_AXN=1 1. axis of the current plane
_AXN=2 2. 2nd axis of the current plane
_AXN=3 3. axis of the current plane.

For example, to machine a center drilling (in Z) in plane G18, you program:

```
G18
_AXN=1
```



_PTAB (evaluation of pitch PIT)

Parameter _PTAB determines the unit of measurement of the pitch.

- 0=according to programmed system of units inch/metric
- 1=pitch in mm
- 2=pitch in thread starts per inch
- 3=pitch in inches/revolution.

This parameter is needed in connection with the option for selecting different thread tables in the cycles support.

_TECHNO (technological settings)

Parameter _TECHNO can be set to select technological characteristics for tapping operations.

Possible settings are:

Units digit (exact stop behavior):

- 0=as programmed before cycle call
- 1=(G601)
- 2=(G602)
- 3=(G603)

Tens digit (feedforward control):

- 0=as programmed before cycle call
- 1=with feedforward control (FFWON)
- 2=without feedforward control (FFWOF)

Hundreds digit (brake application point):

- 0=without calculation
- 1=with calculation

© Siemens AG 2004 All rights reserved.

As of SW 6.4:
With input field "Technology" "Yes", both the machine manufacturer and the operator/programmer can customize the technology for tapping.

Adaptation by the machine manufacturer (SW 6.4 and later)

- Preconditions: The manufacturer password is set, input field "Technology" = "Yes"
- When the interactive cycle screen form CYCLE840 is opened, the parameters are preassigned with the value of GUD7 variable _SC_MASK[1]. When the parameters are changed, the values are written directly to this GUD7 variable.
- This enables the machine manufacturer to customize the default settings to the characteristics of his machine.

Adaptation by the operator/programmer (SW 6.4 and later)

- Preconditions: The manufacturer password is reset, input field "Technology" = "Yes"
- The modified data is used to generate CYCLE840. The settings of the machine manufacturer will take effect the next time a tapping cycle is called.

Input field "Technology" and the following input fields are not shown if the keyswitch is in position 0 or 1.

Other Information

Depending on the setting in machine data NUM_ENCS, the cycle selects whether tapping is to performed with or without encoder.

The direction of rotation for the spindle must be programmed with M3 or M4 before the cycle is called.

In thread blocks with G63, the values of the feedrate override switch and spindle speed override switch are frozen at 100 percent.

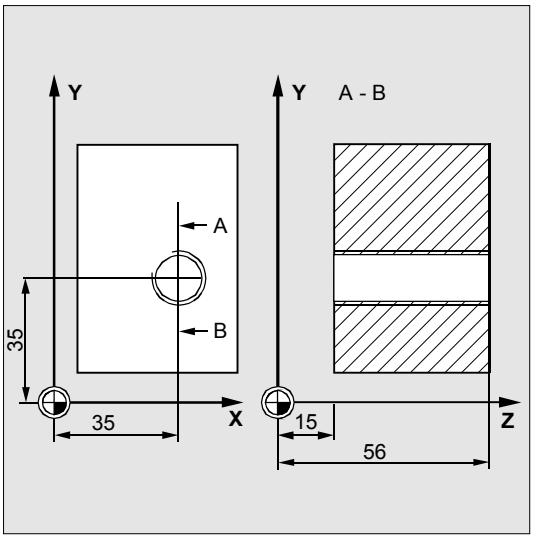
A longer compensating chuck is usually required for tapping without encoder.

© Siemens AG 2004 All rights reserved.

Programming example

Thread without encoder

In this program a thread is tapped without encoder at position X35 Y35 in the XY plane, the drilling axis is the Z axis. Parameters SDR and SDAC for the direction of rotation must be assigned, parameter ENC is assigned the value 1, the value for the depth is absolute. Pitch parameter PIT can be omitted. A compensating chuck is used in machining.

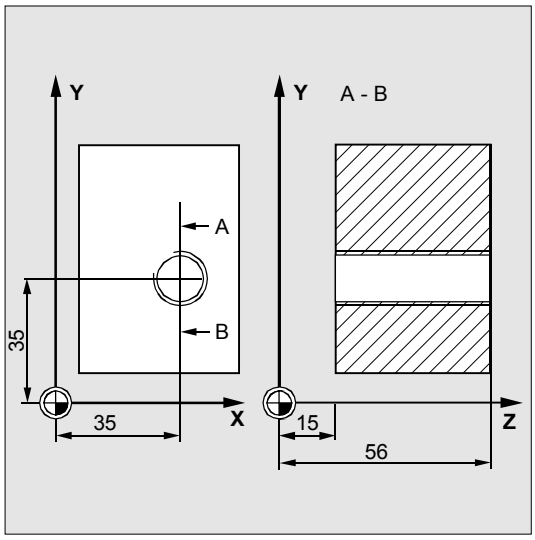



```
N10 G90 G0 D2 T2 S500 M3                ;Specification of technology values
N20 G17 X35 Y35 Z60                     ;Traverse to drilling position
N30 G1 F200                             ;Setting the path feedrate
N40 CYCLE840 (59, 56, , 15, , 1, 4, 3, 1) ;Cycle call, dwell time 1 sec, SDR=4,
                                        ;SDAC=3, no safety distance,
                                        ;parameters MPIT, PIT are not programmed,
                                        ;i.e. the lead results from the correlation
                                        ;between the freely programmable values F
                                        ;and S.
N50 M30                                 ;End of program
```


© Siemens AG 2004 All rights reserved.

Thread with encoder

In this program a thread is tapped with encoder at position X35 Y35 in the XY plane; the boring axis is the Z axis. The pitch parameter must be defined, automatic reversal of the direction of rotation is programmed. A compensating chuck is used in machining.



DEF INT SDR=0 DEF REAL PIT=3.5	;Definition of parameters with ;value assignments
N10 G90 G0 D2 T2 S500 M4	;Specification of technology values
N20 G17 X35 Y35 Z60	;Traverse to drilling position
N30 CYCLE840 (59, 56, , 15, , , , , , -> ->, PIT)	;Cycle call, without safety distance, value ;for depth programmed as an absolute ;value, SDAC, ENC, MPIT are omitted ;(i.e., are assigned the value zero)
N40 M30	;End of program

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

2.1.7 Boring 1 – CYCLE85

Programming

```
CYCLE85 (RTP, RFP, SDIS, DP, DPR, DTB, FFR, RFF)
```

Parameters

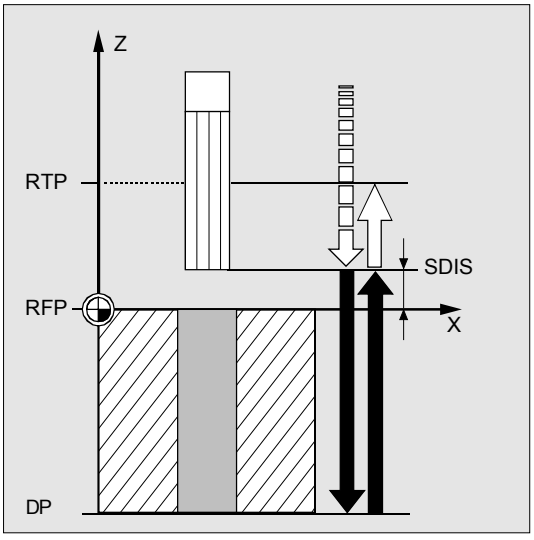
RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Final drilling depth (absolute)
DPR	real	Final drilling depth relative to reference plane (enter without sign)
DTB	real	Dwell time at final drilling depth (chip breaking)
FFR	real	Feed
RFF	real	Retraction feedrate

Function

The tool drills at the programmed spindle speed and feedrate to the programmed final drilling depth.
The inward and outward movement is performed at the feedrate that is assigned to FFR and RFF respectively.
This cycle can be used to ream drill holes.

Sequence

Position reached before the beginning of the cycle:
The drilling position is the position in the two axes of the selected plane.



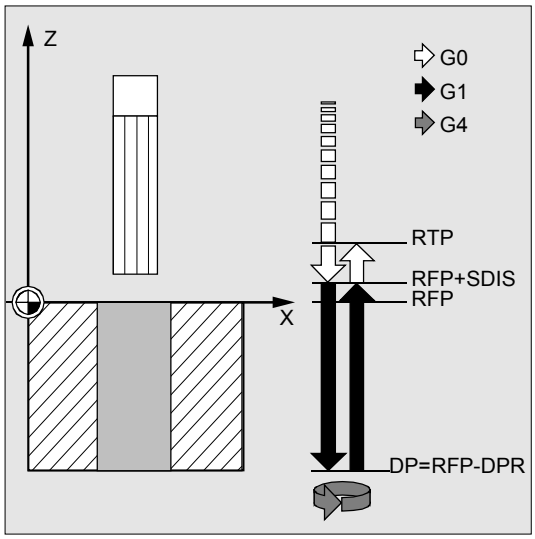
© Siemens AG 2004 All rights reserved.

The cycle implements the following motion sequence:

- Approach of the reference plane brought forward by the safety distance with G0.
- Traverse to final drilling depth with G1 and at the feedrate programmed under parameter FFR
- Dwell time at final drilling depth
- Retraction to the reference plane brought forward by the safety distance with G1 and the retraction feedrate defined under parameter RFF
- Retraction to retraction plane with G0.

Explanation of parameters

See Subsection 2.1.2 (Drilling, Centering – CYCLE81)



DTB (dwell time)
Parameter DTB is the dwell time at the final drilling depth (chip breaking) in seconds.

FFR (feedrate)
The feedrate value assigned to FFR is active for boring.

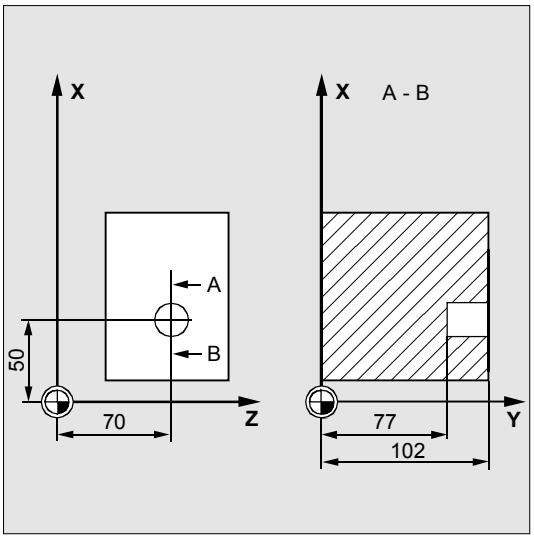
RFF (retraction feedrate)
The feedrate value assigned to RFF is active for retraction from the plane.

© Siemens AG 2004 All rights reserved.

Programming example

First boring pass

Cycle CYCLE85 is called at position Z70 X50 in the ZX plane. The boring axis is the Y axis. The value for the final drilling depth in the cycle call is programmed as a relative value, no dwell time is programmed. The top edge of the workpiece is positioned at Y102.




```
DEF REAL FFR, RFF, RFP=102, DPR=25,          ;Definition of parameters and
SDIS=2                                       ;value assignments
N10 G0 FFR=300 RFF=1.5*FFR S500 M4           ;Specification of technology values
N20 G18 T1 D1 Z70 X50 Y105                   ;Traverse to drilling position
N21 M6
N30 CYCLE85 (RFP+3, RFP, SDIS, , DPR, ,->    ;Cycle call, no dwell time programmed
-> FFR, RFF)
N40 M30                                      ;End of program
```

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

2.1.8 Boring 2 – CYCLE86



Programming

```
CYCLE86 (RTP, RFP, SDIS, DP, DPR, DTB, SDIR, RPA, RPO, RPAP, POSS)
```



Parameters

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Final drilling depth (absolute)
DPR	real	Final drilling depth relative to reference plane (enter without sign)
DTB	real	Dwell time at final drilling depth (chip breaking)
SDIR	int	Direction of rotation Value: 3 (for M3) 4 (for M4)
RPA	real	Retraction path in abscissa of the active plane (incremental, enter with sign)
RPO	real	Retraction path in ordinate of the active plane (incremental, enter with sign)
RPAP	real	Retraction path in applicate of the active plane (incremental, enter with sign)
POSS	real	Spindle position for oriented spindle stop in the cycle (in degrees)

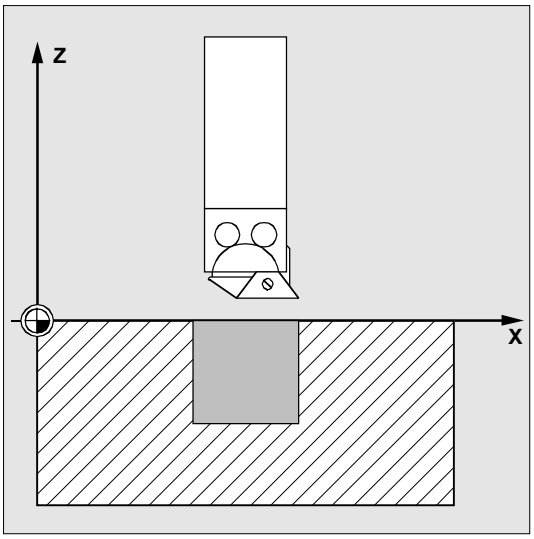


Function

The tool drills at the programmed spindle speed and feedrate to the programmed final drilling depth.
With Boring 2, oriented spindle stop is activated with the SPOS command once the drilling depth has been reached. Then, the programmed retraction positions are approached in rapid traverse and, from there, the retraction plane.

© Siemens AG 2004 All rights reserved.

Sequence

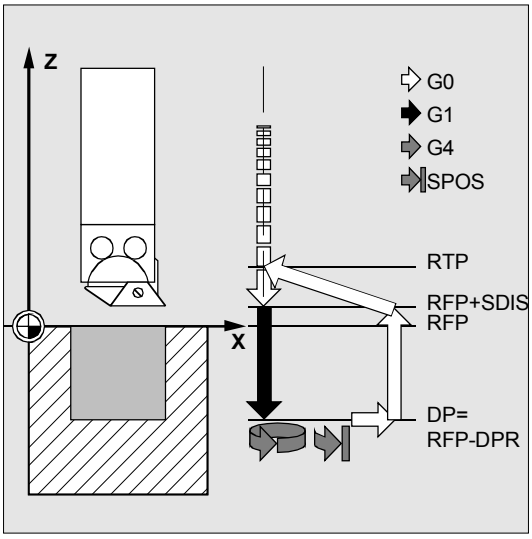


Position reached before the beginning of the cycle:
The drilling position is the position in the two axes of the selected plane.

The cycle implements the following motion sequence:

- Approach of the reference plane brought forward by the safety distance with G0.
- Traverse to final drilling depth with G1 and the feedrate programmed before the program call
- Dwell time at final drilling depth
- Oriented spindle stop at the spindle position programmed under POSS
- Traverse retraction path in up to three axes with G0
- Retraction to the reference plane brought forward by the safety distance with G0
- Retraction to the retraction plane with G0 (initial drilling position in both axes on the plane).

Explanation of parameters



See Subsection 2.1.2 (Drilling, Centering – CYCLE81)

DTB (dwell time)
Parameter DTB is the dwell time at the final drilling depth (chip breaking) in seconds.

SDIR (direction of rotation)
With this parameter you determine the direction of rotation with which boring is performed in the cycle. If values other than 3 or 4 (M3/M4) are generated, alarm 61102 "No spindle direction programmed" is output and the cycle is not executed.

RPA (retraction path, in abscissa)
Under this parameter you define a retraction movement in the abscissa, which is executed after the final drilling depth has been reached and oriented spindle stop has been performed.

RPO (retraction path, in ordinate)
Under this parameter you define a retraction movement in the ordinate which is executed after the final drilling has been reached and oriented spindle stop has been performed.

© Siemens AG 2004 All rights reserved.

RPAP (retraction path, in applicate)
Under this parameter you define a retraction movement in the boring axis which is executed after the final drilling has been reached and oriented spindle stop has been performed.

POSS (spindle position)
Under POSS the spindle position for the oriented spindle stop which is performed after the final drilling depth has been reached is programmed in degrees.



Other Information

With the SPOS command you can perform an oriented spindle stop of the active master spindle. The angular value is programmed with a transfer parameter.



Cycle CYCLE86 can be used if the spindle to be used for the boring operation is technically able to go into position-controlled spindle operation.

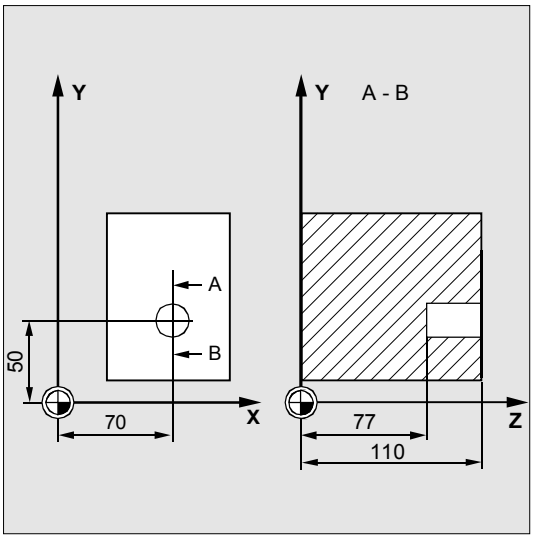
Turning machine without Y axis
You can now use the cycle CYCLE86 for turning machines without Y axis. Retraction to drilling depth is then traversed in two axes only. If a retraction path was programmed for the 3rd axis, it would be ignored.
If you call the cycle without Y axis in the G18 plane, the following alarm appears: 61005 "3. geometry axis not available", as the Y axis would then be the boring axis.

© Siemens AG 2004 All rights reserved.

Programming example

Second boring pass

Cycle CYCLE86 is called at position X70 Y50 in the ZX plane. the boring axis is the Z axis. The final drilling depth is programmed as an absolute value, a safety distance is not defined. The dwell time at the final drilling depth is 2 s. The top edge of the workpiece is positioned at Z110. In the cycle, the spindle is turned with M3 and stops at 45 degrees.



`DEF REAL DP, DTB, POSS`	;Definition of parameters
`N10 DP=77 DTB=2 POSS=45`	;Value assignments
`N20 G0 G17 G90 F200 S300`	;Specification of technology values
`N30 D1 T3 Z112`	;Traverse to retraction plane
`N40 X70 Y50`	;Traverse to drilling position
`N50 CYCLE86 (112, 110, , DP, , DTB, 3,-> -> -1, -1, +1, POSS)`	;Cycle call with absolute drilling depth
`N60 M30`	;End of program

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

2.1.9 Boring 3 – CYCLE87

Programming

```
CYCLE87 (RTP, RFP, SDIS, DP, DPR, SDIR)
```

Parameters

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Final drilling depth (absolute)
DPR	real	Final drilling depth relative to reference plane (enter without sign)
SDIR	int	Direction of rotation Value: 3 (for M3) 4 (for M4)

Function

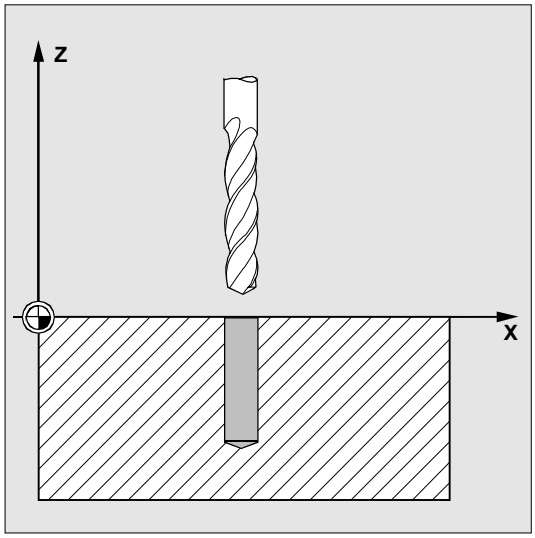
The tool drills at the programmed spindle speed and feedrate to the programmed final drilling depth.
With Boring 3, a spindle stop without orientation M5 and then a programmed stop M0 are generated when the final drilling depth is reached. The NC START key is pressed to continue the retraction movement in rapid traverse mode until the retraction plane is reached.

Sequence

Position reached before the beginning of the cycle:
The drilling position is the position in the two axes of the selected plane.

The cycle implements the following motion sequence:

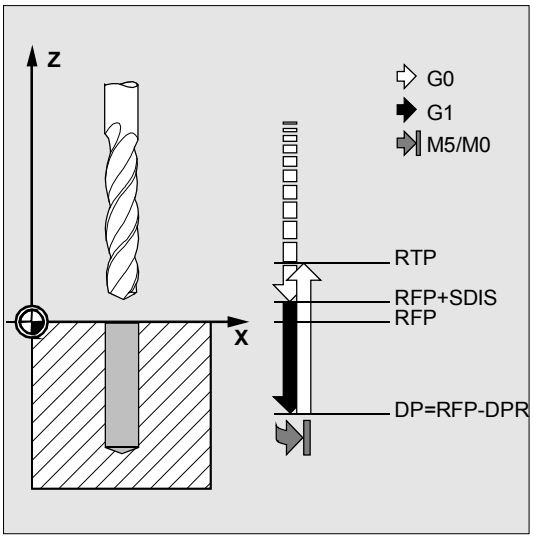
- Approach of the reference plane brought forward by the safety distance with G0.
- Traverse to final drilling depth with G1 and the feedrate programmed before the program call
- Spindle stop with M5
- Press NC START key
- Retraction to retraction plane with G0.



© Siemens AG 2004 All rights reserved.

Explanation of parameters

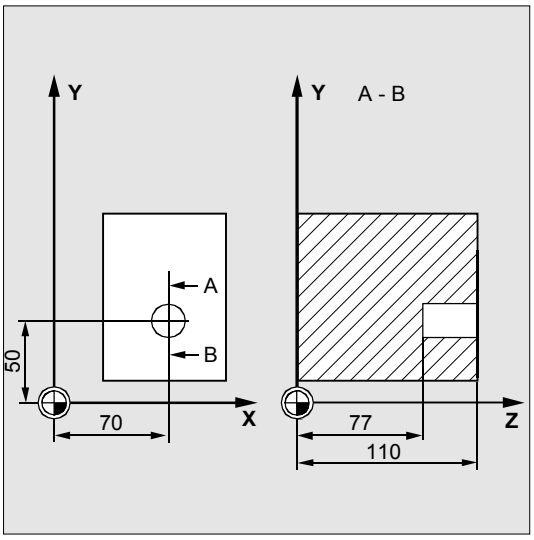
See Subsection 2.1.2 (Drilling, Centering – CYCLE81)



SDIR (direction of rotation)
With this parameter you determine the direction of rotation with which boring is performed in the cycle. If values other than 3 or 4 (M3/M4) are generated, alarm 61102 "No spindle direction programmed" is output and the cycle is aborted.

Programming example

Third boring pass
Cycle CYCLE87 is called at position X70 Y50 in the ZX plane. the boring axis is the Z axis. The final drilling depth is programmed as an absolute value. The safety distance is 2mm.



`DEF REAL DP, SDIS`	;Definition of parameters
`N10 DP=77 SDIS=2`	;Value assignments
`N20 G0 G17 G90 F200 S300`	;Specification of technology values
`N30 D1 T3 Z113`	;Traverse to retraction plane
`N40 X70 Y50`	;Traverse to drilling position
`N50 CYCLE87 (113, 110, SDIS, DP, , 3)`	;Cycle call with programmed spindle ;direction M3
`N60 M30`	;End of program

© Siemens AG 2004 All rights reserved.

2.1.10 Boring 4 – CYCLE88

Programming

```
CYCLE88 (RTP, RFP, SDIS, DP, DPR, DTB, SDIR)
```

Parameters

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Final drilling depth (absolute)
DPR	real	Final drilling depth relative to reference plane (enter without sign)
DTB	real	Dwell time at final drilling depth
SDIR	int	Direction of rotation Values: 3 (for M3) 4 (for M4)

Function

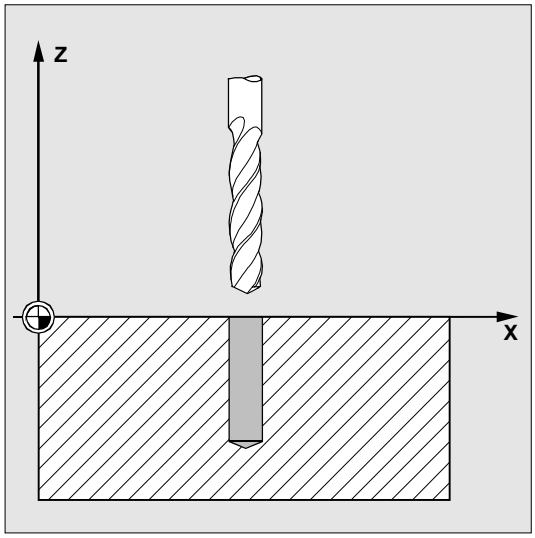
The tool drills at the programmed spindle speed and feedrate to the programmed final drilling depth. With Boring 4, a dwell time, a spindle stop without orientation M5 and a programmed stop M0 are generated when the final drilling depth is reached. Pressing the NC START key continues the retraction movement in rapid traverse mode until the retraction plane is reached.

Sequence

Position reached before the beginning of the cycle:
The drilling position is the position in the two axes of the selected plane.

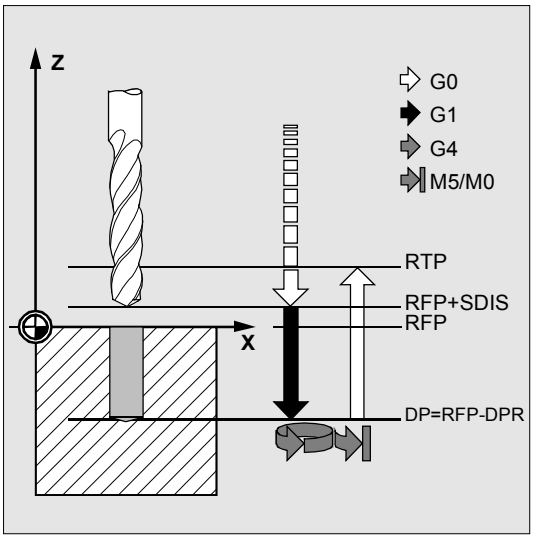
The cycle implements the following motion sequence:

- Approach of the reference plane brought forward by the safety distance with G0.
- Traverse to final drilling depth with G1 and the feedrate programmed before the program call
- Dwell time at final drilling depth
- Spindle stop with M5 (_ZSD[5]=1) or
- spindle and program stop with M5 M0 (_ZSD[5]=0). Press the NC START key after program stop.
- Retraction to retraction plane with G0.



© Siemens AG 2004 All rights reserved.

Explanation of parameters

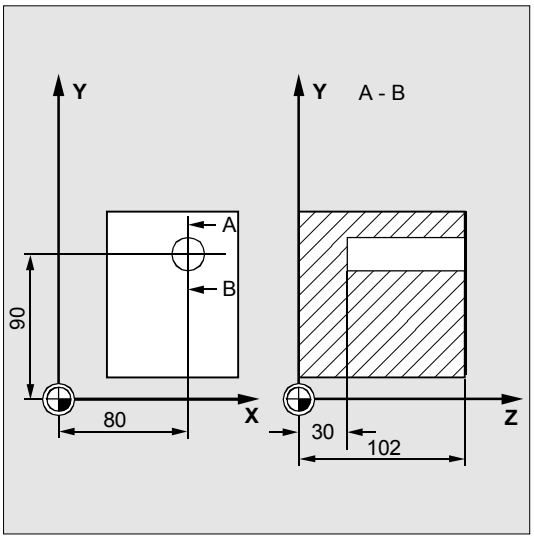


See Subsection 2.1.2 (Drilling, Centering – CYCLE81)
See Section 3.2 for cycle setting data _ZSD[5].

DTB (dwell time)
Parameter DTB is the dwell time at the final drilling depth (chip breaking) in seconds.

SDIR (direction of rotation)
The programmed direction of rotation is active for the movement to the final drilling depth.
If values other than 3 or 4 (M3/M4) are programmed, alarm 61102 "No spindle direction programmed".
is output and the cycle is aborted.

Programming example



Fourth boring pass
Cycle CYCLE88 is called at position X80 Y90 in the ZX plane. The boring axis is the Z axis. The safety distance is programmed as 3mm. The final drilling depth is defined as a value relative to the reference plane.
M4 is active in the cycle.

```
DEF REAL RFP, RTP, DPR, DTB, SDIS          ;Definition of parameters
N10 RFP=102 RTP=105 DPR=72 DTB=3 SDIS=3    ;Value assignments
N20 G17 G90 T1 D1 F100 S450                ;Specification of technology values
N21 M6
N30 G0 X80 Y90 Z105                        ;Traverse to drilling position
N40 CYCLE88 (RTP, RFP, SDIS, , DPR, ->     ;Cycle call with programmed
-> DTB, 4)                                 ;spindle direction M4
N50 M30                                    ;End of program
```

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

2.1.11 Boring 5 – CYCLE89

Programming

```
CYCLE89 (RTP, RFP, SDIS, DP, DPR, DTB)
```

Parameters

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Final drilling depth (absolute)
DPR	real	Final drilling depth relative to reference plane (enter without sign)
DTB	real	Dwell time at final drilling depth (chip breaking)

Function

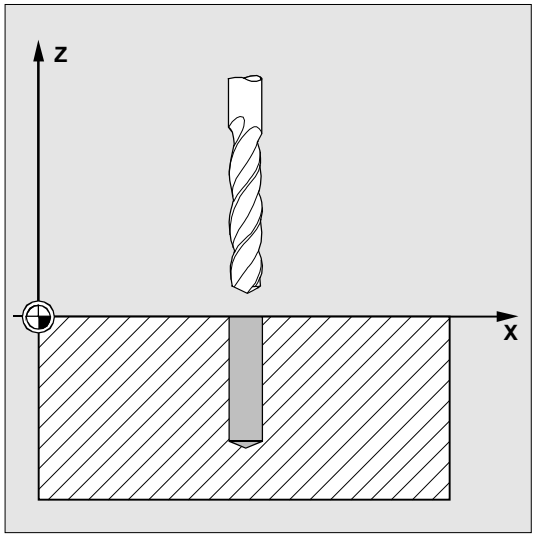
The tool drills at the programmed spindle speed and feedrate to the programmed final drilling depth. Once the final drilling depth has been reached a dwell time can be programmed.

Sequence

Position reached before the beginning of the cycle:
The drilling position is the position in the two axes of the selected plane.

The cycle implements the following motion sequence:

- Approach of the reference plane brought forward by the safety distance with G0.
- Traverse to final drilling depth with G1 and the feedrate programmed before the program call
- Dwell time at final drilling depth
- Retraction to the reference plane brought forward by the safety distance with G1 and the same feedrate value
- Retraction to retraction plane with G0.

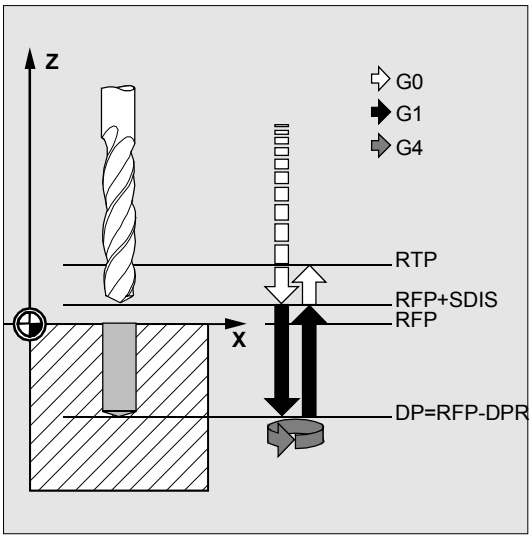


© Siemens AG 2004 All rights reserved.

Explanation of parameters

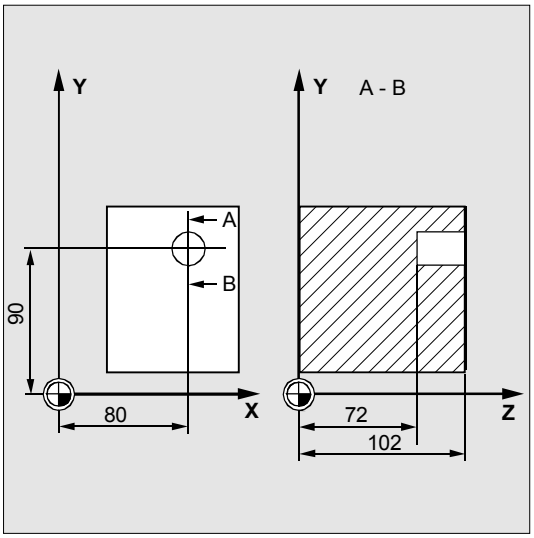
See Subsection 2.1.2 (Drilling, Centering – CYCLE81)

DTB (dwell time)
Parameter DTB is the dwell time at the final drilling depth (chip breaking) in seconds.



Programming example

Fifth boring pass
Boring cycle CYCLE89 is called at position X80 Y90 in the XY plane with a safety distance of 5mm and the final drilling depth specified as an absolute value. the boring axis is the Z axis.




```
DEF REAL RFP, RTP, DP, DTB                ;Definition of parameters
RFP=102 RTP=107 DP=72 DTB=3               ;Value assignments
N10 G90 G17 F100 S450 M4                  ;Specification of technology values
N20 G0 T1 D1 X80 Y90 Z107                 ;Traverse to drilling position
N21 M6
N30 CYCLE89 (RTP, RFP, 5, DP, , DTB)      ;Cycle call
N40 M30                                   ;End of program
```


© Siemens AG 2004 All rights reserved.

2.2 Modal call of drilling cycles

With NC programming it is possible to call any subroutine modally.
This feature is of special importance for drilling cycles.

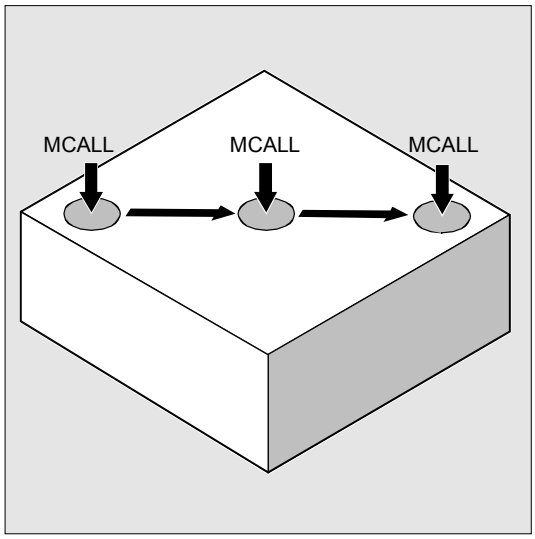
Programming

Modal call of a subroutine
```
MCALL
```

with drilling cycle (for example)
```
MCALL CYCLE81 (RTP, RFP, SDIS, DP, DPR)
```

Function

In NC programming, subroutines and cycles can be called modally, also i.e. maintaining the parameters previous values.
You generate a modal subroutine call by programming the keyword MCALL (modal subroutine call) in front of the subroutine name. This function causes the subroutine to be called and executed automatically after each block that contains traversing movement. The function is deactivated by programming MCALL without a subroutine name or by a new modal call of another subroutine.



Nesting of modal calls is not permissible, i.e., subroutines that are called modally cannot contain any further modal subroutine calls.

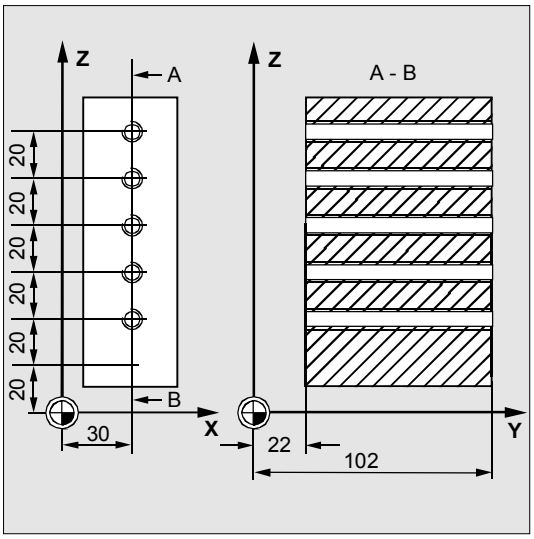
Any number of modal drilling cycles can be programmed, the number is not limited to a certain number of G functions reserved for this purpose.

© Siemens AG 2004 All rights reserved.

Programming example

Row of holes_5

With this program you can machine a row of 5 thread holes positioned parallel to the Z axis in the ZX plane. The distance between each of the holes is 20mm. The row of holes starts at Z20 and X30, the first hole in the row being 20mm from this point. In this example, the geometry of the row of holes has been programmed without using a cycle. First of all, drilling is performed with cycle CYCLE81 and then with CYCLE84 tapping (rigid). The holes are 80mm deep. This is the difference between the reference plane and the final drilling depth.



`DEF REAL RFP=102, DP=22, RTP=105, ->`	;Definition of parameters with
`-> PIT=4.2, SDIS`	;Value assignments
`DEF INT COUNT=1`	
`N10 SDIS=3`	;Value for safety distance
`N20 G90 F300 S500 M3 D1 T1`	;Specification of technology values
`N30 G18 G0 Y105 Z20 X30`	;Approach start position
`N40 MCALL CYCLE81 (RTP, RFP, SDIS, DP)`	;Modal call of the drilling cycle
`N50 MA1: G91 Z20`	;Traverse to next position (ZX plane) ;Cycle is executed
`N60 COUNT=COUNT+1` `N70 IF COUNT<6 GOTOB MA1`	;Loop for drilling positions along the row of ;holes
`N80 MCALL`	;Deselect modal call
`N90 G90 Y105 Z20`	;Approach starting position again
`N100 COUNT=1`	;Set counter to zero
`N110 ...`	;Change tool
`N120 MCALL CYCLE84 (RTP, RFP, SDIS, ->` `-> DP , , , 3, , PIT, , 400)`	;Modal call of tapping cycle
`N130 MA2: G91 Z20`	;Next drilling position
`N140 COUNT=COUNT+1` `N150 IF COUNT<6 GOTOB MA2`	;Loop for drilling position of the row of ;holes
`N160 MCALL`	;Deselect modal call
`N170 G90 X30 Y105 Z20`	;Approach starting position again
`N180 M30`	;End of program

Must be programmed in a single block

© Siemens AG 2004 All rights reserved.



Other Information

Explanation of this example

The modal call must be deselected in block N80 because in the next block the tool is traversed to a position where no drilling is to be performed.
It is advisable to store the drilling positions for a machining task of this type in a subroutine which is then called at MA1 or MA2.



The program given in the description of the drilling pattern cycles has been adapted and greatly simplified using these cycles.
The drilling pattern cycles are based on the call principle.
MCALL DRILLING CYCLE (...)
DRILLING PATTERN (...).

© Siemens AG 2004 All rights reserved.

2.3 Drilling pattern cycles

The drilling pattern cycles only describe the geometry of an arrangement of holes on a plane. The link to a drilling cycle is established via the modal call (see Section 2.2) of this drilling cycle before the drilling pattern cycle is programmed.

2.3.1 Preconditions

Drilling pattern cycles without drilling cycle call
Drilling pattern cycles can also be used for other applications without the drilling cycle first being called modally because the drilling pattern cycles can be parameterized without reference to the drilling cycle used.
If there was no modal call of the subroutine prior to calling the drilling pattern cycle, error message 62100 "No drilling cycle active" appears.
You can acknowledge this error message with the error acknowledgment key and continue program processing by pressing the NC Start key. The drilling pattern cycle then approaches each of the positions calculated from the input data one after the other without calling a subroutine at these points.

Behavior when quantity parameter is zero
The number of holes in a drilling pattern must be parameterized. If the value of the quantity parameter is zero when the cycle is called (or if this parameter is omitted from the parameter list), alarm
61103 "Number of holes is zero" is output
and the cycle is aborted.

Checks in the case of limited ranges of input parameter values
Generally there are no plausibility checks for defining parameters in the drilling pattern cycles if they are not expressly declared for a parameter with a description of the response.

© Siemens AG 2004 All rights reserved.

2.3.2 Row of holes – HOLES1

Programming

```
HOLES1 (SPCA, SPCO, STA1, FDIS, DBH, NUM)
```

Parameters

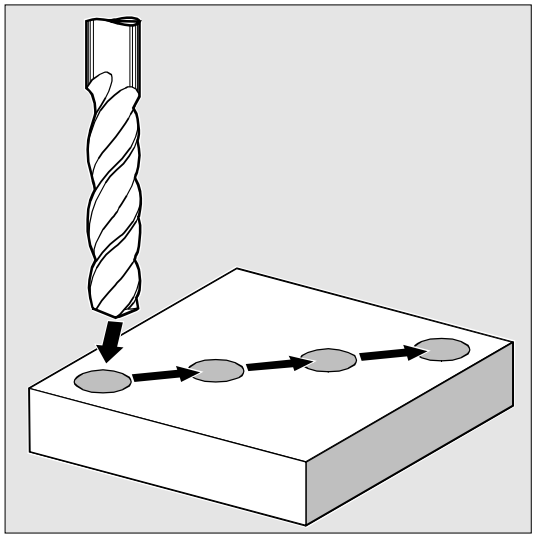
SPCA	real	Abscissa of a reference point on the straight line (absolute)
SPCO	real	Ordinate of this reference point (absolute)
STA1	real	Angle to abscissa Value range: –180<STA1<=180 degrees
FDIS	real	Distance between the first hole and the reference point (enter without sign)
DBH	real	Distance between the holes (enter without sign)
NUM	int	Number of drill holes

Function

With this cycle you can program a row of holes, i.e. a number of holes that lie along a straight line or a grid of holes. The type of hole is determined by the drilling cycle that has already been called modally.

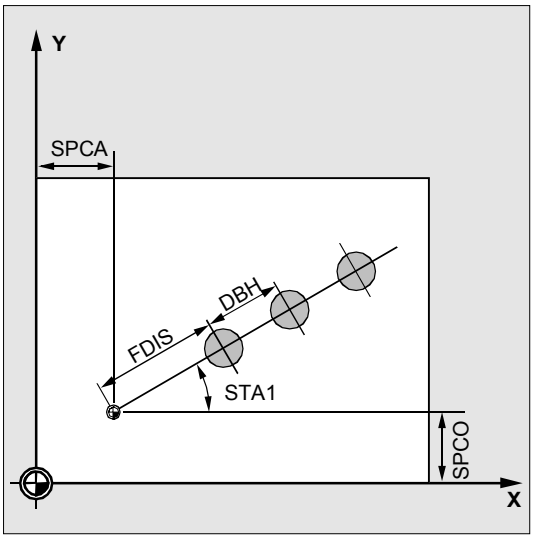
Sequence

To avoid unnecessary travel, the cycle calculates whether the row of holes is machined starting from the first hole or the last hole from the actual position of the plane axes and the geometry of the row of holes. The drilling positions are then approached one after the other in rapid traverse.



© Siemens AG 2004 All rights reserved.

Explanation of parameters



SPCA and SPCO (reference point abscissa and ordinate)
One point along the straight line of the row of holes is defined as the reference point for determining the distances between the holes. The distance to the first hole FDIS is defined from this point.

STA1 (angle)
The straight line can be in any position on the plane. It is specified both by the point defined by SPCA and SPCO and by the angle contained by the straight line and the abscissa of the workpiece coordinate system that is active when the cycle is called. The angle is entered under STA1 in degrees.

FDIS and DBH (distance)
Under FDIS you enter the distance between the first hole and the reference point defined under SPCA and SPCO. The parameter DBH contains the distance between any two holes.

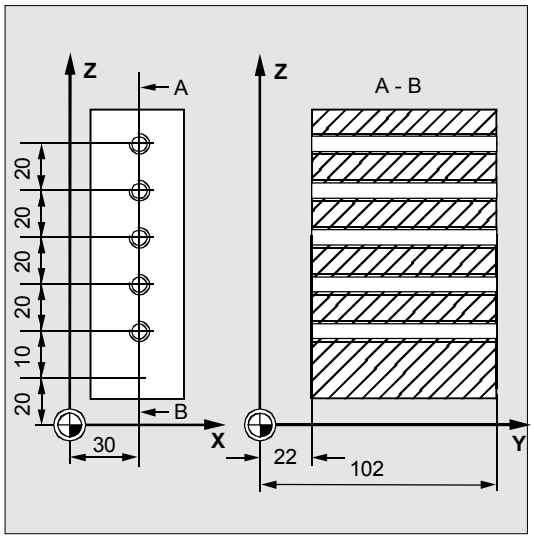
NUM (number)
You determine the number of holes with the parameter NUM.

© Siemens AG 2004 All rights reserved.

Programming example

row of holes

With this program you can machine a row of holes of five tapped holes positioned in parallel to the Z axis on the ZX plane, with a distance between each hole of 20mm. The row of holes starts at Z20 and X30, the first hole in the row being 10mm from this point. The geometry of the row of holes is described by the cycle HOLES1. First of all, drilling is performed with cycle CYCLE81 and then with CYCLE84 tapping (rigid). The holes are 80 mm deep. This is the difference between the reference plane and the final drilling depth.



`DEF REAL RFP=102, DP=22, RTP=105`
`DEF REAL SDIS, FDIS`
`DEF REAL SPCA=30, SPCO=20, STA1=0, ->`
`-> DBH=20`
`DEF INT NUM=5`
	;Definition of parameters with
	;value assignments

`N10 SDIS=3 FDIS=10`	;Value for safety distance and distance of ;the first hole to the reference point
`N20 G90 F30 S500 M3 D1 T1`	;Specification of technology values for the ;machining section
`N30 G18 G0 Z20 Y105 X30`	;Approach start position
`N40 MCALL CYCLE81 (RTP, RFP, SDIS, DP)`	;Modal call of drilling cycle
`N50 HOLES1 (SPCA, SPCO, STA1, FDIS, ->` `-> DBH, NUM)`	;Call of row of holes cycle, the cycle starts ;with the first hole. Only the drilling positions ;are approached in this cycle
`N60 MCALL`	;Deselect modal call
`...`	;Change tool
`N70 G90 G0 Z30 Y75 X105`	;Traverse to position next to 5th hole
`N80 MCALL CYCLE84 (RTP, RFP, SDIS, DP, ->` `-> , 3, , 4.2, , , 400)`	;Modal call of tapping cycle
`N90 HOLES1 (SPCA, SPCO, STA1, FDIS, ->` `-> DBH, NUM)`	;Call of row of holes cycle starting with ;the 5th hole in the row
`N100 MCALL`	;Deselect modal call
`N110 M30`	;End of program

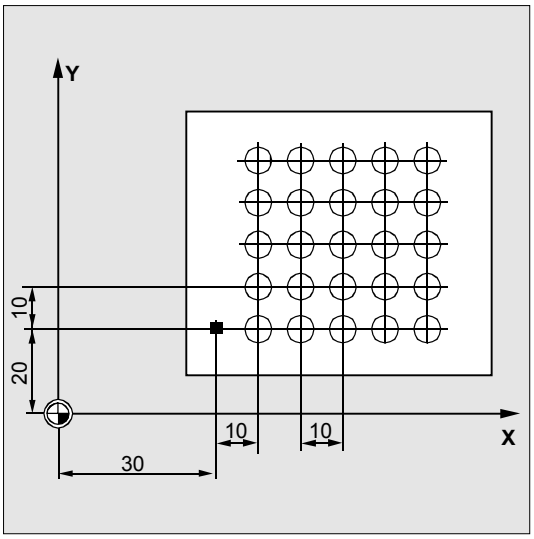
-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

Programming example

Grid of holes

With this program you can machine a grid of holes consisting of five rows of five holes each that lie in the XY plane at a distance of 10mm from one another. The starting point of the grid is X30 Y20.




```
DEF REAL RFP=102, DP=75, RTP=105, SDIS=3      ;Definition of parameters with
DEF REAL SPCA=30, SPCO=20, STA1=0, ->         ;value assignments
-> DBH=10, FDIS=10
DEF INT NUM=5, LINENUM=5, COUNT=0
```

```
DEF REAL LINEDIST
```

```
N10 LINEDIST=DBH                              ;Distance between rows = distance between
                                              ;holes
N20 G90 F300 S500 M3 D1 T1                    ;Specification of technology values
N30 G17 G0 X=SPCA-10 Y=SPCO Z105              ;Approach start position
N40 MCALL CYCLE81 (RTP, RFP, SDIS, DP)        ;Modal call of drilling cycle
N50 LABEL1: HOLES1 (SPCA, SPCO, STA1, ->      ;Call of row of holes cycle
-> FDIS, DBH, NUM)
N60 SPCO=SPCO+LINEDIST                        ;Ordinate of reference point for the next line
N70 COUNT=COUNT+1                             ;Jump back to LABEL1 if the condition is
N80 IF COUNT<LINENUM GOTOB LABEL1             ;fulfilled
N90 MCALL                                     ;Deselect modal call
N100 G90 G0 X=SPCA-10 Y=SPCO Z105             ;Approach start position
N110 M30                                      ;End of program
```

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

2.3.3 Circle of holes – HOLES2



Programming

```
HOLES2 (CPA, CPO, RAD, STA1, INDA, NUM)
```



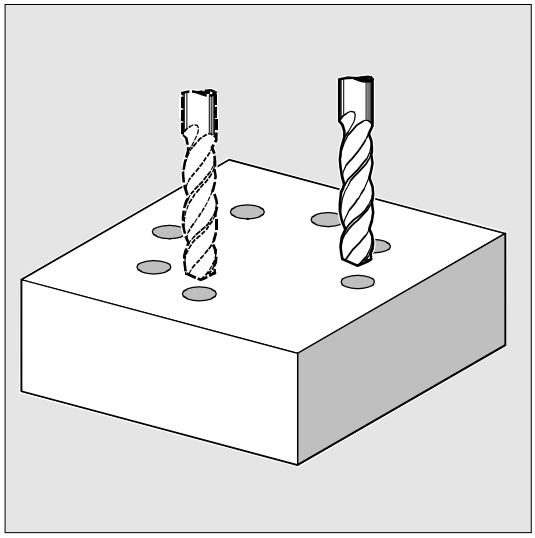
Parameters

CPA	real	Center point of hole circle, abscissa (absolute)
CPO	real	Center point of hole circle, ordinate (absolute)
RAD	real	Radius of circle of holes (enter without sign)
STA1	real	Initial angle Value range: –180<STA1<=180 degrees
INDA	real	Indexing angle
NUM	int	Number of drill holes



Function

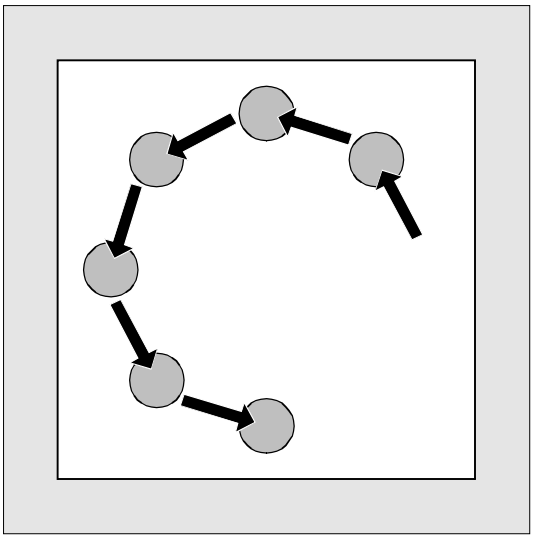
A circle of holes can be machined with this cycle. The machining plane must be defined before the cycle is called.
The type of hole is determined by the drilling cycle that has already been called modally.



© Siemens AG 2004 All rights reserved.

Sequence

In the cycle, the drilling positions are approached one after the other on the plane with G0.

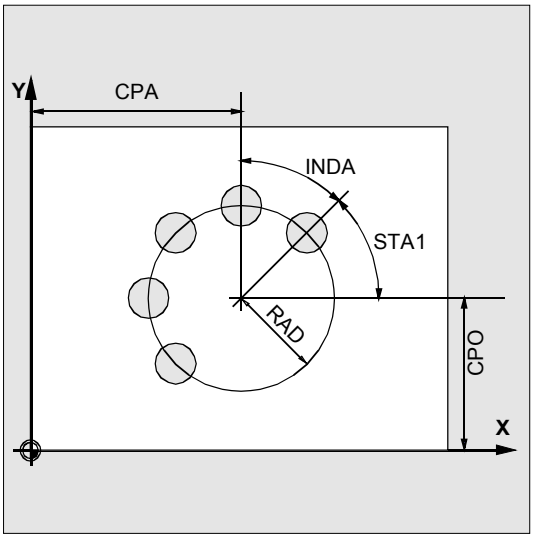


Explanation of parameters

CPA, CPO and RAD (center point and radius abscissa, ordinate)
The position of the circle of holes in the machining plane is defined by the center point (parameters CPA and CPO) and the radius (parameter RAD). Only positive values are permissible for the radius.

STA1 and INDA (start angle and indexing angle)
The arrangement of the holes in the circle is defined by these parameters.
Parameter STA1 defines the angle of rotation between the positive direction of the abscissa in the coordinate system active before the cycle was called and the first hole. Parameter INDA contains the angle of rotation from one hole to the next.
If parameter INDA is set to zero, the indexing angle is calculated internally from the number of holes so as to space them equally in a circle.

NUM (number)
You determine the number of holes with the parameter NUM.

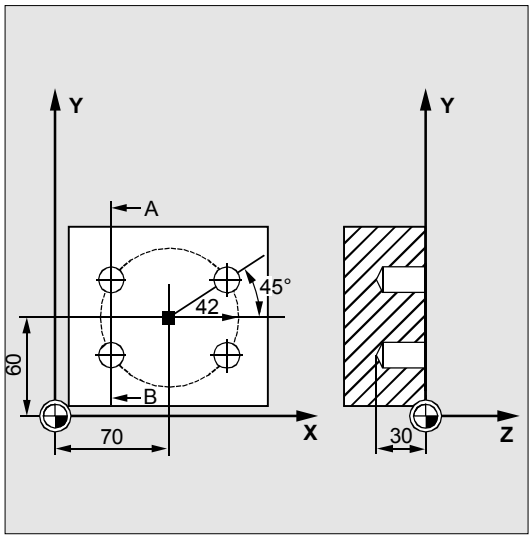


© Siemens AG 2004 All rights reserved.

Programming example

Hole circle

The program uses CYCLE82 4 to produce holes with a depth of 30mm. The final drilling depth is defined as a value relative to the reference plane. The circle is defined by the center point X70 Y60 and the radius 42mm in the XY plane. The initial angle is 45 degrees. The safety distance in the drilling axis Z is 2mm.



`DEF REAL CPA=70,CPO=60,RAD=42,STA1=45` `DEF INT NUM=4`	;Definition of parameters with ;value assignments
`N10 G90 F140 S710 M3 D1 T40`	;Specification of technology values
`N20 G17 G0 X50 Y45 Z2`	;Approach start position
`N30 MCALL CYCLE82 (2, 0,2, , 30)`	;Modal call of drilling cycle, without dwell ;time, DP is not programmed
`N40 HOLES2 (CPA, CPO, RAD, STA1, , NUM)`	;Call of circle of holes cycle, the indexing ;angle is calculated internally by the cycle as ;parameter INDA has been omitted
`N50 MCALL`	;Deselect modal call
`N60 M30`	;End of program

© Siemens AG 2004 All rights reserved.

2.3.4 Point grid – CYCLE801



Programming

```
CYCLE801 (_SPCA, _SPCO, _STA, _DIS1,
_DIS2, _NUM1, _NUM2)
```



Parameters

_SPCA	real	Reference point for grid of holes in the 1st axis, abscissa (absolute)
_SPCO	real	Reference point for grid of holes in the 2nd axis, abscissa (absolute)
_STA	real	Angle to abscissa
_DIS1	real	Distance between columns (without sign)
_DIS2	real	Distance between rows (without sign)
_NUM1	int	Number of columns
_NUM2	int	Number of rows



Function

Cycle CYCLE801 can be used to machine a "grid of holes". The type of hole is determined by the drilling cycle that has already been called modally.

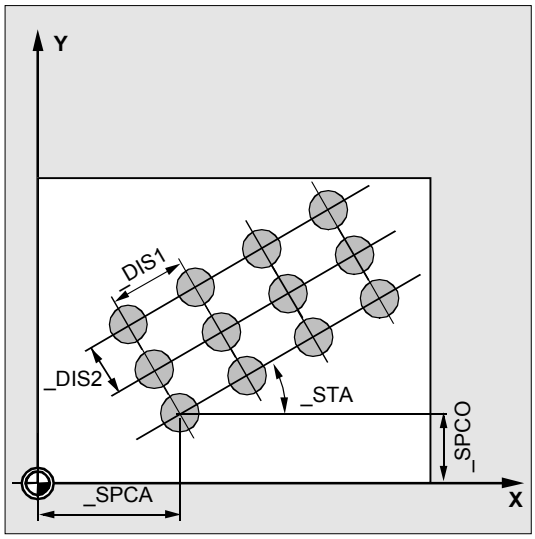


Sequence

The cycle calculates the sequence of holes such that the empty paths between them are kept as short as possible. The starting position of the machining operation is defined according to the last position reached in the plane prior to the cycle call.
Starting positions are one of the four possible corner positions in each case.

© Siemens AG 2004 All rights reserved.

Explanation of parameters



_SPCA and _SPCO (reference point abscissa and ordinate)
These two parameters determine the first point of the hole grid. The row and column distances are specified in relation to this point.

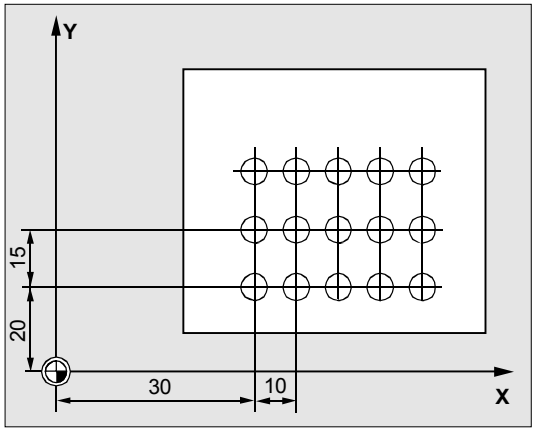
_STA (angle)
The grid of holes can positioned at any angle in the plane. This angle is programmed in degrees in _STA and refers to the abscissa of the workpiece coordinate system active as the cycle is called.

_DIS1 and _DIS2 (column and row distances)
The distances must be entered without sign. To avoid unnecessary empty travel, the dot matrix is machined line by line or column by column based on a comparison of distance measurements.

_NUM1 and _NUM2 (number)
This parameter determines the number of columns or lines.

Programming example

Cycle CYCLE801 is used to machine a dot matrix, consisting of 15 holes in three lines
and five columns. The associated drilling program is called modally beforehand.




```
N10 G90 G17 F900 S4000 M3 T2 D1          ;Specification of technology values
N15 MCALL CYCLE82(10,0,1,-22,0,0)        ;Modal call of a drilling cycle
N20 CYCLE801(30,20,0,10,15,5,3)          ;Call dot matrix
N25 M30                                  ;End of program
```

■

© Siemens AG 2004 All rights reserved.

Notes

© Siemens AG 2004 All rights reserved.

Milling Cycles

3.1 General notes ... 3-106
3.2 Preconditions ... 3-107
3.3 Thread milling - CYCLE90 ... 3-109
3.4 Elongated holes on a circle - LONGHOLE ... 3-116
3.5 Slots on a circle - SLOT1 ... 3-121
3.6 Circumferential slot - SLOT2 ... 3-129
3.7 Milling rectangular pockets - POCKET1 ... 3-135
3.8 Milling circular pockets - POCKET2 ... 3-139
3.9 Milling rectangular pockets - POCKET3 ... 3-143
3.10 Milling circular pockets - POCKET4 ... 3-152
3.11 Face milling - CYCLE71 ... 3-157
3.12 Path milling - CYCLE72 ... 3-163
3.13 Milling rectangular spigots - CYCLE76 ... 3-173
3.14 Milling circular spigots - CYCLE77 ... 3-178
3.15 Pocket milling with islands - CYCLE73, CYCLE74, CYCLE75 ... 3-182
3.15.1 Transfer pocket edge contour - CYCLE74 ... 3-183
3.15.2 Transfer island contour - CYCLE75 ... 3-185
3.15.3 Contour programming ... 3-186
3.15.4 Pocket milling with islands - CYCLE73 ... 3-188
3.16 Swiveling - CYCLE800 ... 3-210
3.16.1 Operation, parameter assignment, input screen form ... 3-213
3.16.2 Operating instructions ... 3-217
3.16.3 Parameters ... 3-218
3.16.4 Starting up CYCLE800 ... 3-222
3.16.5 User cycle TOOLCARR.spf ... 3-239
3.16.6 Error messages ... 3-245
3.17 High speed settings - CYCLE832 (SW 6.3 and later) ... 3-246
3.17.1 Call CYCLE 832 in HMI menu tree ... 3-249
3.17.2 Parameters ... 3-252
3.17.3 Customizing the technology ... 3-253
3.17.4 Customizing additional program parameters CYC_832T ... 3-255
3.17.5 Interfaces ... 3-257
3.17.6 Error messages ... 3-258
3.18 Engraving cycle CYCLE60 (SW 6.4 and later) ... 3-259

© Siemens AG 2004 All rights reserved.

3.1 General notes

The following sections describe how milling cycles are programmed.

This section is intended to guide you in selecting cycles and assigning them with parameters. In addition to a detailed description of the function of the individual cycles and the corresponding parameters, you will also find a sample program at the end of each section to familiarize you with the use of cycles.

The Sections are structured as follows:

- **Programming**
- **Parameters**
- **Function**
- **Sequence**
- **Explanation of parameters**
- **Other Information**
- **Programming example**

"Programming" and "Parameters" explain the use of cycles sufficiently for the experienced user, whereas beginners can find all the information they need for programming cycles under "Function", "Sequence of operations", "Explanation of parameters", "Additional notes" and the "Sample program".

© Siemens AG 2004 All rights reserved.

3.2 Preconditions

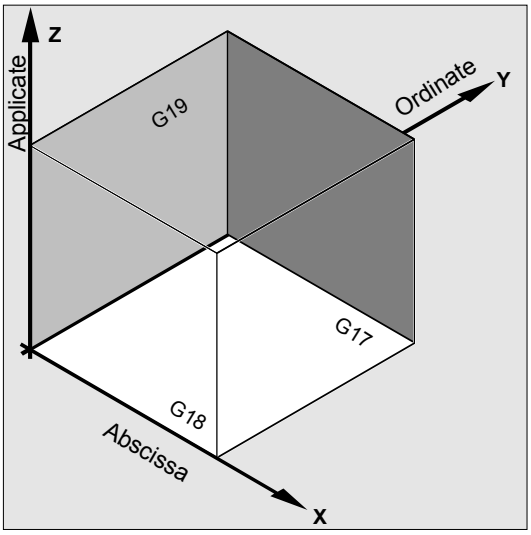
Programs required in the control
The milling cycles call the programs

- MESSAGE.SPF and
- PITCH.SPF

internally as subroutines. Moreover, you need the data block GUD7.DEF
and the macro definition file SMAC.DEF.
Load them in the parts program memory of the control unit before executing the milling cycles.

Call and return conditions
Milling cycles are programmed independently of the actual axis names. You must activate a tool offset before you call the milling cycles.
The required values for the feedrate, spindle speed and spindle direction of rotation must be programmed in the parts program if no parameters are available for these in the milling cycle.
The center point coordinates of the milling pattern or the pocket to be machined are programmed in the right-handed coordinate system.
The G functions and current programmable frame active before the cycle was called remain active beyond the cycle.

Level definition
Milling cycles generally assume that the current workpiece coordinate system has been defined by selecting a plane (G17, G18 or G19) and activating a programmable frame (if necessary). The infeed axis is always the 3rd axis of the coordinate system



References: /PG/, Programming Guide Fundamentals
/PGA/, Programming Guide Job planning

© Siemens AG 2004 All rights reserved.

Spindle handling

The spindle commands in the cycles always refer to the active master spindle of the control.
If you want to use a cycle on a machine with several spindles, you must first define the spindle that is to be used as the master spindle with the command SETMS



References: /PG/, Programming Guide Fundamentals
/PGA/, Programming Guide Job planning

Machining status messages

Status messages are displayed on the control monitor during the processing of milling cycles.
The following messages can be displayed:

- "Elongated hole <No.>(first figure) being machined“
- "Slot <No.>(other figure) being machined“
- "Circumferential slot <No.>(last figure) being machined“.

In each case <No.> stands for the number of the figure that is currently being machined.

These messages do not interrupt program processing and continue to be displayed until the next message is displayed or the cycle is completed.

Cycle setting data

Some of the parameters and the behavior milling cycles can be altered by cycle setting.
The cycle setting data are defined in data block GUD7.DEF.

The following new cycle setting data are introduced:

_ZSD[x]	Value	Meaning	Cycles affected
_ZSD[1]	0	Depth computation in the new cycles is made between the reference plane + safety distance and depth (_RFP + _SDIS - _DP)	POCKET1 to POCKET4, LONGHOLE, CYCLE71, SLOT1, CYCLE72, SLOT2
	1	Depth computation is made without including safety distance	
_ZSD[2]	0	Dimension of rectangular pocket or rectangular spigot from the center point	POCKET3 CYCLE76
	1	Dimension of rectangular pocket or rectangular spigot from a corner	
_ZSD[5]	0	Execute at drilling depth M5 M0	CYCLE88
	1	Execute at drilling depth M5	

© Siemens AG 2004 All rights reserved.

3.3 Thread milling - CYCLE90

Programming

```
CYCLE90 (RTP, RFP, SDIS, DP, DPR, DIATH, KDIAM, PIT, FFR, CDIR, TYPTH,
CPA, CPO)
```

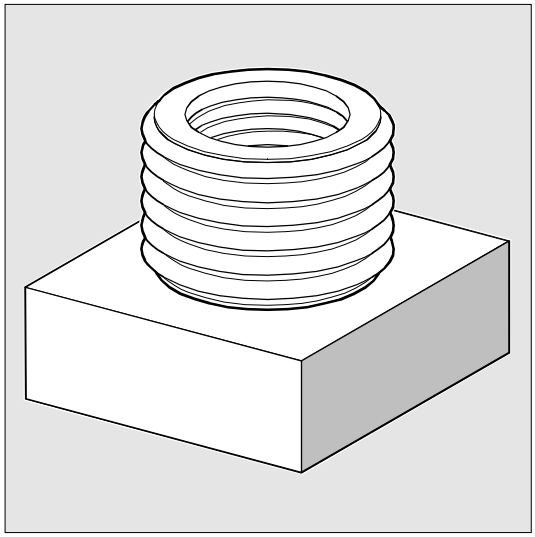
Parameters

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Final drilling depth (absolute)
DPR	real	Final drilling depth relative to reference plane (enter without sign)
DIATH	real	Nominal diameter, outside diameter of thread
KDIAM	real	Core diameter, inside diameter of thread
PIT	real	Thread pitch; Value range: 0.001 ... 2000.000 mm
FFR	real	Feedrate for thread milling (enter without sign)
CDIR	int	Direction of rotation for thread milling Values: 2 (for thread milling with G2) 3 (for thread milling with G3)
TYPTH	int	Thread type: Values: 0= inside thread 1=outside thread
CPA	real	Center point of circle, abscissa (absolute)
CPO	real	Center point of circle, ordinate (absolute)

Function

You can produce inside and outside threads with cycle CYCLE90. The path in thread milling is based on helical interpolation. All three geometrical axes of the current plane which you define before calling the cycle are involved in this movement.

The programmed feedrate F depends on the axis grouping defined in the FGROUP instruction before the cycle call



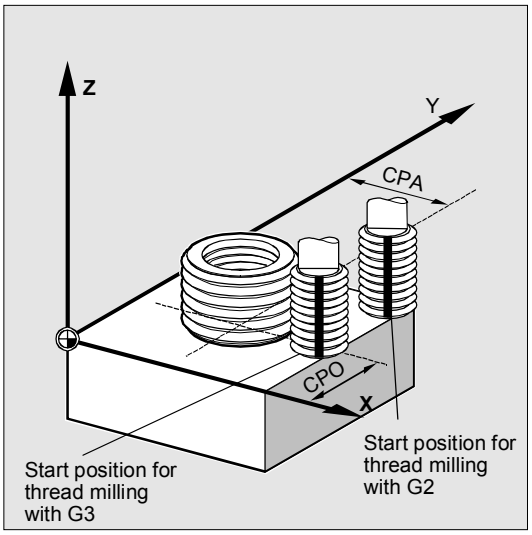
References: /PG/, Programming Guide Fundamentals
/PGA/, Programming Guide Job planning

© Siemens AG 2004 All rights reserved.

Sequence

Outside threads

Position reached before the beginning of the cycle:
This can be any position from which the starting position on the outside diameter of the thread at the retraction plane level can be reached without collision. This start position for thread milling with G2 lies between the positive abscissa and the positive ordinate in the current level (i.e. in the 1st quadrant of the coordinate system). For thread milling with G3, the start position lies between the positive abscissa and the negative ordinate (i.e. in the 4th quadrant of the coordinate system).
The distance from the thread diameter depends on the thread size and the tool radius used.



The cycle implements the following motion sequence:

- Travel to the starting point with G0 at the retraction plane level in the applicate of the current plane
- Infeed to the reference plane brought forward by the safety distance with G0
- Movement to the thread diameter along a circular path in the direction G2/G3 opposite to that defined in CDIR
- Thread milling along a helical path with G2/G3 and feedrate FFR
- Travel-out movement along a circular path in the opposite direction G2/G3 and the reduced feedrate FFR
- Retraction to retraction plane in the applicate with G0

© Siemens AG 2004 All rights reserved.

Inside threads

Position reached before the beginning of the cycle:
This can be any position from which the starting position on the center point of the thread at the retraction plane level can be approached without collision.

The cycle implements the following motion sequence:

- Travel to the center point of the thread with G0 at the retraction plane level in the applicate of the current plane
- Infeed to the reference plane brought forward by the safety distance with G0
- Approach with G1 and the reduced feedrate FFR along an approach circle calculated in the cycle
- Movement to the thread diameter along a circular path in the direction G2/G3 defined in CDIR
- Thread milling along a helical path with G2/G3 and feedrate FFR
- Travel-out movement along a circular path with the same direction of rotation and the reduced feedrate FFR
- Retraction to the center point of the thread with G0
- Retraction to retraction plane in the applicate with G0

Thread from bottom to top
For technological reasons, it may be preferable to machine the thread from the bottom to the top. The retraction plane RTP is then below the thread depth DP.
This machining operation is possible, the depth data must be programmed as absolute values and before cycle start, the machine must be positioned on the retraction plane or one position behind the retraction plane.

© Siemens AG 2004 All rights reserved.

Programming example
(thread from bottom to top)

A thread must be cut starting from -20 up to 0 with a 3 mm pitch. The retraction plane is at 8.

```
N10 G17 X100 Y100 S300 M3 T1 D1 F1000
N20 Z8
N30 CYCLE90 (8, -20, 0, 0, 0, 46, 40, 3, 800, 3, 0, 50, 50)
N40 M2
```

The hole must have at least a depth of -21.5 (half pitch in excess).

Overshoot in the thread longitudinal direction

For thread milling, the travel-in and travel-out movements occur along all three axes concerned. This means that the travel-out movement includes a further step in the vertical axis, beyond the programmed thread depth.
The overshoot is calculated:

$$z \quad \frac{p}{4} * \frac{2*WR \quad RDIFF}{DIATH}$$

ΔZ	Overshoot, internal
p	Pitch
WR	Tool radius
DIATH	Outside diameter of the thread
RDIFF	Radius difference for travel-out

For inside threads RDIFF = DIATH/2 - WR,
For outside threads RDIFF = DIATH/2 + WR.

Function extension on SW 6.3 and later
The feedrate is converted to the tool cutting edge. For the inside thread, the travel-in and travel-out movement now takes a technologically more favorable semicircular path whose radius is calculated internally by the cycle to match the tool. This results in calculation within the cycle of the overshoot by the following formula:

© Siemens AG 2004 All rights reserved.

- For tool diameters <2/3 nominal diameter

$$\Delta z = \frac{p}{2} * \frac{WR + DIATH/2}{DIATH}$$

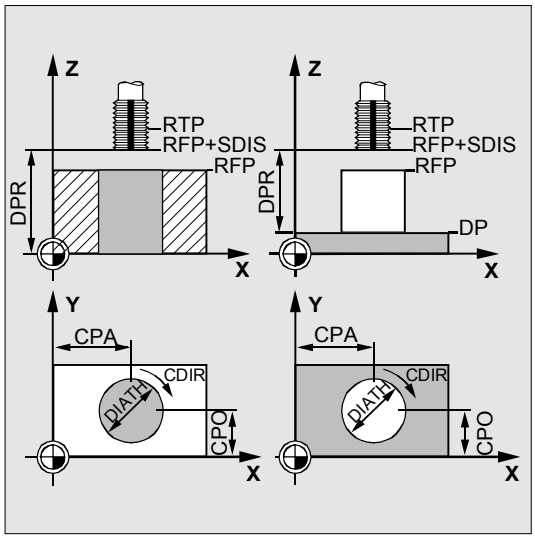
- For tool diameters ≥ 2/3 nominal diameter

$$\Delta z = \frac{p/2 * KDIAM}{DIATH}$$

ΔZ	Overshoot, internal
p	Pitch
WR	Tool radius
DIATH	Outside diameter of the thread
KDIAM	Core diameter of the thread

Explanation of parameters

See Subsection 2.1.2 (Drilling, Centering – CYCLE81)



DIATH, KDIAM and PIT (nominal diameter, core diameter and thread pitch)
With these parameters you define the thread data such as, nominal diameter, core diameter and pitch. Parameter DIATH is the outside diameter and KDIAM the inside diameter of the thread. The travel-in and travel-out movements are generated by the cycle based on these parameters.

FFR (feedrate)
The value of parameter FFR is defined as the current feedrate value for thread milling. In thread milling it is active for the movement along the helical path.
This value is reduced in the cycle for the travel-in and travel-out movements. Retraction is performed outside the helical path with G0.

CDIR (direction of rotation)
You define the value for the machining direction of the thread in this parameter.
If the parameter is assigned an invalid value, the following alarm is output:
"Wrong mill direction: G3 will be generated"
In this case the cycle is continued and G3 is automatically generated.

© Siemens AG 2004 All rights reserved.

TYPTH (thread type)
With parameter TYPTH you determine whether an outside or inside thread is to be machined.

CPA and CPO (center point)
With these parameters you define the center point of the hole or spigot on which the thread is to be machined.



Other Information

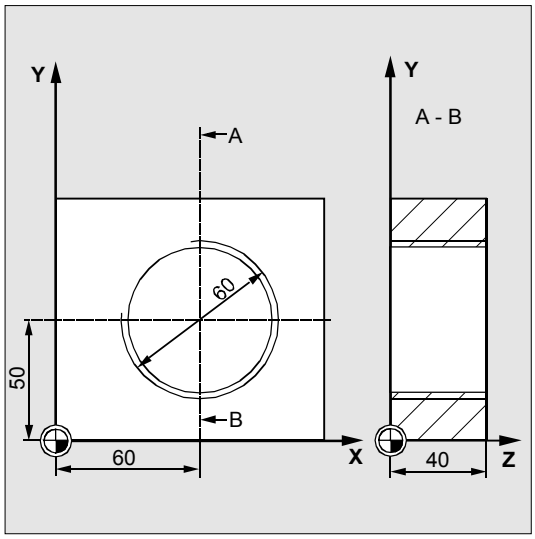
The milling cutter radius is taken into account by the cycle. A tool offset must therefore be programmed before the cycle is called. Otherwise, the following alarm is output:
61000 "No tool offset active"
and the cycle is aborted.
When the tool radius equals zero or a negative value, the cycle is also aborted with this alarm.
With inside threads, the tool radius is monitored and alarm 61105 "Cutter radius too large"
is output and the cycle is aborted.

© Siemens AG 2004 All rights reserved.

Programming example

Internal thread

With this program you can machine an inside thread at position X60 Y50 on the G17 plane.




```
DEF REAL RTP=48, RFP=40, SDIS=5, ->          ;Definition of variables with
-> DPR=40, DIATH=60, KDIAM=50                ;value assignments
DEF REAL PIT=2, FFR=500, CPA=60,CPO=50
DEF INT CDIR=2, TYPTH=0
N10 G90 G0 G17 X0 Y0 Z80 S200 M3             ;Approach start position
N20 T5 D1                                    ;Specification of technology values
N30 CYCLE90 (RTP, RFP, SDIS, DPR, ->         ;Cycle call
-> DIATH, KDIAM, PIT, FFR, CDIR, ->
-> TYPTH, CPA, CPO)
N40 G0 G90 Z100                              ;Approach position after cycle
N50 M02                                      ;End of program
```

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

3.4 Elongated holes on a circle - LONGHOLE

Programming

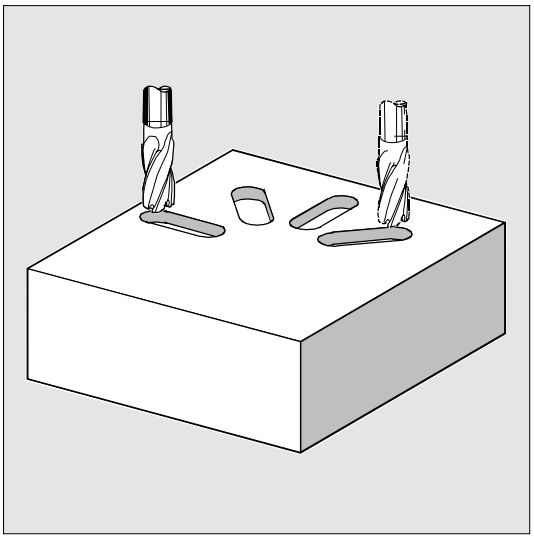
```
LONGHOLE (RTP, RFP, SDIS, DP, DPR, NUM, LENG, CPA, CPO, RAD, STA1, INDA,
FFD, FFP1, MID)
```

Parameters

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Elongated hole final drilling depth (absolute)
DPR	real	Elongated hole final drilling depth relative to reference plane (enter without sign)
NUM	integer	Number of elongated holes
LENG	real	Length of elongated hole (enter without sign)
CPA	real	Center point of circle, abscissa (absolute)
CPO	real	Center point of circle, ordinate (absolute)
RAD	real	Radius of circle (enter without sign)
STA1	real	Initial angle
INDA	real	Indexing angle
FFD	real	Feedrate for depth infeed
FFP1	real	Feedrate for surface machining
MID	real	Maximum infeed depth for infeed (enter without sign)

The cycle requires a milling cutter with an "end tooth cutting over center" (DIN 844).

Function

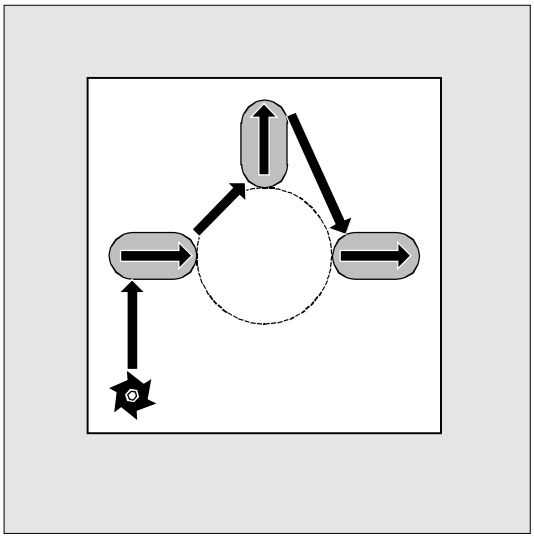


Elongated holes arranged on a circle can be machined with this cycle. The longitudinal axis of the elongated holes is arranged radially.
Unlike the slot, the width of the elongated hole is determined by the diameter of the tool.
To avoid unnecessary travel, the cycle calculates the most optimum path. If several depth infeed movements are required to machine an elongated hole, the infeed is performed at alternate end points. The path to be traversed in the plane along the longitudinal axis of the elongated hole changes direction after every infeed.
The cycle automatically looks for the shortest path when changing to the next elongated hole.

© Siemens AG 2004 All rights reserved.

Sequence

Position reached before the beginning of the cycle:
The starting position can be any position from which each of the elongated holes can be approached without collision.

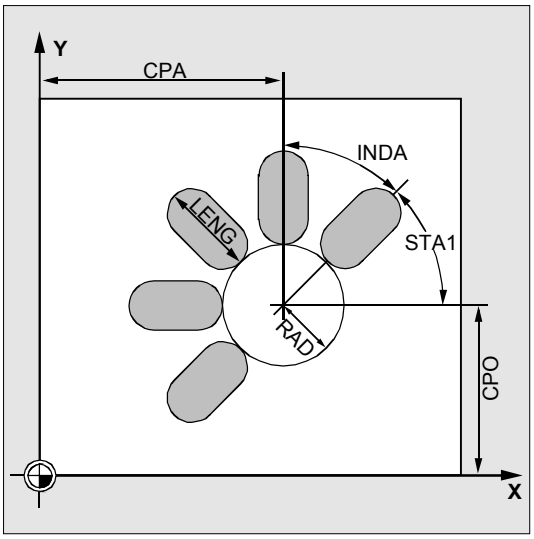


The cycle implements the following motion sequence:

- The starting position of a cycle is approached with G0. The nearest end point of the first elongated hole to be machined is approached in both axes of the current plane at the retraction plane level in the applicate of this plane and then lowered in the applicate to the reference plane brought forward by the safety distance.
- Each elongated hole is milled in a reciprocating movement. Machining is performed in the plane with G1 and the feedrate programmed under FFP1. At each reversal point, the infeed to the next machining depth calculated by the cycle is performed with G1 and the feedrate FFD until the final depth is reached.
- Retraction to the retraction plane with G0 and approach to the next elongated hole along the shortest path.
- When the last elongated hole has been machined, the tool is traversed from the last position reached in the machining plane to the retraction plane with G0 and the cycle is terminated.

© Siemens AG 2004 All rights reserved.

Explanation of the parameters



See Section 2.1.2. (Drilling, Centering – CYCLE81) for a description of parameters RTP, RFP, SDIS.
See Section 3.2 for cycle setting data _ZSD[1].

DP and DPR (elongated hole depth)
The elongated hole depth can be defined as either absolute (DP) or relative (DPR) to the reference plane. If it is entered as a relative value, the cycle automatically calculates the correct depth on the basis of the positions of the reference and retraction planes.

NUM (number)
The number of elongated holes us determined with the parameter NUM.

LENG (elongated hole length)
The elongated hole length is programmed under LENG. If it is detected during the cycle run that this length is less than the cutter diameter, then the cycle is aborted with alarm 61105 "Cutter radius too large".

MID (infeed depth)
The maximum infeed depth is defined with this parameter.
The depth infeed is performed by the cycle in equally sized infeed steps.
Using MID and the total depth, the cycle automatically calculates this infeed which lies between 0.5x maximum infeed depth and the maximum infeed depth.
The minimum possible number of infeed steps is used as the basis. _MID=0 means that the cut to pocket depth is made with one infeed.
The depth infeed commences at the reference plane moved forward by the safety distance (as a function of _ZSD[1]).

FFD and FFP1 (feedrate depth and plane)
Feedrate FFP1 is active for all traversing movements performed in the plane at feedrate. FFD is active for infeeds that are perpendicular to this plane.

CPA, CPO and RAD (center point and radius)
The position of the circle in the machining plane is defined by the center point (parameters CPA and CPO) and the radius (parameter RAD). Only positive values are permissible for the radius.

© Siemens AG 2004 All rights reserved.

STA1 and INDA (start angle and indexing angle)
The arrangement of the elongated holes around the circle is defined by these parameters.
If INDA=0 the indexing angle is calculated from the number of elongated holes so that they are equally distributed around the circle.



Other Information

A tool offset must be activated before the cycle is called. The cycle is otherwise aborted with alarm 61000 "No tool offset active".

If incorrect values are assigned to the parameters that determine the arrangement and size of the elongated holes and thus cause mutual contour violation of the elongated holes, the cycle is not started. The cycle is aborted after the error message
61104 "Contour violation of slots/elongated holes".

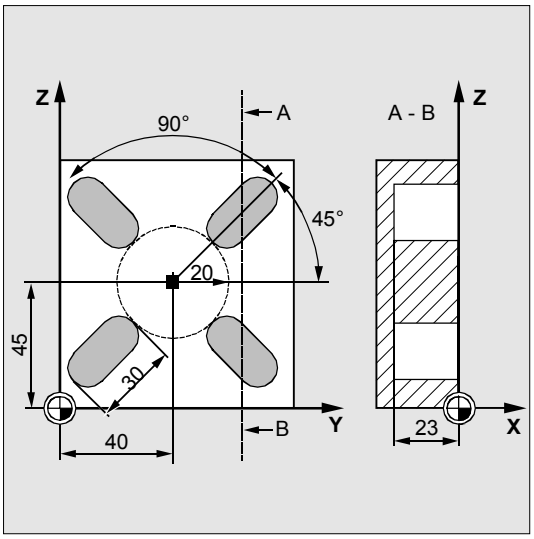
During the cycle, the workpiece coordinate system is shifted and rotated. The values in the workpiece coordinate system are shown on the actual value display as if the longitudinal axis of the elongated hole being machined were positioned on the first axis of the current machining plane.
When the cycle is completed, the workpiece coordinate system is again in the same position as it was before the cycle was called.

© Siemens AG 2004 All rights reserved.

Programming example

Machining elongated holes

With this program you can machine four elongated holes 30mm in length and with a relative depth of 23mm (difference between the reference plane and the base of the elongated hole) that lie in a circle with the center point Z45 Y40 and a radius of 20mm in the YZ plane. The initial angle is 45 degrees, the indexing angle is 90 degrees. The maximum infeed depth is 6mm, the safety distance is 1mm.




```
N10 G19 G90 S600 M3                    ;Specification of technology values
T10 D1
M6
N20 G0 Y50 Z25 X5                      ;Approach starting position
N30 LONGHOLE (5, 0, 1, , 23, 4, 30, ->  ;Cycle call
-> 40,  45, 20, 45, 90, 100 ,320, 6)
N40 M30                                ;End of program
```

Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

3.5 Slots on a circle - SLOT1

Programming

```
SLOT1 (RTP, RFP, SDIS, DP, DPR, NUM, LENG, WID, CPA, CPO, RAD, STA1, INDA,
FFD, FFP1, MID, CDIR, FAL, VARI, MIDF, FFP2, SSF, _FALD, _STA2)
```

Parameters

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Slot depth (absolute)
DPR	real	Slot depth relative to the reference plane (enter without sign)
NUM	integer	Number of slots
LENG	real	Slot length (enter without sign)
WID	real	Slot width (enter without sign)
CPA	real	Center point of circle, abscissa (absolute)
CPO	real	Center point of circle, ordinate (absolute)
RAD	real	Radius of circle (enter without sign)
STA1	real	Initial angle
INDA	real	Indexing angle
FFD	real	Feedrate for depth infeed
FFP1	real	Feedrate for surface machining
MID	real	Maximum infeed depth for infeed (enter without sign)
CDIR	integer	Milling direction for machining the slot Values: 0...Down-cut milling (as spindle rotation) 1...Up-cut milling 2...with G2 (independent of spindle direction) 3...with G3
FAL	real	Final machining allowance on slot edge (enter without sign)
VARI	integer	Machining type (enter without sign) UNITS DIGIT: Values: 0...Complete machining 1...Roughing 2...Finishing TENS DIGIT: Values: 0...Perpendicular with G0 1...Perpendicular with G1 3...Oscillation with G1
MIDF	real	Maximum infeed depth for finishing
FFP2	real	Feedrate for finishing
SSF	real	Speed for finishing

© Siemens AG 2004 All rights reserved.

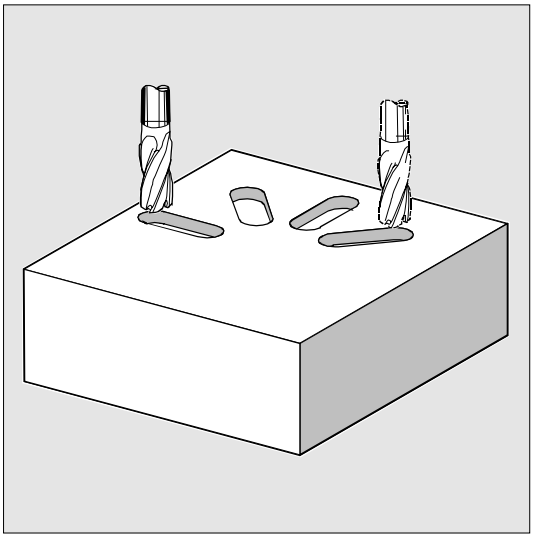
_FALD	real	Final machining allowance on the base of slot
_STA2	real	Maximum insertion angle for oscillation movement

The cycle requires a milling cutter with an "end tooth cutting over center" (DIN 844).

Function

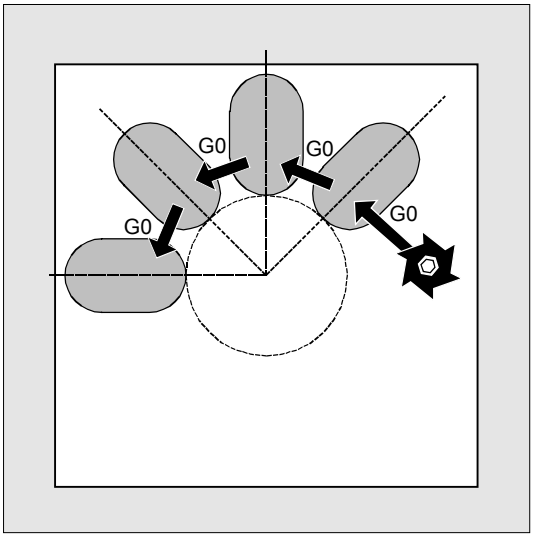
Cycle SLOT1 is a combined roughing-finishing cycle.

With this cycle you can machine slots arranged on a circle. The longitudinal axis of the slots is arranged radially. Unlike the elongated hole, a value is defined for the slot width.



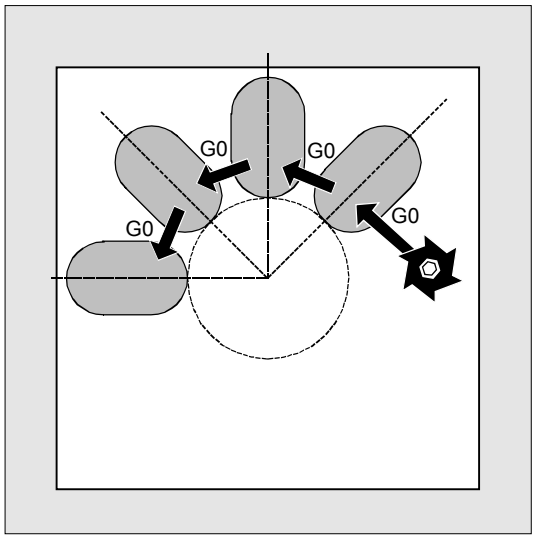
Sequence

Position reached before the beginning of the cycle:
The starting position can be any position from which each of the slots can be approached without collision.



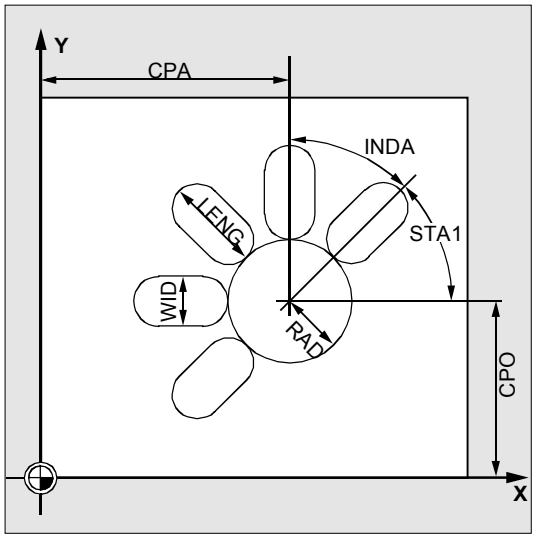
© Siemens AG 2004 All rights reserved.

The cycle implements the following motion sequence:



- Travel to the position marked in the figure on the right at the beginning of the cycle with G0
- Complete machining of a slot is performed in the following stages:
  - Approach to reference plane brought forward by the safety distance with G0.
  - Infeed to the next machining depth as programmed under VAR1 and at feed value FFD.
  - Solid machining of the slot to the final machining allowance on slot base and slot edge at feed value FFP1.
    Subsequent finishing at feed value FFP2 and spindle speed SSF along the contour according to the machining direction programmed under CDIR.
  - The vertical depth infeed of G0/G1 always takes place at the same position in the machining plane until the final depth of the slot is reached.
  - With reciprocating machining, select the starting point so that the end point always reaches the same position in the machining plane.
- Retract tool to retraction plane and move to next slot with G0.
- At the end of machining the last slot the tool moves to the retraction plane with G0 and the cycle ends.

Explanation of parameters



See Section 2.1.2. (Drilling, Centering – CYCLE81) for a description of parameters RTP, RFP, SDIS.
See Section 3.2 for cycle setting data _ZSD[1].

DP and DPR (slot depth)
The slot depth can be defined as either absolute (DP) or relative (DPR) to the reference plane.
If it is entered as a relative value, the cycle automatically calculates the correct depth on the basis of the positions of the reference and retraction planes.

© Siemens AG 2004 All rights reserved.

NUM (number)
The number of slots is determined with the parameter NUM.

LENG and WID (slot length and slot width)
The shape of a slot in the plane is determined with parameters LENG and WID. The milling cutter diameter must be smaller than the slot width. Otherwise alarm 61105 "Cutter radius too large" is activated and the cycle aborted.
The milling cutter diameter must not be smaller than half of the slot width. This is not checked.

CPA, CPO and RAD (center point and radius)
The position of the circle of holes in the machining plane is defined by the center point (parameters CPA and CPO) and the radius (parameter RAD). Only positive values may be entered for the radius.

STA1 and INDA (start angle and indexing angle)
The arrangement of the slot on the circle is defined by these parameters.
STA1 defines the angle between the positive direction of the abscissa of the workpiece coordinate system active before the cycle was called and the first slot.
Parameter INDA contains the angle from one slot to the next.
If INDA=0 the indexing angle is calculated from the number of slots so that they are equally distributed around the circle.

FFD and FFP1 (feedrate depth and plane)
Feedrate FFD is operative for vertical infeed to the machining plane with G1 and for insertion with oscillation motion.
Feedrate FFP1 is active for all movements in the plane traversed at feedrate when roughing.

MID (infeed depth)
The maximum infeed depth is defined with this parameter. The depth infeed is performed by the cycle in equally sized infeed steps.
Using MID and the total depth, the cycle automatically calculates this infeed which lies between 0.5x maximum infeed depth and the maximum infeed depth. The minimum possible number of infeed steps is used as the basis. MID=0 means that the cut to slot depth is made with one infeed.

© Siemens AG 2004 All rights reserved.

The depth infeed commences at the reference plane moved forward by the safety distance (as a function of _ZSD[1]).

CDIR (milling direction)
You define the slot machining direction in this parameter.
Under parameter CDIR, the mill direction

- directly "2 for G2" and "3 for G3" or
- alternatively "up-cut milling" or "down-cut milling"

can be programmed. Up-cut milling or down-cut milling is determined within the cycle via the spindle direction activated prior to the cycle call.

Down-cut	**Up-cut**
M3 → G3	M3 → G2
M4 → G2	M4 → G3

FAL (final machining allowance at slot edge)
With this parameter you can program a final machining allowance on the slot edge. FAL does not affect the depth infeed. If the value of FAL is greater than allowed for the specified width and the milling cutter used, FAL is automatically reduced to the maximum possible value. In the case of rough machining, milling is performed with a reciprocating movement and depth infeed at both end points of the slot.

VARI, MIDF, FFP2 and SSF (machining type, infeed depth, feedrate and speed)
You can define the type of machining with parameter VARI.

Possible settings are:
UNITS DIGIT

- 0=Complete machining in two parts
  - Machining of the slot (SLOT1, SLOT2) or pocket (POCKET1, POCKET2) to the final machining allowance is performed at the spindle speed programmed before the cycle was called and with feedrate FFP1. Depth infeed is defined with MID.
  - Removing stock for the remaining final machining allowance is performed at the spindle speed preset by SSF and feed FFP2. With parameter MIDF it is possible to program a depth infeed other than that for roughing. However, this is only effective for finishing at the edge. If MIDF=0, the infeed is equal to the final depth.
    If FFP2 is not programmed, feed FFP1 applies. Similarly, if there is no message from SSF, i.e. the speed

© Siemens AG 2004 All rights reserved.

programmed before cycle start applies.

- 1=Roughing
  The slot (SLOT1, SLOT2) or pocket (POCKET1, POCKET2) is solid machined up to the final machining allowance at the speed programmed before the cycle call and feedrate FFP1. The depth infeed is programmed in MID.
- 2=Finishing
  The cycle requires that the slot (SLOT1, SLOT2) or pocket (POCKET1, POCKET2) already have been machined to a remaining final machining allowance, so it is only necessary to machine the final machining allowance. If FFP2 and SSF are not programmed, the feedrate FFP1 or the speed programmed before the cycle call is active. For finishing at the edge, it is possible to program a value for the depth infeed in parameter MIDF.
  In machining mode VARI=30 the edge finishing cut is performed at the last roughing depth.

TENS POINT (infeed)

- 0=Perpendicular with G0
- 1=Vertical with G1
- 3=Oscillation with G1

If another value is programmed for parameter VARI, the cycle aborts after outputting alarm 61102 "Operating mode not defined correctly".

Cutter diameter=slot width (WID)

- With complete machining, finishing machining only takes place on the base.
- In machining mode VARI=32 positioning is paraxial in Z with G1 followed by finish-cutting (infeed possible via MIDF).

_FALD (final machining allowance on slot base)
A separate final machining allowance on the base is taken into account in roughing operations.

_STA2 (insertion angle)
Parameter _STA2 defines the maximum insertion angle for the oscillation motion.

- **Vertical insertion (VARI=0X, VARI=1X)**
  Vertical depth insertion is always performed at the same position on the machining plane down to the final depth of the slot.
- **Insertion with oscillation on the center axis of**

© Siemens AG 2004 All rights reserved.

the slot (VARI=3X)
means that the mill center point oscillates along an oblique linear path until it has reached the next current depth. The maximum insertion angle is programmed under _STA2, the length of the oscillation path is calculated from LENG-WID.
The oscillating depth infeed ends at the same point as with vertical depth infeed motions, the starting point in the plane is calculated accordingly. The roughing operation begins in the plane once the current depth is reached. The feedrate is programmed under _FFD.

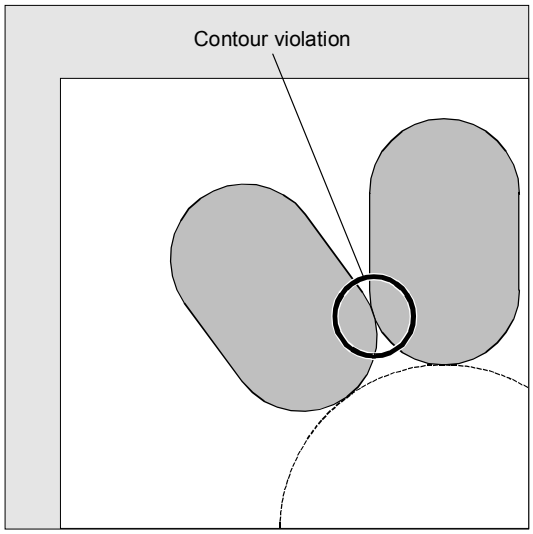
Other Information

A tool offset must be activated before the cycle is called. Otherwise the cycle is aborted with Alarm 61000 "No tool offset active".

If incorrect values are assigned to the parameters that determine the arrangement and size of the slots and thus cause mutual contour violation of the slots, the cycle is not started. The cycle is aborted after the error message
61104 "Contour violation of slots/elongated holes".

During the cycle, the workpiece coordinate system is shifted and rotated. The values in the workpiece coordinate system displayed on the actual value display are such that the longitudinal axis of the slot that has just been machined corresponds to the first axis of the current machining plane.

When the cycle is completed, the workpiece coordinate system is again in the same position as it was before the cycle was called.

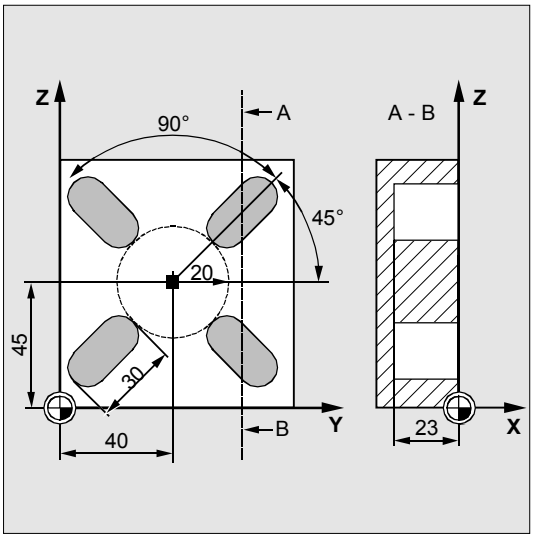


© Siemens AG 2004 All rights reserved.

Programming example

Slots

This program produces the same arrangement of four slots on a circle as the program for elongated hole machining (see Section 3.4).
The slots have the following dimensions: Length 30mm, width 15mm and depth 23mm. The safety distance is 1 mm, the final machining allowance is 0.5 mm, the milling direction is G2, the maximum infeed in the depth is 10 mm.
The slots must be machined completely with an oscillating insertion motion.



`N10 G19 G90 S600 M3`	;Specification of technology values
`N15 T10 D1`	
`N17 M6`	
`N20 G0 Y20 Z50 X5`	;Approach start position

`N30 SLOT1 (5, 0, 1, -23, , 4, 30, 15, ->`
`->40, 45, 20, 45, 90, 100, 320, 10, ->`
`->2, 0.5, 30, 10, 400, 1200, 0.6, 5)`
	;Cycle call, parameters VARI, MIDF,
	;FFP2 and SSF omitted

`N40 M30`	;End of program

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

3.6 Circumferential slot - SLOT2

Programming

```
SLOT2 (RTP, RFP, SDIS, DP, DPR, NUM, AFSL, WID, CPA, CPO, RAD, STA1, INDA,
FFD, FFP1, MID, CDIR, FAL, VARI, MIDF, FFP2, SSF, _FFCP)
```

Parameters

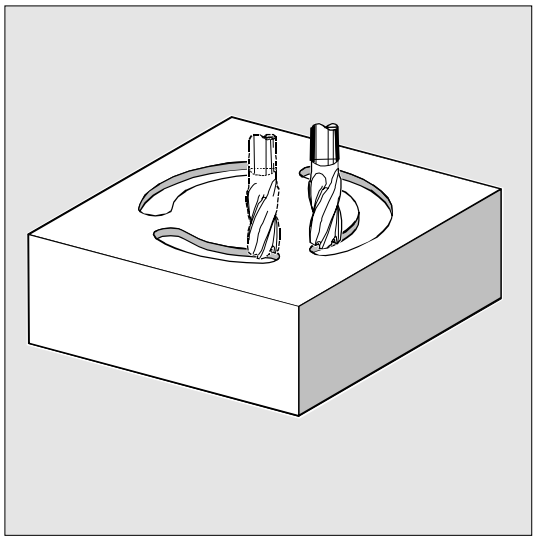
RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Slot depth (absolute)
DPR	real	Slot depth relative to the reference plane (enter without sign)
NUM	integer	Number of slots
AFSL	real	Angle for the slot length (enter without sign)
WID	real	Circumferential slot width (enter without sign)
CPA	real	Center point of circle, abscissa (absolute)
CPO	real	Center point of circle, ordinate (absolute)
RAD	real	Radius of circle (enter without sign)
STA1	real	Initial angle
INDA	real	Indexing angle
FFD	real	Feedrate for depth infeed
FFP1	real	Feedrate for surface machining
MID	real	Maximum infeed depth for infeed (enter without sign)
CDIR	integer	Milling direction for machining the circumferential slot Values: 2 (for G2) 3 (for G3)
FAL	real	Final machining allowance on slot edge (enter without sign)
VARI	integer	Machining type UNITS DIGIT: Values: 0=Complete machining 1=Roughing 2=Finishing TENS DIGIT (SW 6.3 and higher) Values: 0=positioning from slot to slot in a straight line with G0 1=positioning from slot to slot in a circular path with feedrate
MIDF	real	Maximum infeed depth for finishing
FFP2	real	Feedrate for finishing
SSF	real	Speed for finishing
_FFCP (SW 6.3 and higher)	real	Feedrate for intermediate positioning on a circular path, in mm/min

The cycle requires a milling cutter with an "end tooth cutting over center" (DIN 844).

© Siemens AG 2004 All rights reserved.

Function

Cycle SLOT2 is a combined roughing-finishing cycle. With this cycle you can machine circumferential slots arranged on a circle.

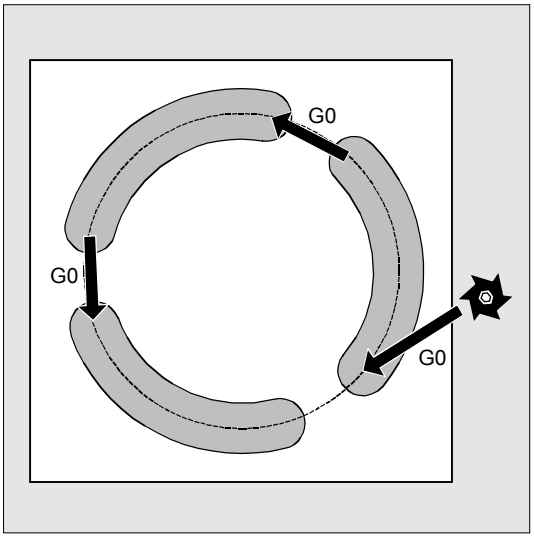


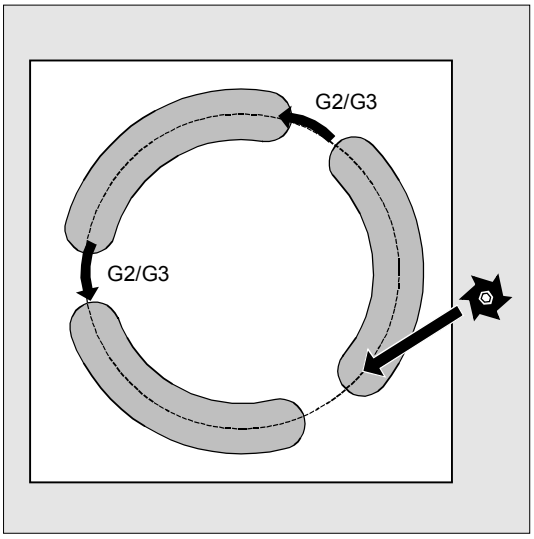
Sequence

Position reached before the beginning of the cycle:
The starting position can be any position from which each of the slots can be approached without collision.

The cycle implements the following motion sequence:

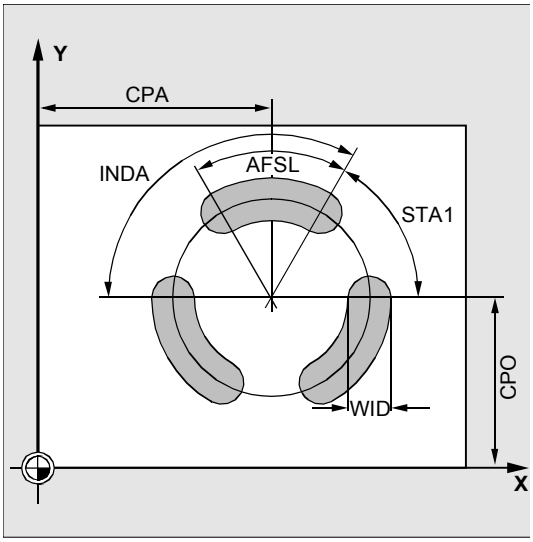
- Travel to the position marked in the figure on the right at the beginning of the cycle with G0.
- The circumferential slot is machined in the same steps as an elongated hole.
- After finishing a circumferential slot, the tool is withdrawn to the retraction plane and the transition to the next slot is made, either on a straight line with G0 or on a circular path at the feedrate programmed in _FFCP.
- When the last slot has been machined, the tool is traversed to the end position reached in the machining plane specified in the display to the retraction plane with G0 and the cycle is terminated.





© Siemens AG 2004 All rights reserved.

Explanation of the parameters



See Section 2.1.2. (Drilling, Centering – CYCLE81) for a description of parameters RTP, RFP, SDIS.

See Section 3.5 (SLOT1) for a description of parameters DP, DPR, FFD, FFP1, MID, CDIR, FAL, VARI, MIDF, FFP2, SSF.
See Section 3.2 for cycle setting data _ZSD[1].

NUM (number)
The number of slots is determined with the parameter NUM.

AFSL and WID (angle and circumferential slot width)
With parameters AFSL and WID you define the shape of a slot in the plane. The cycle checks whether the slot width is violated with the active tool. If this is the case, alarm
61105 "Cutter radius too large"
and the cycle is aborted.

CPA, CPO and RAD (center point and radius)
The position of the circle in the machining plane is defined by the center point (parameters CPA and CPO) and the radius (parameter RAD). Only positive values are permissible for the radius.

STA1 and INDA (start angle and indexing angle)
The arrangement of circumferential slots on the circle is defined by these parameters.

STA1 defines the angle between the positive direction of the abscissa of the workpiece coordinate system active before the cycle was called and the first circumferential slot.

The INDA parameter contains the angle from one circumferential slot to the next.
If INDA=0 the indexing angle is calculated from the number of circumferential slots so that they are equally distributed around the circle.

© Siemens AG 2004 All rights reserved.

New machining types in SW 6.3 and higher:

Finishing the edge only (VARI=x3)

- There is a new selection "Finishing the edge". This is the only machining type where the diameter of the cutter is allowed to be less than half the slot width. There is no checking of whether it is sufficiently large to execute the final machining allowance FAL.
- A number of depth infeeds are possible. These are programmed as usual by parameter MID. The slot is circumnavigated once at each depth.
- To approach and return from the contour, the cycle generates a smooth approach to the segment of the circular path.

Intermediate positioning on the circular path (VARI=1x)

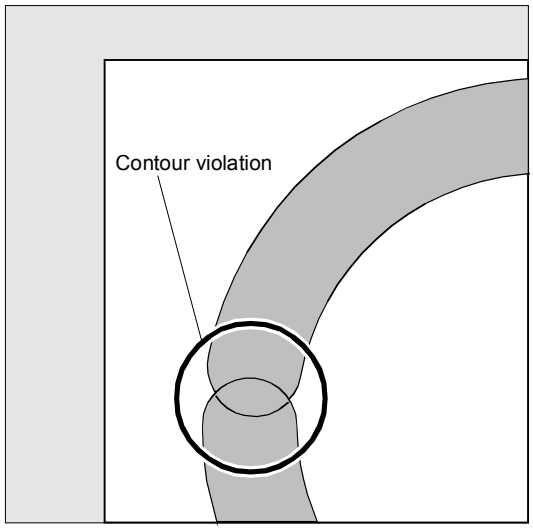
- Especially when used on turning machines, it can happen that there will be a spigot in the middle of the circle on which the slots lie, that does not allow direct positioning from one slot to the next with G0.
- The circular path is taken to be the circle on which the slots lie (determined by parameters CPA, CPO and RAD). Positioning takes place at the same level as intermediate positioning on a straight line with G0. The positioning federate for the circular path is programmed under the parameter in mm/min.

© Siemens AG 2004 All rights reserved.

Other Information

A tool offset must be activated before the cycle is called. The cycle is otherwise aborted with alarm 61000 "No tool offset active".

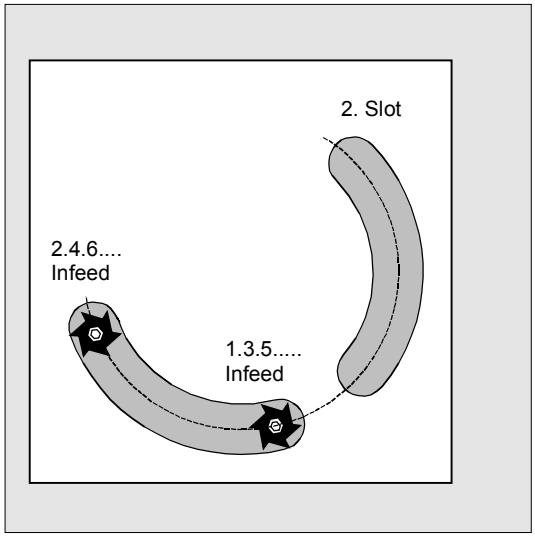
If incorrect values are assigned to the parameters that determine the arrangement and size of the slots and thus cause mutual contour violation of the slots, the cycle is not started.
The cycle is aborted after the error message 61104 "Contour violation of slots/elongated holes"
.



During the cycle, the workpiece coordinate system is shifted and rotated. The actual-value display in the workpiece coordinate system is always shown such that the circumferential slot currently being machined starts on the 1st axis of the current processing level and the zero point of the workpiece coordinate system is in the center of the circle.
When the cycle is completed, the workpiece coordinate system is again in the same position as it was before the cycle was called.

Special case: Slot width = cutter diameter

- The machining case of slot width = cutter diameter is allowed for roughing and finishing. This machining case occurs if slot width WID – 2 * final machining allowance FAL = cutter diameter.
- The traversing strategy is then as for the LONGHOLE cycle, i.e. depth infeed takes place alternately at the reversal points, see graphic.

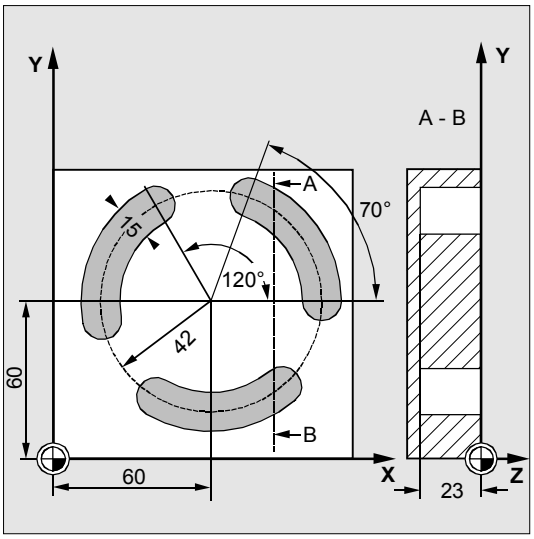


© Siemens AG 2004 All rights reserved.

Programming example

Slots2

With this program you can machine three circumferential slots arranged on a circle whose center point is X60 Y60 and radius 42mm in the XY plane. The circumferential slots have the following dimensions: Width 15mm, angle for slot length 70 degrees, depth 23mm. The initial angle is 0 degrees, the indexing angle is 120 degrees. The slot contours are machined to a final machining allowance of 0.5mm, the safety distance in infeed axis Z is 2mm, the maximum depth infeed is 6mm. The slots are to be completely machined. The same speed and feedrate are used for finishing. Infeed during finishing is performed straight to the base of the slot.



`DEF REAL FFD=100`	;Definition of variables with value ;assignment
`N10 G17 G90 S600 M3`	;Specification of technology values
`N15 T10 D1`	
`N17 M6`	
`N20 G0 X60 Y60 Z5`	;Approach starting position

`N30 SLOT2 (2, 0, 2, -23, , 3, 70, ->`
`-> 15, 60, 60, 42, , 120, FFD, ->`
`-> FFD+200, 6, 2, 0.5)`
	;Cycle call
	;Reference plane+SDIS=retraction plane
	;means: ;Lower in infeed axis with G0 to
	;reference plane+SDIS no longer applicable,
	;parameters VAR, MIDF, ;FFP2 and SSF
	;omitted

`N40 M30`	;End of program

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

3.7 Milling rectangular pockets - POCKET1

Programming

```
POCKET1 (RTP, RFP, SDIS, DP, DPR, LENG, WID, CRAD, CPA, CPD, STA1, FFD,
FFP1, MID, CDIR, FAL, VARI, MIDF, FFP2, SSF)
```

Parameters

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Pocket depth (absolute)
DPR	real	Pocket depth relative to the reference plane (enter without sign)
LENG	real	Pocket length (enter without sign)
WID	real	Pocket width (enter without sign)
CRAD	real	Corner radius (enter without sign)
CPA	real	Pocket center point, abscissa (absolute)
CPO	real	Pocket center point, ordinate (absolute)
STA1	real	Angle between longitudinal axis and abscissa Value range: 0<=STA1<180 degrees
FFD	real	Feedrate for depth infeed
FFP1	real	Feedrate for surface machining
MID	real	Maximum infeed depth for infeed (enter without sign)
CDIR	integer	Milling direction for machining the pocket Values: 2 (for G2) 3 (for G3)
FAL	real	Final machining allowance on pocket edge (enter without sign)
VARI	integer	Machining type Values: 0=Complete machining 1=Roughing 2=Finishing
MIDF	real	Maximum infeed depth for finishing
FFP2	real	Feedrate for finishing
SSF	real	Speed for finishing

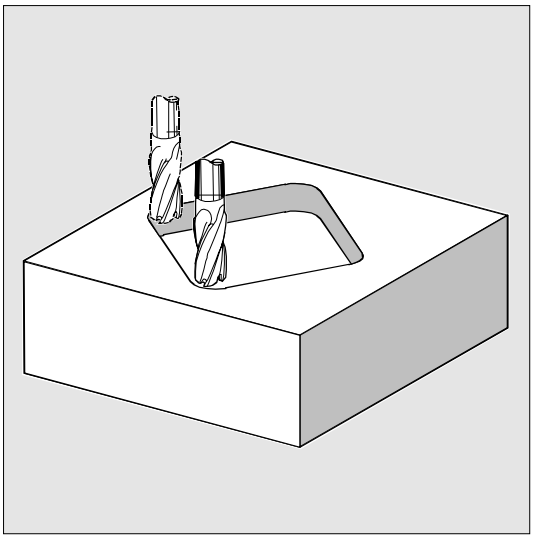
The cycle requires a milling cutter with an "end tooth cutting over center" (DIN 844).

The pocket milling cycle POCKET3 can be performed with any tool.

© Siemens AG 2004 All rights reserved.

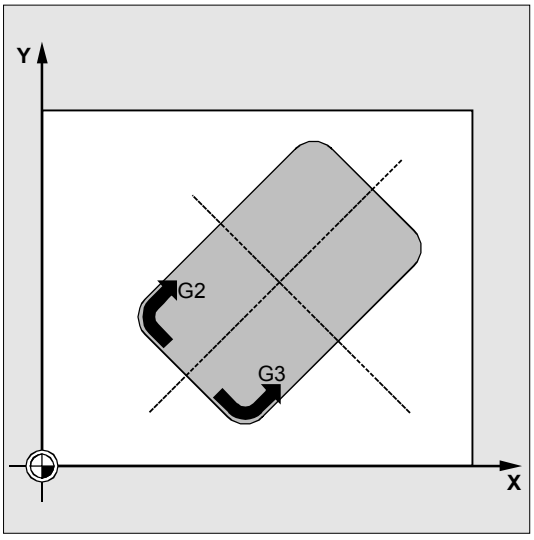
Function

The cycle is a combined roughing-finishing cycle.
With this cycle you can machine rectangular pockets in any position in the machining plane.



Sequence

Position reached before the beginning of the cycle:
This can be any position from which the starting position on the center point of the pocket at the retraction plane level can be approached without collision.

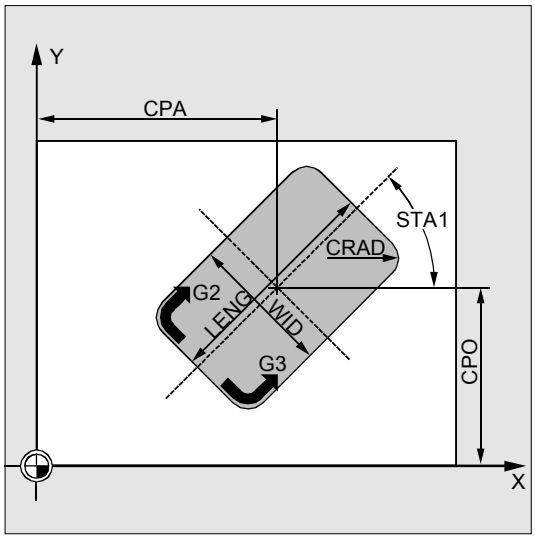


The cycle implements the following motion sequence:

- With G0, the pocket center point is approached at the retraction plane level and then, from this position, with G0 the reference plane brought forward by the safety distance is approached. Complete machining of the pocket is performed in the following stages:
  - Infeed to the next machining depth with G1 and feedrate FFD.
  - Pocket milling up to the final machining allowance with feedrate FFP1 and the spindle speed that was active before the cycle was called.
- After roughing is completed:
  - Infeed to the machining depth defined by MIDF
  - Final machining along the contour with feedrate FFP2 and speed SSF.
  - The machining direction is defined by CDIR.
- When machining of the pocket is completed the tool is traversed to the pocket center point on the retraction plane and the cycle is terminated.

© Siemens AG 2004 All rights reserved.

Explanation of the parameters



See Section 2.1.2. (Drilling, Centering – CYCLE81) for a description of parameters RTP, RFP, SDIS.

See Section 3.5 (SLOT1) for a description of parameters FFD, FFP1, MID, CDIR, FAL, VARI, MIDF, FFP2, SSF.
See Section 3.2 for cycle setting data _ZSD[1].

DP and DPR (pocket depth)
The pocket depth can be defined as either absolute (DP) or relative (DPR) to the reference plane.
If it is entered as a relative value, the cycle automatically calculates the correct depth on the basis of the positions of the reference and retraction planes.

LENG, WID and CRAD (length, width and radius)
The shape of a pocket in the plane is determined with parameters LENG, WID and CRAD.
If it is not possible to traverse to the programmed corner radius with the active tool because its radius is larger, the corner radius of the completed pocket corresponds to the tool radius. If the milling cutter radius is greater than half the length or width of the pocket, the cycle is aborted and alarm 61105 "Cutter radius too large" is output.

CPA, CPO (center point)
With parameters CPA and CPO you define the center point of the pocket in the abscissa and ordinate.

STA1 (angle)
STA1 defines the angle between the positive abscissa and the longitudinal axis of the pocket.

Other Information

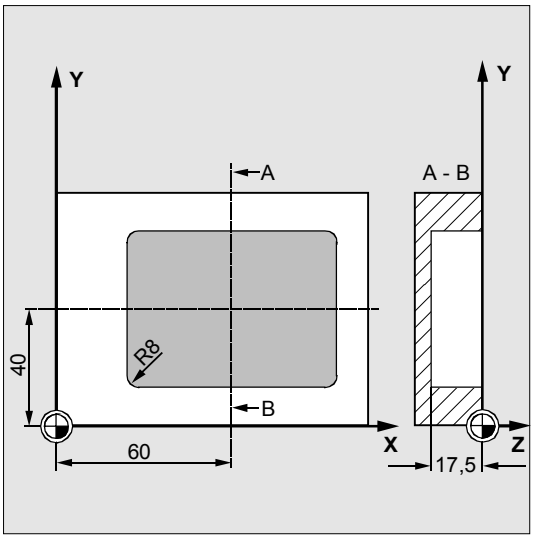
A tool offset must be activated before the cycle is called. The cycle is otherwise aborted with alarm 61000 "No tool offset active".
A new workpiece coordinate system that influences the actual value display is used in the cycle. The zero point of this coordinate system lies on the pocket center point.
The original coordinate system becomes active again after the end of the cycle.

© Siemens AG 2004 All rights reserved.

Programming example

Pocket

With this program you can machine a pocket that is 60 mm long, 40 mm wide, 17.5 mm deep (difference between the reference plane and the base of the pocket) and which has a corner radius of 8 mm in the XY plane. The angle to the X axis is 0 degrees. The final machining allowance of the pocket edges is 0.75mm, the safety distance in the Z axis, which is added to the reference plane, is 0.5mm. The center point of the pocket lies at X60 and Y40, the maximum depth infeed is 4mm.
Only roughing is to be performed.



`DEF REAL LENG, WID, DPR, CRAD` `DEF INT VARI`	;Definition of variables
`N10 LENG=60 WID=40 DPR=17.5 CRAD=8` `N20 VARI=1`	;Value assignments
`N30 G90 S600 M4`	;Specification of technology values
`N35 T20 D2`	
`N37 M6`	
`N40 G17 G0 X60 Y40 Z5`	;Approach start position

`N50 POCKET1 (5, 0, 0.5, , DPR, ->`
`-> LENG, WID, CRAD, 60, 40, 0, ->`
`-> 120, 300, 4, 2, 0.75, VARI)`
	;Cycle call
	;Parameters MIDF, FFP2 and SSF are
	;omitted

`N60 M30`	;End of program

Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

3.8 Milling circular pockets - POCKET2

Programming

```
POCKET2 (RTP, RFP, SDIS, DP, DPR, PRAD, CPA, CPO, FFD, FFP1, MID, CDIR,
FAL, VARI, MIDF, FFP2, SSF)
```

Parameters

RTP	real	Retraction plane (absolute)
RFP	real	Reference plane (absolute)
SDIS	real	Safety distance (enter without sign)
DP	real	Pocket depth (absolute)
DPR	real	Pocket depth relative to the reference plane (enter without sign)
PRAD	real	Pocket radius (enter without sign)
CPA	real	Pocket center point, abscissa (absolute)
CPO	real	Pocket center point, ordinate (absolute)
FFD	real	Feedrate for depth infeed
FFP1	real	Feedrate for surface machining
MID	real	Maximum infeed depth for infeed (enter without sign)
CDIR	integer	Milling direction for machining the pocket Values: 2 (for G2) 3 (for G3)
FAL	real	Final machining allowance on pocket edge (enter without sign)
VARI	integer	Machining type Values: 0=Complete machining 1=Roughing 2=Finishing
MIDF	real	Maximum infeed depth for finishing
FFP2	real	Feedrate for finishing
SSF	real	Speed for finishing

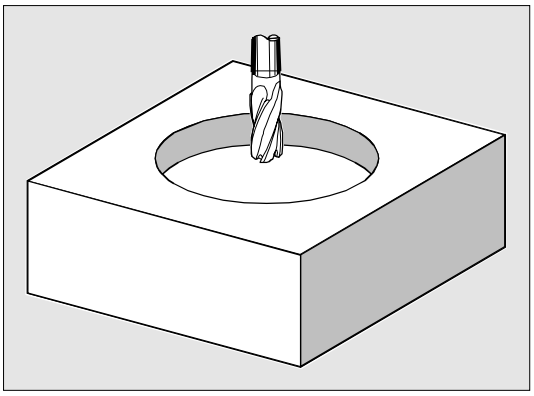
The cycle requires a milling cutter with an "end tooth cutting over center" (DIN 844).

The pocket milling cycle POCKET4 can be performed with any tool.

© Siemens AG 2004 All rights reserved.

Function

The cycle is a combined roughing-finishing cycle. With this cycle you can machine circular pockets in the machining plane.



Sequence

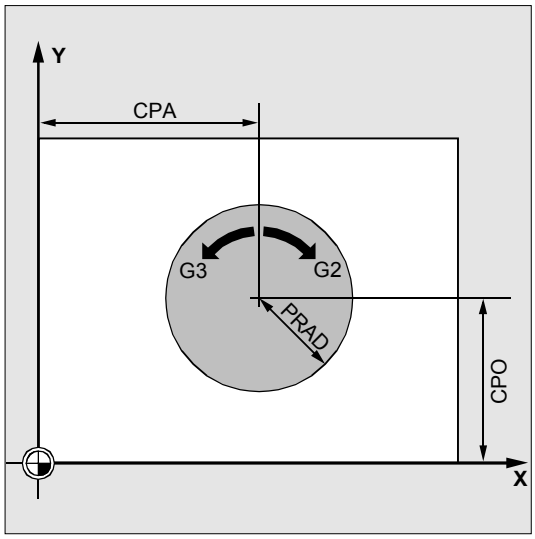
Position reached before the beginning of the cycle:
This can be any position from which the starting position on the center point of the pocket at the retraction plane level can be approached without collision.

The cycle implements the following motion sequence:

- With G0, the pocket center point is approached at the retraction plane level and then, from this position, with G0 the reference plane brought forward by the safety distance is approached. Complete machining of the pocket is performed in the following stages:
  - Infeed perpendicular to the pocket center to the next machining depth with feedrate FFD.
  - Pocket milling up to the final machining allowance with feedrate FFP1 and the spindle speed that was active before the cycle was called.
- After roughing is completed:
  - Infeed to the next machining depth defined by MIDF.
  - Final machining along the contour with feedrate FFP2 and speed SSF.
  - The machining direction is defined by CDIR.
- When machining is completed the tool is traversed to the pocket center point in the retraction plane and the cycle is terminated.

© Siemens AG 2004 All rights reserved.

Explanation of the parameters



See Section 2.1.2. (Drilling, Centering – CYCLE81) for a description of parameters RTP, RFP, SDIS.
See Section 3.7 for a description of parameters DP, DPR.

See Section 3.5 (SLOT1) for a description of parameters FFD, FFP1, MID, CDIR, FAL, VARI, MIDF, FFP2, SSF.
See Section 3.2 for cycle setting data _ZSD[1].

PRAD (pocket radius)
The shape of the circular pocket is determined by the radius only.
If the radius is less than the tool radius of the active tool, the cycle is aborted after alarm 61105 "Milling cutter radius too large" is output.

CPA, CPO (pocket center point)
With parameters CPA and CPO you define the center point of the circular pocket in the abscissa and ordinate.

Other Information

A tool offset must be activated before the cycle is called. The cycle is otherwise aborted with alarm 61000 "No tool offset active".

The depth infeed is always made in the pocket center point. It can be useful to drill there beforehand.

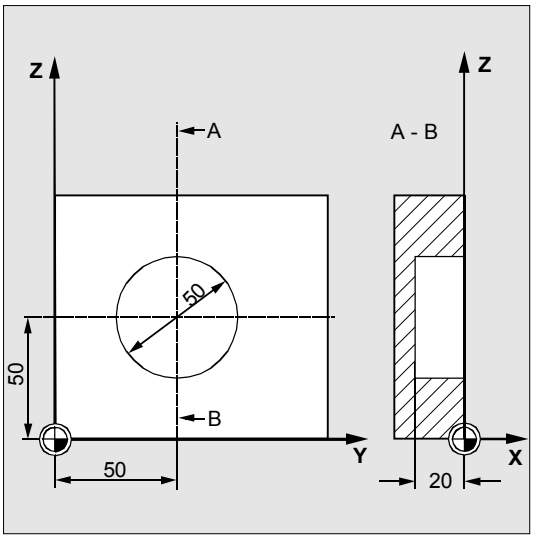
A new workpiece coordinate system that influences the actual value display is used in the cycle. The zero point of this coordinate system lies on the pocket center point.
The original coordinate system becomes active again after the end of the cycle.

© Siemens AG 2004 All rights reserved.

Programming example

Circular pocket

With this program you can machine a circular pocket in the YZ plane. The center point is defined by Y50 Z50. The infeed axis for the depth infeed is the X axis, the pocket depth is entered as an absolute value. Neither a final machining allowance nor a safety distance is defined.




```
DEF REAL RTP=3, RFP=0, DP=-20,->          ;Definition of variables with
-> PRAD=25, FFD=100, FFP1, MID=6          ;value assignments
N10 FFP1=FFD*2
N20 G19 G90 G0 S650 M3                    ;Specification of technology values
N25 T10 D1
N27 M6
N30 Y50 Z50                               ;Approach start position
N40 POCKET2 (RTP, RFP, , DP, , PRAD, ->   ;Cycle call
-> 50, 50, FFD, FFP1, MID, 3, )           ;Parameters FAL, VARI, MIDF, FFP2,
                                          ;SSF are omitted
N50 M30                                   ;End of program
```

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

3.9 Milling rectangular pockets - POCKET3

The POCKET3 cycle is available in SW 4 and higher.

Programming

POCKET3 (_RTP, _RFP, _SDIS, _DP, _LENG, _WID, _CRAD, _PA, _PO, _STA, _MID, _FAL, _FALD, _FFP1, _FFD, _CDIR, _VARI, _MIDA, _AP1, _AP2, _AD, _RAD1, _DP1)

Parameters

The following input parameters are always required:

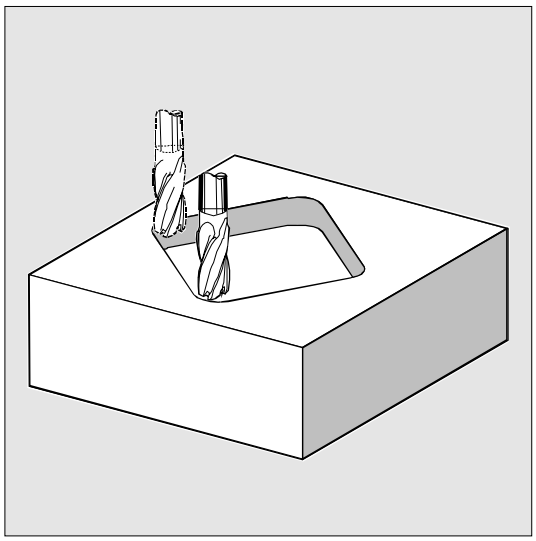
_RTP	real	Retraction plane (absolute)
_RFP	real	Reference plane (absolute)
_SDIS	real	Safety distance (to be added to the reference plane, enter without sign)
_DP	real	Pocket depth (absolute)
_LENG	real	Pocket length for dimensioning from the corner with sign
_WID	real	Pocket width for dimensioning from the corner with sign
_CRAD	real	Pocket corner radius (enter without sign)
_PA	real	Pocket reference point, abscissa (absolute)
_PO	real	Pocket reference point, ordinate (absolute)
_STA	real	Angle between the pocket longitudinal axis and the first axis of the plane (abscissa, enter without sign); Value range: $0° \leq$ _STA $< 180°$
_MID	real	Maximum infeed depth (enter without sign)
_FAL	real	Final machining allowance on pocket edge (enter without sign)
_FALD	real	Final allowance at base (enter without sign)
_FFP1	real	Feedrate for surface machining
_FFD	real	Feedrate for depth infeed
_CDIR	integer	Milling direction: (enter without sign) Values: 0...Down-cut milling (as spindle rotation) 1...Up-cut milling 2...with G2 (independent of spindle direction) 3...with G3
_VARI	integer	Type of machining: (enter without sign) UNITS DIGIT: Values: 1...Roughing 2...Finishing TENS DIGIT: Values: 0...Perpendicular to the pocket center with G0 1...Perpendicular to the pocket center with G1 2...Along a helix 3...Oscillation along the pocket longitudinal axis

© Siemens AG 2004 All rights reserved.

The other parameters can be selected as options. They define the insertion strategy and overlapping for solid machining: (enter without sign)

_MIDA	real	Maximum infeed width during solid machining in the plane
_AP1	real	Basic size pocket length
_AP2	real	Basic size pocket width
_AD	real	Basic pocket depth from reference plane
_RAD1	real	Radius of the helical path on insertion (relative to the tool center point path) or maximum insertion angle for oscillating motion
_DP1	real	Insertion depth per 360° revolution on insertion along helical path

Function



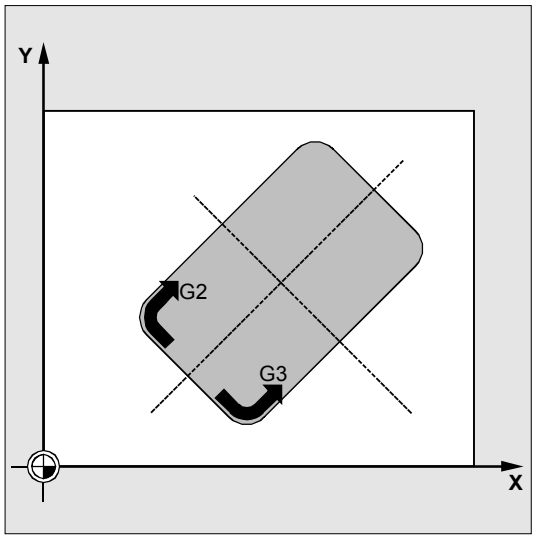
The cycle can be applied to roughing and finishing.
For finishing, a face cutter is needed.
The depth infeed will always start at the pocket center point and be performed vertically from there; thus predrill can be suitably performed in this position.

New functions compared to POCKET1:

- The milling direction can be defined with a G instruction (G2/G3) or up-cut milling or down-cut from the spindle direction
- For solid machining, the maximum infeed width in the plane is programmable
- Final machining allowance for the pocket base
- Three different insertion strategies:
  - Vertically at the pocket center point
  - Along a helical path around the pocket center
  - Oscillating around the pocket central axis
- Shorter approach paths in the plane for finishing
- Consideration of a blank contour in the plane and a basic size at the base (optimum processing of pre-formed pockets possible)

© Siemens AG 2004 All rights reserved.

Sequence



Position reached before the beginning of the cycle:
This can be any position from which the starting position on the center point of the pocket at the retraction plane level can be approached without collision.

Motion sequence when roughing (VARI=X1):
With G0, the pocket center point is approached at the retraction plane level and then, from this position, with G0 the reference plane brought forward by the safety distance is approached. Pocket machining is then performed according to the selected insertion strategy and considering the programmed base size.

Insertion strategies:

- **Vertical insertion to pocket center (VARI=0X, VARI=1X)** means that the current infeed depth internally calculated in the cycle (≤ programmed maximum infeed depth through _MID) is executed in one block with G0 or G1.
- **Insertion along helical path (VARI=2X)**
  means that the milling center point travels on the helical path determined by radius _RAD1 and depth per revolution _DP1. The feedrate is always programmed through _FFD. The sense of rotation of this helical path corresponds to the direction to be used for machining the pocket.
  The depth programmed under _DP1 on insertion is calculated as the maximum depth and is always calculated as a whole number of revolutions of the helical path.
  When the current depth for the infeed (these may be several revolutions on the helical path) has been calculated, a full circle is made to remove the slope on insertion.
  Then pocket solid machining starts in this plane and continues until reaching the final machining allowance.
  The starting point of the helical path described is on the pocket longitudinal axis in the "plus direction" and reached with G1.

© Siemens AG 2004 All rights reserved.

- **Oscillating insertion on center axis of pocket (VARI=3X)**
  means that the mill center point oscillates along an oblique linear path until it has reached the next current depth. The maximum insertion angle is programmed under _RAD1, the position of the oscillation path is calculated within the cycle. When the current depth has been reached, the path is traversed again without depth infeed in order to remove the slope caused by insertion. The feedrate is programmed through _FFD.

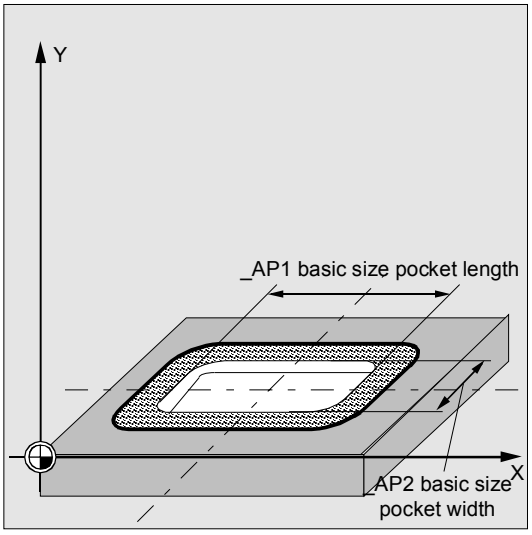
Accounting for blank dimensions

During solid machining, it is possible to take blank dimensions (for example, in the machining of precast workpieces) into account.

The basic size for the length and width (_AP1 and _AP2) are programmed without sign and their symmetrical positions around the pocket center computed in the cycle. They define the part of the pocket that does not have to be solid-machined. The basic size for the depth (_AD) is also programmed without a sign and computed in the direction of the pocket depth from the reference plane.

Depth infeed to account for workpiece sizes is carried out according to the programmed type (helical path, oscillating, vertical). If the cycle detects that the blank contour and the radius of the active tool would provide enough room in the pocket center, infeed is continued vertically downward to the pocket center for as long as possible to avoid time-consuming approach paths outside the material.

The pocket is solid machined beginning from the top and proceeding in the downward direction.

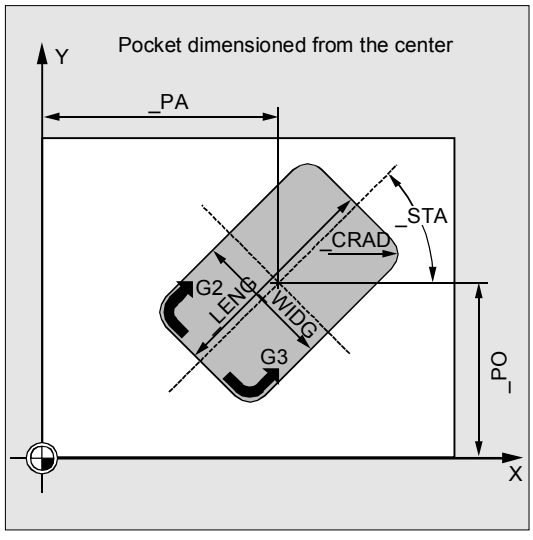


© Siemens AG 2004 All rights reserved.

Motion sequence when finishing (VARI=X2)
Finishing is performed in sequence from the edge until reaching the final machining allowance on the base, then the base is finished. If one of the final machining allowances is equal to zero, this part of the finishing process is skipped.

- Finishing the edge
  While finishing on the edge, the pocket is only machined once.
  For finishing on the edge the path includes one quadrant reaching the corner radius. The radius of this path is normally 2 mm or, if "less room" is available, equals the difference between the corner radius and the mill radius.
  If the final machining allowance on the edge is larger than 2mm, the approach radius is increased accordingly.
  The depth infeed is performed with G0 in the open towards the pocket center and the starting point of the approach path is also reached with G0.
- Finishing on the base
  During finishing on the base, the machine performs G0 towards the pocket center until reaching a distance equal to pocket depth + final machining allowance + safety distance. From this point onward, the tool is always fed in **vertically** at the depth infeed feedrate (since a tool with a front cutting edge is used for base finishing).
  The base surface of the pocket is machined once.

Explanation of the parameters

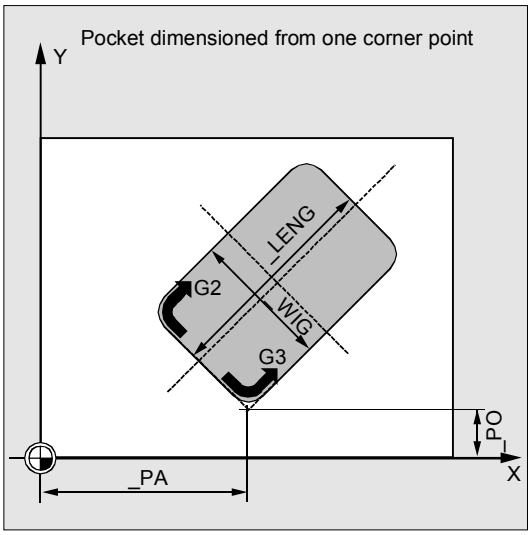


See Subsection 2.1.2. (Drilling, Centering – CYCLE81) for a description of parameters _RTP, _RFP, _SDIS.
See Section 3.7 for a description of parameter _DP.
See Section 3.2 for cycle setting data _ZSD[1], _ZSD[2].

_LENG, _WID and _CRAD (pocket length, pocket width and corner radius)
The shape of a pocket in the plane is determined with parameters _LENG, _WID and _CRAD.
The pocket can be dimensioned from the center or from one corner point. When dimensioning from a corner point, use _LENG and _WID with sign.
If it is not possible to traverse to the programmed

© Siemens AG 2004 All rights reserved.

corner radius with the active tool because its radius is larger, the corner radius of the completed pocket corresponds to the tool radius.
If the milling cutter radius is greater than half the length or width of the pocket, the cycle is aborted after outputting alarm
61105 "Cutter radius too large".



_PA, _PO (reference point)
The center point of the pocket in the abscissa and ordinate is defined with parameters _PA and _PO. This is either the pocket center point or a corner point. The value of this parameter depends on cycle setting data bit _ZSD[2]:

- 0 means pocket center point
- 1 means corner point.

When dimensioning the pocket from a corner, the length and width parameters must be entered with sign (_LENG, _WID), thus completely defining the position of the pocket.

_STA (angle)
_STA indicates the angle between the 1st axis of the plane (abscissa) and the longitudinal axis of the pocket.

_MID (infeed depth)
With this parameter you determine the maximum infeed depth when roughing.
The depth infeed is performed by the cycle in equally sized infeed steps.
The cycle automatically calculates this infeed using _MID and the total depth. The minimum possible number of infeed steps is used as the basis.
_MID=0 means that the cut to pocket depth is made with one infeed.

_FAL (final machining allowance at the edge)
The final machining allowance only affects machining of the pocket in the plane at the edge.
When the final machining allowance ≥ tool diameter, the pocket will not necessarily be machined completely. The following message appears:
"Caution: Final machining allowance ≥ tool diameter"
is output but the cycle is continued.

© Siemens AG 2004 All rights reserved.

_FALD (final machining allowance on the base)
For roughing, a separate final machining allowance is considered on the base (POCKET1 does not normally consider any final machining allowance).

_FFD and _FFP1 (infeed depth and plane)
Feedrate _FFD is used for insertion into the material. Feedrate FFP1 is used for all movements in the plane traversed at feedrate when machining.

_CDIR (milling direction)
The value for the machining direction of the pocket is defined in this parameter.
Under parameter _CDIR the mill direction

- directly "2 for G2" and "3 for G3" or
- alternatively "up-cut milling" or "down-cut milling"

can be programmed. Up-cut milling or down-cut milling is determined within the cycle via the spindle direction activated prior to the cycle call.

Down-cut	**Up-cut**
M3 → G3	M3 → G2
M4 → G2	M4 → G3

_VARI (machining type)
You can define the type of machining with parameter _VARI.
Possible settings are:
Unit's place:

- 1=Roughing
- 2=Finishing

Tens digit (infeed):

- 0=Perpendicular to the pocket center with G0
- 1=Perpendicular to the pocket center with G1
- 2=Along a helical path
- 3=Oscillating along the pocket longitudinal axis

If another value has been programmed for parameter _VARI, the cycle is aborted after alarm 61002 "Machining type incorrectly defined" is output.

_MIDA (max. infeed width)
With this parameter you define the maximum infeed width for solid machining in the plane. In the same way as the known calculation of the infeed depth (equal distribution of the overall depth using the largest possible value), the width is evenly divided, using the value programmed in _MIDA as a maximum value.
If this parameter is not programmed, or if its value is 0,

© Siemens AG 2004 All rights reserved.

the cycle uses 80 percent of the mill diameter as maximum infeed width.



Other Information

Applies if the width infeed determined from edge machining is recalculated on reaching the full pocket depth; otherwise, the width infeed calculated at the start is retained for the full cycle.

_AP1, _AP2, _AD (blank dimension)
With the parameters _AP1, _AP2 and _AD you define the blank dimension (incremental) of the pocket in the horizontal and vertical planes.

_RAD1 (radius)
With the parameter _RAD1 you define the radius of the helical path (i.e. the tool center point path) or the maximum insertion angle for oscillation.

_DP1 (insertion depth)
With the parameter _DP1 you define the infeed depth for insertion on the helical path.

© Siemens AG 2004 All rights reserved.

Other Information

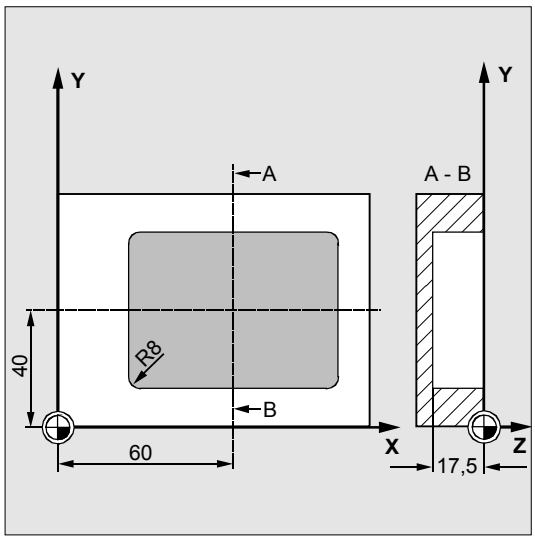
A tool offset must be activated before the cycle is called. The cycle is otherwise aborted with alarm 61000 "No tool offset active".

A new workpiece coordinate system that influences the actual value display is used in the cycle. The zero point of this coordinate system lies on the pocket center point.
The original coordinate system becomes active again after the end of the cycle.

Programming example

Pocket

With this program you can machine a pocket that is 60mm long, 40mm wide, 17.5mm deep in the XY plane, and which has a corner radius of 8mm. The angle in relation to the X axis is 0 degrees. The final machining allowance of the pocket edges is 0.75mm, 0.2mm at the base, the safety distance in the Z axis, which is added to the reference plane, is 0.5mm. The center point of the pocket lies at X60 and Y40, the maximum depth infeed is 4mm.
Up-cut milling uses the spindle rotation direction as direction of machining.
Only roughing is to be performed.



`N10 G90 S600 M4`	;Specification of technology values
`N15 T10 D1`	
`N17 M6`	
`N20 G17 G0 X60 Y40 Z5`	;Approach start position
`N25 _ZSD[2]=0`	;Dimensioning the pocket via the center ;point
`N30 POCKET3 (5, 0, 0.5, -17.5, 60, -> -> 40, 8, 60, 40, 0, 4, 0.75, 0.2, -> -> 1000, 750, 0, 11, 5)`	;Cycle call
`N40 M30`	;End of program

Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

3.10 Milling circular pockets - POCKET4

The cycle POCKET4 is available with SW 4.

Programming

POCKET4 (_RTP, _RFP, _SDIS, _DP, _PRAD, _PA, _PO, _MID, _FAL, _FALD, _FFP1, _FFD, _CDIR, _VARI, _MIDA, _AP1, _AD, _RAD1, _DP1)

Parameters

The following input parameters are always required:

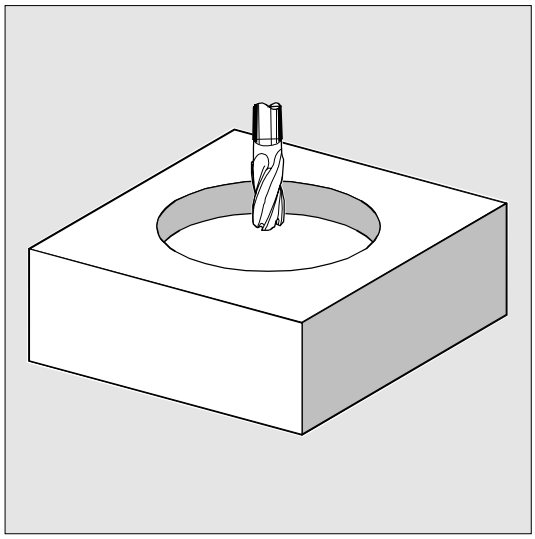
_RTP	real	Retraction plane (absolute)
_RFP	real	Reference plane (absolute)
_SDIS	real	Safety distance (to be added to the reference plane, enter without sign)
_DP	real	Pocket depth (absolute)
_PRAD	real	Pocket radius
_PA	real	Pocket center point, abscissa (absolute)
_PO	real	Pocket center point, ordinate (absolute)
_MID	real	Maximum infeed depth (enter without sign)
_FAL	real	Final machining allowance on pocket edge (enter without sign)
_FALD	real	Final allowance at base (enter without sign)
_FFP1	real	Feedrate for surface machining
_FFD	real	Feedrate for depth infeed
_CDIR	integer	Milling direction: (enter without sign) Values: 0...Down-cut milling (as spindle rotation) 1...Up-cut milling 2...with G2 (independent of spindle direction) 3...with G3
_VARI	integer	Type of machining: (enter without sign) UNITS DIGIT: Values: 1...Roughing 2...Finishing TENS DIGIT: Values: 0...Perpendicular to the pocket center with G0 1...Perpendicular to the pocket center with G1 2...Along a helix

The other parameters can be selected as options. They define the insertion strategy and overlapping for solid machining: (enter without sign)

_MIDA	real	Maximum infeed width during solid machining in the plane
_AP1	real	Basic size pocket radius
_AD	real	Basic pocket depth from reference plane
_RAD1	real	Radius of the helical path during insertion related to the tool center point path)
_DP1	real	Insertion depth per 360° revolution on insertion along helical path

© Siemens AG 2004 All rights reserved.

Function

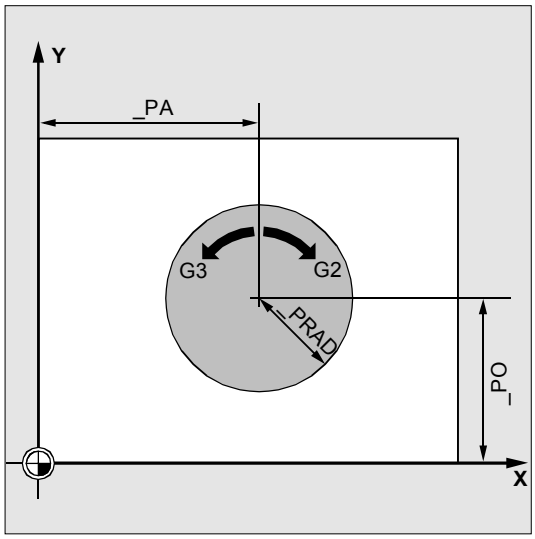


With this cycle you can machine circular pockets in the machining plane.
For finishing, a face cutter is needed.
The depth infeed will always start at the pocket center point and be performed vertically from there; thus predrill can be suitably performed in this position.

New functions compared to POCKET2:

- The milling direction can be defined with a G instruction (G2/G3) or up-cut milling or down-cut from the spindle direction
- For solid machining, the maximum infeed width in the plane is programmable
- Final machining allowance for the pocket base
- Two different insertion strategies:
  - Vertically from the pocket center point
  - Along a helical path around the pocket center
- Shorter approach paths in the plane for finishing
- Consideration of a blank contour in the plane and a basic size at the base (optimum processing of pre-formed pockets possible)
- _MIDA is recalculated when machining the edge.

Sequence



Position reached before the beginning of the cycle:
This can be any position from which the starting position on the center point of the pocket at the retraction plane level can be approached without collision.

Motion sequence when roughing (VARI=X1):
With G0, the pocket center point is approached at the retraction plane level and then, from this position, with G0 the reference plane brought forward by the safety distance is approached. Pocket machining is then performed according to the selected insertion strategy and considering the programmed base size.

© Siemens AG 2004 All rights reserved.

Insertion strategies:
see Section 3.9 (POCKET3)

Accounting for blank dimensions
During solid machining, it is possible to take blank dimensions (for example, in the machining of precast workpieces) into account.
For circular pockets, the basic size _AP1 at the edge is also circular (with a smaller radius than the pocket radius).

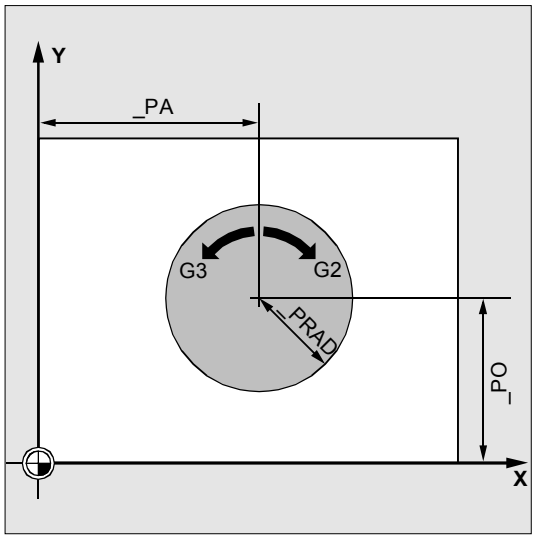
For additional explanations see Section 3.9 (POCKET3)

Motion sequence when finishing (VARI=X2):
Finishing is performed in sequence from the edge until reaching the final machining allowance on the base, then the base is finished. If one of the final machining allowances is equal to zero, this part of the finishing process is skipped.

- Finishing the edge
  While finishing on the edge, the pocket is only machined once.
  For finishing on the edge the path includes one fourth of circle which reaches the pocket radius. The radius of this path is less than or equal to 2mm or, if "less room" is available, equals the difference between the pocket radius and the mill radius.
  The depth infeed is performed with G0 in the open towards the pocket center and the starting point of the approach path is also reached with G0.
- Finishing on the base
  During finishing on the base, the machine performs G0 towards the pocket center until reaching a distance equal to pocket depth + final machining allowance + safety distance. From this point onward, the tool is always fed in **vertically** at the depth infeed feedrate (since a tool with a front cutting edge is used for base finishing).
  The base surface of the pocket is machined once.

© Siemens AG 2004 All rights reserved.

Explanation of the parameters



See Subsection 2.1.2. (Drilling, Centering – CYCLE81) for a description of parameters _RTP, _RFP, _SDIS.
See Section 3.7 (POCKET1) for a description of parameter _DP.

See Section 3.9 (POCKET3) for a description of parameters _MID, _FAL, _FALD, _FFP1, _FFD, _CDIR, _MIDA, _AP1, _AD, _RAD1, _DP1.
See Section 3.2 for cycle setting data _ZSD[1].

_PRAD (pocket radius)
The shape of the circular pocket is determined by the radius only.

If the radius is less than the tool radius of the active tool, the cycle is aborted after alarm 61105 "Milling cutter radius too large" is output.

_PA, _PO (pocket center point)
With parameters _PA and _PO you define the center point of the pocket. Circular pockets are always measured from the center.

_VARI (machining type)
You can define the type of machining with parameter _VARI.
Possible settings are:
Unit's place:
- 1=Roughing
- 2=Finishing

Tens digit (infeed):
- 0=Perpendicular to the pocket center with G0
- 1=Perpendicular to the pocket center with G1
- 2=Along a helical path

If another value has been programmed for parameter _VARI, the cycle is aborted after alarm 61002 "Machining type incorrectly defined" is output.

Edge-only roughing with POCKET4
POCKET4 (circular pocket) can now machine only the edge when rough-cutting.
Define the rough dimension in the depth (parameter _AD) so that it is at least the same size as the pocket depth (DP) minus the final machining allowance depth (_FALD).

© Siemens AG 2004 All rights reserved.

Example:
Requirement: depth calculation without including safety distance (_ZSD[1]=1)

_RTP=0	Reference plane
_SDIS=2	Safety distance
_DP=-21	pocket depth
_FALD=1.25	depth allowance
→_AD≥19.75	rough dimension for depth must be greater than or equal to pocket depth incremental minus allowance depth, i.e. 21-1.25=19.75

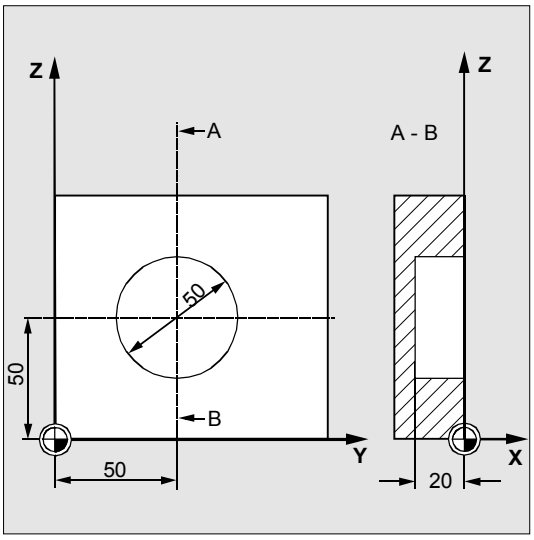
Other Information

A tool offset must be activated before the cycle is called. Otherwise the cycle is aborted with Alarm 61000 "No tool offset active".
A new workpiece coordinate system that influences the actual value display is used in the cycle. The zero point of this coordinate system lies on the pocket center point.
The original coordinate system becomes active again after the end of the cycle.

Programming example

Circular pocket

With this program you can machine a circular pocket in the YZ plane. The center point is defined by Y50 Z50. The infeed axis for the depth infeed is the X axis. Neither a final machining allowance nor a safety distance is defined. The pocket will be machined using down-cut milling. Infeed occurs along a helical path.




```
N10 G19 G90 G0 S650 M3                  ;Specification of technology values
N15 T20 D1
N17 M6
N20 Y50 Z50                             ;Approach start position
N30 Pocket4(3, 0, 0, -20, 25, 50, ->    ;Cycle call
50, 6, 0, 0, 200, 100, 1, 21, 0, ->
0, 0, 2, 3)
N40 M30                                 ;End of program
```

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

3.11 Face milling - CYCLE71



The cycle CYCLE71 is available in SW 4 and higher.



Programming

CYCLE71 (_RTP, _RFP, _SDIS, _DP, _PA, _PO, _LENG, _WID, _STA,_MID, _MIDA, _FDP, _FALD, _FFP1, _VARI, _FDP1)



Parameters

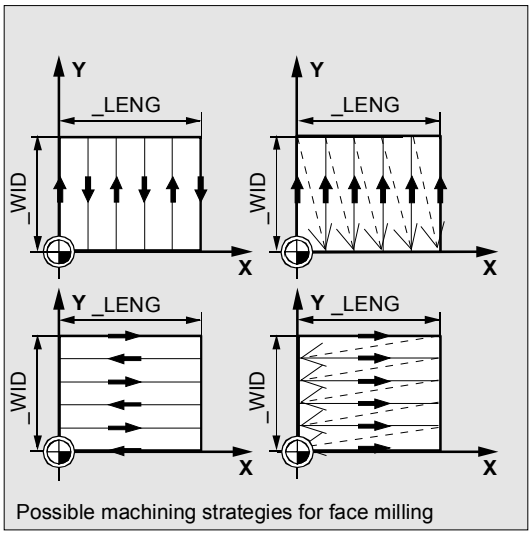
The following input parameters are always required:

_RTP	real	Retraction plane (absolute)
_RFP	real	Reference plane (absolute)
_SDIS	real	Safety distance (to be added to the reference plane, enter without sign)
_DP	real	Depth (absolute)
_PA	real	Starting point, abscissa (absolute)
_PO	real	Starting point, ordinate (absolute)
_LENG	real	Rectangle length along the 1st axis, incremental. The corner from which dimensions are measured is given by the plus/minus sign.
_WID	real	Rectangle length along the 2nd axis, incremental. The corner from which dimensions are measured is given by the plus/minus sign.
_STA	real	Angle between the longitudinal axis of the rectangle and the first axis of the plane (abscissa, enter without sign); Value range: $0° \leq$ _STA $< 180°$
_MID	real	Maximum infeed depth (enter without sign)
_MIDA	real	Maximum infeed width value for solid machining in the plane (enter without sign)
_FDP	real	Retraction travel in cutting direction (incremental, enter without sign)
_FALD	real	Final machining allowance in depth (incremental, enter without sign) In the roughing mode, _FALD refers to the remaining material on the surface.
_FFP1	real	Feedrate for surface machining
_VARI	integer	Type of machining: (enter without sign) UNITS DIGIT: Values: 1...Roughing 2...Finishing TENS DIGIT: Values: 1...Parallel to the abscissa, in one direction 2...Parallel to the ordinate, in one direction 3...Parallel to the abscissa, in alternate directions 4...Parallel to the ordinate, in alternate directions
_FDP1	real	Overrun travel in direction of plane infeed (incremental, enter without sign)

© Siemens AG 2004 All rights reserved.

Function

With cycle CYCLE71, you can face mill any rectangular surface. The cycle differentiates between roughing (machining the surface in several steps until reaching the final machining allowance) and finishing (end milling the surface in one step). Maximum infeed can be defined in width and depth.
The cycle operates without cutter radius compensation. The depth infeed is programmed in the open.



Possible machining strategies for face milling

Sequence

Position reached before the beginning of the cycle:
This can be any position from which the starting position on the infeed point at the retraction plane level can be reached without collision.

The cycle implements the following motion sequence:

- G0 is applied to approach the infeed point on the current position plane. The reference plane, shifted forward by the safety distance, is then also approached with G0 from this position. Then, also with G0, infeed to machining plane. G0 is possible, since infeed occurs in the open.
  There are several roughing strategies (paraxial in one direction or back and forth).
- Motion sequence when roughing (VARI=X1):
  Roughing is possible on several planes according to the programmed values _DP, _MID and _FALD. Machining will be performed in the downward direction, i.e. by removing stock on one plane at a time, and then executing the next depth infeed in open space (parameter_FDP).
  The traversing paths for stock removal on the plane are determined by the settings in parameters _LENG, _WID, _MIDA, _FDP, _FDP1 and the cutter radius of the active tool.
  The first path to be milled is always selected so that the infeed width is exactly _MIDA, so that no width exceeds the maximum possible value. The tool

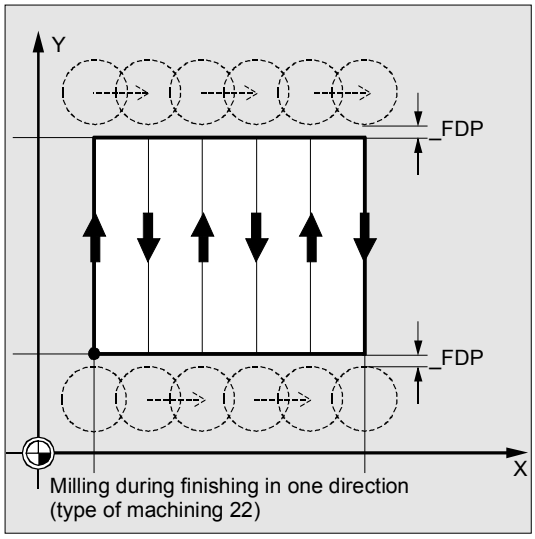
© Siemens AG 2004 All rights reserved.

center point thus does not always travel exactly to the edge (only if _MIDA = mill radius). The dimension by which the tool traverses outside the edge always equals

cutter diameter - _MIDA,

even when only 1 surface cut is performed, i.e. surface width + overrun less than _MIDA. The other paths for width infeed are calculated internally so as to produce a uniform path width (<=_MIDA).

- Motion sequence when finishing (VARI=X2):
  When finishing, the surface is once milled in the plane. The final machining allowance for roughing must also be selected so that the remaining depth can be machined in one pass with the finishing tool. After each surface milling pass in the plane, the tool retracts completely. The retraction travel is programmed by the parameter _FDP.
- Machining in one direction stops at the final machining allowance + safety distance and the next starting point is reached at rapid traverse. Roughing in one directions stops when reaching the calculated infeed depth + safety distance. The infeed depth is performed towards the same point as for roughing.
  After finishing has been completed, the tool retracts from the last position reached to the retraction plane _RTP.



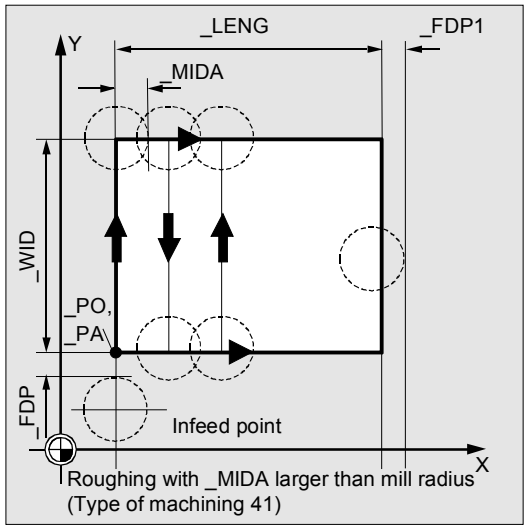
Milling during finishing in one direction
(type of machining 22)

Explanation of the parameters

See Subsection 2.1.2. (Drilling, Centering – CYCLE81) for a description of parameters _RTP, _RFP, _SDIS
See Section 3.9 (POCKET3) for a description of parameters _STA, _MID, _FFP1.
See Section 3.2 for cycle setting data _ZSD[1].

_DP (depth)
The depth can be defined as an absolute value (_DP) in relation to the reference plane.

_PA, _PO (starting point)
With parameters _PA and _PO you define the starting point of the surface in the abscissa and ordinate.



Roughing with _MIDA larger than mill radius
(Type of machining 41)

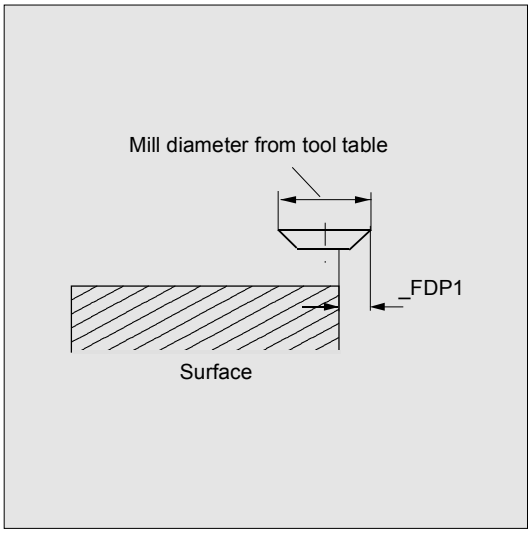
© Siemens AG 2004 All rights reserved.

_LENG, _WID (length)
With parameters _LENG and _WID you determine the length and width of the rectangle in the plane. The sign determines the position of the rectangle relative to _PA and _PO.

_MIDA (max. infeed width)
With this parameter, you define the maximum infeed width for solid machining in the plane. In the same way as the known calculation of the infeed depth (equal distribution of the overall depth using the largest possible value), the width is evenly divided, using the value programmed in _MIDA as a maximum value.
If this parameter is not programmed, or if its value is 0, the cycle uses 80 percent of the mill diameter as maximum infeed width.

_FDP (retraction travel)
This parameter defines the dimension for retraction travel in the plane. This parameter must be programmed with a value greater than zero.

_FDP1 (overrun travel)
By means of this parameter an overrun travel in the direction of the plane infeed may be defined (_MIDA) allowing the deviation between the current cutter radius and the cutting edge (e.g. tool nose radius or inclined cutting tips) to be compensated. The last cutter center point path therefore always corresponds to _LENG (or _WID) + _FDP1 tool radius (from correction table).



_FALD (final machining allowance)
During roughing, the depth final machining allowance used is defined by this parameter.
The residual material designated as the final machining allowance must always be specified for finish cutting so as to ensure that the tool can be lifted and inserted at the starting point of the next cut without risk of collision.

© Siemens AG 2004 All rights reserved.

_VARI (machining type)
You can define the type of machining with parameter _VARI.
Possible settings are:

Unit's place:
- 1=Roughing to final machining allowance
- 2=Finishing

Ten's place:
- 1=Parallel to the abscissa, in one direction
- 2=Parallel to the ordinate, in one direction
- 3=Parallel to the abscissa, with changing direction
- 4=Parallel to the ordinate, with changing direction

If another value has been programmed for parameter _VARI, the cycle is aborted after alarm 61002 "Machining type incorrectly defined" is output.



Other Information

A tool offset must be activated before the cycle is called. The cycle is otherwise aborted with alarm 61000 "No tool offset active".

© Siemens AG 2004 All rights reserved.

Programming example

Face milling

Parameters for cycle call:

- Retraction plane: 10 mm
- Reference plane: 0 mm
- Safety distance: 2 mm
- Milling depth: -11 mm
- Max. infeed depth 6 mm
- No final machining allowance -
- Starting point of the rectangle X = 100mm
  Y = 100 mm
- Rectangle dimensions X = +60 mm
  Y = +40 mm
- Angle of rotation in the plane 10 degrees
- Max. infeed width 10 mm
- Retract travel at end of milling path: 5 mm
- Feedrate for surface machining 4000 mm/min.
- Type of machining: roughing parallel to the X axis with changing direction
- Overrun on last cut as determined by the cutting edge geometry 2 mm

```
%_N_TSTCYC71_MPF                                ;Program for face milling with CYCLE71
;$PATH=/_N_MPF_DIR
;*
$TC_DP1[1,1]=120                                ;Tool type
$TC_DP6 [1,1] = 10                              ;Tool radius
N100 T1
N102 M06
N110 G17 G0 G90 G54 G94 F2000 X0 Y0 Z20         ;Approach start position
;
CYCLE71(  10,  0,   2,-11, 100, 100, ->         ;Cycle call
-> 60, 40, 10, 6, 10, 5, 0, 4000, 31, 2)
N125 G0 G90 X0 Y0
N130 M30                                        ;End of program
```

Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

3.12 Path milling - CYCLE72

The cycle CYCLE72 is available with SW 4
(not for FM-NC).

Programming

CYCLE72 (_KNAME, _RTP, _RFP, _SDIS, _DP, _MID, _FAL, _FALD, _FFP1, _FFD, _VARI, _RL, _AS1, _LP1, _FF3, _AS2, _LP2)

Parameters

The following input parameters are always required:

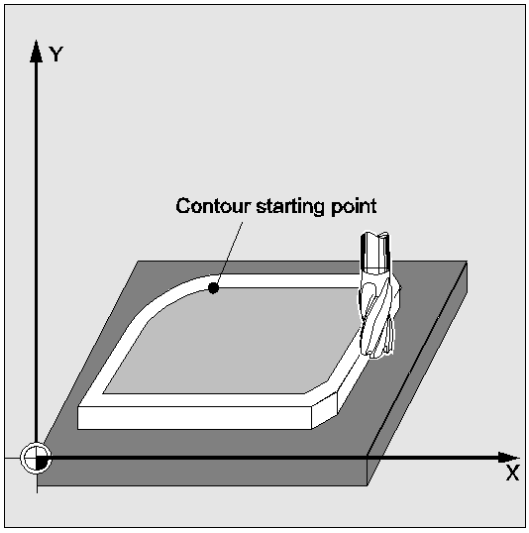
_KNAME	string	Name of the contour subroutine
_RTP	real	Retraction plane (absolute)
_RFP	real	Reference plane (absolute)
_SDIS	real	Safety distance (to be added to the reference plane, enter without sign)
_DP	real	Depth (absolute)
_MID	real	Maximum infeed depth (incremental, enter without sign)
_FAL	real	Final machining allowance at the edge contour (enter without sign)
_FALD	real	Final machining allowance at the base (incremental, enter without sign)
_FFP1	real	Feedrate for surface machining
_FFD	real	Feedrate for depth infeed (enter without sign)
_VARI	integer	Type of machining: (enter without sign) UNITS DIGIT: Values: 1...Roughing 2...Finishing TENS DIGIT: Values: 0...Intermediate paths with G0 1...Intermediate paths with G1 HUNDREDS DIGIT: Values: 0...Retraction at end of contour to _RTP 1...Retraction at end of contour to _RFP + _SDIS 2...Retraction by _SDIS at end of contour 3...No retraction at end of contour
_RL	integer	Contouring is centric, on right or left (with G40, G41 or G42, enter without sign) Values: 40...G40 (approach and return, straight line only) 41...G41 42...G42

© Siemens AG 2004 All rights reserved.

_AS1	integer	Specification of approach direction/path: (enter without sign) UNITS DIGIT: Values: 1...Straight tangential line 2...Quadrant 3...Semicircle TENS DIGIT: Values: 0...Approach to the contour in the plane 1...Approach to the contour in a spatial path
_LP1	real	Length of the approach travel (along a straight line) or radius of the mill center path of the arc of approach (along a circle) (enter without sign)
The other parameter can be preset optionally (enter without sign).		
_FF3	real	Return feedrate and feedrate for intermediate positioning in the plane (when retracting)
_AS2	integer	Specification of return direction/path: (enter without sign) UNITS DIGIT: Values: 1...Straight tangential line 2...Quadrant 3...Semicircle TENS DIGIT: Values: 0...Retraction from the contour in the plane 1...Retraction from the contour in a spatial path
_LP2	real	Length of the return travel (along a straight line) or radius of the return arc (along a circle) (enter without sign)

© Siemens AG 2004 All rights reserved.

Function



With the cycle CYCLE72 it is possible to mill along any contour defined in a subroutine. The cycle operates with or without cutter radius compensation.
The contour does not need to be closed; internal or external machining is defined by the position of the cutter radius compensation (center, on left or right of contour).
The contour must be programmed in the direction of milling and must be in the same plane. In addition, it must consist of at least 2 contour blocks (start and end points) since the contour subroutine is invoked within the cycle.

Cycle functions:

- Selection of roughing (single-pass parallel to the contour considering a final machining allowance if necessary at several depths until reaching the final machining allowance) and finishing (single-pass of final contour, if necessary at several depths)
- Flexible approach and retraction to/from the contour either tangentially or radially (quadrant or semicircle)
- Programmable depth infeed
- Intermediate motions either with rapid traverse or at feedrate.

The requirement for executing a cycle is NC SW 4.3. or higher including the function "Soft approach and return".

Sequence

Position reached before the beginning of the cycle:
The starting position can be any position from which the start of the contour at the retraction plane level can be reached without collision.

© Siemens AG 2004 All rights reserved.

The cycle creates the following motion sequence when roughing (VARI=XX1):

The depth infeeds are divided evenly using the highest possible value according to the preset parameter.

- Travel to starting point for initial cut with G0/G1 (and _FF3). This point is calculated internally in the control and depends on
  - the contour starting point (first point in subroutine),
  - the direction of the contour at the starting point,
  - the approach mode and corresponding
  - parameters and the tool radius.

  The cutter radius path compensation is activated in this block.
- Depth infeed to first or next machining depth plus programmed safety distance (parameter _SDIS) with G0/G1. The first processing depth results from
  - the overall depth,
  - the final machining allowance and
  - the maximum possible depth infeed.
- Approach the contour vertically with down feed and then in the plane at the programmed feedrate for surface machining or 3D at the feedrate programmed under _FAD corresponding to the programming for flexible approach.
- Milling along the contour with G40/G41/G42.
- Soft retraction from the contour with G1 and still with the feedrate for surface machining by a lift calculated within the cycle.
- Retraction with G0 /G1 (and feedrate for intermediate travel _FF3) depending on program.
- Return to depth infeed point with G0/G1 (and _FF3).
- This operating sequence is repeated on the next machining plane, until reaching the final machining allowance in depth.

When roughing is over, the tool lies on the contour starting point (calculated within the control unit) at the retraction plane level.

© Siemens AG 2004 All rights reserved.

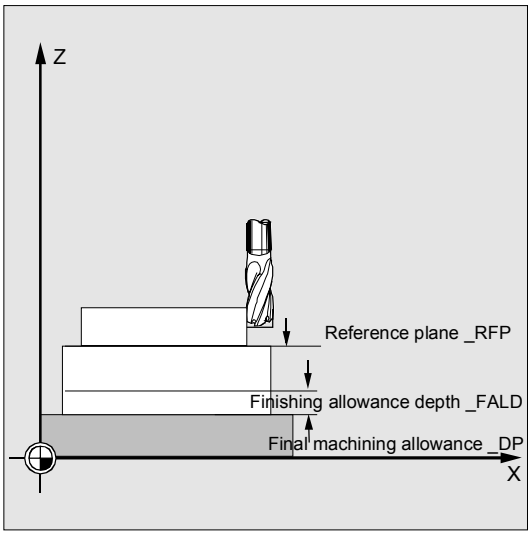
The cycle creates the following motion sequence when finishing (VARI=XX2):
During finishing, milling is performed at the relevant infeed along the base of the contour until the final dimension is reached.
Approaching and retraction to/from the contour is performed in a flexible way according to the corresponding preset parameters. The corresponding path is calculated within the control unit.
At the end of the cycle, the tool is positioned at the contour retraction point at the retraction plane level.

Contour programming
For programming the contour, please note the following:

- In the subroutine no programmable frame (TRANS, ROT, SCALE, MIRROR) may be selected before the first programmed position.
- The first block of the contour subroutine is a straight line block containing G90, G0 and defines the contour start.
- The cutter radius compensation is selected and deselected from the upper level cycle; then the contour subroutine has no G40, G41, G42 programmed.

© Siemens AG 2004 All rights reserved.

Explanation of the parameters



See Subsection 2.1.2. (Drilling, Centering – CYCLE81) for a description of parameters _RTP, _RFP, _SDIS

See Section 3.9 for a description of parameters _MID, _FAL, _FALD, _FFP1, _FFD. See Section 3.11 for a description of parameter _DP.
See Section 3.2 for cycle setting data _ZSD[1].

_KNAME (name)
The contour to be milled is completely programmed in a subroutine. With parameter _KNAME you can define the name of the contour subroutine.

The milling contour can also be a section of the calling routine or of any other program. The section is identified by start or end labels or by block numbers. In this case, the program name and labels/block number are identified by an ":".

Examples:

_KNAME=“CONTOUR_1“	The milling contour is the complete program "Contour_1".
_KNAME=“START:END“	The milling contour is defined as the section starting from the block labeled START to the block labeled END in the calling routine.
_KNAME= “/_N_SPF_DIR/_N_CONTOUR_1_SPF:N130:N210“	The milling contour is defined in blocks N130 to N210 in program CONTOUR_1. The program name must be entered complete with path and extension, see description of call in References: /PGA/ Programming Guide Advanced

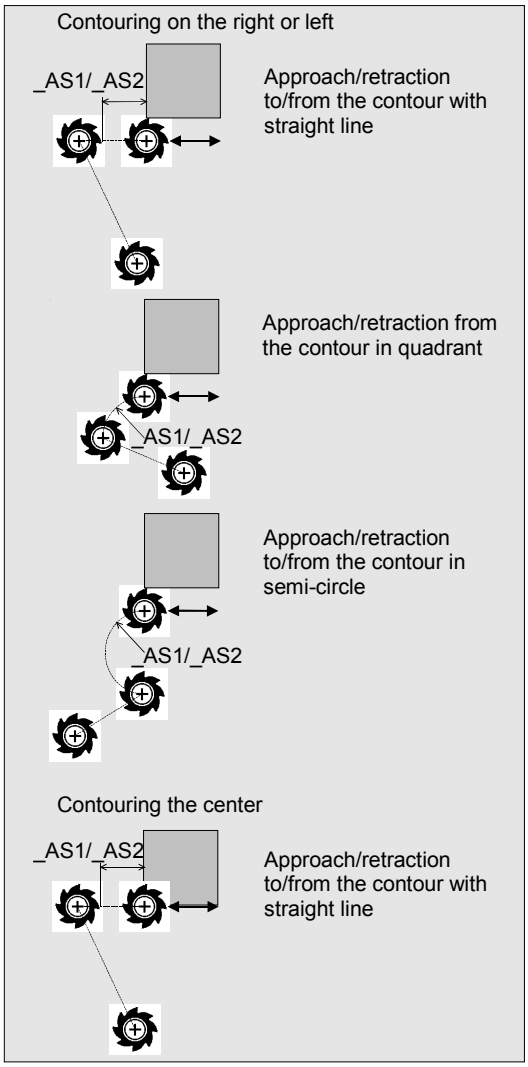
If the section is defined by block numbers, it must be noted that these block numbers for the section in _KNAME must be adjusted if the program is modified and subsequently renumbered.

© Siemens AG 2004 All rights reserved.

_VARI (machining type)
You can define the type of machining with parameter _VARI. For possible values, see "Parameter CYCLE72".
If another value has been programmed for parameter _VARI, the cycle is aborted after alarm 61002 "Machining type incorrectly defined" is output.

_RL (travel around the contour)
Parameter _RL is set to define how the tool must travel around the contour, i.e. along the center path or on the left or right-hand side with G40, G41 or G42. For possible values, see "Parameter CYCLE72".

_AS1, _AS2 (direction of approach/approach travel, direction of retraction/retraction travel)
With the parameter _AS1 you can specify the approach travel and with _AS2 the retraction travel. For possible values, see "Parameter CYCLE72". If _AS2 is not programmed, then the behavior programmed for the approach path will apply to the return path.
The flexible approach of the contour along a 3-D path (helix or straight line) should be programmed only if the tool is suitable and not yet engaged.



With center path travel (G40), tool must approach and return along a straight line.

_LP1, _LP2 (length, radius)
Parameter _LP1 is set to program the approach path or approach radius (distance between tool outer edge and contour starting point) and _LP2 to program the return path or return radius (distance between tool outer edge and end point of contour).
Parameters _LP1, _LP2 must be set to >0. A setting of zero generates error message 61116 "Approach or retract path=0."

When G40 is programmed, the approach or retract path corresponds to the distance between the tool center point and the starting or end point of the contour.

© Siemens AG 2004 All rights reserved.

_FF3 (retraction feedrate)
Parameter _FF3 is used to define a retraction feedrate for intermediate positioning in the plane (in the open) when intermediate motions are to be performed with feed (G01). If no feedrate is programmed, the intermediate motions are carried out with surface feed for G01.

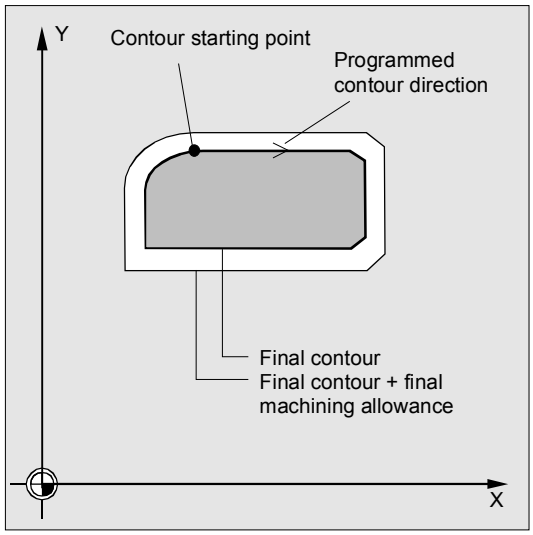


Other Information

A tool offset must be activated before the cycle is called. The cycle is otherwise aborted with alarm 61000 "No tool offset active".

© Siemens AG 2004 All rights reserved.

Programming example 1



Milling a closed contour externally

This program is used to mill a contour as shown in the figure.

Parameters for cycle call:

- Retraction plane 250 mm
- Reference plane 200
- Safety distance 3 mm
- Depth 175 mm
- Maximum depth infeed 10 mm
- Finishing allowance in depth 1.5 mm
- Feedrate depth infeed 400 mm/min.
- Finishing allowance in plane 1 mm
- Feedrate in the plane 800 mm/min.
- Machining: Roughing up to the finishing
  Final machining allowance,
  intermediate paths
  with G1, for intermediate paths
  Return in Z to _RFP + _SDIS

Parameters for the approach:

- G41 - to the left of the contour, i.e. external machining
- Approach and return on quadrant
  in plane 20 mm radius
- Retraction feedrate 1000 mm/min

```
%_N_EDGECONTOUR1_MPF                    ;Program for re-milling a contour with
;$PATH=/_N_MPF_DIR                      ;CYCLE72
N10 T20 D1                              ;T20: milling cutter with radius 7
N15 M6                                  ;Changing tool T20,
N20 S500 M3 F3000                       ;Program feedrate and spindle speed
N25 G17 G0 G90 X100 Y200 Z250 G94       ;Approach start position
N30 CYCLE72 ("MYCONTOUR", 250, 200, ->  ;Cycle call
-> 3, 175, 10,1, 1.5, 800, 400, 111, ->
-> 41, 2, 20, 1000, 2, 20)
N90 X100 Y200
N95 M02                                 ;End of program
```

Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

```
%_N_MYCONTOUR_SPF                    ;Subroutine for contour milling (for example)
;$PATH=/_N_SPF_DIR
N100 G1 G90 X150 Y160                ;Start point of contour
N110 X230 CHF=10
N120 Y80 CHF=10
N130 X125
N140 Y135
N150 G2 X150 Y160 CR=25
N160 M17
```



Programming example 2

Milling round the outside of a closed contour as described in sample program 1, with the contour defined in the calling program

```
$TC_DP1[20,1]=120 $TC_DP6[20,1]=7
N10 T20 D1                           ;T20: milling cutter with radius 7
N15 M6                               ;Changing tool T20,
N20 S500 M3 F3000                    ;Program feedrate and spindle speed
N25 G17 G0 G90 G94 X100 Y200 Z250 -> ;Approach starting position, ;cycle call
CYCLE72 ("START:END", 250, 200, ->
-> 3, 175, 10,1, 1.5, 800, 400, 11, ->
-> 41, 2, 20, 1000, 2, 20)
N30 G0 X100 Y200
N35 GOTOF END
START:
N100 G1 G90 X150 Y160
N110 X230 CHF=10
N120 Y80 CHF=10
N130 X125
N140 Y135
N150 G2 X150 Y160 CR=25
END:
N160 M02
```


© Siemens AG 2004 All rights reserved.

3.13 Milling rectangular spigots - CYCLE76



Programming

CYCLE76 (_RTP, _RFP, _SDIS, _DP, _DPR, _LENG, _WID, _CRAD, _PA, _PO, _STA, _MID, _FAL, _FALD, _FFP1, _FFD, _CDIR, _VARI, _AP1, _AP2)



Parameters

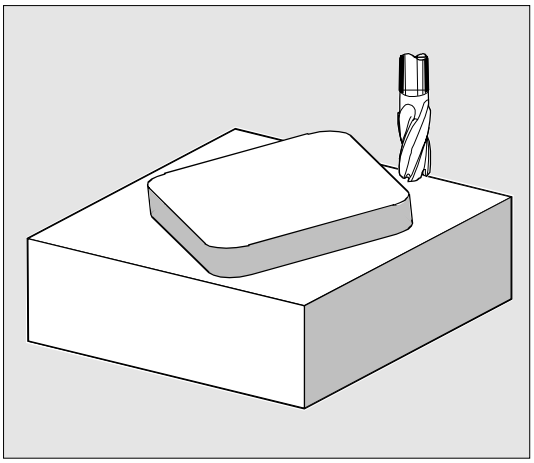
The following input parameters are always required:

_RTP	real	Retraction plane (absolute)
_RFP	real	Reference plane (absolute)
_SDIS	real	Safety distance (to be added to the reference plane, enter without sign)
_DP	real	Depth (absolute)
_DPR	real	Depth relative to the reference plane (enter without sign)
_LENG	real	Spigot length, for dimensioning from corner with sign
_WID	real	Spigot width, for dimensioning from corner with sign
_CRAD	real	Spigot corner radius (enter without sign)
_PA	real	Spigot reference point, abscissa (absolute)
_PO	real	Spigot reference point, ordinate (absolute)
_STA	real	Angle between longitudinal axis and 1st axis of plane
_MID	real	Maximum depth infeed (incremental, enter without sign)
_FAL	real	Final machining allowance on edge contour (incremental)
_FALD	real	Final machining allowance at the base (incremental, enter without sign)
_FFP1	real	Feedrate on contour
_FFD	real	Feedrate for depth infeed
_CDIR	integer	Milling direction: (enter without sign) Values: 0...Down-cut milling 1...Up-cut milling 2...with G2 (independent of spindle direction) 3...with G3
_VARI	integer	Type of machining: Values: 1...Roughing to final machining allowance 2...Finishing (allowance X/Y/Z=0)
_AP1	real	Length of blank spigot
_AP2	real	Width of blank spigot

© Siemens AG 2004 All rights reserved.

Function

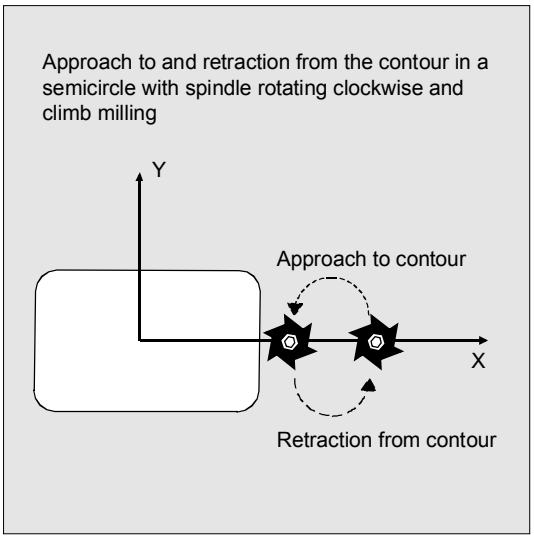
With this cycle you can machine rectangular spigots in the machining plane. For finishing, a face cutter is needed. Depth infeed is always performed in the position reached prior to semicircular positioning on the contour.



Sequence

Position reached before the beginning of the cycle:
The starting point is a position in the positive range of the abscissa with integrated approach semicircle and allowance for programmed, abscissa-related blank dimension.

Sequence of motions for roughing (_VARI=1)
Approach to and exit from contour:
The retraction plane (_RTP) is approached in rapid traverse so that the tool can be positioned from there on the starting point in the machining plane. The starting point is defined as being 0 degrees in relation to the abscissa.
The tool is fed in at rapid traverse to the safety distance (_SDIS) and then traverses to machining depth at normal feedrate.
The tool approaches the spigot contour along a semicircular path.
The milling direction can be defined as up-cut or down-cut milling in relation to the spindle direction.
If the spigot has been circumnavigated once, the tool lifts off the contour in the plane along a semicircular path and is then fed in to the next machining depth.
The contour is then approached again along a semicircle and the spigot circumnavigated once. It repeats the process until the spigot depth has been reached. The tool then approaches the retraction plane (_RTP) in rapid traverse.



© Siemens AG 2004 All rights reserved.

Depth infeed:

- Infeed to safety distance
- Insertion to machining depth.

The first machining depth is the product of:

- the total depth,
- the final machining allowance and
- the maximum possible depth infeed.

Sequence of motions for finishing (_VARI=2)

Depending on the setting of parameters _FAL and _FALD, a finishing operation is performed on the spigot surface or base or both. The approach strategy matches the motions in the plane executed for roughing operations.

Explanation of the parameters

See Subsection 2.1.2. (Drilling, Centering – CYCLE81) for a description of parameters RTP, RFP, SDIS, DP, DPR.
See Section 3.9 for a description of parameters _MID, _FAL, _FALD, _FFP1, _FFD.
See Section 3.2 for cycle setting data _ZSD[2].

_LENG, _WID and _CRAD (spigot length, spigot width and corner radius)

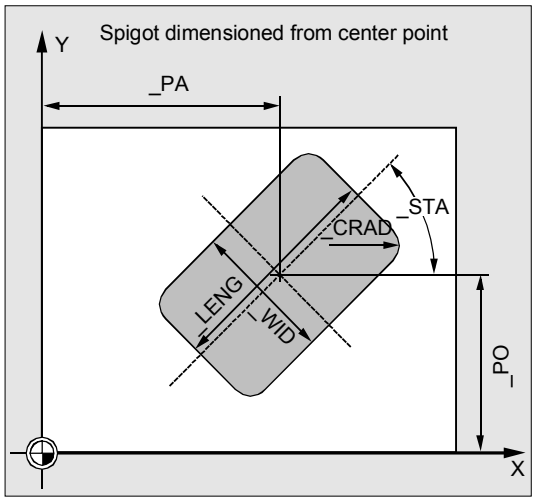
The shape of a spigot in the plane is determined with parameters _LENG, _WID and _CRAD.
The spigot can be dimensioned from the center or from one corner point. When dimensioning from a corner point, use _LENG and _WID with sign.
The absolute length value (_LENG) always refers to the abscissa (with a plane angle of zero degrees).

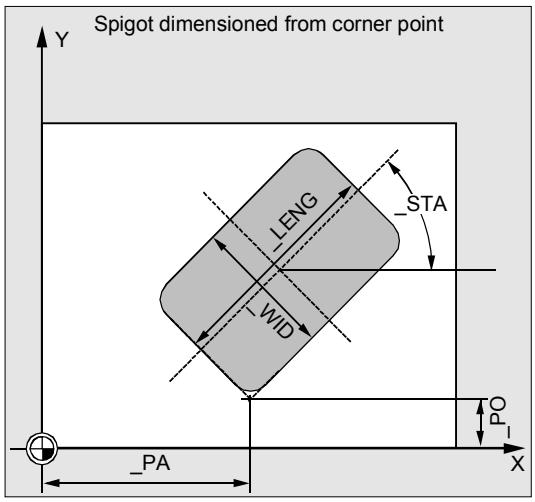
_PA, _PO (reference point)

Parameters _PA and _PO are set to define the reference point of the spigot in abscissa and ordinate. This is either the spigot center point or a corner point. The value of this parameter depends on cycle setting data bit _ZSD[2]:

- 0 means spigot center point
- 1 means corner point.

When the spigot is dimensioned from a corner, the length and width parameters must be entered with sign (_LENG, _WID) so that a unique position for the spigot is defined.





© Siemens AG 2004 All rights reserved.

_STA (angle)
_STA specifies the angle between the 1st axis of the plane (abscissa) and the longitudinal axis of the spigot.

_CDIR (milling direction)
The machining direction of the spigot is defined in this parameter.
Under parameter _CDIR the mill direction

- directly "2 for G2" and "3 for G3" or
- alternatively "up-cut milling" or "down-cut milling"

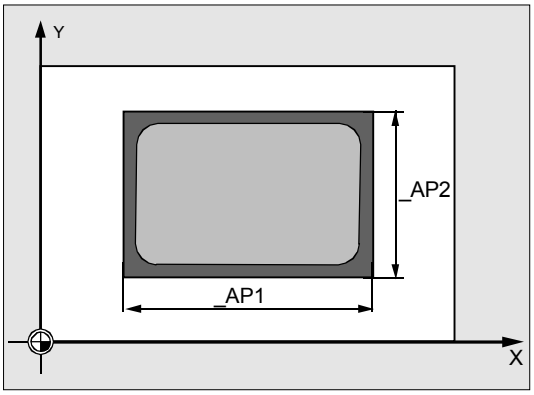
can be programmed. Up-cut milling or down-cut milling is determined within the cycle via the spindle direction activated prior to the cycle call.

Down-cut	**Up-cut**
M3 → G3	M3 → G2
M4 → G2	M4 → G3

_VARI (machining type)
You can define the type of machining with parameter _VARI.
Possible settings are:

- 1=Roughing
- 2=Finishing

_AP1, _AP2 (blank dimensions)
Blank dimensions (e.g. in the case of precast workpieces) can be taken into account in machining of the spigot.
The basic size for the length and width (_AP1 and _AP2) are programmed without sign and their symmetrical positions around the spigot center computed in the cycle. The internally calculated radius of the approach semicircle is dependent on this dimension.



Other Information

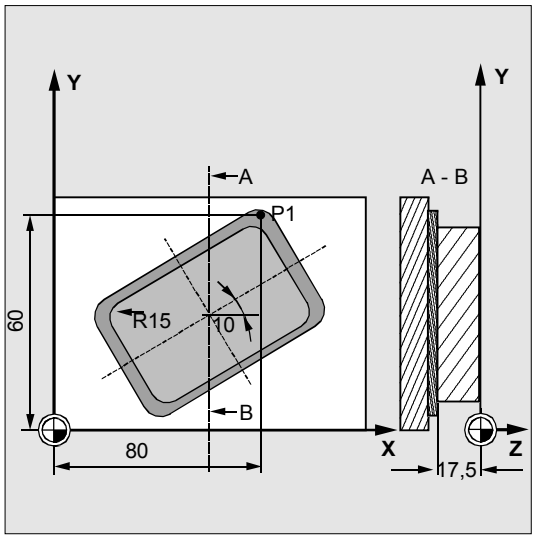
A tool offset must be activated before the cycle is called. The cycle is otherwise aborted with alarm 61009 "Active tool number=0".
A new workpiece coordinate system that influences the actual value display is used in the cycle. The zero point of this coordinate system lies on the pocket center point.
The original coordinate system becomes active again after the end of the cycle.

© Siemens AG 2004 All rights reserved.

Programming example

Spigots

This program allows you to machine a spigot that is 60mm long, 40mm wide, 15mm deep in the XY plane and with a corner radius of 15mm. The spigot has an angle of 10 degrees in relation to the X axis and is programmed from a corner point P1. When a spigot is dimensioned with reference to corners, the length and width must be entered with a sign to define a unique position for the spigot. The spigot is premachined with an allowance of 80mm in its length and 50mm in its width.




```
N10 G90 G0 G17 X100 Y100 T20 D1 S3000 M3   ;Specification of technology values
N11 M6
N20 _ZSD[2]=1                              ;Dimensioning of spigot referred to ;corners
N30 CYCLE76 (10, 0, 2, -17.5, , -60, ->    ;Cycle call
-> -40, 15, 80, 60, 10, 11, , , 900, ->
-> 800, 0, 1, 80, 50)
N40 M30                                    ;End of program
```

Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

3.14 Milling circular spigots - CYCLE77



Programming

CYCLE77 (_RTP, _RFP, _SDIS, _DP, _DPR, _PRAD, _PA, _PO, _MID, _FAL, _FALD, _FFP1, _FFD, _CDIR, _VARI, _AP1)



Parameters

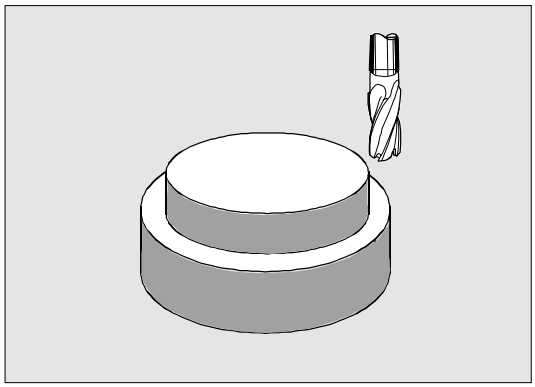
The following input parameters are always required:

_RTP	real	Retraction plane (absolute)
_RFP	real	Reference plane (absolute)
_SDIS	real	Safety distance (to be added to the reference plane, enter without sign)
_DP	real	Depth (absolute)
_DPR	real	Depth relative to the reference plane (enter without sign)
_PRAD	real	Diameter of spigot (enter without sign)
_PA	real	Spigot center point, abscissa (absolute)
_PO	real	Spigot center point, ordinate (absolute)
_MID	real	Maximum depth infeed (incremental, enter without sign)
_FAL	real	Final machining allowance on edge contour (incremental)
_FALD	real	Final machining allowance at the base (incremental, enter without sign)
_FFP1	real	Feedrate on contour
_FFD	real	Feedrate for depth infeed (or spatial infeed)
_CDIR	integer	Milling direction: (enter without sign) Values: 0...Down-cut milling 1...Up-cut milling 2...with G2 (independent of spindle direction) 3...with G3
_VARI	integer	Machining type Values: 1...Roughing to final machining allowance 2...Finishing (allowance X/Y/Z=0)
_AP1	real	Diameter of blank spigot

© Siemens AG 2004 All rights reserved.

Function

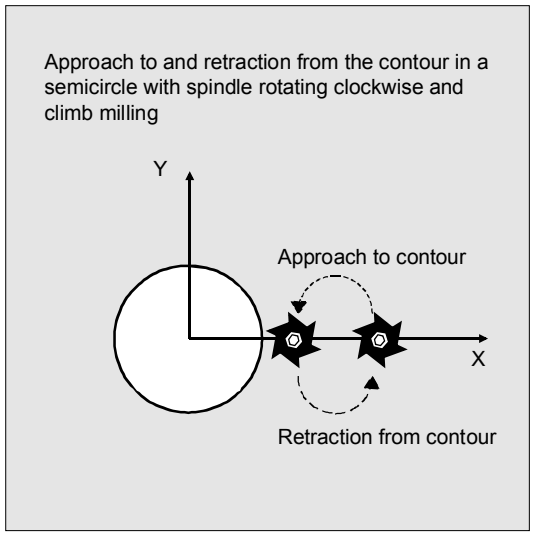
With this cycle you can machine circular spigots in the machining plane. For finishing, a face cutter is needed. Depth infeed is always performed in the position reached prior to semicircular positioning on the contour.



Sequence

Position reached before the beginning of the cycle:
The starting point is a position in the positive range of the abscissa with integrated approach semicircle and allowance for programmed blank dimension.

Sequence of motions for roughing (_VARI=1)
Approach to and exit from contour:
The retraction plane (_RTP) is approached in rapid traverse so that the tool can be positioned from there on the starting point in the machining plane. The starting point is defined as being 0 degrees in relation to the abscissa axis.
The tool is fed in at rapid traverse to the safety distance (_SDIS) and then traverses to machining depth at normal feedrate. The spigot contour is approached along a semicircular path, making allowance for the programmed blank spigot.
The milling direction can be defined as up-cut or down-cut milling in relation to the spindle direction.
If the spigot has been circumnavigated once, the tool lifts off the contour in the plane along a semicircular path and is then fed in to the next machining depth.
The contour is then approached again along a semicircle and the spigot circumnavigated once. It repeats the process until the spigot depth has been reached.
The tool then approaches the retraction plane (_RTP) in rapid traverse.



© Siemens AG 2004 All rights reserved.

Depth infeed:

- Infeed to safety distance
- Insertion to machining depth.

The first machining depth is the product of:

- the total depth,
- the final machining allowance and
- the maximum possible depth infeed.

Sequence of motions for finishing (_VARI=2)

Depending on the setting of parameters _FAL and _FALD, a finishing operation is performed on the spigot surface or base or both. The approach strategy matches the motions in the plane executed for roughing operations.

Explanation of the parameters

See Subsection 2.1.2 (Drilling, Centering – CYCLE81) for a description of parameters RTP, RFP, SDIS, DP, DPR.
See Section 3.9 for a description of parameters _MID, _FAL, _FALD, _FFP1, _FFD.

_PRAD (diameter of spigot)

The diameter must be entered without a sign.

_PA, _PO (spigot center point)

With parameters _PA and _PO you define the reference point of the spigot.
Circular spigots are always measured from the center.

_CDIR (milling direction)

The machining direction of the spigot is defined in this parameter.
Under parameter _CDIR the mill direction

- directly "2 for G2" and "3 for G3" or
- alternatively "up-cut milling" or "down-cut milling"

can be programmed. Up-cut milling or down-cut milling is determined within the cycle via the spindle direction activated prior to the cycle call.

Down-cut	**Up-cut**
M3 → G3	M3 → G2
M4 → G2	M4 → G3

_VARI (machining type)

You can define the type of machining with parameter _VARI.
Possible settings are:

- 1=Roughing
- 2=Finishing

© Siemens AG 2004 All rights reserved.

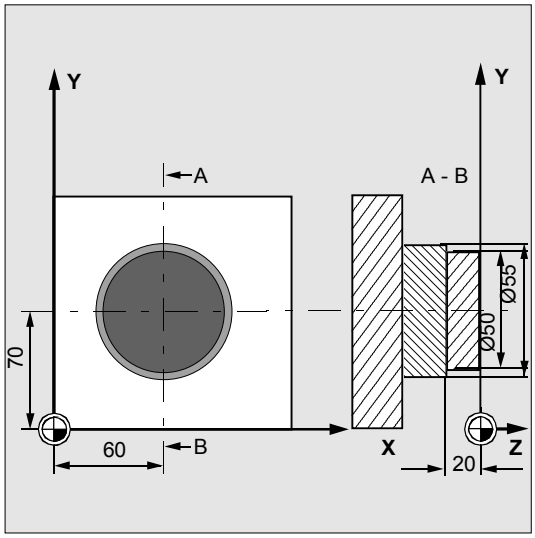
_AP1 (diameter of blank spigot)
This parameter defines the blank dimension of the spigot (without sign). The internally calculated radius of the approach semicircle is dependent on this dimension.

Other Information

A tool offset must be activated before the cycle is called. The cycle is otherwise aborted with alarm 61009 "Active tool number=0".
A new workpiece coordinate system that influences the actual value display is used in the cycle. The zero point of this coordinate system lies on the pocket center point.
The original coordinate system becomes active again after the end of the cycle.

Programming example

Circular spigot
Machine a spigot from a blank with a diameter of 55mm and a maximum infeed of 10mm per cut. Enter a final machining allowance for finishing the spigot surface. The entire spigot is machined in a down-cut milling operation.




```
N10 G90 G17 G0 S1800 M3 D1 T1          ;Specification of technology values
N11 M6
N20 CYCLE77 (10, 0, 3, -20, ,50, 60, ->  ;Roughing cycle call
-> 70, 10, 0.5, 0, 900, 800, 1, 1, 55)
N30 D1 T2 M6                           ;Change tool
N40 S2400 M3                           ;Specification of technology values
N50 CYCLE77 (10, 0, 3, -20, , 50, 60, -> ;Finishing cycle call
-> 70, 10, 0, 0, 800, 800, 1, 2, 55)
N40 M30                                ;End of program
```

Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

3.15 Pocket milling with islands - CYCLE73, CYCLE74, CYCLE75

Pocket milling with islands is an option and requires SW6 in both the NCK and HMI Advanced .

Precondition

To use the pocket milling cycle with islands, you need specific machine data settings.

References: for current infos see:

- the file "siemensd.txt" from the delivery software (standard cycles) or
- in the case of HMI Advanced F:\dh\cst.dir\HLP.dir\siemensd.txt

Function

Cycles CYCLE73, CYCLE74 and CYCLE75 enable you to machine pockets with islands.
The contours of the pocket and islands are defined in DIN code in the same program as the pocket machining operation or as a subroutine.
Cycles CYCLE74 and CYCLE75 transfer the pocket edge contour or island contours to CYCLE73, the actual pocket milling cycle.
CYCLE73 uses a geometry processor to create a machining program which it then executes. To ensure correct program processing, it is important to program cycle calls in the proper sequence.

- CYCLE74( ) ;Transfer edge contour
- CYCLE75( ) ;Transfer island contour 1
- CYCLE75( ) ;Transfer island contour 2
- ...
- CYCLE73( ) ;Machining pocket

© Siemens AG 2004 All rights reserved.

3.15.1 Transfer pocket edge contour - CYCLE74



Pocket milling with islands is an option and requires SW6 in both the NCK and HMI Advanced .



Programming

CYCLE74 (_KNAME, _LSANF, _LSEND)



Parameters

_KNAME	string	Name of contour subroutine of pocket edge contour
_LSANF	string	Block number/label identifying start of contour definition
_LSEND	string	Block number/label identifying end of contour definition



Function

Cycle CYCLE74 transfers the pocket edge contour to pocket milling cycle CYCLE73. This is done by creating a temporary internal file in the standard cycles directory and storing the transferred parameter values in it.

If a file of this type already exists, it is deleted and set up again.

For this reason, a program sequence for milling pockets with islands must always begin with a call for CYCLE74.

© Siemens AG 2004 All rights reserved.

Explanation of parameters

The edge contour can be programmed either in a separate program or in the main program that calls the routine. The contour is transferred to the cycle by parameter _KNAME, name of program or _LSANF, _LSEND and the program section from ... to identified by block numbers or labels.

So there are three options for contour programming:

- **Contour is defined in a separate program,**
  in which case only _KNAME needs to be programmed;
  e.g. CYCLE74 ("EDGE","","")
- **Contour is defined in the calling program,**
  in which case only _LSANF and _LSEND need to be programmed;
  e.g. CYCLE74 ("","N10","N160")
- **The edge contour is part of a program but not part of the program that calls the cycle,**
  in which case all three parameters need to be programmed.
  e.g. CYCLE74("EDGE","LABEL_START", "LABEL_END")

The program name is describable by its path name and program type.
Example:
_KNAME="/N_WKS_DIR/_N_EXAMPLE3_WPD/_N_EDGE_MPF"

© Siemens AG 2004 All rights reserved.

3.15.2 Transfer island contour - CYCLE75



Pocket milling with islands is an option and requires SW6 in both the NCK and HMI Advanced .



Programming

CYCLE75 (_KNAME, _LSANF, _LSEND)



Parameters

_KNAME	string	Name of contour subroutine of island contour
_LSANF	string	Block number/label identifying start of contour definition
_LSEND	string	Block number/label identifying end of contour definition



Function

Cycle CYCLE75 transfers the island contours to pocket milling cycle CYCLE73. The cycle is called once per island contour. It does not need to be called if no island contours are programmed.

The transferred parameter values are written to the temporary file opened by CYCLE74.



Explanation of parameters

The number and meaning of parameters are the same as for CYCLE74.



(see CYCLE74)

© Siemens AG 2004 All rights reserved.

3.15.3 Contour programming



Pocket edge and island contours must always be closed, i.e. the start and end points are identical.

The start point, i.e. first point on a contour must always be programmed with G0, and all other contour elements via G1 to G3.

When the contour is programmed, the last contour element (block with label or block number at end of contour) must not contain a radius or chamfer.

The tool may not stand at a starting position of the programmed contour element before CYCLE73 is invoked.

The necessary programs must always be stored in one directory (workpiece or parts program). It is permissible to use the subroutine memory for pocket edge or island contours.

Workpiece-related geometric dimensional data may be programmed in either metric or inches. Switching between these units of measurement within individual contour programs will causes errors in the machining program.

When G90/G91 are programmed alternately in contour programs, care must be taken to program the correct dimensional command at the start of the program in the sequence of contour programs to be executed.

When the pocket machining program is calculated, only the geometries in the plane are taken into account.

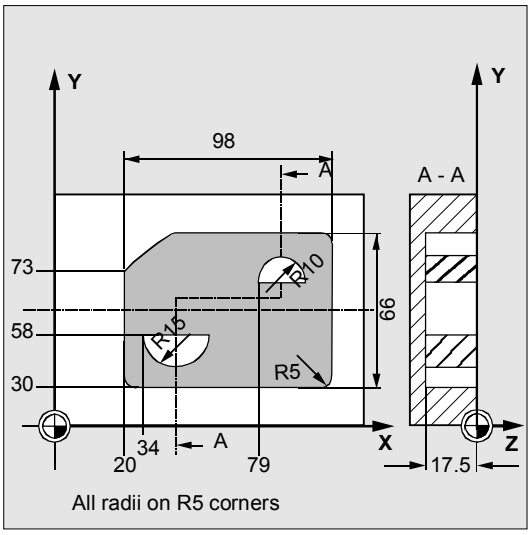
If other axes or functions (T.., D.., S.., M.., etc.) are programmed in the contour sections, they are skipped during contour preparation within the cycle.

All machine-specific program commands (e.g. tool call, speed, M command) must be programmed before the cycle commences.
Feedrates must be set as parameters in CYCLE73.
The tool radius must be greater than zero.
It is not possible to repeat island contours by offsets implemented by suitable control commands (e.g. zero offset, frames, etc.).
Every island to be repeated always has to be programmed again with the offsets calculated in the coordinates.

© Siemens AG 2004 All rights reserved.

Programming example

Sample program 1.mpf (pocket with islands)




```
%_N_SAMPLE1_MPF
;$PATH=/_N_MPF_DIR
; Example_1: Pocket with islands
;
$TC_DP1[5,1]=120 $TC_DP6[5,1]=6 $TC_DP3[5,1]=111    ;Tool offset mill T5 D1
$TC_DP1[2.2]=120 $TC_DP6[2.2]=5 $TC_DP3[2.2]=130
N100 G17 G40 G90                                    ;Initial conditions G code
N110 T5 D1                                          ;Load milling tool
N120 M6
N130 S500 M3 F2000 M8
GOTOF _MACHINE
;
N510 _EDGE:G0 G64 X25 Y30 F2000                     ;Define edge contour
N520 G1 X118 RND=5
N530 Y96 RND=5
N540 X40 RND=5
N545 X20 Y75 RND=5
N550 Y35
N560 _ENDEDGE:G3 X25 Y30 CR=5
;
N570 _ISLAND1:G0 X34 Y58                            ;Define bottom island
N580 G1 X64
N590 _ENDISLAND1:G2 X34 Y58 CR=15
;
N600 _ISLAND2:G0 X79 Y73                            ;Define top island
N610 G1 X99
N620 _ENDISLAND2:G3 X79 Y73 CR=10
;
_MACHINE:
;Programming of contours
SAMPLE_CONT:
CYCLE74 ("EXAMPLE1","_EDGE","_ENDEDGE")             ;Transfer edge contour
CYCLE75 ("EXAMPLE1","_ISLAND1","_ENDISLAND1")       ;Transfer island contour 1
CYCLE75 ("EXAMPLE1","_ISLAND2","_ENDISLAND2")       ;Transfer island contour 2
ENDLABEL:
M30
```

© Siemens AG 2004 All rights reserved.

3.15.4 Pocket milling with islands - CYCLE73

Pocket milling with islands is an option and requires SW6 in both the NCK and HMI Advanced .

Programming

CYCLE73 (_VARI, _BNAME, _PNAME, _TN, _RTP,
_RFP, _SDIS, _DP, _DPR, _MID, _MIDA, _FAL, _FALD,
_FFP1, _FFD, _CDIR, _PA, _PO, _RAD, _DP1)

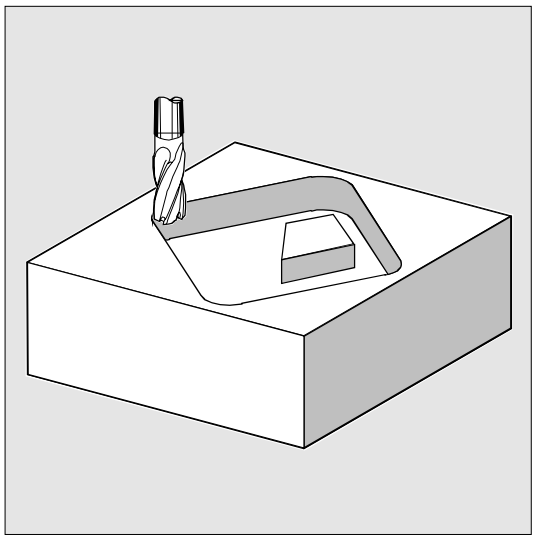
Parameters

_VARI	integer	Type of machining: (enter without sign) UNITS POSITION (select machining): Values: 1...Rough cut (remove stock) from solid material 2...Rough cut residual material 3...Finish edge 4...Finish base 5...Rough drill TENS DIGIT (select insertion strategy): Values: 1...Perpendicular with G1 2...Along a helix 3...Oscillate HUNDREDS DIGIT (select liftoff mode): Values: 0...to retraction plane (_RTP) 1...By safety distance (_SDIS) via reference plane (_RFP) THOUSANDS DIGIT (select start point): Values: 1...automatic 2…manual
_BNAME	string	Name for program of drill positions
_PNAME	string	Name for pocket milling machining program
_TN	string	Name of stock removal tool
_RTP	real	Retraction plane (absolute)
_RFP	real	Reference plane (absolute)
_SDIS	real	Safety distance (to be added to the reference plane, enter without sign)
_DP	real	Pocket depth (absolute)
_DPR	real	Pocket depth (incremental)
_MID	real	Maximum infeed depth for infeed (enter without sign)
_MIDA	real	Maximum infeed depth in the plane (enter without sign)
_FAL	real	Final machining allowance in the plane (enter without sign)
_FALD	real	Final machining allowance on base (enter without sign)
_FFP1	real	Feedrate for surface machining
_FFD	real	Feedrate for depth infeed

© Siemens AG 2004 All rights reserved.

_CDIR	integer	Milling direction for machining the pocket: (enter without sign) Values: 0...Down-cut milling (as spindle rotation) 1...Up-cut milling 2...with G2 (independent of spindle direction) 3...with G3
_PA	real	Start point in first axis (only with manual selection of start point)
_PO	real	Start point in second axis (only with manual selection of start point)
_RAD	real	Radius center-point path on insertion along helical path or max. insertion angle for oscillating insertion motion
_DP1	real	Insertion depth per 360° revolution on insertion along helical path

Function



Cycle CYCLE73 enables you to machine pockets with or without islands. It supports complete machining of this type of pocket and offers the following machining operations:

- Predrilling
- Solid machine pocket
- Machine residual material
- Edge finishing
- Finish base.

Pocket and island contours are freely programmed in DIN code supported, for example, by the geometry processor.
The cycle is executed once for each operation according to the programmed machining type (_VARI). In other words, in applications requiring roughing and finishing, or an additional rough-cut residual material operation, CYCLE73 must be called a second time.

Solid machine pocket
When a pocket is solid machined, it is machined with the active tool down to the programmed finishing dimensions. The insertion strategy for milling can be selected. The cutting operation is segmented in the pocket depth direction (tool axis) in accordance with the specified values.

Machine residual material
The cycle allows material to be removed with a smaller milling tool. The traversing motions defined by the residual material of the last milling operation and the current tool radius are output in the generated program. The residual material technology can be programmed repeatedly with a succession of decreasing tool radii. No check is made on completion of the cycle for any

© Siemens AG 2004 All rights reserved.

further residual material in the pocket.

Edge/base finishing
Another function of the cycle is to finish the pocket base or circumnavigate the pocket and individual islands in a finish operation.

Predrilling
Depending on the milling tool used, it may be necessary to drill before solid machining the workpiece. The cycle automatically calculates the rough drilling positions as a function of the solid machining operation to be performed afterwards. The drilling cycle called modally beforehand is executed at each of these positions. Rough drilling can be executed in a number of technological machining operations (e.g. 1. centering, 2. drilling).

© Siemens AG 2004 All rights reserved.

Rough drilling sequence

In the first machining section of the rough drilling operation, a REPEAT command must be used after a modal call for the drilling cycle to call a sequence of machining steps with the contents of CYCLE73 and the contour repetition. The drilling cycle must be deselected modally before the next tool change. Other drilling technologies can be programmed subsequently.
The next program section contains CYCLE73 which contains all necessary parameters as well as the programs for solid machining and drilling.
Parameter _VARI is the only one to define all solid machining parameters and it must always be programmed for this reason.
The cycle now generates the solid machining and drilling position programs for the pocket. It then calls the drilling position program and executes it.
If the operation involves several different pockets, it will be necessary to call the associated contours again in this section. This block can be omitted if there is only one pocket.
This entire machining section must be marked by a skip command to the following "Solid machine pocket" section.

Example

Rough drill, with solid machining

ACCEPTANCE4_CONT:	;Marker with name for beginning of pocket ;contour
CYCLE74("EDGEA01", ,)	;Definition of contour for pocket edge
CYCLE75("INS11A01", ,)	;Definition of contour for 1st island
CYCLE75("INS1A01", ,)	
CYCLE75("INS2A01", ,)	
CYCLE75("INS3A01", ,)	
ENDLABEL:	;Marker for end of a pocket contour
T4 M6	
D1 M3 F1000 S4000	
MCALL CYCLE81(10,0,1,-3)	;Modal call of drilling cycle
REPEAT ACCEPTANCE4_MACH ACCEPTANCE4_MACH_END	;Execute drilling position program
MCALL	;Deselect drilling cycle modally
GOTOF ACCEPTANCE4_MACH_END	;Branch to Solid machine pocket
ACCEPTANCE4_MACH:	;Start of section Generate programs
;REPEAT ACCEPTANCE4_CONT ENDLABEL	;Required only if there is more than one pocket ;contour

© Siemens AG 2004 All rights reserved.

```
CYCLE73(1015,“ACCEPTANCE4_DRILL“,“ACCEPTANCE4_M
ILL1“,“3“,10,0,1,-12,0,,2,0.5,,9000,400,0,,,,)
ACCEPTANCE4_MACH_END:                          ;End of section Generate programs
T3 M6
D1 M3 S2000
;REPEAT ACCEPTANCE4_CONT ENDLABEL              ;Required only if there is more than one
                                               pocket ;contour
CYCLE73(1011,“ACCEPTANCE4_DRILL“,“ACCEPTANCE4_M;Solid machine pocket
ILL1“,“3“,10,0,1,-12,0,,2,0.5,,9000,400,0,,,,)
```

Sequence for roughing, solid machining (_VARI=XXX1)

All parameters must be written to the CYCLE73 command again.

The program performs the following machining steps:

- Approach a manually calculated or automatically generated start point located on the return plane. G0 is then used to traverse the axis to a reference plane brought forward by the safety distance.
- Infeed to the current machining depth according to the selected insertion strategy (_VARI) with feed value _FFD.
- Mill pocket with islands down to final machining allowance with feed _FFP1. The machining direction corresponds to the setting in _CDIR. The pocket can be split if the ratio between mill diameter and clearance between islands or between islands and edge contours is not ideal. For this purpose, the cycle calculates additional start points for mill insertion.
- Lift off in accordance with selected retraction mode and return to start point for next plane infeed.
- When the pocket has been machined, the tool is retracted either to the return plane or by the safety distance via the reference plane, depending on the selected liftoff mode. The tool position in the plane is above the pocket surface as determined by the generated program.

© Siemens AG 2004 All rights reserved.



Sequence of motions for finishing (_VARI=XXX3)

- The pocket and island contours are circumnavigated once each during the edge finishing operation. Vertical insertion with G1 (_VARI) must be programmed as the insertion strategy. Approach and retraction at the start and edge points respectively of the finishing operation are each executed along a tangential circle segment.
- To finish the base, the tool is inserted to pocket depth + final machining allowance + safety distance with G0. From this position the tool is fed in vertically at the feedrate for depth infeed. The base surface of the pocket is machined once.
- Liftoff and retraction as the same as for solid machining.
- Parameters _FAL, _FALD and _VARI=XXX4 must be assigned for simultaneous finishing in the plane and on the base.

© Siemens AG 2004 All rights reserved.

Explanation of the parameters

_VARI (machining type)
You can define the type of machining with parameter _VARI. Possible settings are:
Unit's place:

- 1=Rough cut (solid machine) from solid material
- 2=Rough cut residual material
- 3=Finish edge
- 4=Finish base
- 5=Rough drill.

When "Rough cut from solid material" is set, the machining program solid-machines the pocket completely down to the final machining allowance.
If it is not possible to machine areas of the edge surfaces with the selected mill diameter, then setting "2" can be selected to machine them with a smaller milling tool afterward. To do this, cycle CYCLE73 must be called again.

Ten's place:

- 1=Vertical with G1
- 2=Along a helical path
- 3=Oscillation.

Selection of insertion strategies:

- **Insert vertically (_VARI=XX1X)**
  means that the current infeed depth calculated internally is executed in one block.
- **Insert along helical path (_VARI=XX2X)**
  means that the mill center point traverses along the helical path determined by radius _RAD and depth per revolution _DP1. The feedrate is always programmed through _FFD. The sense of rotation of this helical path corresponds to the direction to be used for machining the pocket.
  The depth programmed under _DP1 on insertion is calculated as the maximum depth and is always calculated as a whole number of revolutions of the helical path.
  When the current depth for the infeed (these may be several revolutions on the helical path) has been calculated, a full circle is made to remove the slope on insertion.
  Then pocket solid machining starts in this plane and continues until reaching the final machining allowance.

© Siemens AG 2004 All rights reserved.

- **Insertion with oscillation (_VARI=XX3X)**
  means that the mill center point oscillates along an oblique linear path until it has reached the next current depth. The maximum insertion angle is programmed under _RAD, the position of the oscillation path is calculated within the cycle. When the current depth has been reached, the path is traversed again without depth infeed in order to remove the slope caused by insertion. The feedrate is programmed through _FFD.

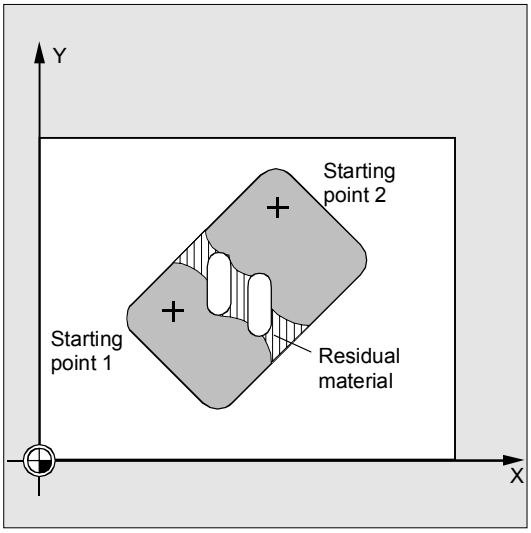
Hundreds digit: (_VARI=X1XX)

- 0=To retraction plane (_RTP)
- 1=By safety distance (_SDIS) via reference plane (_RFP).

Thousands digit: (_VARI=1XXX)

- 1=Start point automatic
- 2=Start point manual.

When the starting point is selected automatically, the cycle internally calculates the starting point for machining itself.



Caution: Manually specified start positions must not be too close to the island surface. Manually specified start positions are not monitored internally.
If the pocket has to be split as a result of the island position and the mill diameter used, then several start points are calculated automatically.
With manual start point selection, parameters _PA and _PO must also be programmed. However, these can only define one start point.
If the pocket has to be split, the required start points are calculated automatically.

© Siemens AG 2004 All rights reserved.

_BNAME (name for drilling position program)
_PNAME (name for pocket milling program)
The pocket milling cycle generates programs with traversing blocks required to rough drill or mill the workpiece. These programs are stored in the same directory as the calling program in the parts program memory, i.e. in the "parts programs" directory (MPF.DIR), if the cycle is called from there or in the relevant workpiece directory. The programs are always main programs (type MPF).
The names of these programs are defined by parameters _BNAME and _PNAME.
A drilling program name is needed only when _VARI=XXX5.
Example: No drilling program name:
CYCLE73(1011,"",ACCEPTANCE4_MILL,...)

_TN (name of solid machining tool)
This parameter must be set to the solid machining tool. Depending on whether the tool management function is active or not, the parameter must be set to a tool name or tool number.
Example:

- with tool management
  CYCLE73(1015,"PART1_DRILL","PART1_MILL", "MILL3",...)
- with tool management
  CYCLE73(1015,"PART1_DRILL","PART1_ MILL","3",...)

Parameter _TN is defined as a compulsory parameter with a maximum length of 16 characters. It must therefore be assigned to the cutting tool in every subsequent CYCLE73 call. When the residual material machining operation is used more than once, the tool from the last residual material removal process must be used.

TOOL AND OFFSET:
It must be ensured that the tool offset is processed exclusively by D1. Replacement tool strategies may not be used.

© Siemens AG 2004 All rights reserved.

_RFP and _RTP (reference plane and retraction plane)
The reference plane (RFP) and retraction plane are generally set to different values. The cycle assumes that the retraction plane is in front of the reference plane. The distance between the retraction plane and the final drilling depth is therefore greater than the distance between the reference plane and the final drilling depth.

_SDIS (safety distance)
The safety distance (SDIS) is effective with regard to the reference plane which is brought forward by the safety distance.
The direction in which the safety distance is active is automatically determined by the cycle.

_DP (absolute pocket depth) and
_DPR (incremental pocket depth)
The pocket depth can be specified as either an absolute value (_DP) or relative value (_DPR) in relation to the reference plane. If the incremental option is selected, the cycle automatically calculates the depth on the basis of the reference and retraction plane positions.

_MID (maximum infeed depth)
The maximum infeed depth is defined with this parameter. The depth infeed is performed by the cycle in equally sized infeed steps.
The cycle calculates this infeed automatically on the basis of _MID and the total depth.
The minimum possible number of infeed steps is used as the basis. _MID=0 means that the cut to pocket depth is made with one infeed.

_MIDA (max. infeed depth in the plane)
With this parameter you define the maximum infeed width for solid machining in the plane. This value is never exceeded.
If this parameter is not programmed, or if its value is 0, the cycle uses 80 percent of the mill radius as the maximum infeed width.
If an infeed width of more than 80 percent of the mill diameter is programmed, the cycle is aborted after output of alarm 61982 "Infeed width in plane too large".

© Siemens AG 2004 All rights reserved.

_FAL (final machining allowance in the plane)
The final machining allowance only affects machining of the pocket in the plane at the edge.
When the final machining allowance ≥ tool diameter, the pocket will not necessarily be machined completely.

_FALD (final machining allowance on the base)
A separate final machining allowance on the base is taken into account in roughing operations.

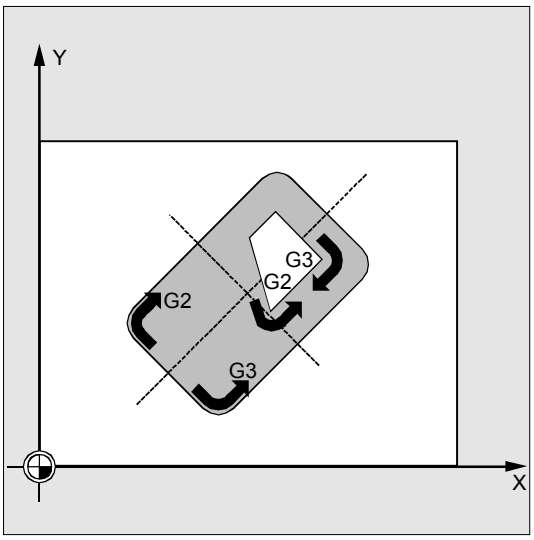
_FFD and _FFP1 (feedrate for depth infeed and surface machining)
Feedrate _FFD is used for insertion into the material.
Feedrate FFP1 is used for all movements in the plane traversed at feedrate when machining.

_CDIR (milling direction)
The value for the machining direction of the pocket is defined in this parameter.
Under parameter _CDIR the mill direction

- directly "2 for G2" and "3 for G3" or
- alternatively "up-cut milling" or "down-cut milling"

can be programmed. Up-cut milling or down-cut milling is determined within the cycle via the spindle direction activated prior to the cycle call.

Down-cut	**Up-cut**
M3 → G3	M3 → G2
M4 → G2	M4 → G3



_PA, _PO (start point for first and second axes)
When the start point is selected manually, the start point must be programmed in these parameters such that it can be approached without risk of collision.
Please note that only one start point can be programmed (see description of parameter _VARI).

_RAD (center-point path or insertion angle)
Parameter _RAD defines the radius of the helical path (referred to tool center point path) or the maximum insertion angle for oscillation.

_DP1 (insertion depth for helical path)
With the parameter _DP1 you define the infeed depth for insertion on the helical path.

© Siemens AG 2004 All rights reserved.

Other Information

Name for pocket machining (NAME)

Pockets are generally machined in several technological machining steps. However, the contours defining the pocket geometries are defined only once.
To ensure that contours can be automatically assigned to the appropriate machining step in the program, the contour definition is marked with labels and this program section then repeated later with the REPEAT instruction.
When programs are written using the cycles support function, a name for the pocket machining program is therefore entered in the relevant screen forms. The name length is restricted to 8 characters.
In sample program 2, this is, e.g., "ACCEPTANCE4".
The T number contains the milling tool for all machining technologies. When residual material is machined more than once, the tool used beforehand must always be entered in the T number.

Explanation of the cycle structure

Cycle CYCLE73 is used to solve very complex problems associated with solid machining of pockets with islands which require a high level of computing capacity in the control. For best timing, the calculation is carried out in the HMI.

The calculation is started from the cycle. Its result contains programs with traversing blocks for drilling or milling operations which are stored in the file system of the control. These are then called by the cycle and executed.
This structure means that it is only necessary to perform the calculation the first time a program is executed with CYCLE73 call. From the second program run onwards, the generated traversing program is available for immediate call by the cycle.
Recalculation is performed when:

- A finished contour has been modified;
- A transfer parameter of the cycle has changed;
- A tool with different tool offset data has been activated prior to the cycle call;
- In the case of different technologies, such as solid machining and residual material, with machining programs generated in different ways.

© Siemens AG 2004 All rights reserved.

Program storage in the file system

If the contours for CYCLE73 are programmed outside the program that makes the call, the following applies for the search in the file system of the control:

- If the calling program is stored in a workpiece directory, then the programs containing the edge or island contours must be stored in the same workpiece directory;
- If the invoked program is located in the "Parts program" directory (MPF.DIR), a search is also made for the programs there.

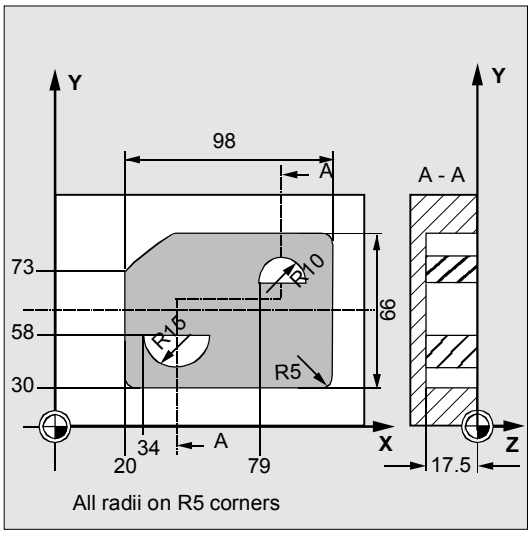
The programs generated by the cycle are also stored in the same directory as the program containing the cycle call, i.e. in the same workpiece directory or in MPF.DIR or SPF.DIR.

Note on simulation

In the pocket milling simulation, the generated programs are saved to the NCU file system. Therefore, only the "NC Active Data" setting is practical since tool offset data are included in program calculation.

Programming example 1

The machining task involves machining a pocket with two islands from solid material and then finishing the pocket in plane X, Y
Sample program 1.mpf (pocket with islands)




```
%_N_SAMPLE1_MPF
;$PATH=/_N_WKS_DIR/_N_CC73BEI1_WPD
; Example_1: Pocket with islands
;Solid machine and finish
```

```
$TC_DP1[5,1]=120 $TC_DP3[5,1]=111 $TC_DP6[5,1]=4
```
;Tool offset mill T5 D1

```
$TC_DP1[2,1]=120 $TC_DP3[2,1]=130 $TC_DP6[2,1]=5
```

```
N100 G17 G40 G90
```
;Initial conditions G code

```
N110 T5 D1
```
;Load milling tool

```
N120 M6
```


© Siemens AG 2004 All rights reserved.

```
N130 M3 F2000 S500 M8
N140 GOTOF _MACHINE
;
N510 _EDGE:G0 G64 X25 Y30                    ;Define edge contour
N520 G1 X118 RND=5
N530 Y96 RND=5
N540 X40 RND=5
N545 X20 Y75 RND=5
N550 Y35
N560 _ENDEDGE:G3 X25 Y30 CR=5
;
N570 _ISLAND1:G0 X34 Y58                     ;Define bottom island
N580 G1 X64
N590 _ENDISLAND1:G2 X34 Y58 CR=15
;
N600 _ISLAND2:G0 X79 Y73                     ;Define top island
N610 G1 X99
N620 _ENDISLAND2:G3 X79 Y73 CR=10
;

;Programming of contours
_MACHINE:
SAMPLE1_CONT:
CYCLE74 ("","_EDGE","_ENDEDGE")
CYCLE75 ("","_ISLAND1","_ENDISLAND1")
CYCLE75 ("","_ISLAND2","_ENDISLAND2")
ENDLABEL:

;Programming Mill Pocket
CYCLE73 (1021,"","SAMPLE1_MILL1","5",10,0,1,
-17.5,0,,2,0.5,,9000,3000,0,,,4,3)

T2 D1 M6
S3000 M3
;Programming Finish Pocket
CYCLE73 (1113,"","SAMPLE1_MILL3","5",10,0,1,
-17.5,0,,2,,,8000,1000,0,,,4,2)
M30
```

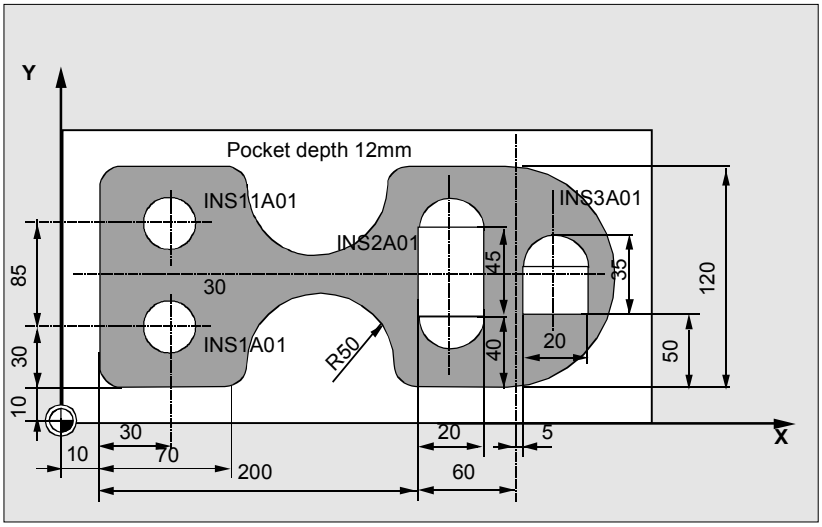
© Siemens AG 2004 All rights reserved.

Programming example 2

Machining task:
Before the pocket is milled, the workpiece must be rough drilled to ensure optimum insertion of the milling tool.

- Center for rough drilling
- Drilling
- Solid machine pocket with islands, mill radius 12mm
- Solid machine residual material, mill radius 6mm
- Finish pocket, mill radius 5mm.

Sketch of machining operation



Machining program:

```
%_N_SAMPLE2_MPF
;$PATH=/_N_WKS_DIR/_N_CC73BEI2_WPD
; Example_2: Pocket with islands
; 2*rough drill, machine, machine resid.  mat. , finish
;
; Tool offset data
$TC_DP1[2,1]=220 $TC_DP6[2,1]=10
$TC_DP1[3,1]=120 $TC_DP6[3,1]=12
$TC_DP1[4,1]=220 $TC_DP6[4,1]=3
$TC_DP1[5,1]=120 $TC_DP6[5,1]=5
$TC_DP1[6.1]=120 $TC_DP6[6.1]=6
TRANS X10 Y10

;Defining machining contours
ACCEPTANCE4_CONT:
CYCLE74("EDGEA01", ,)
CYCLE75("INS11A01", ,)
CYCLE75("INS1A01",,)
CYCLE75("INS2A01",,)
CYCLE75("INS3A01",,)
ENDLABEL:
```


© Siemens AG 2004 All rights reserved.

```
;Program centering
T4 M6
D1 M3 F1000 S4000
MCALL CYCLE81 (10,0,1,-3,)
REPEAT ACCEPTANCE4_MACH ACCEPTANCE4_MACH_END
MCALL

;Program drilling
T2 M6
D1 M3 F2222 S3000
MCALL CYCLE81 (10,0,1,-12,)
REPEAT ACCEPTANCE4_MACH ACCEPTANCE4_MACH_END
MCALL

GOTOF ACCEPTANCE4_MACH_END
ACCEPTANCE4_MACH:
REPEAT ACCEPTANCE4_CONT ENDLABEL
CYCLE73(1015,"ACCEPTANCE4_DRILL","ACCEPTANCE4_MILL1",
"3",10,0,1,-12,0,,2,0.5,,2000,400,0,,,,)
ACCEPTANCE4_MACH_END:

;Program solid machining
T3 M6
D1 M3 S4000
REPEAT ACCEPTANCE4_CONT ENDLABEL
CYCLE73(1011,"","ACCEPTANCE4_MILL1","3",10,0,1,
-12,0,,2,0.5,,2000,400,0,,,,)

;Program solid machining of residual material
T6 M6
D1 M3 S4000
REPEAT ACCEPTANCE4_CONT ENDLABEL
CYCLE73(1012,"","ACCEPTANCE4_2_MILL4","3",10,0,1,
-12,0,,2,0.5,,1500,800,0,,,,)

;Program finishing
T5 M6
D1 M3 S4500
REPEAT ACCEPTANCE4_CONT ENDLABEL
CYCLE73(1013,"","ACCEPTANCE4_MILL3","3",10,0,1,
-12,0,,2,,,3000,700,0,,,,)
M30
```

© Siemens AG 2004 All rights reserved.

Edge contour programming example 2:

```
%_N_EDGEA01_MPF
;$PATH=/_N_WKS_DIR/_N_CC73BEI2_WPD
;Ste 17.05.99
;Edge contour programming example 2

N5 G0 G90 X260 Y0
N7 G3 X260 Y120 CR=60
N8 G1 X170 RND=15
N9 G2 X70 Y120 CR=50
N10 G1 X0 RND=15
N11 Y0 RND=15
N35 X70 RND=15
N40 G2 X170 Y0 CR=50
N45 G1 X260 Y0
N50 M30
```

Island contour sample program 2

```
%_N_ISL1A01_MPF
;$PATH=/_N_WKS_DIR/_N_CC73BEI2_WPD
;Ste 18.06.99
;Island contour sample program 2

N5 G90 G0 X30 Y15
N10 G91 G3 X0 Y30 CR=15
N12 X0 Y-30 CR=15
N15 M30

%_N_ISL11A01_MPF
;$PATH=/_N_WKS_DIR/_N_CC73BEI2_WPD
;Ste 18.06.99
;Island contour sample program 2

N5 G90 G0 X30 Y70
N10 G91 G3 X0 Y30 CR=15
N12 X0 Y-30 CR=15
N15 M30

%_N_ISL2A01_MPF
;$PATH=/_N_WKS_DIR/_N_CC73BEI2_WPD
;Ste 18.06.99
;Island contour sample program 2

N5 G90 G0 X200 Y40
N10 G3 X220 Y40 CR=10
N15 G1 Y85
N20 G3 X200 Y85 CR=10
N25 G1 Y40
N30 M30
```


© Siemens AG 2004 All rights reserved.

```
%_N_ISL3A01_MPF
;$PATH=/_N_WKS_DIR/_N_CC73BEI2_WPD
;Ste 18.06.99
;Island contour sample program 2

N5 G0 G90 X265 Y50
N10 G1 G91 X20
N15 Y25
N20 G3 X-20 I-10
N25 G1 Y-25
N30 M30
```

© Siemens AG 2004 All rights reserved.

Programming example 3

Machining task:
Shows the program sequence of a machining task, illustrated by two different pockets with islands. The machining process is tool-oriented, i.e. each time a new tool becomes available, all machining tasks requiring this particular tool are performed complete on both pockets before the next tool is used.

- Predrilling
- Solid machine pocket with islands
- Remove residual material

```
%_N_SAMPLE3_MPF
;$PATH=/_N_WKS_DIR/_N_CC73BEI3_WPD
; Sample3
; 07.04.2000

; Tool offset data
$TC_DP1[2,1]=220 $TC_DP3[2,1]=330 $TC_DP6[2,1]=10
$TC_DP1[3,1]=120 $TC_DP3[3,1]=210 $TC_DP6[3,1]=12
$TC_DP1[6.1]=120 $TC_DP3[6.1]=199 $TC_DP6[6.1]=6

;Workpiece zero
;G54
$P_UIFR[1,X,TR]=620
$P_UIFR[1,Y,TR]=50
$P_UIFR[1,Z,TR]=-320
;G55
$P_UIFR[2,X,TR]=550
$P_UIFR[2,Y,TR]=200
$P_UIFR[2,Z,TR]=-320
;
N10 G0 G17 G54 G40 G90
N20 T2
M6
D1 M3 F2000 S500 M8
N30 G0 Z20

;Machining contours pocket 1
GOTOF ENDLABEL
POCKET1_CONT:
CYCLE74("EDGE","","")
CYCLE75("ISLAND1","","")
CYCLE75("ISLAND2","","")
ENDLABEL:
```


© Siemens AG 2004 All rights reserved.

```
;Machining contours pocket 2
GOTOF ENDLABEL
SAMPLE2_CONT:
CYCLE74("EDGEA01", ,)
CYCLE75("INS11A01", ,)
CYCLE75("INS1A01",,)
CYCLE75("INS2A01",,)
CYCLE75("INS3A01",,)
ENDLABEL:

;Drill
T2 M6
D1 M3 F6000 S4000
MCALL CYCLE81 (10,0,1,-8,)
REPEAT POCKET1_MACH POCKET1_MACH_END
MCALL

G55
MCALL CYCLE81 (10,0,1,-8,)
REPEAT SAMPLE2_MACH SAMPLE2_MACH_END
MCALL

;Removing pocket 1
T3 M6
G54 D1 M3 S3300
GOTOF POCKET1_MACH_END
POCKET1_MACH:
REPEAT POCKET1_CONT ENDLABEL
CYCLE73(1025,"TASCHE1_DRILL","TASCHE1_MILL1","3",10,0,1,-8,0,0,2,0,0,2000,400,0,0,0,3,4)
POCKET1_MACH_END:
REPEAT POCKET1_CONT ENDLABEL
CYCLE73(1021,"POCKET1_DRILL","POCKET1_MILL1","3",10,0,1,-8,0,0,2,0,0,2000,400,0,0,0,3,4)

;Removing pocket 2
G55
GOTOF SAMPLE2_MACH_END

SAMPLE2_MACH:
REPEAT SAMPLE2_CONT ENDLABEL
CYCLE73(1015,"SAMPLE2_DRILL","SAMPLE2_MILL1","3",10,0,1,-8,0,0,2,0,0,2000,400,0,0,0,3,4)
SAMPLE2_MACH_END:
REPEAT SAMPLE2_CONT ENDLABEL
CYCLE73(1011,"SAMPLE2_DRILL","SAMPLE2_MILL1","3",10,0,1,-8,0,0,2,0,0,2000,400,0,0,0,3,4)
```

© Siemens AG 2004 All rights reserved.

```
;Removing residual material pocket 1 and pocket 2
T6 M6
D1 G54 M3 S222
REPEAT POCKET1_CONT ENDLABEL
CYCLE73(1012,"","POCKET1_3_MILL2","3",10,0,1,-8,0,,2,,,2500,800,0,,,,)

G55
REPEAT SAMPLE2_CONT ENDLABEL
CYCLE73(1012,"","SAMPLE2_3_MILL2","3",10,0,1,-8,0,,2,,,2500,800,0,,,,)
G0 Z100
M30

;Edge and island contours
;Pocket 2 corresponds to programming example 2
Pocket 1:
%_N_Edge_MPF
;$PATH=/_N_WKS_DIR/_N_CC73BEI3_WPD
;29.03.99

N1 G0 X0 Y0 G90
N3 G1 X200 Y0
N5 X200 Y100
N10 X0 Y100
N20 X0 Y0
M30

%_N_ISLAND1_MPF
;$PATH=/_N_WKS_DIR/_N_CC73BEI3_WPD
;29.03.99
N100 G0 X130 Y30 Z50 G90
N110 G1 X150 Y30
N120 X150 Y60
N130 X130 Y60
N200 X130 Y30
M30

%_N_ISLAND2_MPF
;$PATH=/_N_WKS_DIR/_N_CC73BEI3_WPD
;29.03.99
N12 G0 X60 Y20
N13 G1 X90 Y20
N14 X90 Y50
N30 X60 Y50
N40 X60 Y20
M30
```

© Siemens AG 2004 All rights reserved.

Explanation

Alarms source CYCLE73...CYCLE75

Alarm number	Alarm text	Explanation, remedy
61703	"Internal cycle error while deleting file"	
61704	"Internal cycle error while writing file"	
61705	"Internal cycle error while reading file"	
61706	"Internal cycle error during checksum formation"	
61707	"Error in ACTIVATE on MMC"	
61708	"Error in READYPROG on MMC"	
61900	"No contour"	
61901	"Contour is not closed"	
61902	"No more free memory"	
61903	"Too many contour elements"	
61904	"Too many intersections"	
61905	"Cutter radius too small"	
61906	"Too many contours"	
61907	"Circle without center point measurement"	
61908	"No starting point specified"	
61909	"Helical radius too small"	
61910	"Helix violates contour"	
61911	"More insertion points required"	
61912	"No path generated"	
61913	"No residual material generated"	
61914	"Programmed helix violates contour"	
61915	"Approach/liftoff motion violates contour"	
61916	"Ramp path too short"	
61917	"Residual corners might be left with less than 50 percent overlap"	
61918	"Cutter radius too large for residual material"	
61980	"Error in island contour"	
61981	"Error in edge contour"	
61982	"Infeed width in plane too large"	
61983	"Pocket edge contour missing"	
61984	"Tool parameter _TN not defined"	
61985	"Name of drilling position program missing"	
61986	"Machine pocket program missing"	
61987	"Drilling position program missing"	
61988	"Name of program for machining pocket missing"	
61989	"Not D1 programmed as active tool cutting edge"	

© Siemens AG 2004 All rights reserved.

3.16 Swiveling - CYCLE800

Calling swiveling – CYCLE800

Entry area Programs / **Milling**

Softkey [>>] → [Swiveling]

is displayed if a swivel data set has been set up (MD 18088: MM_NUM_TOOL_CARRIER >0).

Swiveling is not an option and is available for NC SW 6.3 (CCU SW 6.3).

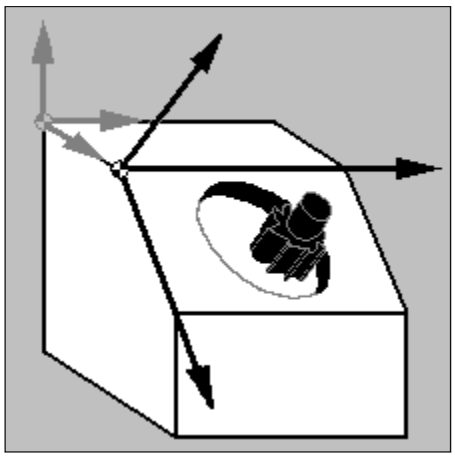
The functions

- 3/2 axes inclined machining and
- Orientational toolholder

are available in the basic version.

References: Description of Functions 840D/840Di/810D
/W1/ "Tool Offset"
/R2/ "Rotary Axes"
/K2/ "System Frames"

Function



The cycle is used during milling to swivel on any type of surface so that it can be machined and/or measured. By calling the appropriate NC functions, the cycle converts the active workpiece zeroes and tool offsets to refer to the inclined surface, taking account of the kinematic chain on the machine, and positions the rotary axes (optional). Swiveling can be implemented either axially, as a projection angle or as a solid angle.
Before the rotary axes are positioned, the linear axes can be retracted if desired.

Machine kinematics

1. Swivel-mounted toolholder (swivel head) → type T
2. Swivel-mounted workholder (swivel table) → type P
3. Mixed kinematics from 1 and 2 → type M

© Siemens AG 2004 All rights reserved.

Important
A tool (tool cutting edge D>0) and the zero offset (ZO) with which the workpiece was scratched or measured must be programmed before the swivel cycle is first called in the main program. This zero offset is converted to the appropriate machining plane in the swivel cycle. The ZO value remains unchanged. Translational and rotational components are saved in system frames (swivel frame), tool reference (TOOLFRAME), toolholder (PARTFRAME) and workpiece reference (WPFRAME) (see HMI → parameters, active ZO).

The current machining plane (G17, G18, G19) is taken into account by the swivel cycle.

Swiveling onto a machining or auxiliary surface always involves three steps:

- Offset of reference point prior to rotation (corresponds to TRANS or ATRANS)
- Rotation (corresponds to AROT or AROTS)
- Offset of zero point after rotation (corresponds to ATRANS)

The offsets or rotations are not machine-specific, but refer to the coordinate system X, Y, Z of the workpiece. No programmable frames are used in the swivel cycle. The frames programmed by the user are taken into account in additive swiveling. Swiveling to a **new** swivel plane clears the programmable frames (TRANS). After a program reset or if there is a power failure, the last swivel plane remains active, with the option to set via machine data. Any type of machining operation can be performed on the swivel plane, e.g. through calling standard or measuring cycles.

Note about calling the 5-axis transformation
If a program activating the 5-axis transformation (TRAORI) is to be executed on the swiveled machining plane, the system frames for the swivel head/swivel table must be deactivated before TRAORI is called (see example).

© Siemens AG 2004 All rights reserved.

Example (machine with swivel table)

```
N1 G54
N2 T="MILL_10mm"
N3 M6
N4 CYCLE800(1,"",0,57,0,40,0,-45,0,0,0,0,0,-1)          ;Swivel cycle
N5 CYCLE71(50,24,2,0,0,0,80,60,0,4,10,5,0,2000,31,5)    ;Face milling
N6 TCARR=0                                               ;swivel data record
                                                          deselection
N7 PAROTOF
N8 TOROTOF                ;(machine kinematic type "T" and "M" only)
N9 TRAORI
N10 G54                   ;Calculation of new zero offset
N11 EXTCALL "WALZ"        ;5-axis machining program with direction vectors
                          ;(A3, B3, C3)
N12 M2
```



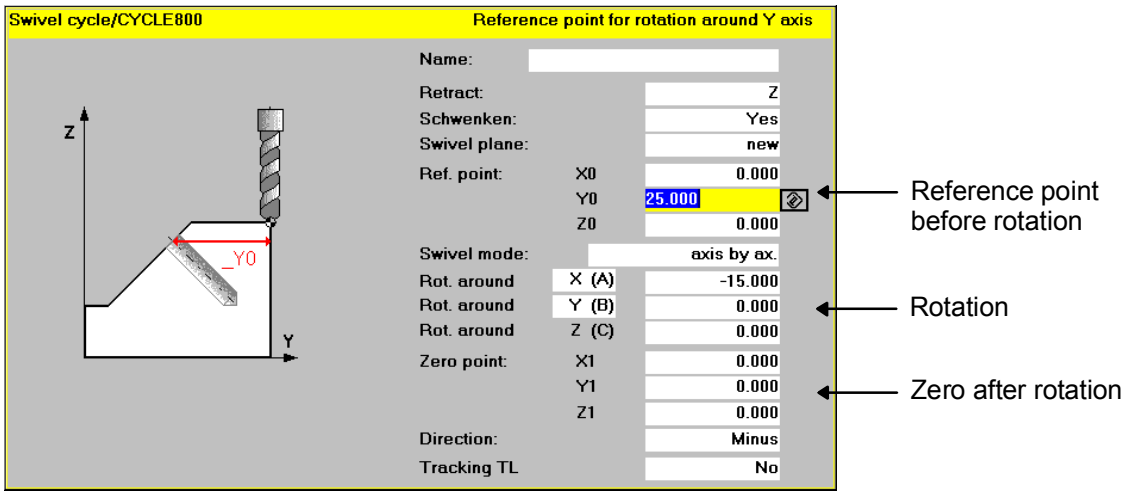
If the 5-axis transformation with the "high speed settings" cycle CYCLE832 has been activated, blocks N6 to N10 are not required.

© Siemens AG 2004 All rights reserved.

3.16.1 Operation, parameter assignment, input screen form

Explanation of the parameters

Interactive screen form CYCLE800 on standard interface



_TC (name of swivel data record)
The existing swivel data records (see Start-Up of CYCLE800) can be selected (toggled).

Each swivel data record has a name. No name needs to be declared if there is only one swivel data record.
“0“ → Swivel data record deselection.

_FR (retraction)

- Do not retract
- Traverse axis Z
- Traverse axis Z, XY (only when CYCLE800 is active in the start-up menu).

The retraction positions can be entered in the start-up menu display.

The retraction positions are approached absolutely. If a different retraction sequence or incremental positioning is desired, the process can be modified accordingly in user cycle TOOLCARR during start-up.

© Siemens AG 2004 All rights reserved.

Note:
When programming with standard cycles and high settings for the retraction plane and large swivel angles (through 90 degrees with multiface machining), it is possible that the traversing area of the machine may be too small (software end position violation), as the order of approach is always the machining plane (for G17 X, Y) first, followed by the infeed axis (Z). The retraction plane can be reduced to optimize this behavior.

_DIR (swivel, direction)

- **Swivel yes**
  Rotary axes are positioned or the operator can turn the manual rotary axes.
- **Swivel no (calculation only)**
  If rotary axes are not to be traversed after activating the swivel cycle, the "Swivel no" selection applies. Application: Auxiliary swivel levels according to workpiece drawing
- **Direction minus/plus**
  Reference to rotary axis 1 or 2 when direction of travel is selected in the swivel cycle interactive screen form. Due to the angular range of the rotary axes of machine kinematics, the NCU calculates two possible solutions. One of these solutions is normally appropriate technologically. The selection as to which rotary axis (1st or 2nd rotary axis) the two solutions must refer to is made in the start-up menu CYCLE800. The selection of the solution to be applied is made by selecting the direction "minus" or "plus".

_ST (swivel plane)

- **New**
  Previous swivel frames and programmed frames are deleted and a new swivel frame formed according to the values specified in the input screen.
  Every main program must begin with a swivel cycle with the **new** swivel plane to ensure that a swivel frame from another program is not active.
- **Additive**
  The swivel frame is added to the swivel frame of the last swivel cycle.
  If several swivel cycles are programmed in a program and programmable frames are also active between them (e.g. AROT ATRANS), these are taken into account in the swivel frame.

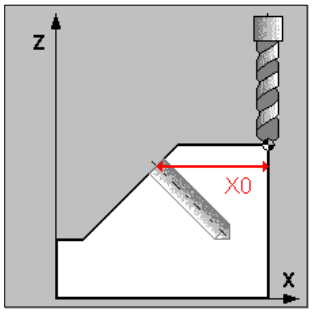
If the currently active ZO contains rotations, e.g. due to previous workpiece gauging, they will be taken into account in the swivel cycle.

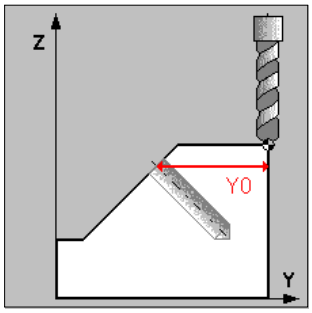
© Siemens AG 2004 All rights reserved.

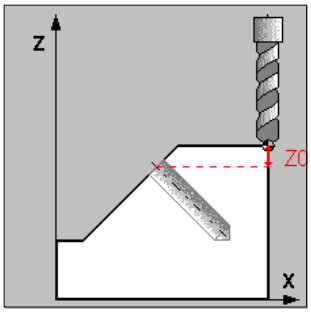
The following help displays relate to machining plane G17 (tool axis Z).

X0, Y0, Z0 (reference points prior to rotation)

Reference points







_MODE (swivel mode)

This parameter defines the swivel mode for the axis.

- Axial
- Projection angle[1)2)]
- Solid angle[1)]

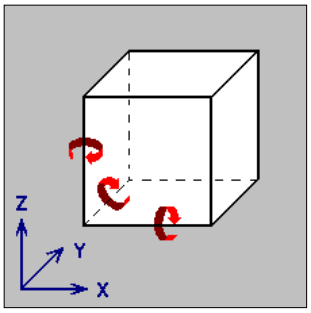
Swivel mode always refers to the coordinate system of the workpiece and is therefore not dependent on the machine.

You can set which swivel modes are available in the CYCLE800 start-up menu.

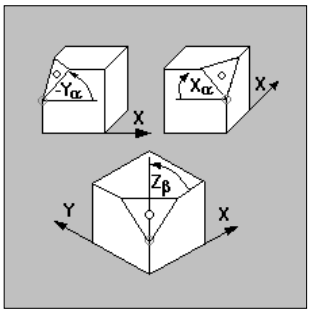
- With the axial swiveling variant, the tool is rotated about the individual axes in succession, with each rotation starting from the previous rotation. The axis sequence can be selected freely.
- When swiveling using the angle of projection, the angle value of the swiveled surface is projected onto the 1st two axes of the coordinate system. The 3rd rotation starts from the previous rotation. The axis sequence can be selected freely.
- With the solid angle variant, the tool is rotated first about the Z axis and then the Z axis. The second rotation starts from the first.

The positive direction of each rotation for the different swivel variants is shown in the help displays.

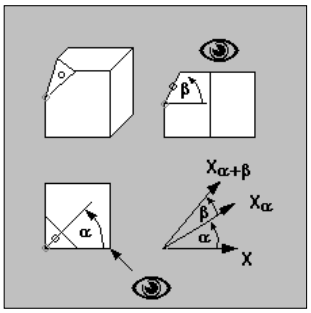
Axial



Projection angle



Solid angles

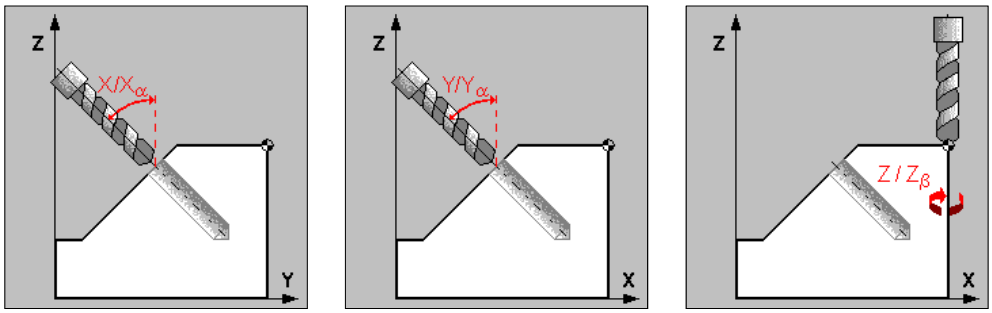


1) Only available if machine manufacturer is selected in the start-up screen form.
2) When programming X and Y the new axis lies in the old Z-X plane.
When programming Z and X the new Z-axis lies in the old Y–Z plane.
When programming Y and Z the new Y-axis lies in the old X–Y plane.
The 3rd rotation starts from the swiveled plane.

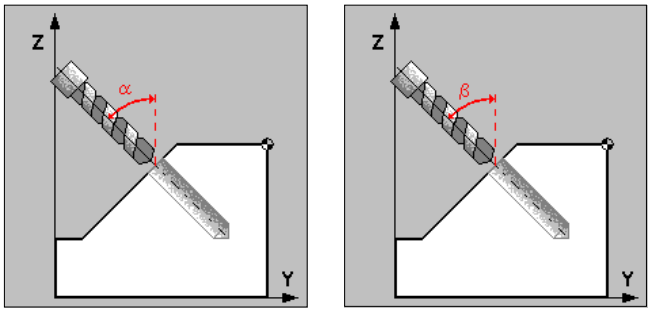
© Siemens AG 2004 All rights reserved.

A, B, C (rotations)

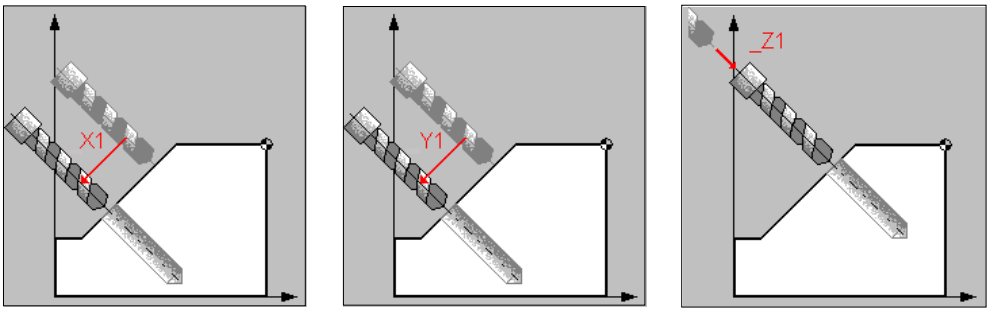
- Rotations (axial, projection angle)



- Rotation (solid angle)



X1, Y1, Z1 (zero point after rotation)



Correction tool (_TC_N_WZ)

- **yes / non**
  Display can be deactivated via the CYCLE800 start-up menu.
- **Yes:** When swiveling onto a machining plane, the rotary axes can be corrected to prevent risk of collision.
  Preconditions:
  1. TRAORI option is required.
  2. The machine manufacturer has adapted user cycle TOOLCARR.spf appropriately.

© Siemens AG 2004 All rights reserved.
SINUMERIK 840D/840Di/810D Programming Guide Cycles (PGZ) - 03.04 Edition

3.16.2 Operating instructions

- If the rotary axes of machine kinematics are defined as the manual axes (CYCLE800 start-up menu), the swivel angle to be set is displayed in cancel alarm 62180/62181.
  After rotation to the swivel angle, the NC program is continued with NC Start.
- It is possible to traverse the axes in the active swivel plane in JOG mode if key WCS is active on the machine control panel. This ensures that the geometry axes are traversed and not the machine axes.
- It is possible (without operator support) to deselect the swivel data record and cancel the swivel frame (WPFRAME, PARTFRAME, TOOLFRAME) by programming **CYCLE800()**.
- In CYCLE800, parameters can also be transferred as input values (e.g. result variable of measuring cycles _OVR[19]).
- If rotary axis values are entered in the active ZO (basic rotation of the workpiece), they will be taken into account in swiveling. Swiveling to the **home position** for machine kinematics (…,0,0,0…) [rotation=0] can cause rotations in the swivel frame (WPFRAME).

© Siemens AG 2004 All rights reserved.

3.16.3 Parameters

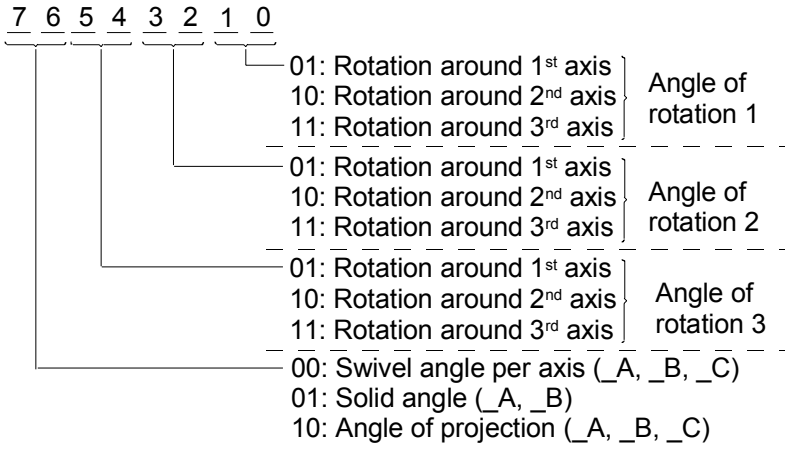


Programming

CYCLE800(_FR, _TC, _ST, _MODE, _XO, _YO, _ZO, _A, _B, _C, _X1, _Y1, _Z1, _DIR)



Parameters

_FR	integer	Move clear Values: 0...Do not retract 1...Retract axis Z (default)[1)] 2...Retract axis Z, X, Y[1)] 1) can be adjusted in user cycle TOOLCARR
_TC	String[20]	Name of swivel data record „“ Swivel data record 1 (default) “HEAD1“ Name of swivel data record “0“ Deselection of data record
_ST	integer	Swivel plane UNITS DIGIT: Values: 0... new 1... Additive TENS DIGIT[2)]: Values: 0x...Do not correct tool tip 1x... Correct tool tip HUNDREDS DIGIT: Values: Reserved 2) Precondition: Option TRAORI must be available
_MODE	integer	Swivel mode Values: 0x...axial (default) 4x...Solid angle 8x...Projection angle Evaluation of angle:  Note: Bits 0 to 5 have no meaning for solid angle
_X0, _Y0, _Z0	real	Reference point prior to rotation

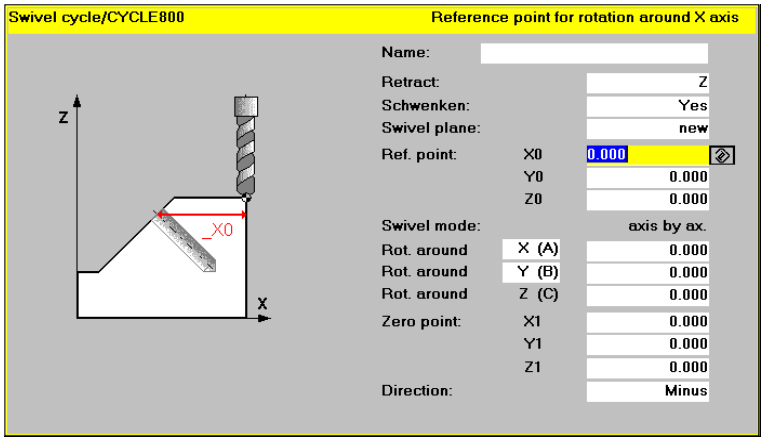
© Siemens AG 2004 All rights reserved.

_A	real	1. Axis angle (axial swivel mode) 2. Angle of rotation in XY plane about the Z axis (swivel mode 'solid angle') 3. Axis angle (swivel mode 'projection angle') sequence of axes
_B	real	1. Axis angle (axial swivel mode) 2. Angle of rotation in space about the Y axis (swivel mode 'solid angle')
_C	real	Axis angle (swivel modes 'axial', 'projection angle')
_X1, _Y1, _Z1	real	Zero point after rotation
_DIR	integer	direction If the NC calculates two solutions when the swivel cycle is called, the operator can select a preferred direction. The machine manufacturer specifies the axis to which the preferred direction refers. Values: -1 (minus)...lower rotary axis value (default) +1 (plus)...higher rotary axis value 0...no movement of rotary axes (calculation only)

Programming example 1

Set swivel plane ZERO

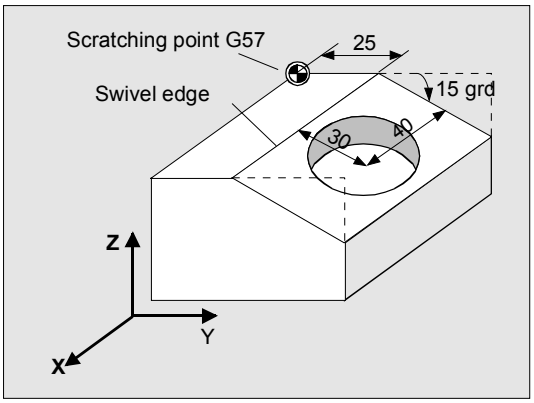
```
%_N_SWIVEL_0_SPF
;$PATH=/_N_WCS_DIR/_N_HAA_SWIVEL_WPD
G54
CYCLE800(1,"",0,57,0,0,0,0,0,0,0,0,-1)
M2
```



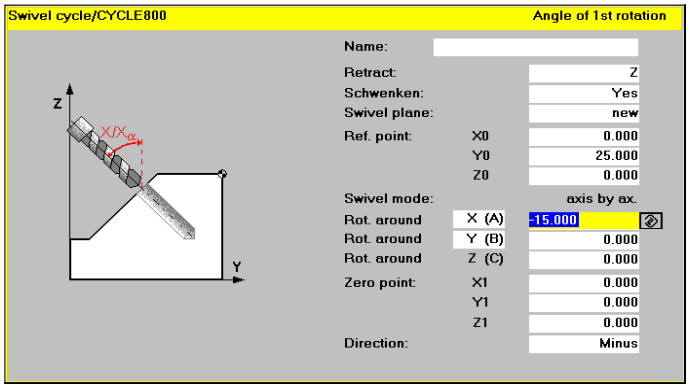
© Siemens AG 2004 All rights reserved.

Programming example 2

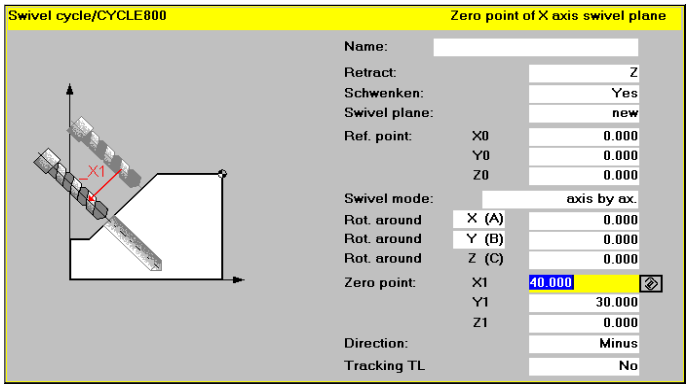
Face milling and milling a circular pocket on a machining plane swiveled through 15 degrees




```
%_N_SWIVEL_CIRCULARPOCKET_SPF
;$PATH=/_N_WCS_DIR/_N_HAA_SWIVEL_WPD
N12 T="MILL_26mm"
N14 M6
N16 G57
N18 CYCLE800(1,"",0,57,0,0,0,0,0,0,0,0,0,1)
N20 M3 S5000
N22 CYCLE71(50,2,2,0,0,0,80,60,0,4,15,5,0,2000,31,5)        ;Face milling
N24 CYCLE800(1,"",0,57,0,25,0,-15,0,0,0,0,0,-1)
```




```
N26 CYCLE71(50,12,2,0,0,0,80,60,0,4,10,5,0,2000,31,5)       ;Face milling
N28 CYCLE800(1,"",1,57,0,0,0,0,0,0,40,30,0,1)
```



© Siemens AG 2004 All rights reserved.

```
N30 T="MILL_10mm"
N32 M6
N34 M3 S5000
N36 POCKET4(50,0,1,-15,20,0,0,4,0.5,0.5,1000,1000,0,11,,,,,);Circular pocket
N38 POCKET4(50,0,1,-15,20,0,0,4,0,0,1000,1000,0,12,,,,,)
N40 M2
```

© Siemens AG 2004 All rights reserved.

3.16.4 Starting up CYCLE800

Access

Softkey [Start-Up] → [>] → [Swivel cycle]

Only available if the IBN screen form is active.

When CYCLE800 is started up, the
data (swivel data record) is set in the tool data $TC_CARR1...40.
This is grouped to form Swivel start-up menus.
→ Operating area "Start-up"; Soft key "Swivel cycle"

References: for current infos see:

- the file "siemensd.txt" from the delivery software (standard cycles) or
- in the case of HMI Advanced F:\dh\cst.dir\HLP.dir\siemensd.txt
- Description of Functions 840D/840Di/810D
  /W1/ "Tool Offset" (3/2 axes inclined machining)

The following cycles must be loaded:

- CYCLE800.SPF, CYCPE_SC.SPF (standard cycles)
- TOOLCARR-SPF (standard cycle)
- PROG_EVENT.SPF (manufacturer cycle).

The GUD variables _TC_FR to _TC_NUM (GUD7)
must be activated.

Activating the start-up screen form

The screen form is activated as follows in the start-up operating area:

- HMI Embedded
  The "Swivel Cycle" softkey in the COMMON.COM file must be activated. First open the COMMON.COM file and then delete ";" in front of SC616.
  The file is in the "User Cycles" directory. Then perform an NC reset.
- HMI Advanced:
  Delete the ";" in front of the following lines in the startup.com file.

```
;HS14=($82084,ac7)
;PRESS(HS14)
;LM("SCHWENK1a","SCHWENK1.com")
;END_PRESS
```

  The file is also in the "User Cycles" directory.
  After these two changes, restart HMI Advanced.

© Siemens AG 2004 All rights reserved.

Machine Data

To use the swiveling cycle, the machine data below must be set as follows (minimum requirement):

- Machine data with an exact value assignment (G)
  → is machine data that must not be changed
- Machine data with a variable value assignment (V)
  → is machine data where the default value can be set to a higher or lower value.

MD number	MD identifier	Value	Comment	Modifiable
10602	$MN_FRAME_GEOAX_CHANGE_MODE	1	1)	V
11450	$MN_SEARCH_RUN_MODE	Bit 1=1	1)	G
18088	$MN_MM_NUM_TOOL_CARRIER	n>0	n → Number of swivel data records 1)	G
20108	$MC_PROG_EVENT_MASK	0	-	V
20110	$MC_RESET_MODE_MASK	'H4041'	Bit 14=1	G
20112	$MC_START_MODE_MASK	'H400'	-	G
21100	$MC_ORIENTATION_IS_EULER	0	Angles of rotations are interpreted as RPY	G
21110	$MC_X_AXIS_IN_OLD_X_Z_PLANE	1		G
20126	$MC_TOOL_CARRIER_RESET_VALUE	0...n	is described in CYCLE800	V
20150	$MC_GCODE_RESET_VALUES[41]	1	TCOABS 1)	G
20150	$MC_GCODE_RESET_VALUES[51]	2	PAROT 1)	G
20150	$MC_GCODE_RESET_VALUES[52]	1	TOROTOF 2) (for kinematics type T and M only)	V
20152	$MC_GCODE_RESET_MODE[41]	0	(default) 1)	G
20152	$MC_GCODE_RESET_MODE[51]	0	(default) 1)	V
20152	$MC_GCODE_RESET_MODE[52]	0	(default) 1)	V
20180	$MC_TOCARR_ROT_ANGLE_INCR[0]	0	(default) 1)	G
20180	$MC_TOCARR_ROT_ANGLE_INCR[1]	0	(default) 1)	G
20182	$MC_TOCARR_ROT_ANGLE_OFFSET[0]	0	(default) 1)	G
20182	$MC_TOCARR_ROT_ANGLE_OFFSET[1]	0	(default) 1)	G
20184	$MC_TOCARR_BASE_FRAME_NUMBER	-1	(default) 1)	G
22530	$MC_TOCARR_CHANGE_M_CODE	0	1)	V
24006	$MC_CHSFRAME_RESET_MASK	Bit 4 = 1	if system frame WPFRAME is to remain active after a reset	V
24008	$MC_CHSFRAME_POWERON_MASK	Bit 4, 3, 2=1	if system frames PAROT, TOROT, WPFRAME are to be cleared on power ON	V
28082	$MC_MM_SYSTEM_FRAME_MASK	Bit 4, 3, 2=1	Setting up system frames	G
30455	MISC_FUNCTION_MASK	Bit 2, 0=1	for rotary axes defined as modulo axes 1)	V

1) For note on machine data, see next pages

© Siemens AG 2004 All rights reserved.

Setting Date

SD no.	SD identifier	Value	Comment	Modifiable
42980	$SC_TOFRAME_MODE	1000	See note 2)	V

Changes to the relevant machine data result in reorganization of the buffered memory (data loss!).
Series start-up must be performed after setting the MD and before NCK reset.

References: /IAM/ Installation & Start-Up Guide HMI

1) For note on machine data, see this page and the next

Note about MD 10602:
If after TRAORI, the ZO is not to be reprogrammed. e.g. for tool correction.

Note on MD 11450/MD 20108:
Activate PROGEVENT after block search

Note about MD 18088:
If several channels are defined in the NCU, the number of swivel data records is divided, taking
MD 28085: MM_LINK_TOA_UNIT into account.
Example:
MD 18088 MM_NUM_TOOL_CARRIER =4
Number of channels=2.
Two swivel data records are provided for each channel.

Note on MD 20180/MD 20182:
For rotary axes with the Hirth tooth system, the relevant values are entered in the "Swivel" start-up display.

Note about MD 22530:
If more than one swivel data record is declared per channel, and if machine functions need to be activated on changeover between swivel heads or tables, an M command can be issued in the PLC program on switchover to another swivel data record.
Example: Number of swivel data records in channel 1 =2
MD 22530: TOCARR_CHANGE_M_CODE = **-800**
Programming of swivel data record 1 (TCARR=1) → M801
Programming of swivel data record 2 (TCARR=2) → M802
By outputting special functions, the PLC can, for example, limit or invert the spindle speed.

© Siemens AG 2004 All rights reserved.

Note on MD 30455
This allows the axis to traverse for G90 with DC (shortest path); see user cycle TOOLCARR.spf

2) Note on MD 20150/MD 20152 and SD 42980:
Note on kinematics (swivel head/mixed kinematics) with Hirth tooth system:

Depending on the active plane (G17,G18,G19), the TOROT commen (or TOROTX, TOROTY) is programmed (G-group 53) for calculation (in NCU) of the compensating frame for the Hirth tooth system in CYCLE800.
If the Hirth tooth system causes the programmed rotation to deviate from the possible positions of the rotary axes, a TOOLFRAME is created for swivel head and mixed kinematics
(see HMI active ZO/details tool reference)
Calculation depends on setting data SD 42980: $SC_TOFRAME_MODE and on machine data MD 21110: $MC_X_AXIS_IN_OLD_X_Z_PLANE=1 (default).
For kinematics (swivel head/mixed kinematics) with Hirth tooth system set setting data SD 42980: $SC_TOFRAME_MODE=1000 !
To ensure that the compensating frame (TOOLFRAME) is retained after a reset or end of program, enter the following value in MD 20150: $MC_GCODE_RESET_VALUES[52]:

- for G17 (TOROT) → MD 20150=2
- for G18 (TOROTY) → MD 20150=3
- for G19 (TOROTX) → MD 20150=4

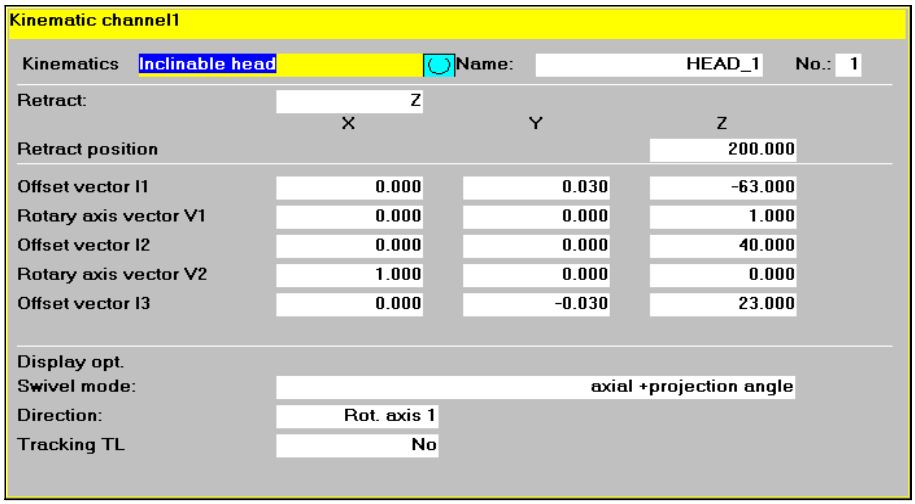
Soft key Save data record
Current swivel data record is saved as the parts program. The parts program corresponds to the name of the swivel data record.

;Soft key Delete data record
Current swivel data record is deleted.

© Siemens AG 2004 All rights reserved.

The CYCLE800 start-up process is supported by the following menu displays:

Start-up of kinematic chain



You must set up a swivel data set for every swivel head, swivel table or combination of both.
Swivel data records can be declared in several channels.
The number of swivel data records is limited by the following machine data:

- MD 18088: MM_NUM_TOOL_CARRIER or
- MD: NUM_CHANNELS (Number of channels option)
- MD 28085: MM_LINK_TOA_UNIT

The swivel data record is assigned to the tool data by parameters $TC_CARR1[n] to $TC_CARR40[n].

The meaning of the parameters in the "Kinematics" screen form is as follows:
Name: Swivel data record $TC_CARR34[n]
n → No. of swivel data record
If more than one swivel data record is declared in each NC channel, then a name is assigned to each record. No name needs to be specified if the swivel-mounted toolholder is not exchangeable (i.e. one swivel data record per channel). The program is advanced to the next swivel data record and the next channel via soft keys (channel +/- swivel data record +/-).

Only characters valid for NC programming must appear in the name!

© Siemens AG 2004 All rights reserved.

Kinematic type $TC_CARR23[n]

- Swivel head (type T)
- Swivel table (type P)
- Swivel head + swivel table (type M)

Retraction/Retraction position
$TC_CARR38[n] X; $TC_CARR39[n] Y; $TC_CARR40[n] Z
n → No. of swivel data record
The start-up engineer determines whether the options 'Retract axis Z' and 'Retract axes Z,X,Y' are possible in the input menu for the swivel cycle.
The mode of retraction can be modified in user cycle TOOLCARR.spf (marker _M41, _M40) if necessary. If user cycle TOOLCARR.spf is not modified, the retraction position is approached as an absolute machine position.



Please note the following when moving the tool axes:
Retract the tool axis in such a way that the tool and workpiece cannot collide when swiveled.

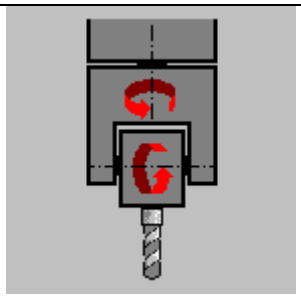
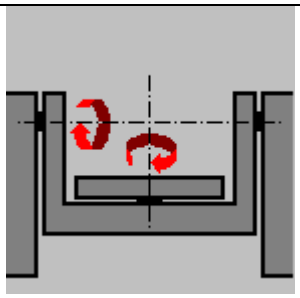
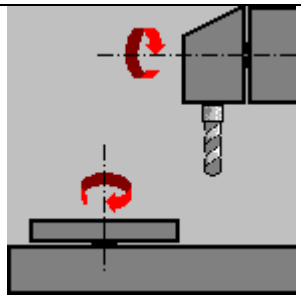
Offset rotary axis vectors (machine kinematics)
$TC_CARR1[n] ... $TC_CARR20[n]
The vectors always contain 3 components that represent the reference to machine axes (X, Y, Z).

The positions in the kinematic chain are measured by the machine manufacturer; they are always relevant with respect to a swivel head table (swivel data record).
Offset vectors I1 to I4 refer to the **non-swiveled state of the rotary axes**.

The machine kinematics used do not need to be fully implemented from the control viewpoint. But then be aware that the traversing range in the swivel planes may be restricted.

If machine kinematics are to be implemented with just one rotary axis, this must always be declared as the 1st rotary axis.
Manually adjustable rotary axes with or without measuring systems are possible and can be used with "plain machines".

© Siemens AG 2004 All rights reserved.

Swivel head (type T)	Swivel table (type P)	Swivel head + swivel table (type M)
		
Offset vector I1	Offset vector I2	Offset vector I1
Rotary axis vector V1	Rotary axis vector V1	Rotary axis vector V1
Offset vector I2	Offset vector I3	Offset vector I2
Rotary axis vector V2	Rotary axis vector V2	Offset vector I3
Offset vector I3	Offset vector I4	Rotary axis vector V2
		Offset vector I4

References:

- Description of Functions 840D/840Di/810D
  - /W1/ "Tool Offset"
  - (3/2 axes inclined machining)
- ISO 841-2001 or DIN 66217

Swivel head

→ I3 distance from the tool holder to the pivot point of the 1st rotary axis
→ I2 distance from the pivot point of the 1st rotary axis Pivot point of the 2nd rotary axis
→ I1 closure of the vector chain I1=-(I2+I3)

Swivel table

→ I2 distance from the tool holder to the Pivot point of the 2nd rotary axis
→ I3 distance from the pivot point of 1st rotary axis to the Pivot point of the 2nd rotary axis
→ I4 closure of the vector chain I4=-(I2+I3)

Swivel head / swivel table (mixed kinematics)

→ I2 distance from the tool holder to the Pivot point of the 1st rotary axis
→ I1 closure of the vector chain I1=-I2
→ I3 distance from the tool holder to the Pivot point of the 2nd rotary axis
→ I4 closure of the vector chain I4=-I3

© Siemens AG 2004 All rights reserved.

The offset vectors do not have to point to the pivot point of the rotary axes. The important thing is that they point to a point on the direction of rotation.
The sign of the offset vectors and the rotary axis vectors result from the stipulates of the axis directions according to ISO 841-2001 or DIN 66217 (right-hand rule).

Relation data toolcarrier ↔ 5-axis transformation

Swivel head (swivel-mounted tool)

$TC_CARR23[1]=**"T"**		$MC_TRAFO_TYPE_1=**24**
I1 $TC_CARR1..3[n]	⟷	$MC_ TRAFO5_PART_OFFSET_1[0..2]
I2 $TC_CARR4..6[n]		$MC_ TRAFO5_JOINT_OFFSET_1[0..2]
I3 $TC_CARR15..17[n]		$MC_ TRAFO5_BASE_TOOL_1 [0..2]

Close vector chain **I1=-(I2+I3)**

Swivel table (swivel-mounted workpiece)

$TC_CARR23[1]="**P"**		$MC_TRAFO_TYPE_1=**40**
I2 $TC_CARR4..6[n]	⟷	$MC_ TRAFO5_BASE_TOOL_1 [0..2]
I3 $TC_CARR15..17[n]		$MC_ TRAFO5_JOINT_OFFSET_1[0..2]
I4 $TC_CARR18..20[n]		$MC_ TRAFO5_PART_OFFSET_1[0..2]

Close vector chain **I4=-(I2+I3)**

Swivel head + swivel table (swivel-mounted tool + workpiece)

$TC_CARR23[1]="**M"**		$MC_TRAFO_TYPE_1=**56**
I1 $TC_CARR1..3[n]		$MC_ TRAFO5_JOINT_OFFSET_1[0..2]
I2 $TC_CARR4..6[n]	⟷	$MC_ TRAFO5_BASE_TOOL_1 [0..2]
I3 $TC_CARR15..17[n]		$MC_ TRAFO5_JOINT_OFFSET_PART_1[0..2]
I4 $TC_CARR18..20[n]		$MC_ TRAFO5_PART_OFFSET_1[0..2]

Close vector chain **I1=-I2 I4=-I3**

© Siemens AG 2004 All rights reserved.

Display options

- **$TC_CARR37[n]** (n → swivel data record)
  If the relevant display options are not set, the value will not be displayed in the input screen form (see subsection 3.16.1).

8 7 6 5 4 3 2 1 0 (Tens digits)

0: Axial
1: Axial + angle of projection
2: Axial + angle of projection + solid angle

Rotary axis 1
0: automatic
1: manual

Rotary axis 2
0: automatic
1: manual

Selection axis feed direction
0: no
1: Reference to rotary axis 1
2: Reference to rotary axis 2

Correct tool tip
0: no
1: yes

reserved

Retraction mode
0: Z axis
1: Z axis or ZXY axis

Automatic compensation with Hirth tooth system
0: no
1: 1st rotary axis
2: 2nd rotary axis
3: 1st and 2nd rotary axis

Swivel data record change / tool change[1)]

0:	no	2)
1:	manual	automatic
2:	automatic	automatic
3:	no	2)
4:	manual	manual
5:	automatic	manual

1) Relevant for ShopMill/ShopTurn only.
2) If a swivel data record change has not been agreed, the setting "Tool change automatic/manual" is not relevant.

© Siemens AG 2004 All rights reserved.

The following display options affect the input screen form for the swivel cycle:

- **Swivel mode**
  → **axial**
  → **axial and projection angle**
  → **axial and projection angle and solid angle**
  Example:
  Selection of swivel mode during start-up:
  axial, projection angle
  In this case, only 'axial' or 'projection angle' is included in the input menu. Programming of a solid angle is not desired or possible in this example.

- **Direction**
  → **Rotary axis 1**
  → **Rotary axis 2**
  → **no**
  When direction of travel is selected in the swivel cycle interactive screen form, reference to **rotary axis** 1 or 2.
  Because of the angular range of **rotary axes** of machine kinematics, the NCU calculates 2 possible solutions. One of these solutions is normally appropriate technologically. The selection as to which **rotary axis** the two solutions must refer to is made in the start-up menu. The selection of the solution to be applied is made in the interactive screen form for the swivel cycle.
  With "no", the direction parameter is not displayed in the operating screen.

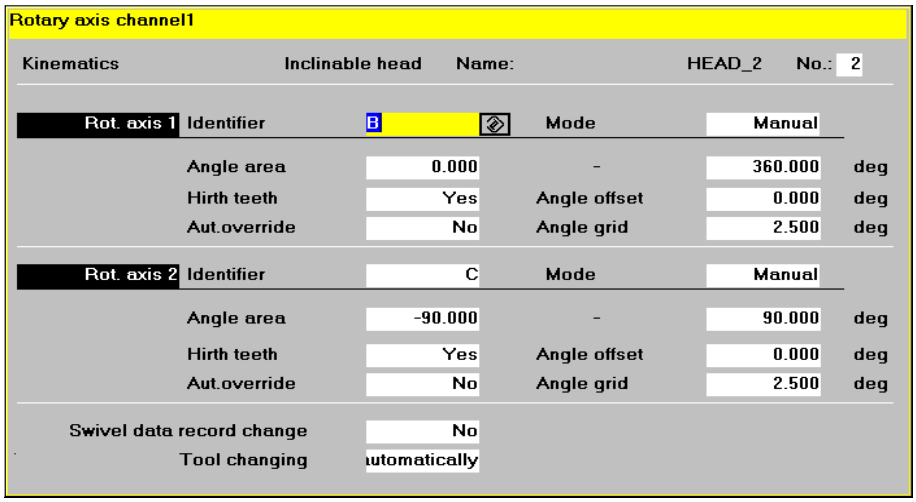
- **Correct tool**
  → **no**
  → **yes**
  "Correct tool" display in the swivel cycle interactive screen form. The function correct tool requires the option 5-axis transformation (TRAORI).
  In the user cycle, query TOOLCARR.spf Variable GUD7 _TC_N_WZ.



See programming example TOOLCARR.spf

© Siemens AG 2004 All rights reserved.

Starting up rotary axis parameters



Input of data for rotary axes 1 and 2 that are relevant to the swivel cycle

Name/kinematics→ see CYCLE800 "Kinematics" start-up menu

Axis identifiers of the rotary axes
$TC_CARR35[n] rotary axis 1
$TC_CARR36[n] rotary axis 2

The following identifiers should be chosen where possible:
Axis rotates about machine axis X --> A
Axis rotates about machine axis Y --> B
Axis rotates about machine axis Z --> C

If the axes of the NCU are known, the same axis identifiers of the relevant NC rotary axes must be selected (see automatic mode).
If the axes of the NCU are unknown, you can use any axis designations (up to 6 letters or numbers).

Mode
$TC_CARR37[n] see display options

- **Automatic**
  NC rotary axes are automatically moved to the appropriate swivel angle
- **Manual**
  Rotary axes are moved manually by the operator to the appropriate position, e.g. "very simple machines" with manually adjustable **rotary axes** (measuring system: steel gauge) The swivel angle to be set is displayed in messages 62180/62181.

© Siemens AG 2004 All rights reserved.

- **Semi-automatic**
  In "semi auto" mode, the messages 62180/62181 (display of the angles to be set) are not displayed.
  Application: Manual rotary axes that are positioned at the calculated angle by an appropriate mechanism. For this purpose, the machine manufacturer can modify user cycle TOOLCARR to pass the swivel angle (local variable _A1, A2) to a subroutine (markers _M21... for manual rotary axes).



Even mixed machine kinematics
(e.g. 1st rotary axis automatic, 2nd rotary axis manual), and "incomplete" machine kinematics
(e.g. 1st rotary axis rotates around axis X) are allowed.
If machine kinematics are to be implemented with just one rotary axis, this must always be declared as the 1st rotary axis.



For the swivel angle display, see display messages CYCLE800 → 62180/62181

Angular range
$TC_CARR30[n] .. $TC_CARR33[n]
A valid angular range must be assigned to each **rotary axis**. This does not have to be the software end position range of the particular rotary axis.
With modulo axes, enter a traversing range of between **0 and 360** degrees.

Offset kinematics (SW 6.3 and later)
In the "Offset Kinematics" input field, it is possible to enter the offset of rotary axis 1 ($TC_CARR24[n]) or rotary axis 2 ($TC_CARR25[n]), if the position of the rotary axes is not equal to 0 in the basic position.

© Siemens AG 2004 All rights reserved.

Hirth tooth system
$TC_CARR26[n]... $TC_CARR29[n]
no
The fields below are concealed.

- **yes**
  → **Angular offset** of Hirth tooth system at beginning of gearing.
  → **Angular grid** of Hirth tooth system
  → **Automatic correction yes/no** (eliminated on SW 6.3 and later)

Swivel data record change (only relevant for ShopMill/ShopTurn)

- **no**
- **automatic**
- **manual**

Tool change (only relevant for ShopMill/ShopTurn)

- **automatic**
- **manual**

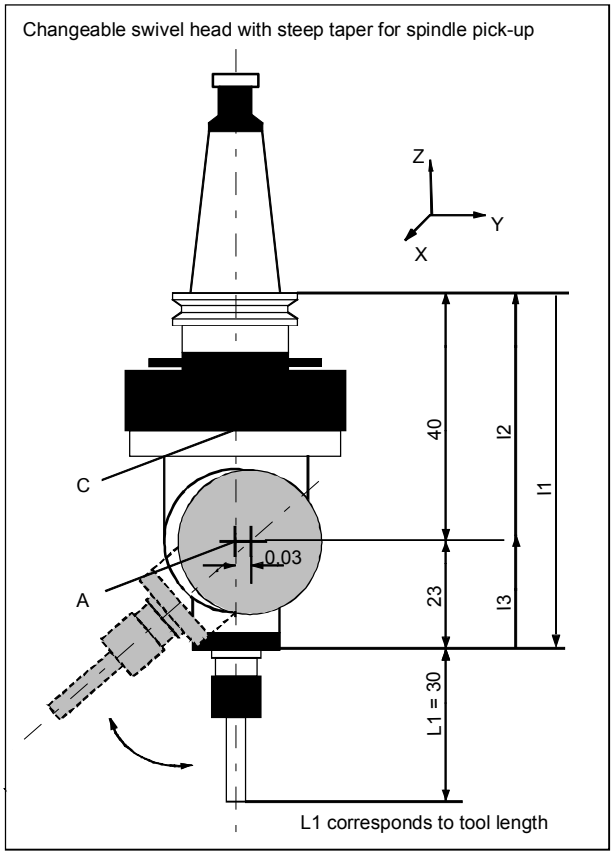
"Tool change" display for kinematic type T and M only

© Siemens AG 2004 All rights reserved.

Start-up examples for machine kinematics

Example 1 Swivel head 1 "HEAD_1"

Rotary axis 1(C) (manual) about Z; rotary axis 2(A) (manual) about X
(drawing not true-to-scale)



Kinematic channel1

Kinematics	Inclinable head		Name: HEAD_1	No.: 1
Retract:	Z			
	X	Y	Z	
Retract position			200.000	
Offset vector l1	0.000	0.030	-63.000	
Rotary axis vector V1	0.000	0.000	1.000	
Offset vector l2	0.000	0.000	40.000	
Rotary axis vector V2	1.000	0.000	0.000	
Offset vector l3	0.000	-0.030	23.000	
Display opt.				
Swivel mode:	axial +projection angle			
Direction:	Rot. axis 1			
Tracking TL	No			

© Siemens AG 2004 All rights reserved.

Example 2 Swivel head 2 "HEAD_2"

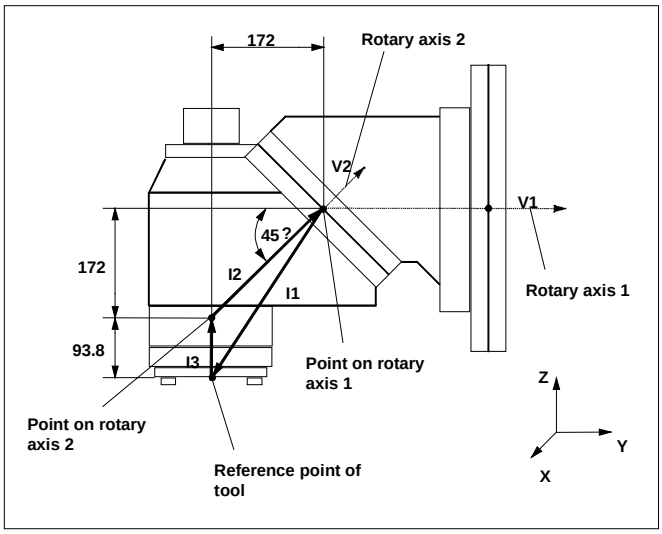
Rotary axis vector V1: Rotary axis B rotates about Y

Rotary axis vector V2: Rotary axis C rotates about Y **and** about Z

Offset vector I1: Closure of the vector chain with fixed mounted swivel head I1=-(I2+I3)

Offset vector I2: distance between pivot point of the **rotary axis 1** and pivot point on **rotary axis 2**

Offset vector I3: distance between the reference point of the tool and pivot point of **rotary axis 2**



Kinematic channel1

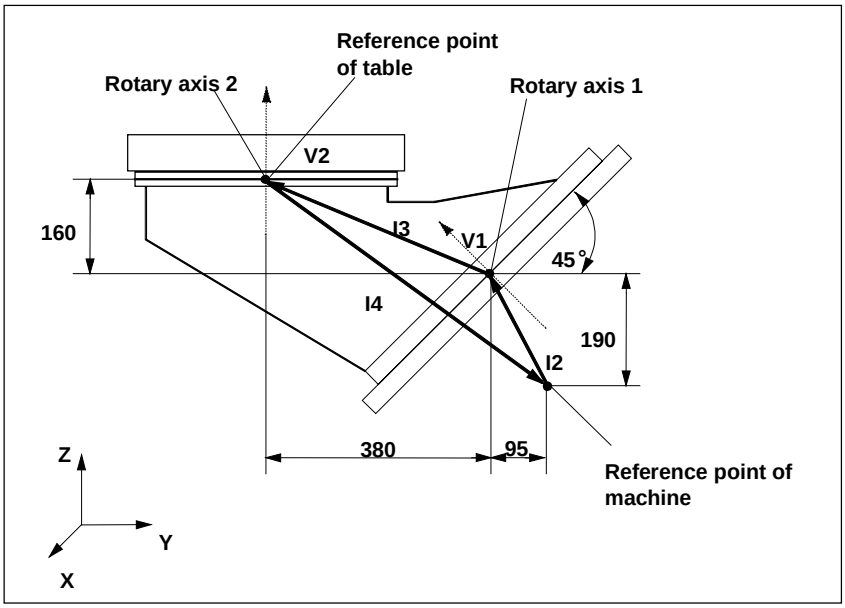
Kinematics	Inclinable head		Name: HEAD_2	No.: 2
Retract:	Z			
	X	Y	Z	
Retract position			200.000	
Offset vector I1	0.000	-172.000	-265.800	
Rotary axis vector V1	0.000	1.000	0.000	
Offset vector I2	0.000	172.000	172.000	
Rotary axis vector V2	0.000	-1.000	1.000	
Offset vector I3	0.000	0.000	93.800	
Display opt.				
Swivel mode:	axial +projection angle			
Direction:	Rot. axis 1			
Tracking TL	No			

The reference point of the pivot point of rotary axes 1 and 2 can be offset on the line of rotation and does not have to coincide with the pivot point.

© Siemens AG 2004 All rights reserved.

Example 3 Table 2

Rotary axis vector V1: Rotary axis B rotates about Y
Rotary axis vector V2: Rotary axis C rotates about Y **and** about Z
Offset vector I2: Distance between machine reference point and pivot point on **rotary axis 1**
Offset vector I3: distance between pivot point of the **rotary axis 1** and pivot point on **rotary axis 2**
Offset vector I4: Closure of vector chain I4=-(I2+I3)



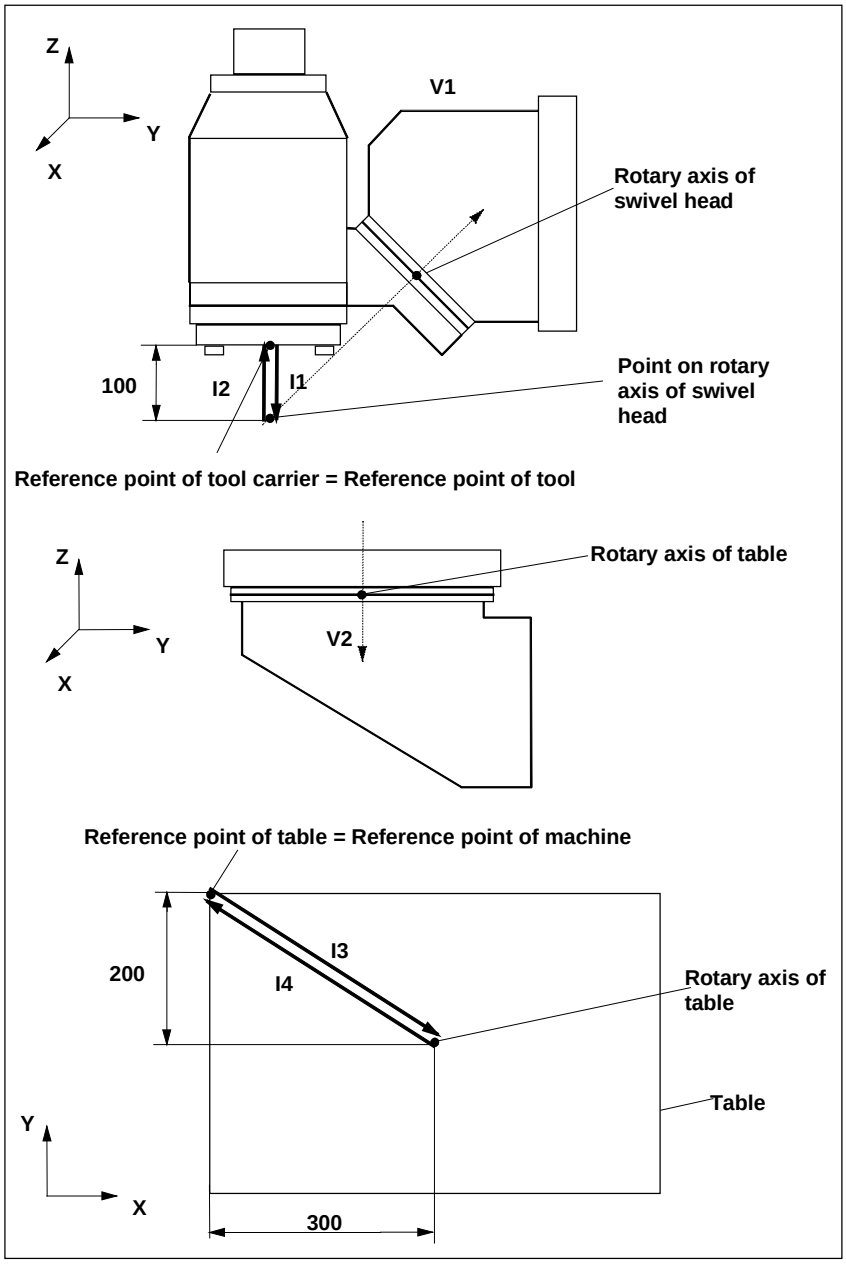
Kinematic channel1

Kinematics	Swivel table		Name:	TABLE_2	No.: 3
Retract:	Z				
		X	Y	Z	
Retract position				100.000	
Offset vector I2		0.000	-95.000	190.000	
Rotary axis vector V1		0.000	-1.000	1.000	
Offset vector I3		0.000	-380.000	160.000	
Rotary axis vector V2		0.000	0.000	1.000	
Offset vector I4		0.000	475.000	-350.000	
Display opt.					
Swivel mode:				axial+projection angle+solid angle	
Direction:		Rot. axis 1			
Tracking TL		No			

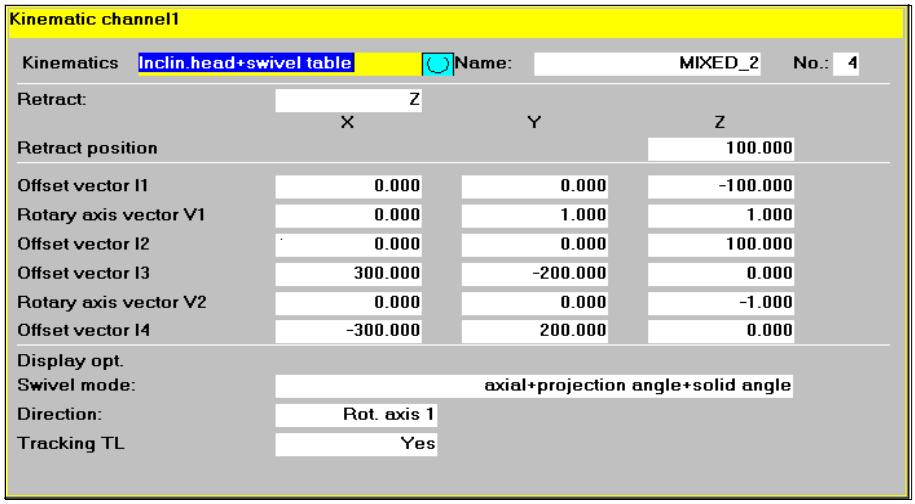
© Siemens AG 2004 All rights reserved.

Example 4 MIXED 2

In this example, the reference points of the toolholder and the tool are identical, as well as the reference points of the table and the machine.
The following thus applies: I1=-I2 and I3=-I4
This results in a closed system.
The advantage of this method is that the position values do not change in the unswiveled state, regardless of whether the swivel head is selected.



© Siemens AG 2004 All rights reserved.



3.16.5 User cycle TOOLCARR.spf

Programming

CYCLE800(_MODE, _TC1, _A1, _A2, _TC2

Parameters

_MODE	Corresponds to the markers of the structure described below
_TC1	Number of the swivel head/table
_A1	Angle of the 1st rotary axis
_A2	Angle of the 2nd rotary axis (if present)
_TC2	Reserve

Note:
The parameters are only valid in conjunction with the relevant jump label in TOOLCARR --> see the program structure below.

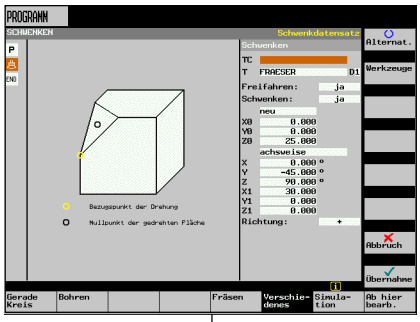
Customization by the machine manufacturer

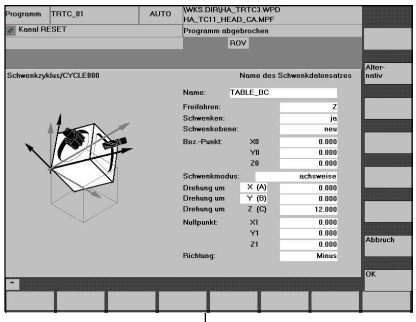
All axis positions during swiveling are traversed by means of cycle TOOLCARR.spf. This is called from the swivel cycle CYCLE800 or E_TCARR (ShopMill) or F_TCARR (ShopTurn). The cycle can be modified by the machine manufacturer during start-up in order to cater to the properties of specific machines.
If the user cycle is not modified, during retraction before swiveling, axis Z (marker _M41) or axis Z is moved first, followed by axes X, Y (marker _M42).
The positions correspond to the start-up menu CYCLE800 "Kinematics" → retraction positions.

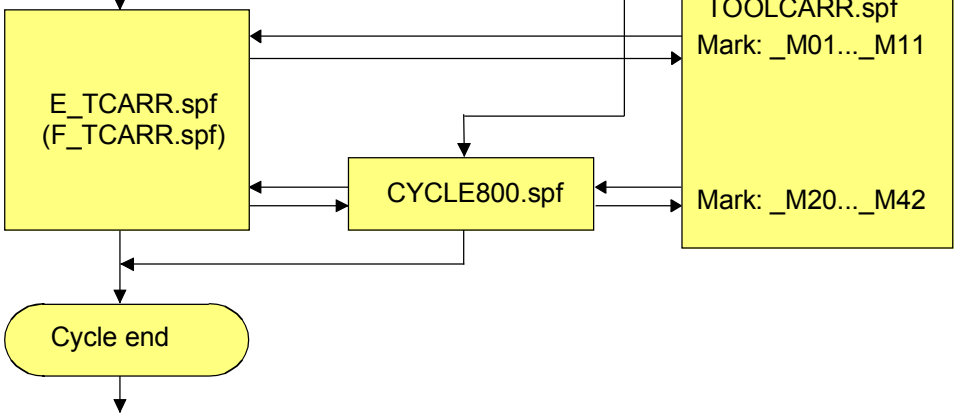
© Siemens AG 2004 All rights reserved.

Structure (coarse) swivel cycles

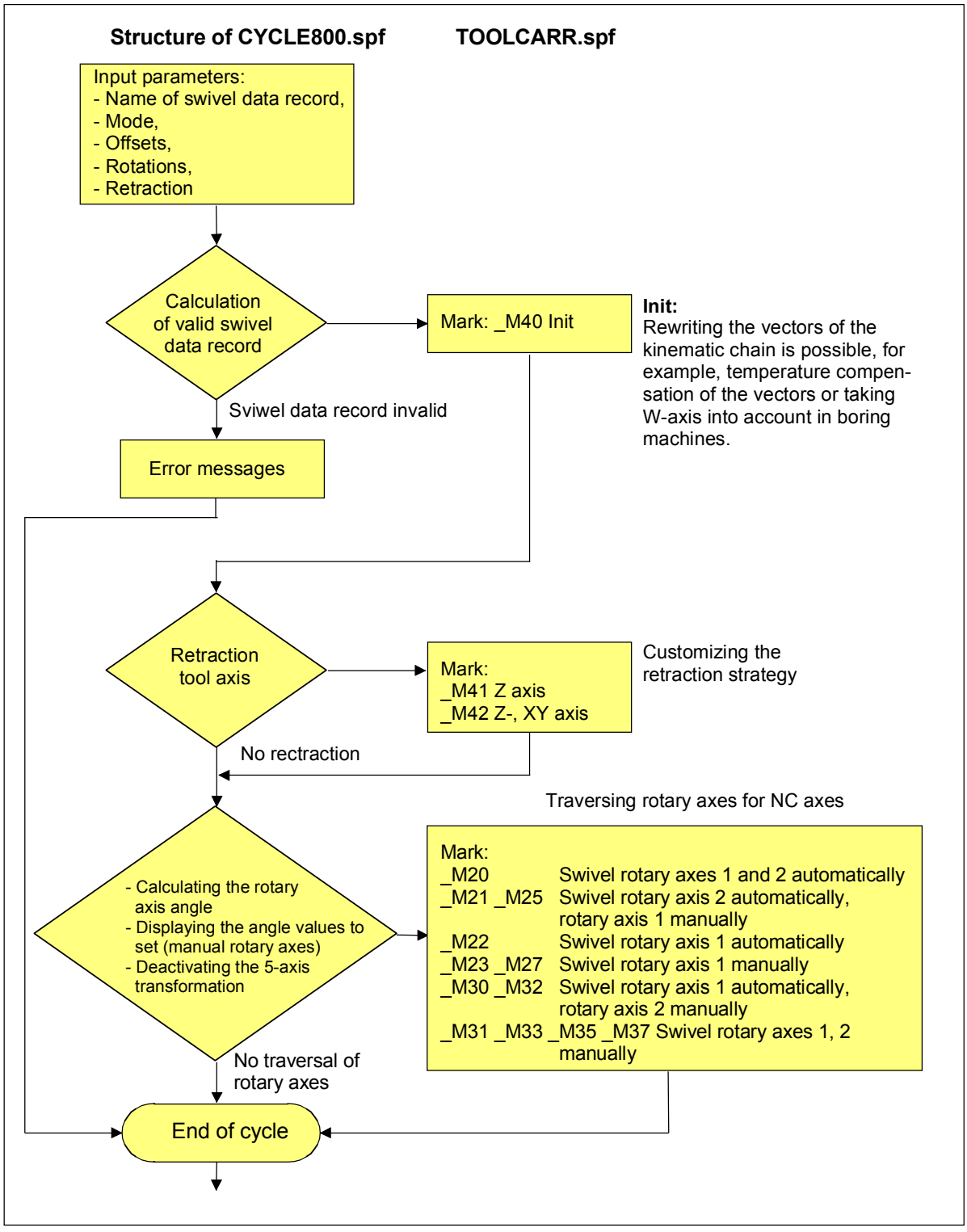
Input screen form ShopMill/ShopTurn Input screen form standard CYCLE800





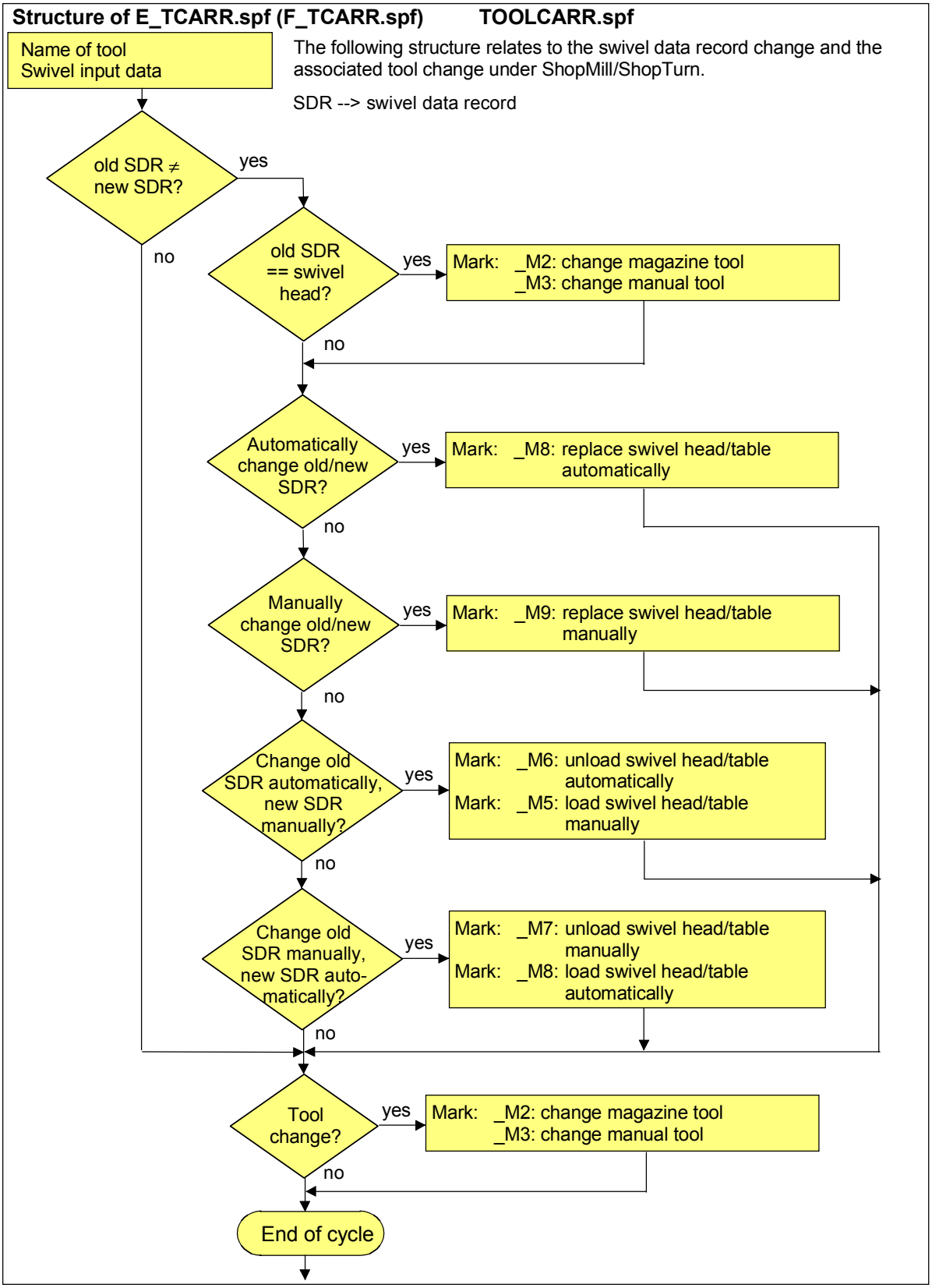


© Siemens AG 2004 All rights reserved.



© Siemens AG 2004 All rights reserved.

Structure of E_TCARR.spf (F_TCARR.spf) TOOLCARR.spf



© Siemens AG 2004 All rights reserved.

Note on markers _M20 to _M37

Markers _M20 to _M37 are distinguished by kinematics with two or one rotary axis. A distinction is also made between automatic rotary axes (known to the NCU) and manual rotary axes.

There is only ever one valid marker for the active swivel data record. Control via parameter/GUD7 variable _TC_ST.

Note on ShopMill/ShopTurn

In the user cycle TOOLCARR.spf, cycle E_SWIV_H or F_SWIV_H is called under ShopMill/ShopTurn (see markers _M2 to _M9).

E_SWIV_H parameters (Par 1, Par 2, Par 3)

- Par 1: Number of swivel data record (_TC1)
- Par 2: Angle 1st rotary axis
- Par 3: Angle 2nd rotary axis

Modification examples:

If the rotary axes (swivel head/table) are **not to be positioned** for swivel data change/tool change, the call of cycle E_SWIV_H can be commented out at the relevant labels.

If the rotary axes are to move to a certain position, an angle value can be transferred in parameters Par 2, Par 3.

© Siemens AG 2004 All rights reserved.

Programming example: switching "correct tool" on/off

This example requires a 5-axis transformation equivalent to the swivel data record to have been set up. The programming part is included in the
label _M20 of manufacturer cycle TOOLCARR.spf

_M20:

```
IF (_TC_N_WZ==1) AND (NOT $P_SEARCH)        ;Correct tool
AND (NOT $P_ISTEST)                         ;yes=1 no=0
TCARR=0
PAROTOF
TRAORI
N820 G0 G90
AX[AXNAME($TC_CARR35[_TC1])]=_A1
AX[AXNAME($TC_CARR36[ _TC1])]=_A2
TRAFOOF
TCARR=_TC1
ELSE                                        ;No tool correction
N820 G0 G90
AX[AXNAME($TC_CARR35[_TC1])]=_A1
AX[AXNAME($TC_CARR36[ _TC1])]=_A2
ENDIF
GOTOF _MEND
```


© Siemens AG 2004 All rights reserved.

3.16.6 Error messages



Explanation

Alarm source CYCLE800

Alarm number	Alarm text	Explanation, remedy
61180	"No name assigned to swivel data record even though machine data $MN_MM_NUM_TOOL_CARRIER > 1"	No name assigned to swivel data record even though several swivel data records exist ($MN_MM_NUM_TOOL_CARRIER>0) or no swivel data record defined ($MN_MM_NUM_TOOL_CARRIER=0)
61181	"NCK software version too old (no TOOLCARRIER functionality)"	TOOLCARRIER functionality as from NCU 6.3xx
61182	"Name of swivel data record unknown"	See Swivel cycle start-up CYCLE800 → Kinematics Name (swivel data record)
61183	"Retraction mode GUD7 _TC_FR outside value range 0..2"	See Swivel cycle start-up CYCLE800 → Retraction; 1st transfer parameter CYCLE800(x,...) is faulty >2
61184	"No solution can be found with current angle inputs"	
61185	"No or incorrect (min>max) angle ranges declared for rotary axes"	Check start-up of swivel cycle CYCLE800
61186	"Invalid rotary axis vectors"	Start-up of swivel cycle CYCLE800: missing or incorrect entry for rotary axis vector V1 or V2
61187	"Block search computation end of block not valid for SWIVEL"	Select block search with calculation contour
61188	"No axis name 1st rotary axis declared"	Start-up of swivel cycle CYCLE800: missing entry under rotary axis 1 identifier
62180	"Set rotary axes x.x [deg]"	Angles to be set for manual rotary axes
62181	"Set rotary axes x.x [deg]"	Angle to be set for manual rotary axis



Typical display of the swivel angle to be set for a manual rotary axis in CYCLE800

62180	"Set rotary axes B: 32.5 [deg]"

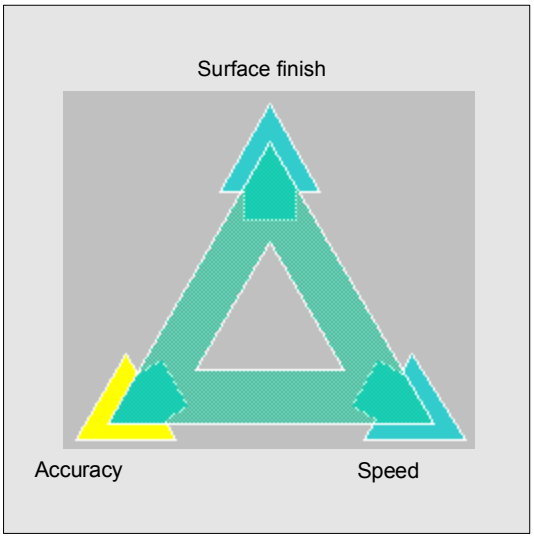
© Siemens AG 2004 All rights reserved.

3.17 High speed settings - CYCLE832 (SW 6.3 and later)

The standard cycle High Speed Settings CYCLE832 is available for the HMI in SW 6.3 and higher and NCU SW 6.3 (CCU SW 4.3).

Application of CYCLE832:

- for technological support when machining free-form contours (surfaces) in the 3- or 5-axis high-speed machining range (High Speed Cutting - HSC)
- used predominantly in the HSC milling range (possibility of use for turning or grinding)
- groups the important G codes and machine and setting data that are needed for HSC machining
- technology / geometry separation by relevant NC program architecture



When executing CAM programs in the HSC range, the control has to process high feedrates with the shortest NC blocks. The user expects a good **surface finish** with great **accuracy** in the 1 mm range with extremely high **machining feedrates** >10 m/min. By applying different machining strategies, the user can fine-tune the program with the aid of CYCLE832.
When **roughing**, the weighting is placed on **speed**.
When **finishing**, the weighting is placed on **accuracy**.
In both cases, specifying a tolerance retains the machining contour in order to achieve the desired surface finish.
When defining the tolerance values for smoothing the contour, the operator must have precise knowledge of the subsequent CAM program. The CYCLE832 cycle supports machine types where a maximum of three linear and two rotary axes are involved in the machining.

The following functions can be defined or enabled/disabled by CYCLE832:

- 4 types of machining: finishing, rough-finishing, roughing, deselection (default setting)
- Tolerance band of the contour to be machined
- Smoothing (G64, G641, G642)
- NC block compressor (COMPCAD, COMPCURV, COMPOF)[1)]
- Feedforward control (FFWON, FFWOF)
- Jerk limitation (SOFT, BRISK)
- 5-axis transformation (TRAORI, TRAFOF)[1)]
- B-SPLINE[1)]

1) only if the relevant option is set.

© Siemens AG 2004 All rights reserved.

Function

The CYCLE832 cycle groups the important G codes and the machine or setting data required for HSC machining.
In CYCLE832, a distinction is made between four technological executions:

- "Finishing"
- "Rough-finishing"
- "Roughing"
- "Deselection" (default setting)

In CAD programs, the four machining types are in the HSC range in direct relation to the accuracy and speed of the path contour (see help display). The operator / programmer uses the **tolerance value** to give relevant weighting.
Different tolerances and settings (customizing technology) can be assigned to the four machining types.

In the input screen form, the relevant G codes are preset (customizing technology) to ensure a smoothing path contour or that the CAM program is executed at optimum speed.
The cycle is located in the main menu before the CAM program (see typical CYCLE832 call).
Different interpretations of the tolerance values are taken into account. For example, for G641, the tolerance value is transferred as ADIS= and for G642, the axis-specific MD 33100 COMPRESS_POS_TOL[AX] is updated.

When activating the "Customize technology" input field, you can enable or disable:
(keyswitch position ≥2).

- compression (COMPCAD, COMPCURV, COMPOF, B-SPLINE),
- continuous-path mode (G64, G641, G642) or
- velocity control (FFWON, FFWOF, SOFT, BRISK).

If 5-axis transformation (TRAORI) is set up, this can be enabled/disabled in the transformation input field.

Never ignore information provided by the machine manufacturer!

© Siemens AG 2004 All rights reserved.

Typical CYCLE832 call

```
T1 D1
G54
M3 S12000
CYCLE832(0.20,1003)          ;Roughing
EXTCALL "CAM_Form_Schrupp"
CYCLE832(0.01,102001)        ;Finishing
EXTCALL "CAM_Form_Schlicht"
CYCLE832(0.1,0)              ;Deselection (default setting)
M02
```

CYCLE832 does not absolve the machine manufacturer from the requisite optimization tasks when commissioning the machine. This involves the optimization of the axes involved in machining and the settings of the NCU (feedforward control, jerk limitation, etc.).

Reduced program call

The following call options of CYCLE832 with a reduced parameter transfer are possible:

- CYCLE832() corresponds to the selection of input screen form "Machining" "Deselection" The G codes (see subsection 3.17.5) used in CYCLE832 are set to the value set in MD 20150: GCODE_RESET_VALUE
- CYCLE832(0.01) Entering the tolerance value. The active G commands are not altered in the cycle.

© Siemens AG 2004 All rights reserved.

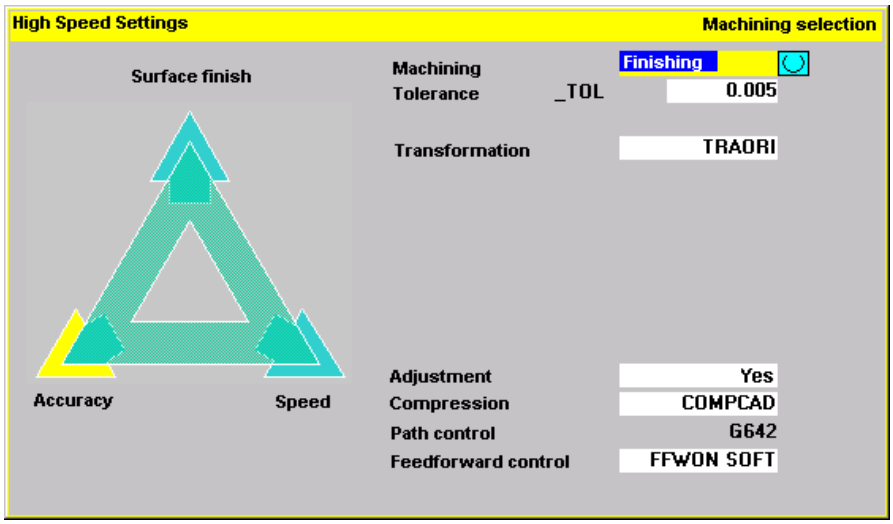
3.17.1 Call CYCLE 832 in HMI menu tree

Explanation of the parameters

Entry area Programs / **Milling**

Softkey [>>] → [High Speed Settings]
is displayed.

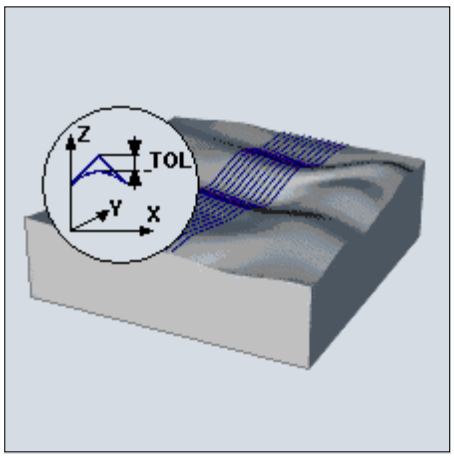
Interactive screen form CYCLE832 on standard interface



Machining (_TOLM)

- Finishing (default)
- Rough-finishing
- Roughing
- Deselection

For variable _TOLM coding see Subsection 3.17.2, Parameters.



Tolerance (_TOL)

The tolerance of axes involved in machining. The tolerance value is written to the relevant machine or setting data depending on the G codes (G642, COMPCAD, COMPCURV,..) (see Subsection 3.1.7.5).

If the machining axis is a rotary axis, the tolerance value is written with a factor (default factor = 8) to the relevant machine or setting data of the rotary axis.

© Siemens AG 2004 All rights reserved.

With G641, the tolerance value corresponds to the ADIS value.
On initial entry, the following values are set for the tolerance:

- **Finishing**: 0.01 (linear axes) 0.08 deg (rotary axes)
- **Rough-finishing**: 0.05 (linear axes) 0.4 deg (rotary axes)
- **Roughing**: 0.1 (linear axes) 0.8 deg (rotary axes)
- **Deselection**: 0.01 (linear axes) 0.1 deg (rotary axes)

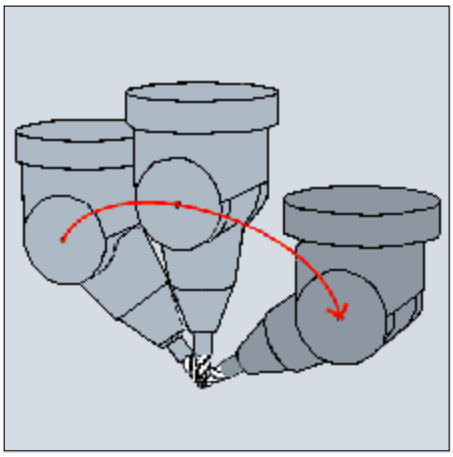
The mm/inch system of units is taken into account.

If the tolerance value is also to be effective for rotary axes, 5-axis transformation must be set up be the machine manufacturer, but not activated in every chase, e.g. machining of CAM programs with resolved rotary axes.

Transformation (_TOLM)

The transformation input field is only displayed when the NC option is set (machining package 5 axes set).

- **No**
- **TRAORI** → 1. Activate 5-axis transformation
- **TRAORI(2)** → 2. Activate 5-axis transformation



Selection of the transformation number or of the manufacturer cycle to call 5-axis transformation.

- In the GUD7 variables _TOLT2, it is possible to store the name of a manufacturer cycle that calls the manufacturer cycle of the transformation. If _TOLT2 is empty ("" default), selection transformation 1,2... calls the 5-axis transformation with TRAORI(1) or TRAORI(2).
- To start a 5-axis transformation program to a swiveled plane (see CYCLE800), the tool holder is deleted and the swivel frame (tool reference) WPFRAME is used by TRAOR after activation.

© Siemens AG 2004 All rights reserved.

Customization, customizing the technology
(see subsection 3.17.3)

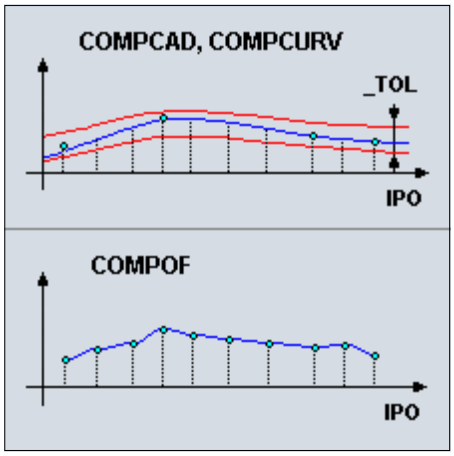
- **yes**
- **no**

The subsequent input parameters can only be changed if customization is set to "Yes".

Input field "Customization" and the following input fields "Compression", "Path control", and "Feedforward control" are not shown if the keyswitch is in position 0 or 1.

Compression, NC block compressor (_TOLM)

- **COMPOF** (default)
- **COMPCAD**
- **COMPCURV**
- **B SPLINE**



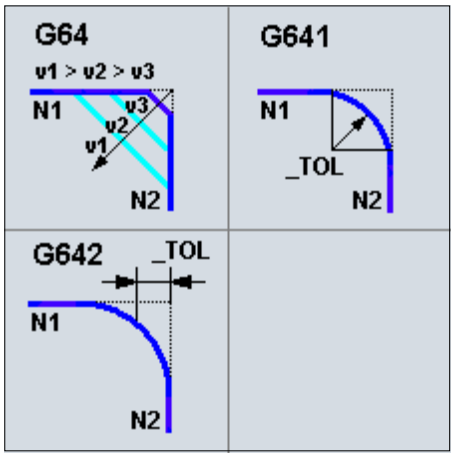
The input field is only displayed if the compressor function option is set.
The selection B spline is only made if the option spline interpolation is set.

Option → A, B and C splines/compressor function

Path control (_TOLM)

- **G642** (default)
- **G641**
- **G64**

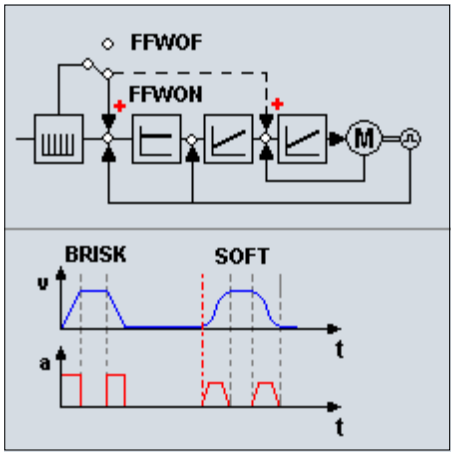
With NC block compressor with COMPCAD, COMPCURV, G642 is always the permanent selection.



Feedforward control, velocity control (_TOLM)

- **FFWOF SOFT** (default)
- **FFWON SOFT**
- **FFWOF BRISK**

The selection of feedforward control (FFWON) and jerk limitation (SOFT) presupposes that the machine manufacturer has optimized the control or the machining axes.



© Siemens AG 2004 All rights reserved.

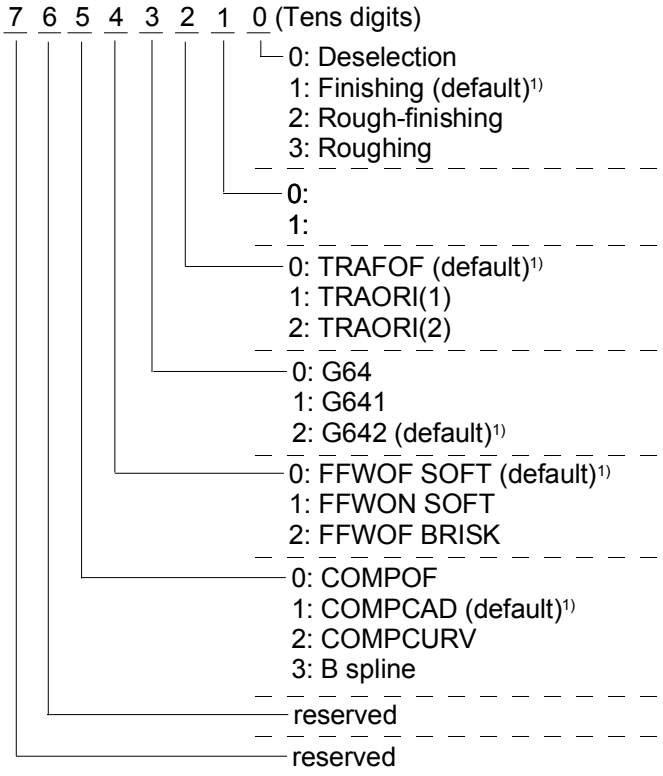
3.17.2 Parameters

Programming

CYCLE832(_TOL, _TOLM)

Parameters

_TOL	real	Tolerance of machining axes → Unit. mm/inch; deg
_TOLM	integer	Tolerance mode



1) Setting may be changed by machine manufacturer, see Section "Customizing technology"

© Siemens AG 2004 All rights reserved.

3.17.3 Customizing the technology

With input field "Customizing technology" "Yes", both the machine manufacturer and the machine setter/programmer can customize the technology for HSC machining.
Always take note of the technology of the subsequent CAM program.

Customization by the machine manufacturer
Preconditions:

- **The manufacturer password is set**,
- Input field "Customizing technology" → "Yes"

When the interactive cycle screen form CYCLE832 is opened, the parameters are preassigned with the values of GUD7 variables _TOLV[n], _TOLT[n].
n → machining: finishing, rough-finishing, roughing, Deselection
When the parameters are changed, the values are written directly to the GUD7 variables _TOLV[n] or _TOLT[n].
This gives the machine manufacturer the opportunity to customize the default settings to his machining tasks.

Example:
Calling cycle CYCL832 roughing with three axes machining axes tolerance 0.1mm with G642 (Siemens default values).
The machine manufacturer can modify the roughing technology with setting:
machining axes tolerance 0.3mm, TRAORI, G641.
Each time the tolerance cycle is called, this setting is then displayed and takes effect during the execution.

Notes for the machine manufacturer

1. In order to optimize the path control response when traversing with G64, in CYCLE832, the velocity jumps overload factor is recalculated in accordance with the following table:

Computation of the velocity jumps overload factor of all the machining axes

IPO [msec]	Overload factor
≥ 12	1.2
9	1.3
6	1.4
4	1.6
3	1.8

© Siemens AG 2004 All rights reserved.

IPO: MD 10071: $MN_IPO_CYCLE_TIME
Overload factor MD 32310:
$MA_MAX_ACCEL_OVL_FACTOR[AX]

Computation of the overload factor by CYCLE832 can be disabled by setting the local variable **_OVL_on=0** in cycle CYCLE832.

2. The tolerance with the NC block compressor (COMPCAP) or smoothing (G642) active is written in CYCLE832 to the MD 33100: $MA_COMPRESS_POS_TOL[AX] (machining axes linear). If rotary axes are involved in the machining (TRAORI) this tolerance is written with **factor 8** to MD 33100: $MA_COMPRESS_POS_TOL[AX] of the rotary axes. If a different factor is to be used, the local variable **FACTOR** in cycle CYCLE832 can be preassigned with the relevant value.

Customization by the machine setter/programmer
Preconditions:

- **The manufacturer password is deleted**
- Keyswitch position 2 or 3
- Protection level 5, 4, 3, 2
- Input variable "Customizing technology" → "Yes"

To customize the technology, the machine setter/ programmer must have precise knowledge of the subsequent CAM machining program.
The modified data is used to generate CYCLE832 and applies to the current call of CYCLE832.
Default settings of the machine manufacturer are not altered.

© Siemens AG 2004 All rights reserved.

3.17.4 Customizing additional program parameters CYC_832T



If the machine manufacturer requires additional responses from CYCLE832 beyond those obtainable by customizing the technology, it is possible to make the necessary changes in cycle CYC_832T. For that purpose CYC_832T should be copied into the CMA directory (HMI manufacturer) and loaded into the NCU.
The CYC_832T functions as a main program.
The machine manufacturer must document the changes.
The cycle CYC_832T is **automatically** called by CYCLE832, if it is loaded in the NCU.
Changes to CYCLE832 are not required because of the customization of CYC_832T.



Parameters

CYC_832T(_ASVS, _FACTOR, _OVL_on)

_ASVS	Machine-specific customization is possible at markers _M0 to _M4. _M4 = Init CYCLE832 call made before the actual program execution of CYCLE832 _M0 = deselection CYCLE832 _M1 = machining finishing _M2 = machining rough-finishing _M3 = machining roughing
_FACTOR[1)]	Rotary axis tolerance factor
_OVL_on[1)]	0=Do not adjust MD $MA_MAX_ACCEL_OVL_FACTOR overload factor

1) The parameters _FACTOR and _OVL_on are only effective at marker _M4(INIT).



Programming example

The machine manufacturer wants to perform the following customization:

1. The rotary axis tolerance is to be 12 times higher than the tolerances of the linear axes
2. In machining modes "finish", "rough-finish", and "rough", the path jerk (MD $MC_MAX_PATH_JERK) is defined as value 15 and the axis jerk (MD $MA_MAX_AX_JERK[AX]) as value 150
3. On deselection of CYCLE832 , the changed machine data must be reset to the default settings (1000,15).

© Siemens AG 2004 All rights reserved.

```
%_N_CYC_832T_SPF
;$PATH=/_N_CST_DIR).
PROC CYC_832T(INT _ASVS,VAR INT
_FACTOR,VAR INT _OVL_on) SAVE DISPLOF
...
N801 CASE _ASVS OF 0 GOTOF _M0 1 GOTOF
_M1 2 GOTOF _M2 3 GOTOF _M3 4 GOTOF _M4
DEFAULT GOTOF _MEND
...
_M4:
_FACTOR=12 ;Init
GOTOF _MEND

_M0: ;Deselection
$MC_MAX_PATH_JERK=1000
$MA_MAX_AX_JERK[X]=15
$MA_MAX_AX_JERK[Y]=15
$MA_MAX_AX_JERK[Z]=15
GOTOF _MEND

_M1: ;Finishing
_M2: ;Rough-finishing
_M3: ;Roughing
$MC_MAX_PATH_JERK=15
$MA_MAX_AX_JERK[X]=150
$MA_MAX_AX_JERK[Y]=150
$MA_MAX_AX_JERK[Z]=150

GOTOF _MEND
_MEND:
RET
```

To permit the machine data in CYC_832T to be overwritten in all protection levels during the program run, they must be redefined with REDEF.
Example:

```
%_N_MGUD_DEF
;$PATH=/_N_DEF_DIR

REDEF $MC_MAX_PATH_JERK APR 7 APW 7
REDEF $MA_MAX_AX_JERK APR 7 APW 7

M30
```


© Siemens AG 2004 All rights reserved.

3.17.5 Interfaces

G codes

List of the G commands programmed in CYCLE832:

- G64, G641, G642
- G601
- FFWON, FFWOF
- SOFT, BRISK
- COMPCAD, COMPCURV,COMPOF,B-SPLINE
- TRAORI, TRAORI(2),TRAOFOF
- UPATH

Note: The G commands should not be generated in the subsequent CAM program. Technology – Geometry separation.

Machine Data

The following machine data are evaluated in cycle CYCLE832 to set the tolerance values.

MD number	MD identifier	Comment
10071	$MN_IPO_CYCLE_TIME	IPO cycle
20480	$MC_SMOOTHING_MODE	
20482	$MC_COMPRESSOR_MODE	
24100 to 24462	Machine data 5-axis transformation	

The following machine data is rewritten in cycle CYCLE832:

MD number	MD identifier	Comment
20490	$MC_IGNORE_OVL_FACTOR_FOR_ADIS	
33100	$MA_COMPRESS_POS_TOL[AX]	Geometry axes 1...3
33100	$MA_COMPRESS_POS_TOL[AX]	Rotary axes 1 and 2 [1)]
32310	$MA_MAX_ACCEL_OVL_FACTOR[AX]	Geometry axes 1...3
32310	$MA_MAX_ACCEL_OVL_FACTOR[AX]	Rotary axes 1 and 2 [1)]

1) Acc. to machine data 5-axis transformation

Setting data

List of setting data rewritten in CYCLE832:

SD no.	SD identifier	Comment
42450	$SC_CONTPREC	For CPRECON and G64
42465	$SC_SMOOTH_CONTUR_TOL	corresponds to the tolerance of the linear axes
42466	$SC_SMOOTH_ORI_TOL	corresponds to the tolerance of the rotary axes
42475	$SC_COMPRESS_CONTUR_TOL	for COMPCURV only
42476	$SC_COMPRESS_ORI_TOL	for COMPCURV only

1) The effectiveness of the setting data $SC_SMOOTH_CONTUR_TOL and $SC_SMOOTH_ORI_TOL depends on MD20480: $MC_SMOOTHING_MODE.
The effectiveness of setting data $SC_COMPRESS_CONTUR_TOL and $SC_COMPRESS_ORI_TOL and depends on MD20482: $MC_COMPRESSOR_MODE

© Siemens AG 2004 All rights reserved.

Channel-specific variable GUD7
The following channel-specific variables must be activated for CYCLE832 functionality (machine manufacturer).
The definitions are a component of the GUD7 definitions of the SIEMENS standard cycle package.

Parameters	Format	Assignment	Comment
`_TOLT2[2]`	STRING[32]	„“ (default)	Program name subroutine for calling the 5-axis transformation
`_TOLT[4]`	integer	**Field (4):** 0: Deselection 1: Finishing 2: Rough-finishing 3: Roughing	Field for saving the technological data settings of the machine manufacturer Coding corresponds to variable _TOLM (see parameters)
`_TOLV[4]`	real	**Field (4):** 0: Deselection 1: Finishing 2: Rough-finishing 3: Roughing	Field for saving the tolerance values of the machining axes through the machine manufacturer settings (see customizing the technology). Default values: 0.01 deselection (GUD7.def) 0.01 finishing 0.05 rough-finishing 0.1 roughing

3.17.6 Error messages

Explanation

Alarm source CYCLE832

Alarm number	Alarm text	Explanation, remedy
61191	"5-axis transformation not set up"	1. Machining package five axes option or multi-axis interpolation not set. 2. Check MD 24100: $MC_TRAFO_TYPE_1 to $MC_TRAFO_TYPE_8 for a valid type of 5-axis transformation
61192	"Second 5-axis transformation not set up"	
61193	"Compressor option not set up"	Set spline interpolation option (A-, B-, and C-splines/compressor function
61194	"Spline interpolation option not set up"	

© Siemens AG 2004 All rights reserved.

3.18 Engraving cycle CYCLE60 (SW 6.4 and later)



Programming

```
CYCLE60 (_TEXT, _RTP, _RFP, _SDIS, _DP, _DPR, _PA, _PO, _STA, _CP1, _CP2,
_WID, _DF, _FFD, _FFP1, _VARI, _CODEP)
```



Parameters

_TEXT	STRING	Text to be engraved (up to 91 characters)
_RTP	real	Retraction plane (absolute)
_RFP	real	Reference plane (absolute)
_SDIS	real	Safety distance (to be added to _RFP, enter without sign)
_DP	real	Depth (absolute)
_DPR	real	Depth relative to the reference plane (enter without sign)
_PA	real	Reference point for text positioning (absolute) • Position of 1st axis (for _VARI = orthogonal), or • Radius of the arc (for _VARI = polar)
_PO	real	Reference point for text positioning (absolute) • Position of 2nd axis (for _VARI = orthogonal), or • Angle with 1st axis (for _VARI = polar)
_STA	real	Angle with 1st axis (for _VARI = linear only)
_CP1	real	Center of the circle (absolute), (for circular arrangement only) • Position of 1st axis (for _VARI = orthogonal), or • Radius of the arc (for _VARI = polar), with reference to the circle center
_CP2	real	Center of the circle (absolute), (for circular arrangement only) • Position of 2nd axis (_VARI = orthogonal), or • Angle with 1st axis (for _VARI = polar)
_WID	real	Character height (enter without sign)
_DF	real	Specification of the character width (equiv. _VARI hundred thousands digit) • Character spacing incremental in mm/inch, or • Overall text width incremental in mm/inch, or • Arc angle in degrees
_FFD	real	Feedrate for depth infeed
_FFP1	real	Feedrate for surface machining

© Siemens AG 2004 All rights reserved.

_VARI	integer	Type of machining: (enter without sign) UNITS DIGIT: Reference point Values: 0...orthogonal (Cartesian) 1...polar TENS DIGIT: Text orientation Values: 0...text in a line 1...text on an upward arc 2...text on a downward arc HUNDREDS DIGIT: Reserved THOUSANDS DIGIT: Reference point of the text horizontal Values: 0...left 1...central 2...right TEN THOUSANDS DIGIT: Reference point of the text vertical Values: 0...bottom 1...central 2...top HUNDRED THOUSANDS DIGIT: Text width Values: 0...character spacing 1…overall text width (only for linear text) 2…arc angle (only for arced text) 7. digit from right (million): circle center point Values: 0...orthogonal (Cartesian) 1...polar
_CODEP	integer	Codepage number for character set entered 1252...Codepage for Western European Languages (value 0 is treated as equivalent to 1252 internally)

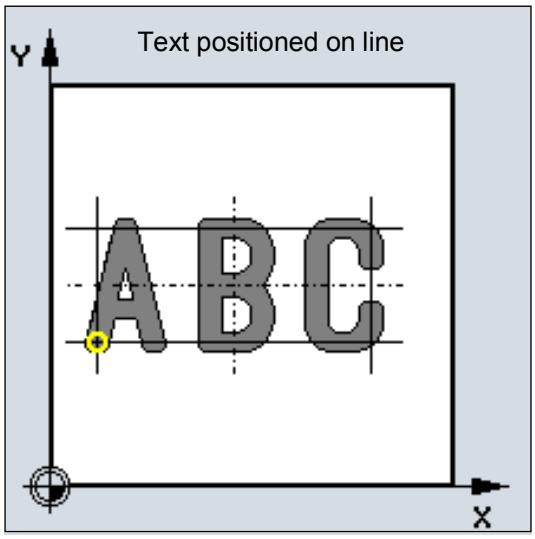
© Siemens AG 2004 All rights reserved.

Function

With engraving cycle CYCLE60, you can mill text positioned on a line or circle. The text may be positioned on an upward or downward circle.

The character height and overall text width, the character spacing or arc angle can be varied for circular text and the orientation of the text can be varied via various parameters.

The form of the characters cannot be influenced. The cycle uses a proportional font, i.e. the individual characters are of different widths. The line width of the characters corresponds to the tool diameter.

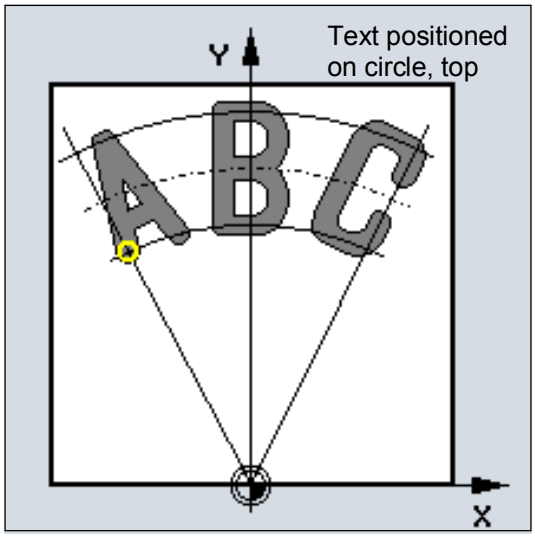


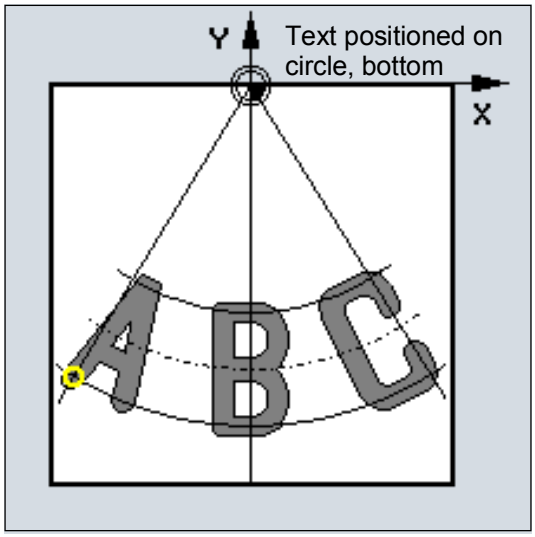
Sequence

Position reached before the beginning of the cycle:
The starting position can be any position from which the starting position of the first character can be approached without collision.

The cycle implements the following motion sequence:

- Rapid traverse positioning on the starting position in the machining plane; then with G0 to the reference plane offset by the safety distance
- Insertion to the programmed depth at feedrate for depth infeed _FFD
- Machining of each character at federate for surface machining _FFP1
- After completion of each character, retraction to the safety plane is performed with G0 followed by positioning on the starting point for the next character with G0
- After execution of all programmed characters, the tool is positioned on the retraction plane with G0





© Siemens AG 2004 All rights reserved.

Explanation of the parameters

See Subsection 2.1.2. (Drilling, Centering – CYCLE81) for a description of parameters _RTP, _RFP, _SDIS.

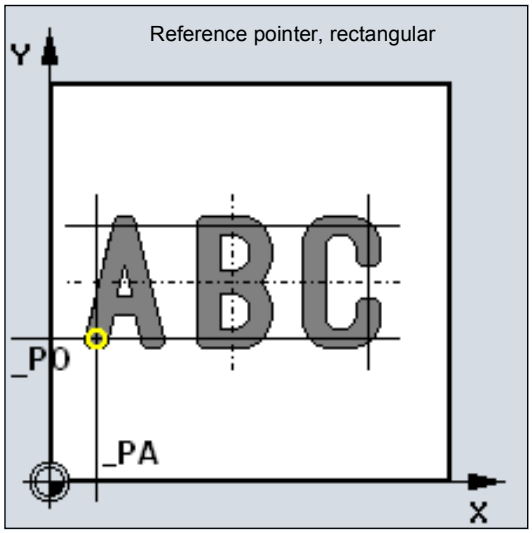
_TEXT (text to be engraved)
The text to be engraved is limited to 91 characters. All uppercase and lowercase letters, numerals, and most special characters in code page 1252 are permitted.

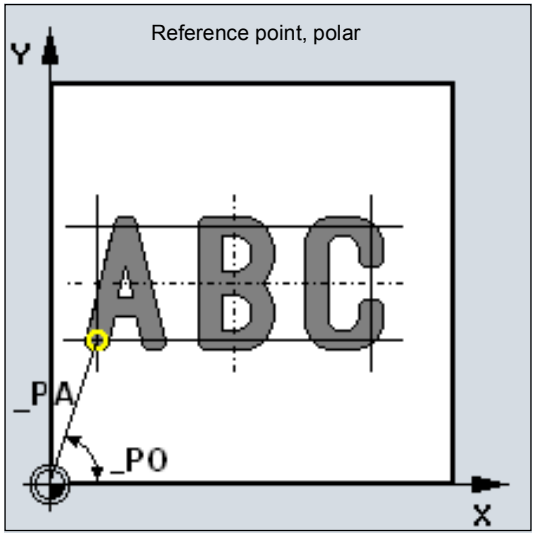
For the permissible special characters, see "Character set" on next page
If the text contains impermissible characters, the cycle terminates with alarm
61179 "Non-existent character".

_DP, _DPR (character depth)
The character depth can be specified as either an absolute value (_DP) or incremental value (_DPR) in relation to the reference plane. If it is entered as a relative value, the cycle automatically calculates the correct depth on the basis of the positions of the reference and retraction planes.
The character depth is reached with infeed; it is not divided into separate infeed steps. Depth infeed for each characters is vertical with G1.

© Siemens AG 2004 All rights reserved.

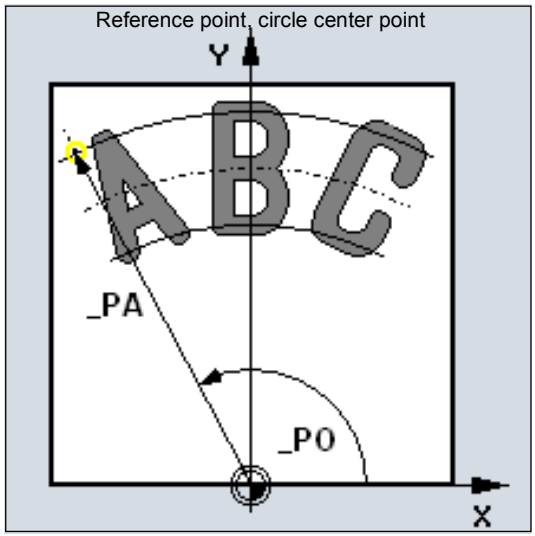
_PA, _PO (reference point for text positioning)
Irrespective of the orientation of the linear or circular orientation of the text, it is possible to program the reference point as either orthogonal (Cartesian) or polar.
If the text is positioned on a line, the reference point is always based on the current workpiece zero.





If the text is positioned on a circle with polar reference point programming, the reference point always refers to the circle center point.

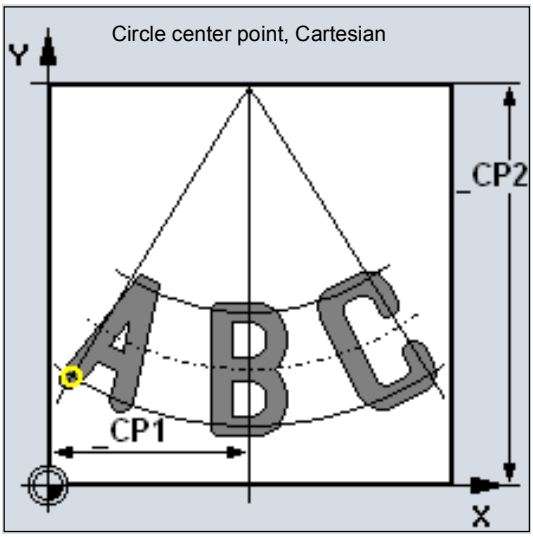
Whether the reference point is orthogonal or polar is specified in parameter _VARI.



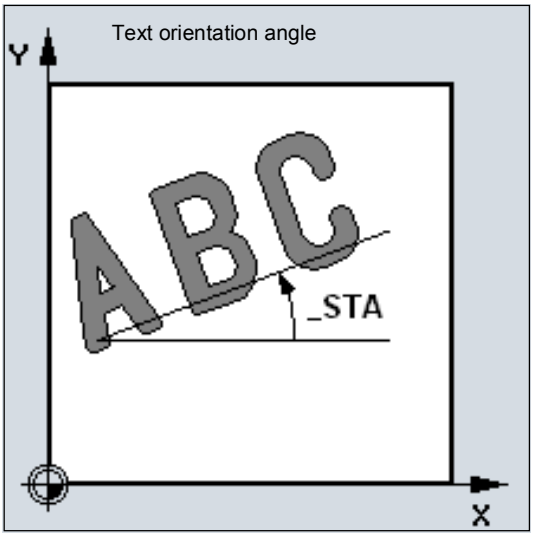
© Siemens AG 2004 All rights reserved.

_CP1, _CP2 (circle center point)

The center point of the circle when positioned on a circle can also be programmed as orthogonal (Cartesian) or polar.
Whether the circle center point is orthogonal or polar is specified in parameter _VARI.
These parameters only apply to circular orientation.



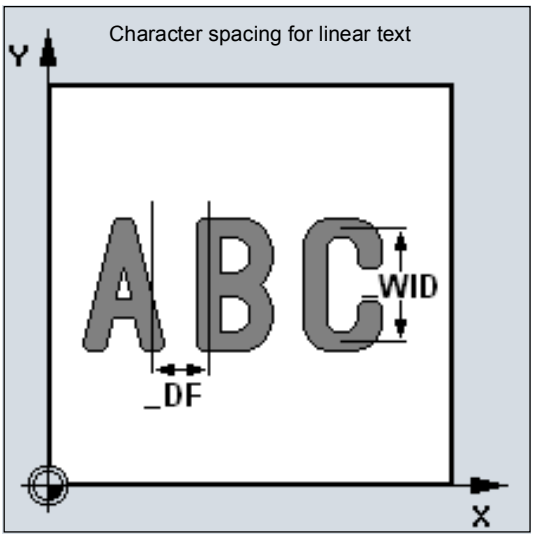
_STA (text orientation angle)
_STA specifies the angle between the 1st axis of the plane (abscissa) and the longitudinal axis (line on which the text is written) of the text to be written. This parameter only applies to linear text orientation.



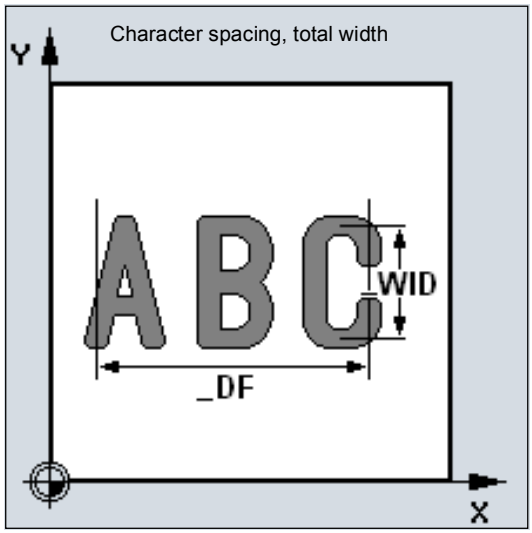
_WID (character height)
The programmed character height corresponds to the height of uppercase letters or numerals minus twice the mill radius.
For special characters, for example (), the dimension 0.15*_WID must be added above and below.

_DF (character spacing)
For linear text, either the character spacing or the overall text width may be specified. These are incremental values.

The cycle monitors whether the programmed overall with is possible, i.e. whether the sum of the character widths exceeds it.
In case of error, the alarm:
61176 "Text length _DF programmed to small“ is output.

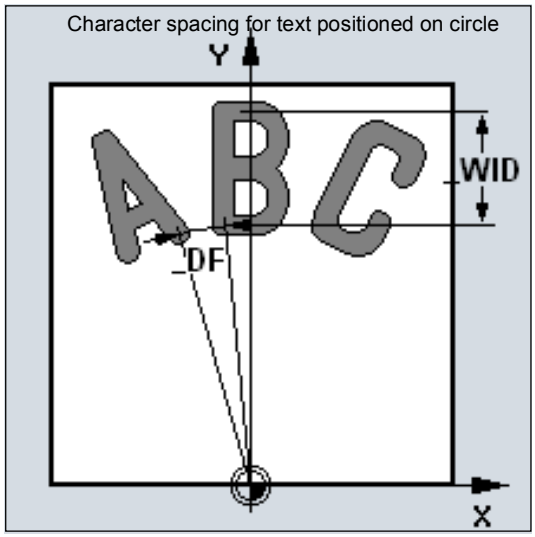


© Siemens AG 2004 All rights reserved.

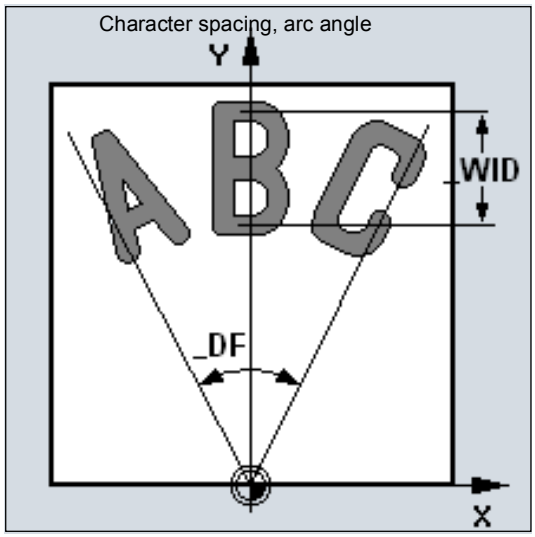


For orientation on a circle, the character spacing or arc angle can be specified between the first and the last characters. The character spacing is not the linear distance between adjacent characters but is applied to the arc as a radian measure.
The arc angle is always specified as a positive value. It refers to the angle between the centerline of the first character, center line of the last character.

The character spacing, overall text width, or arc angle are specified in parameter _VARI.



To distribute the characters evenly in a circle, simply program _DF=360. There is no need to calculate the arc angle between the first and last character.



_FFD, _FFP1 (feedrate)
Feedrate _FFP1 applies to all movements in the plant (milling characters), feedrate _FFD applies to vertical insertion with G1.

© Siemens AG 2004 All rights reserved.

_VARI (text orientation setting)
Parameter _VARI defines the text orientation.
See parameter _VARI in this section for values.
_CODEP (code page number)
CYCLE60 is currently only capable of engraving characters from code page 1252. The parameter therefore always has the value 1252.
If a number is passed that the cycle does not know:
61178 "Channel %1 block %2: character cycle: code page not available"
is output and the cycle is aborted.

Character set
In addition to letters and numerals, the following special characters (character set, parentheses, arithmetic symbols, currency symbols, and other special characters) are possible:

.	,	:	;	!	?	"	'	`	´	^	_
(	)	[	]	{	}	|	#	Ø	ø	°	
+	-	*	/	\	÷	×	<	>	=	~	
€	$	£	§	&	%	@	©	®			
à	á	â	ã	ä	å	À	Á	Â	Ã	Ä	Å
è	é	ê		ë		È	É	Ê		Ë	
ì	í	î		ï		Ì	Í	Î		Ï	
ò	ó	ô	õ	ö		Ò	Ó	Ô	Õ	Ö	
ù	ú	û		ü		Ù	Ú	Û		Ü	
	ý			ÿ			Ý			Ÿ	
ñ	Ñ	š	Š	ž	Ž	ç	Ç	¢	¥		
ß	µ	ð	Ð	þ	Þ	æ	Æ	œ	Œ		

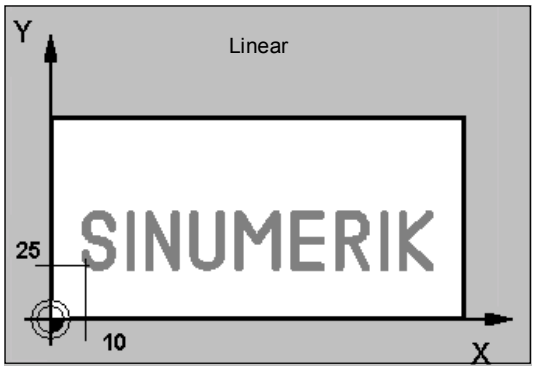
For programming the two special characters: single quotes and double quotes (speech marks), there is a special rule because these characters already have a function in string processing of the NC language. The must be written enclosed in single quotes within the text string.

Example:
To obtain the following sentence:
This is a text with " and ' .
Program the following in _TEXT:
This is a text with ' " ' and ' ' ' .

© Siemens AG 2004 All rights reserved.

Programming example

This program engraves the text "SINUMERIK" on a line. The reference point is at X10 Y25 lower left. The text is 14 mm high with 5 mm spacing between characters.

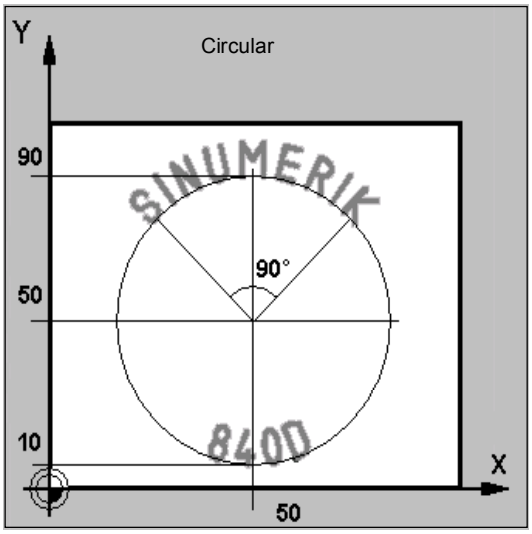



```
N10 G17 DIAMOF F2000 S1500 M3
N20 T1 D1
N30 M6
N40 G0 G90 Z100
N50 CYCLE60("SINUMERIK",100,0,1,-
1.5,0,10,25,0,,,14,5,2500,2000,0,1252)
N60 M30
```

© Siemens AG 2004 All rights reserved.

Programming example 2

This program engraves two texts on a circle "SINUMERIK" upward and "840D" downward.
The reference points are in the center and below at X50, Y90 and at X50 Y10. The radius of the circle 40 mm results from the distance from the reference points and the circle center of X50, Y50.
The text is 9 mm high, the characters spacing results from the programmed arc angle of 90 degrees and 30 degrees respectively.




```
N10 G17 DIAMOF F2000 S800 M3
N20 T1 D1
N30 M6
N40 G0 G90 Z100
N50 CYCLE60("SINUMERIK",100,0,1,-1.5,          ;Text on an upward circle
0,50,90,0,50,50,9,90,2500,2000,
201010,1252)
N60 CYCLE60("840D",100,0,1,-1.5,               ;Text on a downward circle
0,50,10,0,50,50,9,30,2500,2000,201020,125
2)
M30
```

■

© Siemens AG 2004 All rights reserved.

Turning Cycles

4.1 General notes 4-270
4.2 Preconditions 4-271
4.3 Grooving cycle – CYCLE93 4-274
4.4 Undercutting cycle – CYCLE94 4-283
4.5 Stock removal cycle – CYCLE95 4-287
4.6 Thread undercut – CYCLE96 4-300
4.7 Thread cutting – CYCLE97 4-304
4.8 Thread chaining – CYCLE98 4-311
4.9 Thread recutting 4-317
4.10 Extended stock removal cycle - CYCLE950 4-319

© Siemens AG 2004 All rights reserved.

4.1 General notes

The following sections describe how turning cycles are programmed. This section is intended to guide you in selecting cycles and assigning them with parameters. In addition to a detailed description of the function of the individual cycles and the corresponding parameters, you will also find a programming example at the end of each section to familiarize you with the use of cycles.

The Sections are structured as follows:

- **Programming**
- **Parameters**
- **Function**
- **Sequence**
- **Explanation of parameters**
- **Other Information**
- **Programming example**

"Programming" and "Parameters" explain the use of cycles sufficiently for the
experienced user, whereas beginners can find all the information they need for programming cycles under "Function", "Sequence of operations", "Explanation of parameters", "Additional notes" and the "Programming example".

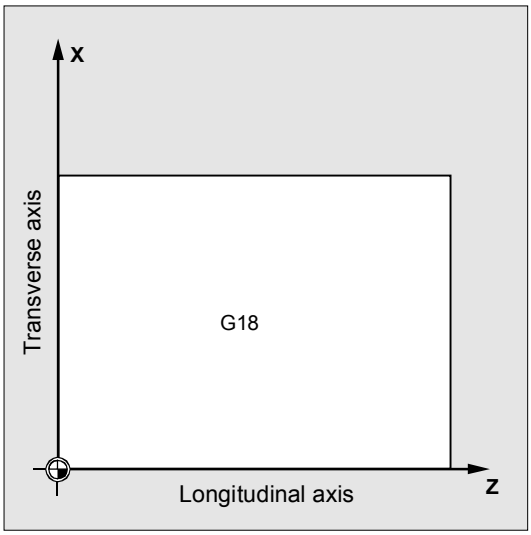
© Siemens AG 2004 All rights reserved.

4.2 Preconditions

Data block for turning cycles
The turning cycles require module GUD7.DEF. It is supplied on diskette together with the cycles.

Call and return conditions
The G functions active before the cycle is called and the programmable frame remain active beyond the cycle.

Level definition
The machining plane must be defined before the cycle is called. In the case of turning, this is usually the G18 (ZX) plane. The two axes of the turning plane are referred to below as the longitudinal axis (first axis of this plane) and the plane axis (second axis of this plane).
If diameter programming is active, the second axis of the plane is always taken as a facing axis:




Reference: /PG/ Programming Guide

Spindle handling
The turning cycles are written in such a way that the spindle commands always refer to the active master spindle of the control.
If you want to use a cycle on a machine with several spindles, the active spindle must first be defined as the master spindle.


References: /PG/ Programming Guide

Machining status messages
Status messages are displayed on the control monitor during processing of the turning cycles. The following messages can be displayed:

- "Thread start <No.> - longitudinal thread machining"
- "Thread start <No.> - face thread machining"

In each case <No.> stands for the number of the figure that is currently being machined.
These messages do not interrupt program processing and continue to be displayed until the next message is displayed or the cycle is completed.

© Siemens AG 2004 All rights reserved.

Cycle setting data

For the stock removal cycle CYCLE95, there is a setting data that is stored in module GUD7.DEF. Cycle setting data _ZSD[0] can be used to vary the calculation of the depth infeed MID in CYCLE95. If it is set to zero, the parameter is calculated as before.

- _ZSD[0]=1 MID is a radius value
- _ZSD[0]=2 MID is a diameter value

For the groove cycle CYCLE93, there is a setting data in module GUD7.DEF. This cycle setting data _ZSD[4] can affect the retraction after the 1st groove.

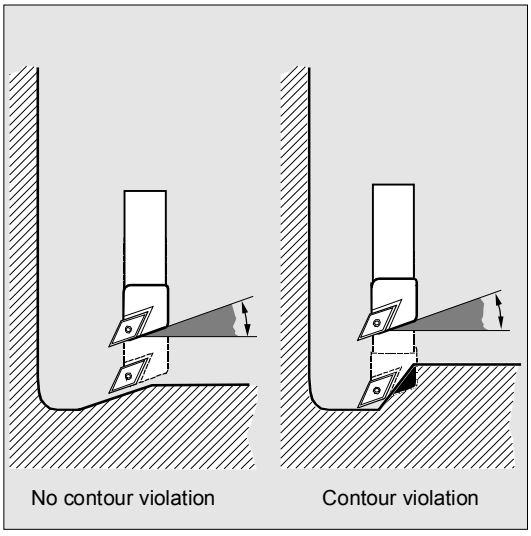
- _ZSD[4[=1 Retraction with G0
- _ZSD[4]=0 Retraction with G1 (as before)

There is an option for setting mirroring behavior in _ZSD[6] for grooving cycle CYCLE93.

- _ZSD[6]=0 The tool offsets are not exchanged in the cycle when mirroring is active (for use without an orientatable toolholder)
- _ZSD[6]=1 The tool offsets are not exchanged in the cycle when mirroring is active (for use with an orientatable toolholder)

Contour monitoring with respect to tool clearance angle

Some turning cycles in which travel movements with relief cutting are generated
monitor the tool clearance angle of the active tool for possible contour violation. This angle is entered as a value in the tool offset (under parameter P24 in the D offset).
An angle between 0 and 90 degrees is entered without a sign.

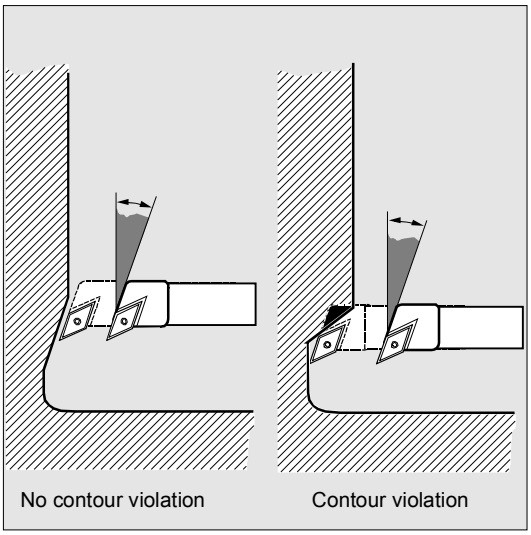


© Siemens AG 2004 All rights reserved.

When entering the tool clearance angle, remember that this depends on whether machining is longitudinal or facing. If a tool is to be used for longitudinal and face machining, two tool offsets must be applied if the tool clearance angles are different.
A check is made in the cycle to determine whether the programmed contour can be machined with the selected tool.
If machining is not possible with this tool, then

- the cycle is terminated with an error message (while cutting) or
- contour machining continues and a message is output (in undercut cycles). The tool nose geometry then determines the contour.

Note that active scale factors or rotations in the current plane modify the relationships at the angles, and that this cannot be allowed for in the contour monitoring that takes place within the cycle.
If the tool clearance angle is specified as zero in the tool offset, this monitoring function is deactivated. The precise reactions are described in the various cycles.



Turning cycles with active adapter transformation
With NCK SW 6.2 and higher, turning cycles can also be executed with an active adapter transformation. The transformed tool offset data for the tool point direction and clearance angle are always read.

© Siemens AG 2004 All rights reserved.

4.3 Grooving cycle – CYCLE93



Programming

```
CYCLE93 (SPD, SPL, WIDG, DIAG, STA1, ANG1, ANG2, RCO1, RCO2, RCI1, RCI2,
FAL1, FAL2, IDEP, DTB, VARI, _VRT)
```



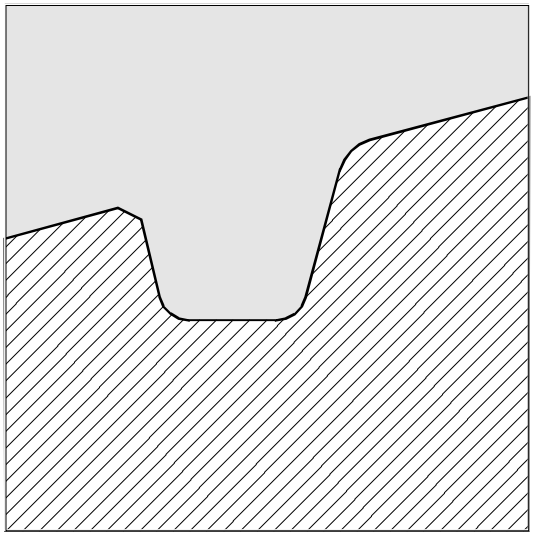
Parameters

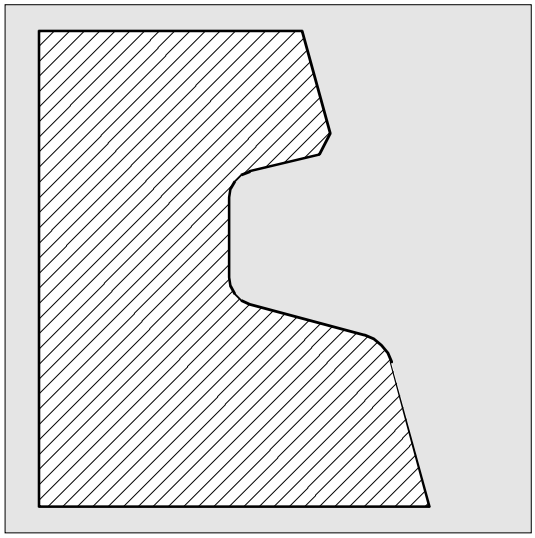
SPD	real	Starting point in the facing axis (enter without sign)
SPL	real	Starting point in the longitudinal axis
WIDG	real	Width of groove (enter without sign)
DIAG	real	Depth of groove (enter without sign)
STA1	real	Angle between contour and longitudinal axis Value range: 0<=STA1<=180 degrees
ANG1	real	Flank angle 1: on the side of the groove defined by the starting point (enter without sign) Value range: 0<=ANG1<89.999 degrees
ANG2	real	Flank angle 2: on the other side (enter without sign) Value range: 0<=ANG2<89.999
RCO1	real	Radius/chamfer 1, outside: on the side defined by the starting point
RCO2	real	Radius/chamfer 2, outside
RCI1	real	Radius/chamfer 1, inside: on the starting point side
RCI2	real	Radius/chamfer 2, inside
FAL1	real	Final machining allowance on the base of groove
FAL2	real	Final machining allowance on the flanks
IDEP	real	Infeed depth (enter without sign)
DTB	real	Dwell time at base of groove
VARI	int	Machining type Value range: 1...8 and 11...18
_VRT	real	Variable retraction distance from contour, incremental (enter without sign)

© Siemens AG 2004 All rights reserved.

Function

With the grooving cycle you can make symmetrical and asymmetrical grooves for longitudinal and traverse machining on straight contour elements. You can machine both external and internal grooves.





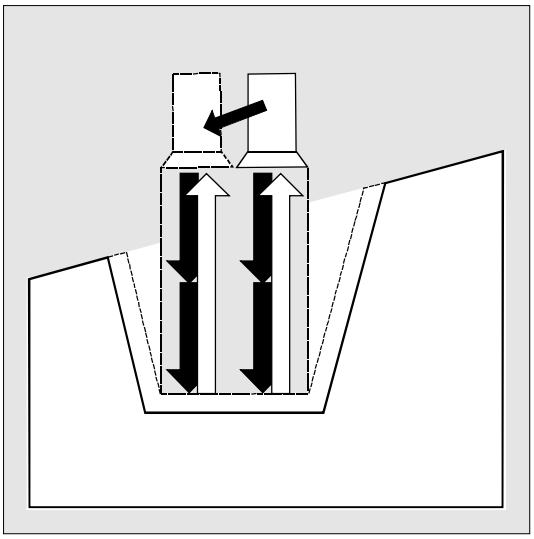
Sequence

The depth infeed (towards the base of the groove) and infeed across the width (from groove to groove) are distributed evenly and with the greatest possible value. If the groove is being machined on an inclined surface, travel from one groove to the next follows the shortest path, i.e. parallel to the cone on which the groove is being machined. The safety distance to the contour is calculated in the cycle.

© Siemens AG 2004 All rights reserved.

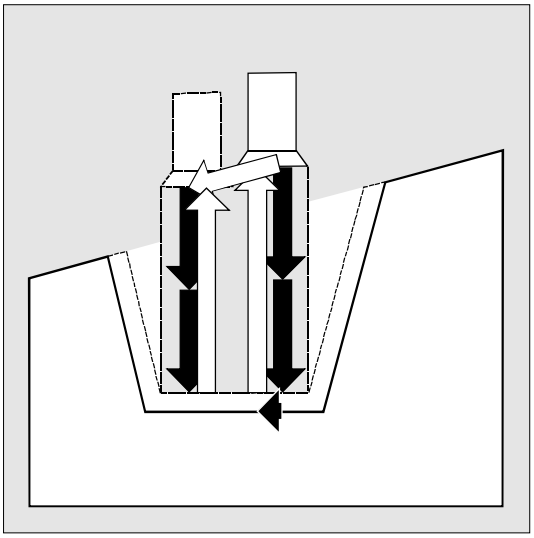
1. Step

Paraxial roughing to the base of the groove in single infeed steps. After each infeed, the tool is retracted for chip breaking.



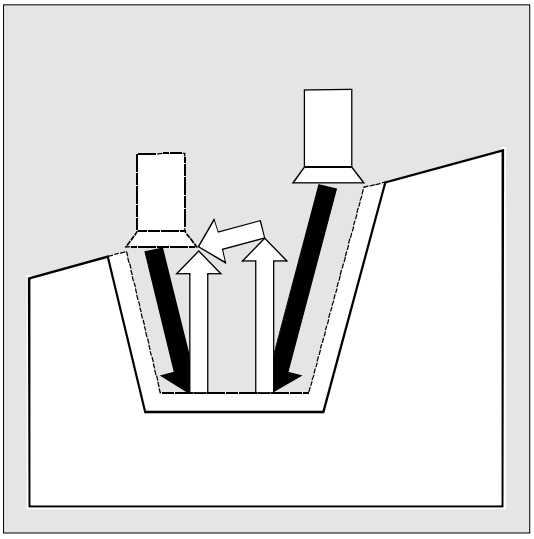
2. Step

The groove is machined perpendicular to the infeed direction in one or more cuts. Each cut is again divided up according to the infeed depth. From the second cut along the groove width the tool is withdrawn by 1 mm before it is fully retracted.



3. Step

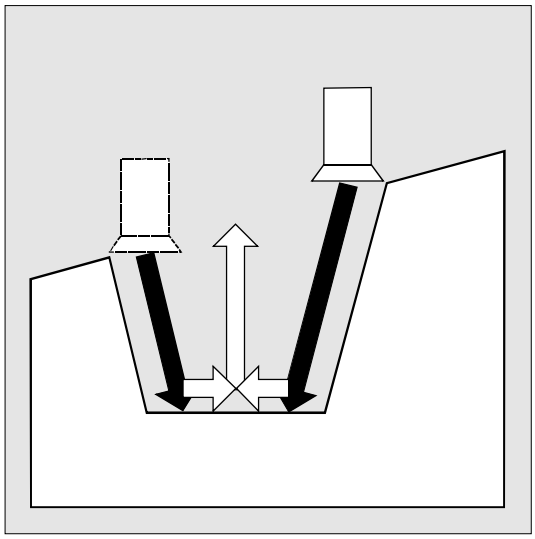
Cutting of the flanks in one step, if angles are programmed under ANG1 or ANG2. The infeed along the groove width is performed in several steps if the flank width is larger.



© Siemens AG 2004 All rights reserved.

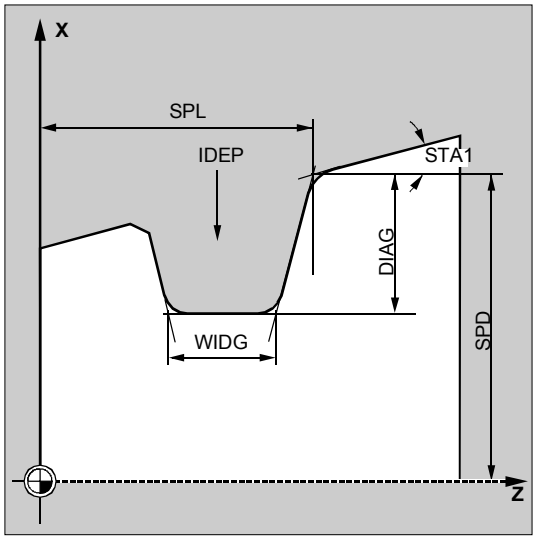
4. Step

Cutting of final machining allowance parallel to the contour from the edge to the center of the groove. The tool radius compensation is automatically selected and deselected by the cycle.

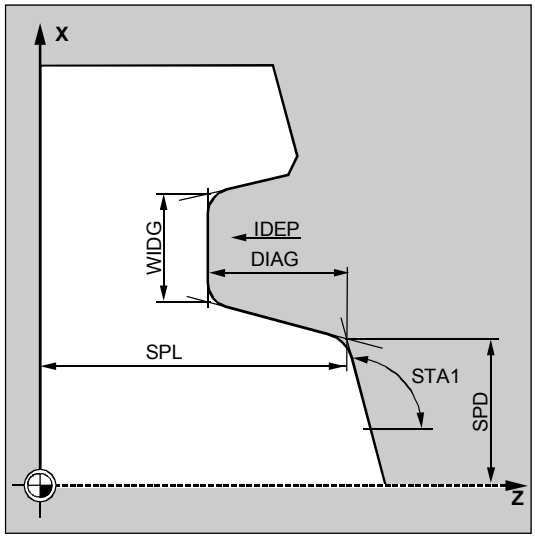


© Siemens AG 2004 All rights reserved.

Explanation of parameters



SPD and SPL (starting point)
You define the starting point of the groove from where the cycle calculates the shape with these coordinates. The cycle determines its starting point itself. For an external groove, movement begins in the direction of the longitudinal axis, for an internal groove in the direction of the facing axis.
Grooves on curved surfaces can be created in a variety of ways. Depending on the shape and radius of the curve, either a paraxial straight line can be placed on the maximum of the curve or a tangential oblique line can be placed on one of the edge points of the groove. Radii and chamfers on the groove edge of a curved surface should only be programmed if the edge point in question is positioned on the straight line defined for the cycle.

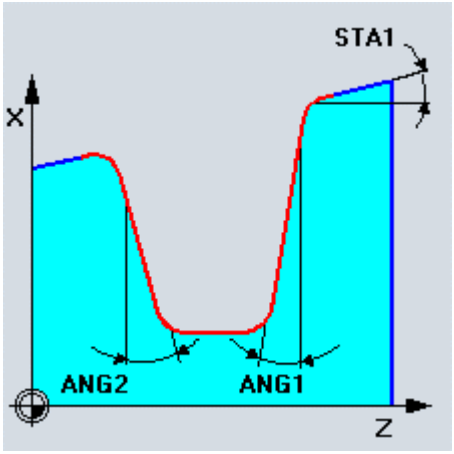


WIDG and DIAG (groove width and groove depth)
The shape of the groove is defined with the parameters groove width (WIDG) and groove depth (DIAG). The cycle always starts its calculation from the point programmed with SPD and SPL.
If the groove is wider than the active tool, the groove is machined in several steps. The total width is divided into equal sections in the cycle. The maximum infeed is 95 percent of the tool width after subtracting the tool nose radii. This ensures a cut overlap.

If the programmed groove width is less than the actual tool width, the following error message is output:
61602 "Tool width incorrectly defined".
The cycle is not started and machining is aborted. The alarm is also output if the value zero has been entered for the tool nose width.

© Siemens AG 2004 All rights reserved.

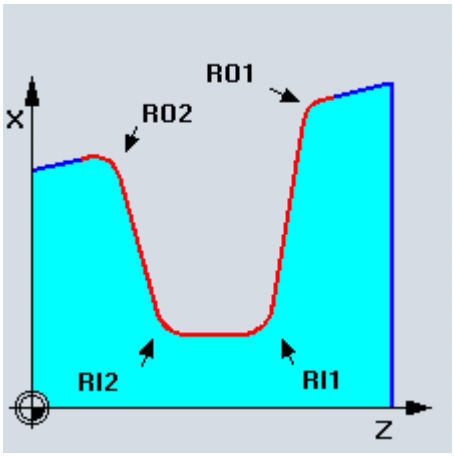
STA1 (angle)
The angle of the oblique surface on which the groove is to be machined is programmed with parameter STA1. The angle can have any value between 0 and 180 degrees and always refers to the longitudinal axis.

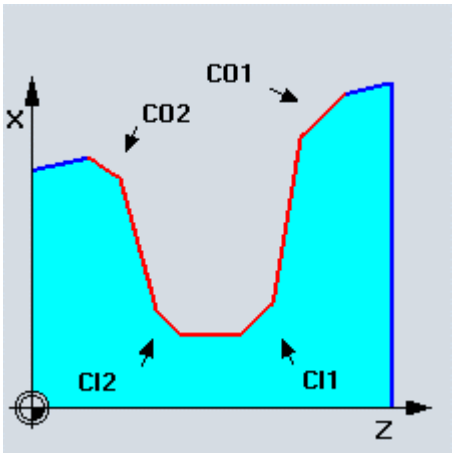


ANG1 and ANG2 (flank angle)
Asymmetrical grooves can be described by separate flank angles. The angles can be assigned any value between 0 and 89.999 degrees.

RCO1, RCO2 and RCI1, RCI2 (radius/chamfer)
The shape of the groove can be modified by entering radii/chamfers for the edge or base of the groove. The values for the radii must always be positive, the values for the chamfers must always be negative.
You can use the tens setting for the VARI parameter to determine the type of calculation for programmed milling.

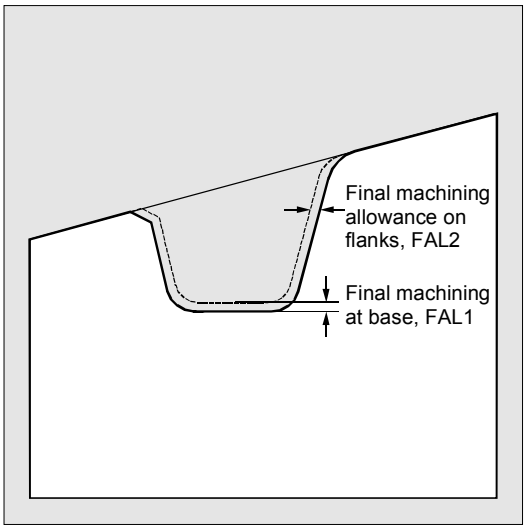
- With VARI<10 (tens=0) the absolute value of this parameter is regarded as chamfer length (chamfering with CHF programming).
- With VARI>10, it is regarded as path length (chamfering with CHR programming).





© Siemens AG 2004 All rights reserved.

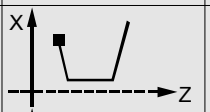
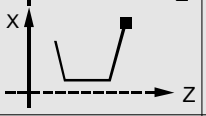
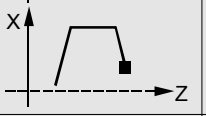
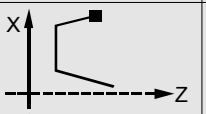
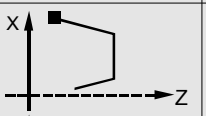
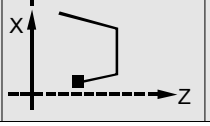
FAL1 and FAL2 (machining allowance)
You can program separate final machining allowances for the groove base and the flanks. Roughing is performed to this final machining allowance. Then, the same tool is used to machine a contour-parallel cut along the final contour.



IDEP (infeed depth)
By programming an infeed depth you can divide the paraxial grooving action into several depth infeeds. After every infeed, the tool is retracted for chipbreaking by 1 mm or by the distance programmed under _VRT. Parameter IDEP, anyway, is to be programmed.

DTB (dwell time)
A dwell time at the base of the groove should be chosen that allows at least one spindle revolution. The dwell time is programmed in seconds.

VARI (machining type)
The units digit of the VARI parameter determines the type of processing for the groove. This parameter can be assigned any of the values shown in the figure.

VARI		Selection of cycle support
1/11		longitudinal, external, left
5/15		longitudinal, external, right
3/13		longitudinal, internal, left
7/17		longitudinal, internal, right
6/16		plane, external, top
8/18		plane, external, bottom
2/12		plane, internal, top
4/14		plane, internal, bottom

The tens value of the VARI determines the type of calculation for the chamfer.
VARI 1...8: Chamfers are calculated as CHF
VARI 11...18: Chamfers are calculated as CHF

Input of the ones position in cycle support is distributed among three selection fields:
1. Field: longitudinal/plane
2. Field: external/internal
3. Field: starting point left/right (for longitudinal) or top/bottom (for plane)

© Siemens AG 2004 All rights reserved.

If the parameter is assigned another value, the cycle is aborted and alarm
61002 "Machining type incorrectly programmed".

The contour monitoring performed by the cycle ensures that a realistic groove contour results. This is not the case if the radii/chamfers touch each other at the base of the groove or overlap or if an attempt at face grooving is made on a section of the contour that runs parallel to the longitudinal axis. In these cases, the cycle is aborted with alarm
61603 "Groove form incorrectly defined".

_VRT (variable retraction path)
The retraction path can be programmed on the basis of the outside or inside diameter of the groove in parameter _VRT.
If _VRT=0 (parameter not programmed),
liftoff is 1mm. The retraction path is always measured in the programmed system of units inch or metric.
The same retraction path is also used for chipbreaking after each depth infeed into the groove.



Other Information

You must activate a double-edged tool before calling the grooving cycle. You must enter the offset values for the two tool edges in two successive D numbers of the tool, the first of which must be activated before the cycle is called. The cycle itself determines which of the two tool offsets it requires for which machining step and activates them automatically. After the cycle is completed, the offset number programmed before the cycle call becomes active again. If no D number has been programmed for a tool offset when the cycle is called, the cycle is aborted with alarm
61000 "No tool offset active"
aborted.

© Siemens AG 2004 All rights reserved.

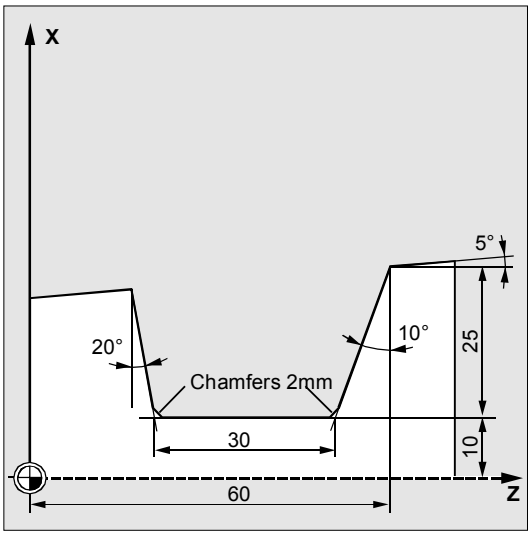
This cycle setting data _ZSD[4] can affect the retraction after the 1st groove.
_ZSD[4]=0 means retraction with G1 as before,
_ZSD[4]=1 means retraction with G0.

Cycle data _ZSD[6] can be set to define how the tool offset is handled in the cycle.
_ZSD[6]=0 TO is exchanged internally in the cycle (without orientatable toolholder)
_ZSD[6]=1 TO is not exchanged internally in the cycle (with orientatable toolholder)

Programming example

Grooving

This program machines a groove on an oblique surface (longitudinal, outside).
The starting point is at X35 Z60.
The cycle uses tool offsets D1 and D2 of tool T1. The grooving tool must be defined accordingly.




```
DEF REAL SPD=35, SPL=60, WIDG=30, ->          ;Definition of parameters with
-> DIAG=25, STA1=5, ANG1=10, ANG2=20, ->      ;value assignments
-> RCO1=0, RCI1=-2, RCI2=-2, RCO2=0, ->
-> FAL1=1, FAL2=1, IDEP=10, DTB=1
DEF INT VARI=5
N10 G0 G18 G90 Z65 X50 T1 D1 S400 M3          ;Starting point before the beginning of the
                                              ;cycle
N20 G95 F0.2                                  ;Specification of technology values
N30 CYCLE93 (SPD, SPL, WIDG, DIAG, ->         ;Cycle call
-> STA1, ANG1, ANG2, RCO1, RCO2, ->
-> RCI1, RCI2, FAL1, FAL2, IDEP, ->
-> DTB, VARI)
N40 G0 G90 X50 Z65                            ; next position
N50 M02                                       ;End of program
```

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

4.4 Undercutting cycle – CYCLE94

Programming

```
CYCLE94 (SPD, SPL, FORM, _VARI)
```

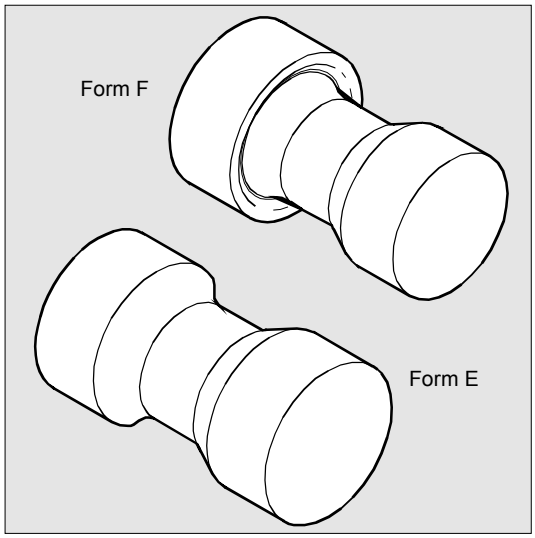
Parameters

SPD	real	Starting point in the facing axis (enter without sign)
SPL	real	Starting point of the contour in the longitudinal axis (enter without sign)
FORM	char	Definition of the form Values: E (for form E) F (for form F)
_VARI	int	Specification of undercut position Values: 0 According to tool point direction 1...4 Define position

Function

With this cycle you can machine undercuts of form E and F in accordance with DIN509 with the usual load on a finished part diameter of >3mm.

Another cycle CYCLE96 exists for producing thread undercuts (see Section 4.6).



Sequence

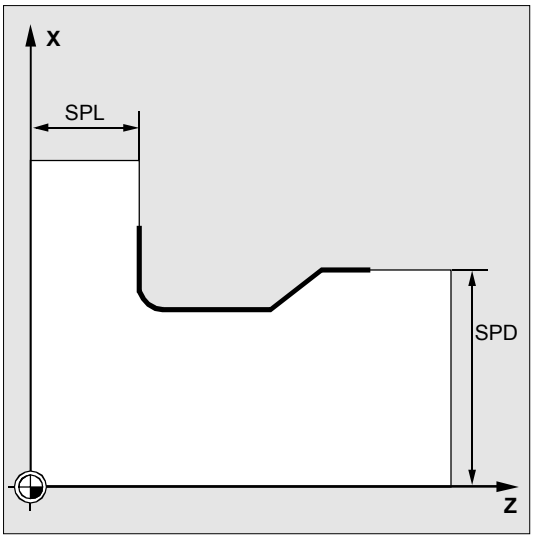
Position reached before the beginning of the cycle:
The starting position can be any position from which the undercut can be approached without collision.

The cycle implements the following motion sequence:

- Approach to the starting point calculated in the cycle with G0
- Selection of tool nose radius compensation according to active tool point direction and traversal of undercut contour at feedrate programmed prior to cycle call
- Retraction to the starting point with G0 and deselection of the tool nose radius compensation with G40

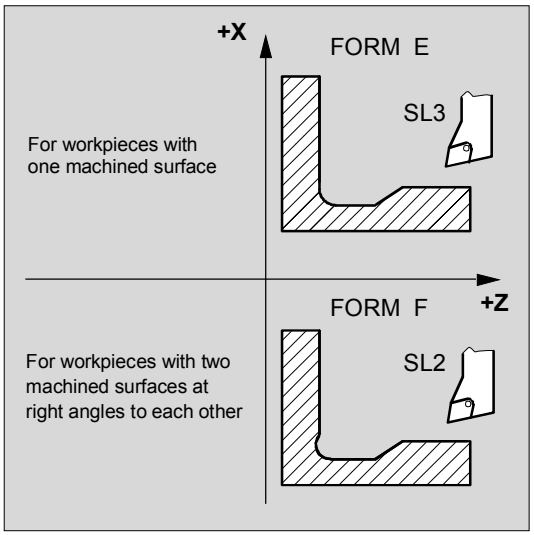
© Siemens AG 2004 All rights reserved.

Explanation of parameters



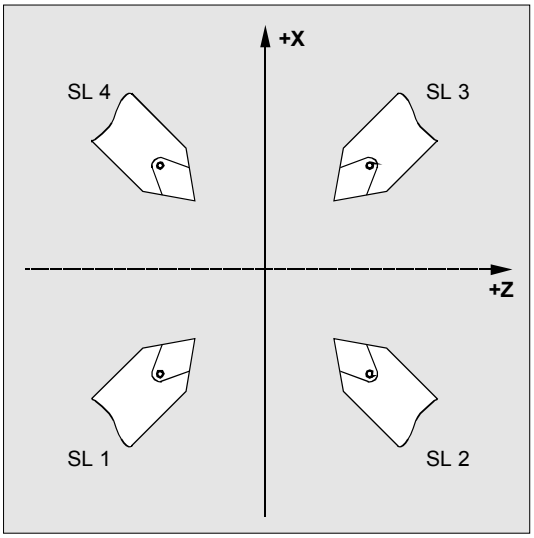
SPD and SPL (starting point)

The finished part diameter for the undercut is entered in parameter SPD. With parameter SPL you define the finished part dimensions in the longitudinal axis.
If the value programmed for SPD results in a final diameter that is <3mm, the cycle is aborted with the alarm
61601 "Finished part diameter too small".



FORM (definition)

Form E and Form F are defined in DIN509 and determined by this parameter.
If the parameter is assigned a value other than E or F, the cycle is aborted after outputting alarm
61609 "Form incorrectly defined".



_VARI (undercut position)

The position of the undercut can be either specified directly or derived from the tool point direction with parameter _VARI.

VARI=0: According to tool point direction

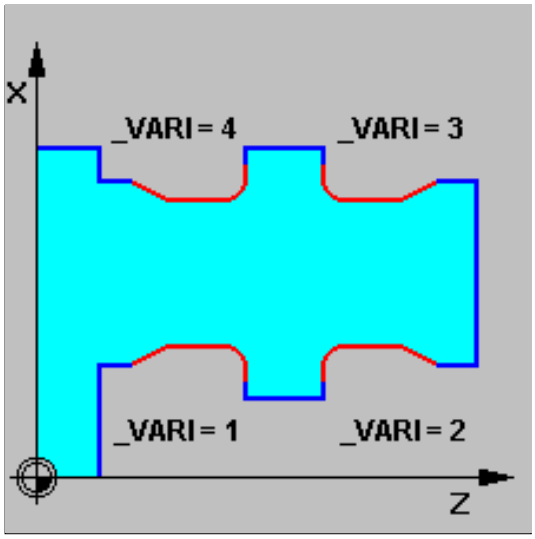
The cycle automatically determines the tool point direction from the active tool offset. The cycle can then only work with tool point
directions 1 to 4.
If the cycle detects a tool point direction of 5 to 9, alarm 61608 "Wrong tool point direction programmed" appears
and the cycle is aborted.

_VARI=1...4: Definition of undercut position
The cycle determines the starting point automatically. This lies 2mm from the final diameter and
10mm from the final dimension in the longitudinal axis.

© Siemens AG 2004 All rights reserved.

The position of this starting point in relation to the programmed coordinate values is determined by the tool point direction of the active tool.
The cycle monitors the clearance angle of the active tool if a value has been assigned to the relevant parameter of the tool offset. If the cycle ascertains that the undercut form cannot be machined with the selected tool because the clearance angle is too small, the messages
"Altered undercut form" is displayed on the control.
However, machining is continued.
When _VARI<>0 the following applies:

- The actual tool point direction is not checked, i.e. all directions can be used if this is meaningful technologically,
- No special consideration is given to functions such as adapter transformation, orientatable toolholder in the cycle. It is assumed that the user is familiar with the conditions on the machine and has specified the position correctly;
- Similarly, mirroring is not specially treated either, the user must be responsible for making the correct settings.



Other Information

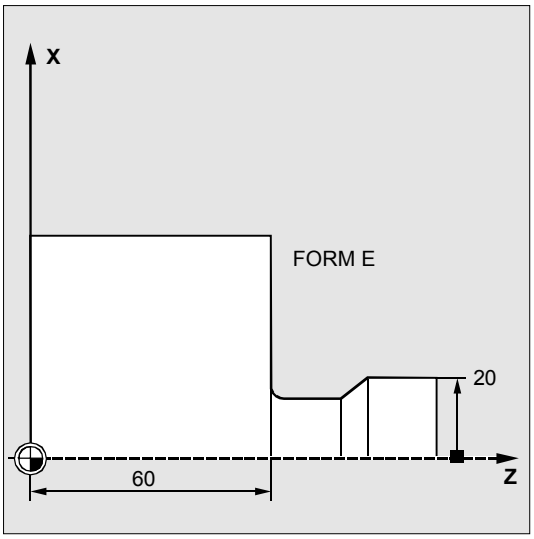
You must activate a tool offset before you call the cycles. Otherwise, the following alarm is output:
61000 "No tool offset active"
and the cycle is aborted.

© Siemens AG 2004 All rights reserved.

Programming example

Undercut_form_E

You can machine an undercut of form E with this program.




```
N10 T25 D3 S300 M3 G18 G95 F0.3   ;Specification of technology values
N20 G0 G90 Z100 X50               ;Selection of starting position
N30 CYCLE94 (20, 60, "E")         ;Cycle call
N40 G90 G0 Z100 X50               ;Approach next position
N50 M02                           ;End of program
```


© Siemens AG 2004 All rights reserved.

4.5 Stock removal cycle – CYCLE95



Programming

```
CYCLE95 (NPP, MID, FALZ, FALX, FAL, FF1, FF2, FF3, VARI, DT, DAM, _VRT)
```

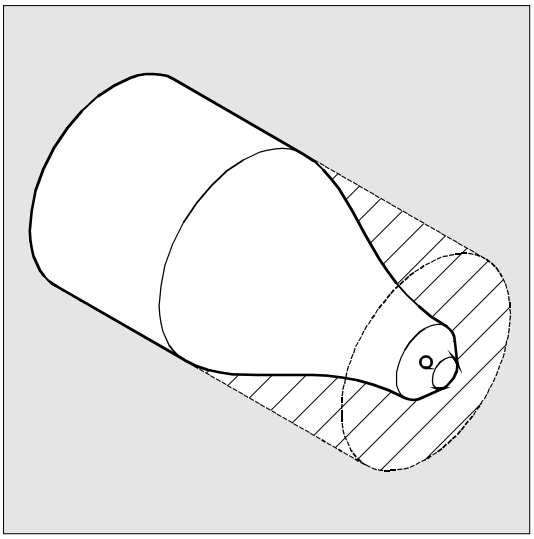


Parameters

NPP	string	Name of the contour subroutine
MID	real	Infeed depth (enter without sign)
FALZ	real	Final machining allowance in the longitudinal axis (enter without sign)
FALX	real	Final machining allowance in the facing axis (enter without sign)
FAL	real	Final machining allowance along contour (enter without sign)
FF1	real	Feedrate for roughing without relief cut
FF2	real	Feedrate for insertion into relief cut elements
FF3	real	Feedrate for finishing
VARI	int	Machining type Value range: 1 ... 12 HUNDREDS DIGIT: Values: 0...with rounding at the contour No residual corners remain, the contour is rounded overlapping. That means the contours are rounded over several points of intersection. 2...with rounding at the contour Machining always continues to the previous roughing point and then stock removal occurs. Depending on the ratio of tool radius to infeed depth (MID), residual corners might be left.
DT	real	Dwell time for chip breaking during roughing
DAM	real	Path length after which each roughing cut is interrupted for chip breaking
_VRT	real	Retraction distance from contour for roughing, incremental (enter without sign)

© Siemens AG 2004 All rights reserved.

Function



With this stock removal cycle you can machine a contour programmed in a subroutine from a blank with paraxial stock removal. Relief cut elements can be included in the contour. With this cycle, contours can be machined in the longitudinal and facing directions, inside and outside. The technology is freely selectable (roughing, finishing, complete machining). During roughing, paraxial cuts are generated from the maximum programmed infeed depth and when a point of intersection with the contour is reached, the residual corners are immediately removed cutting parallel to the contour. Roughing is performed up to the final machining allowance programmed.
Finishing is performed in the same direction as roughing. The tool radius compensation is automatically selected and deselected by the cycle.

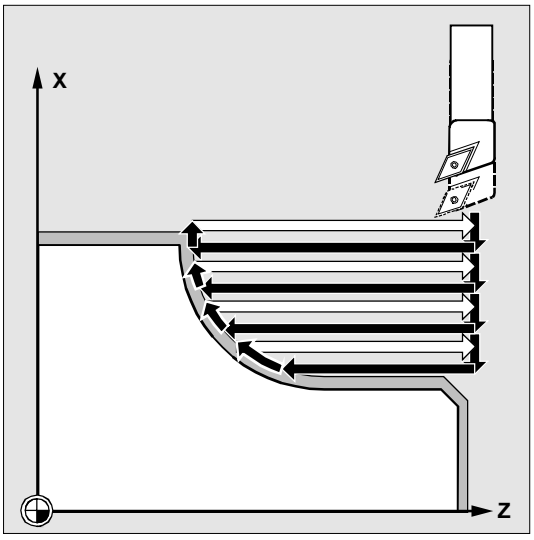
Sequence

Position reached before the beginning of the cycle:
The starting position can be any position from which the starting point of the contour can be approached without collision.

The cycle implements the following motion sequence:

- The cycle starting point is calculated in the cycle and then approached in both axes simultaneously with G0.

Roughing without relief cut elements:

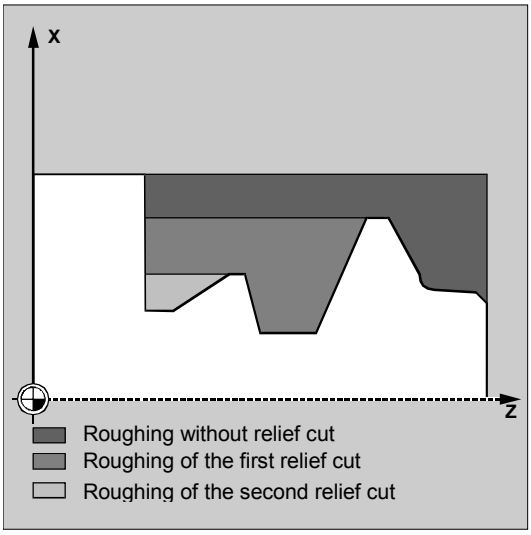


- Paraxial infeed to the actual depth is calculated internally and then approached with G0.
- Approach roughing point paraxially with G1 and feedrate FF1.
- Machine parallel to the contour at contour + final machining allowance to the last roughing intersection point with G1/G2/G3 and FF1.
- Lift off contour by the amount programmed in _VRT in every axis and retraction with G0.
- This procedure is repeated until the total depth of the section to the machining step has been reached.
- When roughing without relief cut elements, retraction to the cycle starting point is effected axis by axis.

© Siemens AG 2004 All rights reserved.

Roughing the relief cut elements:

- Approach the starting point for the next relief cut axis by axis with G0. An additional safety distance is calculated internally.
- Infeed parallel to the contour + final machining allowance with G1/G2/G3 and FF2.
- Approach roughing point paraxially with G1 and feedrate FF1.
- Machine to the last roughing point. Lift and retract as in the first machining section.
- If further relief cut elements are to be machined, repeat the above procedure for each relief cut element.

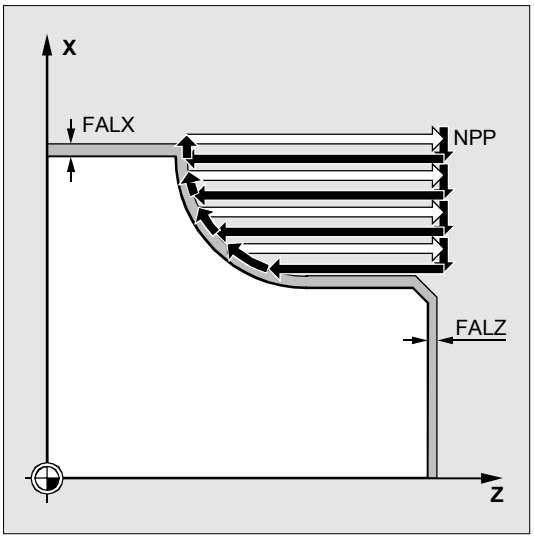


Finishing:

- The calculated cycle starting point is approached in both axes simultaneously with G0 and tool nose radius compensation is selected.
- Both axes then continue to move; with G0 up to an amount final machining allowance + tool nose radius + 1mm safety distance ahead of the contour starting point, and from there with G1 to the contour starting point.
- Finish cutting along the contour with G1/G2/G3 and FF3.
- Retraction to starting point with both axes and G0.

© Siemens AG 2004 All rights reserved.

Explanation of parameters



NPP (name)
Enter the name of the contour subroutine under this parameter. This contour subroutine must not be a subroutine with a parameter list.
Please use the name conventions described in the Programming Guide when naming the contour subroutine.

References: /PG/ Programming Guide

The machining contour can also be a section of the calling routine or from any other program. The section is identified by start or end labels or by block numbers. In this case, the program name and labels/block number are identified by an ":".

Examples:

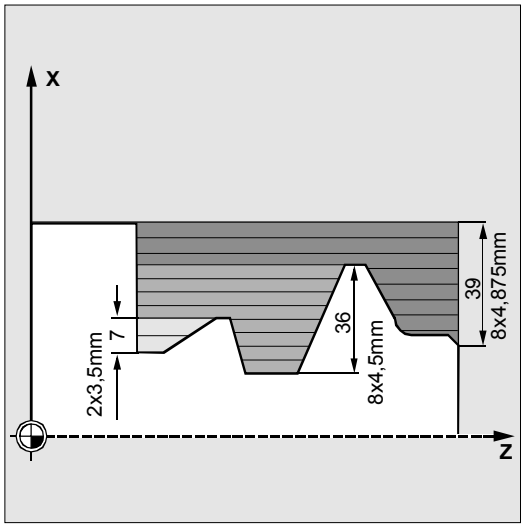
NPP="CONTOUR_1"	The machining contour is the complete program "Contour_1".
NPP="START:END"	The machining contour is defined as the section starting from the block labeled START to the block labeled END in the calling routine.
NPP="/_N_SPF_DIR/_N_CONTOUR_1_SPF:N130: N210"	The machining contour is defined in blocks N130 to N210 in program CONTOUR_1. The program name must be entered complete with path and extension, see description of CALL in **References**: /PGA/ Programming Guide Job planning

If the section is defined by block numbers, it must be noted that these block numbers for the section in NPP must be adjusted if the program is modified and subsequently renumbered.

© Siemens AG 2004 All rights reserved.

MID (infeed depth)
Under parameter MID you define the maximum possible infeed depth for the roughing operation.
The interpretation of this parameter depends on the cycle setting data for software release 4 and higher _ZSD[0] (see Section 4.2).
The cycle automatically calculates the actual infeed depth for roughing.
Where contours with relief cut elements are to be machined, the cycle divides up the roughing operation into single roughing steps. The cycle recalculates the actual infeed depth for every roughing step. This actual infeed depth always lies between the programmed infeed depth and half its value. The number of required roughing cuts is derived from the total depth of a roughing section and the programmed maximum infeed depth. The total depth to be machined is then divided equally amongst these roughing cuts. This method ensures optimum cutting conditions. The machining steps shown in the figure above are used for roughing this contour.

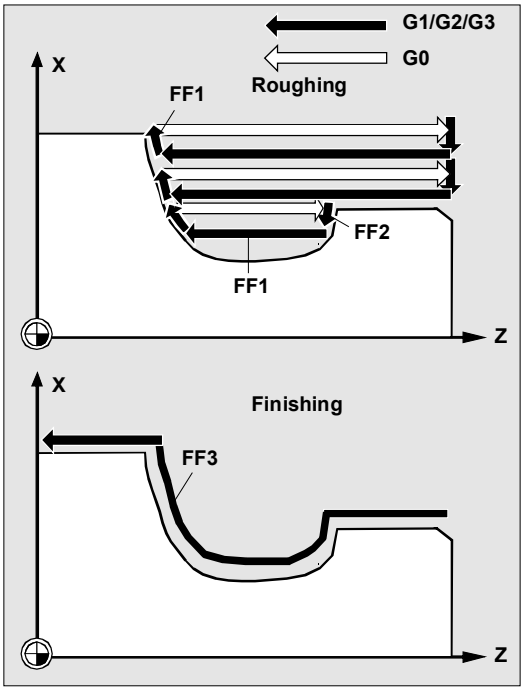
Example for the calculation of the actual infeed depths:
Machining section 1 has a total depth of 39 mm. If the maximum infeed depth is 5mm, eight roughing cuts are required. These are performed with an infeed of 4.875 mm.
In machining section 2, eight roughing cuts, each with an infeed of 4.5 mm are also executed (total difference 36 mm)
Machining section 3 is roughed twice with an actual infeed of 3.5 (total difference 7 mm).



© Siemens AG 2004 All rights reserved.

FAL, FALZ and FALX (final machining allowance)
The final machining allowance for the roughing operation is either defined in parameters FALZ and FALX if you wish to enter different final machining allowances for each axis or in parameter FAL if you wish to enter a final machining allowance that follows a contour. In this case, this value is used for the final machining allowance in both axes.
The programmed values are not subjected to a plausibility check. If all three parameters are assigned values, all the final machining allowances are calculated by the cycle. It is, however, advisable to decide on one or the other form of final machining allowance definition.
Roughing is always performed to these final machining allowances. After each paraxial roughing operation the resulting residual corners are immediately cut parallel to the contour so that these do not have to be removed after the roughing operation is completed. If no final machining allowances have been programmed, roughing is performed to the final contour.
The **finish** cut machining mode does not consider programmed values. Turning always takes place down to finishing dimension.

FF1, FF2 and FF3 (feedrate)
You can define different feedrates for the different machining steps as is shown in the figure on the right.



© Siemens AG 2004 All rights reserved.

VARI (machining type)

You can call the machining types as follows:

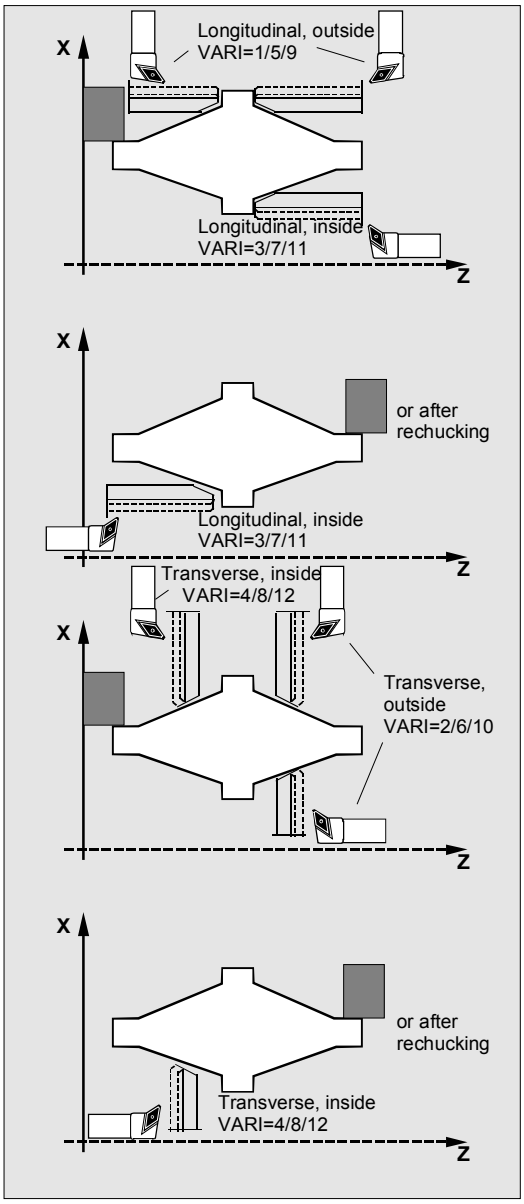
Machining [Roughing]
(Roughing/Finishing/Complete)

Selection [Longitudinal]
(Longitudinal/Transverse)

Selection [Outside]
(Outside/Inside)

The machining types are shown in the table below.

Value	Machining	Selection	Selection
1	Roughing	long.	outside
2	Roughing	transv.	outside
3	Roughing	long.	inside
4	Roughing	transv.	inside
5	Finishing	long.	outside
6	Finishing	transv.	outside
7	Finishing	long.	inside
8	Finishing	transv.	inside
9	Complete machining	long.	outside
10	Complete machining	transv.	outside
11	Complete machining	long.	inside
12	Complete machining	transv.	inside

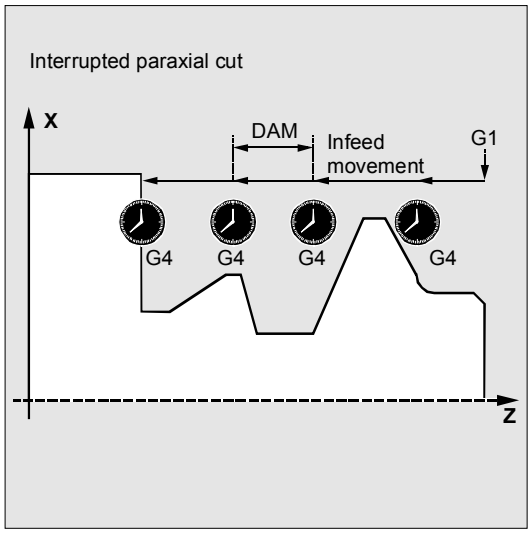


Infeed is always performed in the facing axis with longitudinal machining and in the longitudinal axis with face machining.
Outside machining means that infeed is performed in the direction of the negative axis. In inside machining, infeed is performed in the direction of the positive axis.
When roughing in CYCLE95, you can choose between "with rounding" or "without rounding" at the contour. For doing so, the HUNDREDS DIGIT is implemented in parameter VARI.

A plausibility check is performed on parameter VARI. If you select an invalid value, the cycle is aborted and alarm 61002 "Wrong machining type defined" is put out.

© Siemens AG 2004 All rights reserved.

DT and DAM (dwell time and path length)
With these two parameters you can program an interruption in the individual roughing cuts after a defined path for the purposes of chip breaking. These parameters only apply to roughing. In parameter DAM you define the maximum path after which chip breaking is to be performed. In DT you can also program a dwell time to be included at each of the interruption points. If no path has been specified for cut interruption (DAM = 0), then uninterrupted roughing cuts without dwell times are generated.



_VRT (lift)
The parameter _VRT can be used to program the amount of liftoff during roughing in both axes.
If _VRT=0 (parameter not programmed), liftoff is 1 mm.
The retraction distance is always measured in inches or metric depending on the programmed system, i.e. _VRT=1 for inch →programming produces a tool retract of 1 inch.

Other Information

Contour definition
The contour is programmed in a subroutine whose name is defined as a parameter.
The contour subroutine must contain at least 3 blocks with movements in both axes of the machining plane.
The machining plane (G17, G18, G19) is set in the main program before the cycle is called or applied according to the basic setting of this
G group on the machine. It cannot be altered in the contour subroutine.
If the contour subroutine is shorter, alarms 10933 "The contour subroutine contains too few contour blocks" and 61606 "Error in preprocessing contour" aborted.
Relief cut elements can be programmed consecutively. Blocks without movement in the plane are not subject to any limitations.

© Siemens AG 2004 All rights reserved.

All the traversing blocks for the first two axes in the current plane are processed in the cycle as only these axes are involved in the machining operation. Movements for other axes can be included in the contour subroutine but their travel paths are suppressed during the cycle run.

The only geometry permitted in the contour are straight line and circular programming with G0, G1, G2 and G3. Commands for fillets and chamfers can also be programmed. If any other motion commands are programmed in the contour, it is aborted with alarm 10930 "Illegal interpolation type in the machining contour".

The first block containing a traversing movement in the current machining plane must contain a travel command (G0, G1, G2 or G3), otherwise the cycle is aborted with the alarm
15800 "Wrong starting conditions for CONTPRON".
This alarm is also activated when G41/G42 is active.
The starting point of the contour is the first position on the machining plane programmed in the contour subroutine.

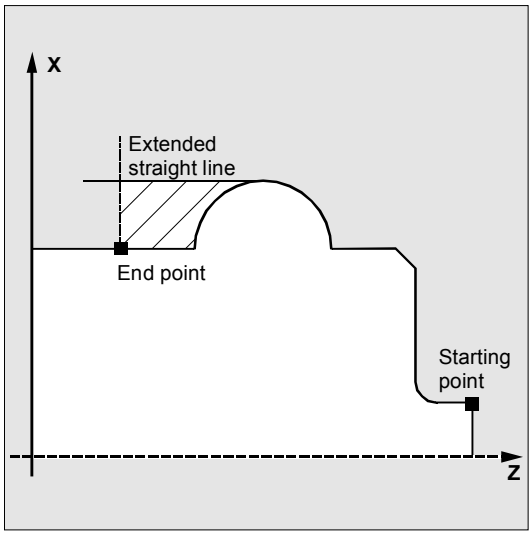
The maximum possible number of blocks for the contour containing travel commands for the current plane depends on the type of contour. In principle, there is no limit to the possible number of relief cuts.

If a contour contains more contour elements than the cycle memory can hold, the cycle is aborted with the alarm
10934 "Overflow contour table".

Machining must then be divided into several machining sections each of which is represented by its own contour subroutine and each cycle called separately.

© Siemens AG 2004 All rights reserved.

If the maximum diameter in one contour subroutine is not within the programmed end point or starting point of the contour, the cycle automatically extends a paraxial straight line to the maximum point of the contour at the end of the machining operation and this part of the contour is then removed as a relief cut.



Programming

- Radius compensation plane with G17/G18/G19
- Frames
- An axis of the plane in which machining is performed is traversed as a positioning axis
- Selection of tool radius compensation with G41/G42

in the contour program causes output of alarm 10931 "Incorrect machining contour" and aborts the cycle.

Contour direction

The direction in which the stock removal contour can be programmed is freely selectable. The machining direction is automatically defined in the cycle. With complete machining operations, the contour is finished in the same direction in which rough cutting took place. If only finishing is selected, the contour is always traversed in the programmed direction.

The first and last programmed contour points are the criteria for selecting the machining direction. For this reason, both coordinates must always be programmed in the first block of the contour subroutine.

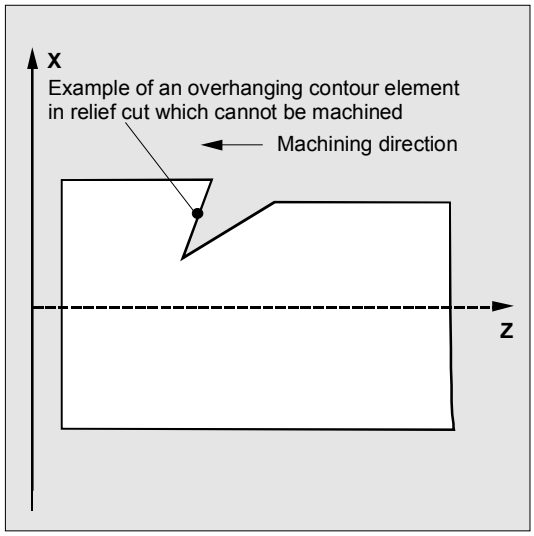
© Siemens AG 2004 All rights reserved.

Contour monitoring

The cycle performs contour monitoring of the following:

- Clearance angles of the active tool
- Programming of arcs with an aperture angle of > 180 degrees

In the case of relief cut elements, the cycle checks whether machining is possible with the active tool. If the cycle detects that this machining is leading to a contour violation, it outputs alarm 61604 "Active tool violates the programmed contour" and machining is aborted.



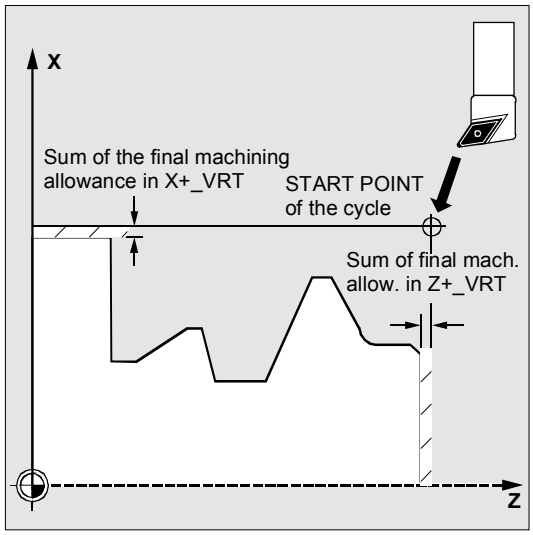
Contour monitoring is not performed if the clearance angle has been defined as zero in the tool offset.

If the arcs in the offset are too large, the following alarm is output:
10931 "Incorrect machining contour".

Overhanging contours cannot be machined by CYCLE95. Contours of this type are not monitored by the cycle and consequently there is no alarm.

Starting point

The cycle determines the starting point of the machining operation automatically. The starting point is positioned on the axis in which infeed is performed at a distance from the contour corresponding to final machining allowance + liftoff distance (parameter _VRT). In the other axis, it is positioned at a distance corresponding to final machining allowance + _VRT in front of the contour starting point.
The tool noise radius compensation is selected internally in the cycle when the starting point is approached.
The last point before the cycle is called must therefore be selected such that it can be approached without risk of collision and adequate space is available for the compensating movement.



Approach strategy of the cycle

The starting point calculated by the cycle is always approached in the two axes simultaneously for roughing and one axis at a time for finishing. In finishing, the infeed axis is the first to travel.

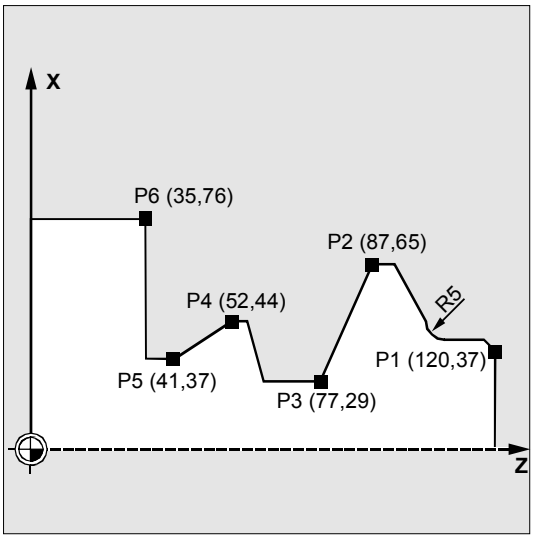
© Siemens AG 2004 All rights reserved.

Programming example 1

Stock removal cycle

The contour illustrated in the figure explaining the assignment parameters must be machined completely (longitudinal, outside). Axis-specific final machining allowances have been defined. No interruption between cuts has been programmed. The maximum infeed is 5mm.

The contour is stored in a separate program.



`DEF STRING[8] UPNAME`	;Definition of a variable for the contour name
`N10 T1 D1 G0 G18 G95 S500 M3 Z125 X81`	;Approach position before cycle call
`UPNAME="CONTOUR_1"`	;Assignment of subroutine name
`N20 CYCLE95 (UPNAME, 5, 1.2, 0.6, , ->` `-> 0 .2, 0.1, 0.2, 9, , , 0.5)`	;Cycle call
`N30 G0 G90 X81`	;Reapproach to starting position
`N40 Z125`	;Traverse in each axis separately
`N50 M30`	;End of program
`PROC CONTOUR_1`	;Beginning of contour subroutine
`N100 G1 Z120 X37` `N110 Z117 X40`	;Traverse in each axis separately
`N120 Z112`	;Rounding with radius 5

`N130 G1 Z95 X65 RND=5`
`N140 Z87`
`N150 Z77 X29`
`N160 Z62`
`N170 Z58 X44`
`N180 Z52`
`N190 Z41 X37`
`N200 Z35`
`N210 G1 X76`
	;Traverse in each axis separately

`N220 M17`	;End of subroutine

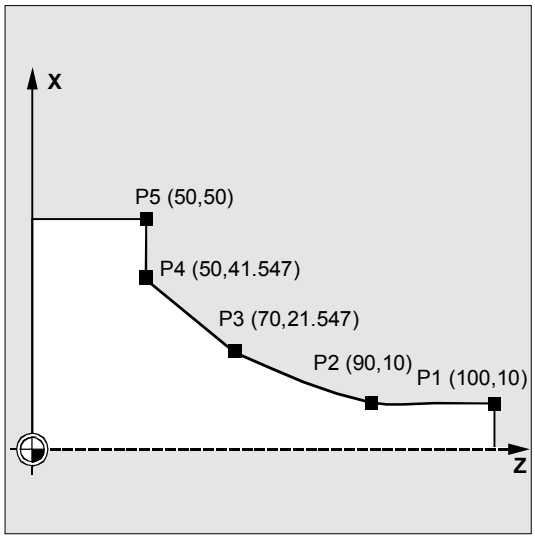
-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

Programming example 2

Stock removal cycle

The stock removal contour is defined in the calling program. After the stock removal cycle, the program is ended.




```
N110 G18 DIAMOF G90 G96 F0.8
N120 S500 M3
N130 T11 D1
N140 G0 X70
N150 Z60
N160 CYCLE95 ("START:END",2.5,0.8, ->      ;Cycle call
-> 0.8,0,0.8,0.75,0.6,1)
N170 M02
START:
N180 G1 X10 Z100 F0.6
N190 Z90
N200 Z=AC(70) ANG=150
N210 Z=AC(50) ANG=135
N220 Z=AC(50) X=AC(50)
END:
N230 M02
```

© Siemens AG 2004 All rights reserved.

4.6 Thread undercut – CYCLE96

Programming

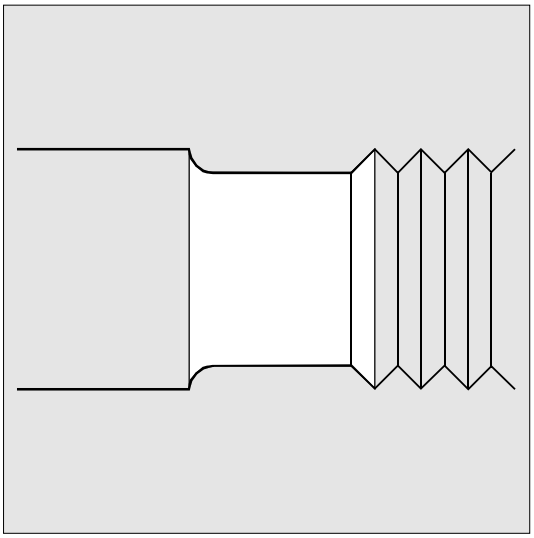
```
CYCLE96 (DIATH, SPL, FORM, _VARI)
```

Parameters

DIATH	real	Nominal diameter of the thread
SPL	real	Starting point on the contour of the longitudinal axis
FORM	char	Definition of the form Values: A (for Form A) B (for Form B) C (for Form C) D (for Form D)
_VARI	int	Specification of undercut position Values: 0 According to tool point direction 1...4 Define position

Function

This cycle is for machining thread undercuts in accordance with DIN 76 on parts with a metric ISO thread.



© Siemens AG 2004 All rights reserved.

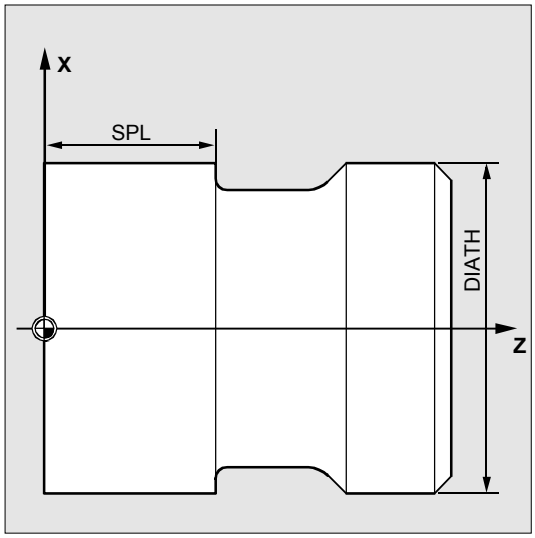
Sequence

Position reached before the beginning of the cycle:
The starting position can be any position from which any thread undercut can be approached without collision.

The cycle implements the following motion sequence:

- Approach to the starting point calculated in the cycle with G0
- Selection of the tool radius compensation for the active tool point direction. Retraction along the undercut contour at the feedrate programmed before cycle call.
- Retraction to the starting point with G0 and deselection of tool radius compensation with G40.

Explanation of parameters



DIATH (nominal diameter)
With this cycle you can machine thread undercuts for metrical ISO threads from M3 to M68.
If the value programmed in DIATH results in a final diameter of <3mm, the cycle is aborted and alarm 61601 "Finished part diameter too small" is output.
If the parameter is assigned a value other than that defined by DIN76 Part 1, again the cycle is aborted and the following alarm is output
61001 "Thread pitch incorrectly defined".

SPL (starting point)
With parameter SPL you define the final dimension in the longitudinal axis.

© Siemens AG 2004 All rights reserved.

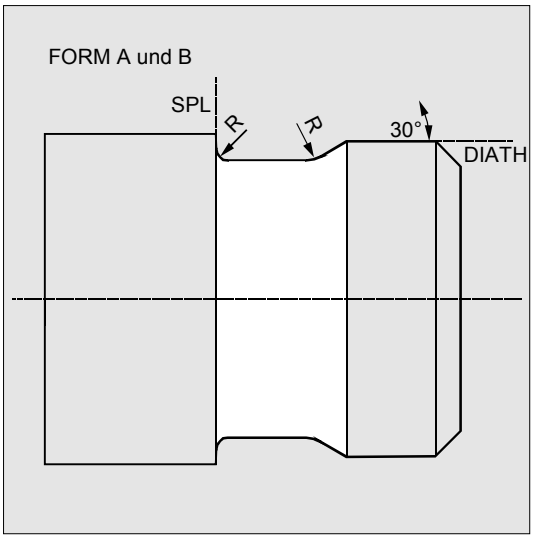
FORM (definition)

Thread undercuts of forms A and B are defined for external threads, form A for normal thread run-outs, form B for short thread run-outs.

Thread undercuts of forms C and D for used for internal threads, form C for normal thread runouts, form D for short thread run-outs.

If the parameter is assigned a value other than A to D, the cycle is aborted after outputting alarm 61609 "Form incorrectly defined".

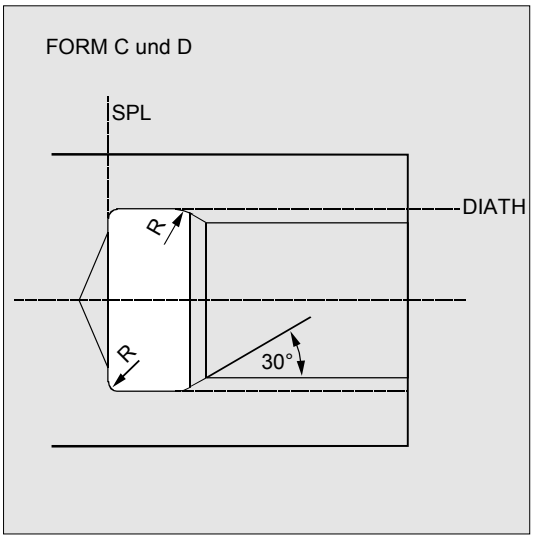
The tool radius compensation is automatically selected by the cycle.



_VARI (undercut position)

The position of the undercut can be either specified directly or derived from the tool point direction with parameter _VARI.

See _VARI with CYCLE94



The cycle automatically determines the starting point that is defined by the tool point direction of the active tool and the thread diameter. The position of this starting point in relation to the programmed coordinate values is determined by the tool point position of the active tool.

The cycle monitors the clearance angle of the active tool if forms A and B are being machined. If the cycle detects that the undercut form cannot be machined with the selected tool, the message "Changed undercut form" is output by the control but machining is continued.

Other Information

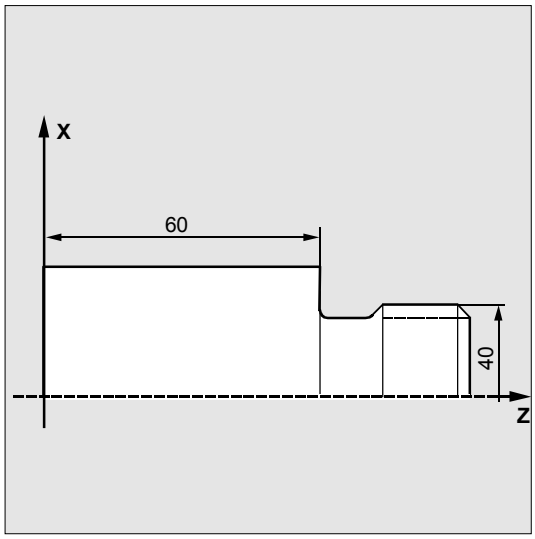
You must activate a tool offset before the cycle is called. Otherwise the following error message is output: 61000 "No tool offset active" the cycle is aborted.

© Siemens AG 2004 All rights reserved.

Programming example

Thread undercut_Form_A

You can machine a thread undercut of form A with this program.




```
N10  D3 T1 S300 M3 G95 F0.3          ;Specification of technology values
N20 G0 G18 G90 Z100 X50              ;Selection of starting position
N30 CYCLE96 (10, 60, "A")            ;Cycle call
N40 G90 G0 X30 Z100                  ;Approach next position
N50 M30                              ;End of program
```

© Siemens AG 2004 All rights reserved.

4.7 Thread cutting – CYCLE97



Programming

```
CYCLE97 (PIT, MPIT, SPL, FPL, DM1, DM2, APP, ROP, TDEP, FAL, IANG, NSP,
NRC, NID, VARI, NUMT, _VRT)
```

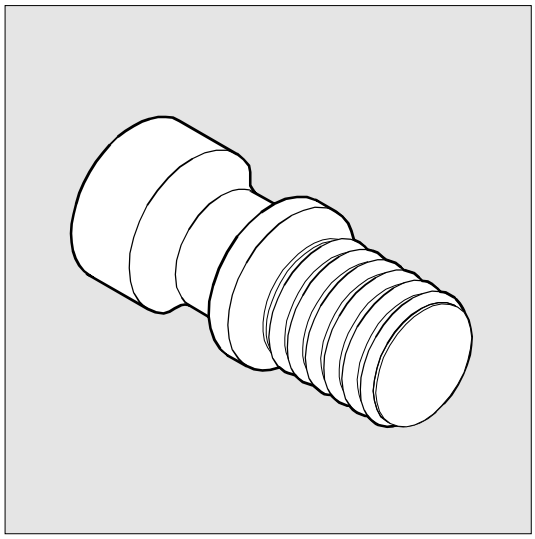


Parameters

PIT	real	Thread pitch as value (enter without sign)
MPIT	real	Thread pitch as thread size Value range: 3 (for M3) ... 60 (for M60)
SPL	real	Starting point of the thread in the longitudinal axis
FPL	real	End point of the thread in the longitudinal axis
DM1	real	Diameter of the thread at the starting point
DM2	real	Diameter of the thread at the end point
APP	real	Arc-in section (enter without sign)
ROP	real	Arc-out section (enter without sign)
TDEP	real	Thread depth (enter without sign)
FAL	real	Final machining allowance (enter without sign)
IANG	real	Infeed angle Value range: "+" (for flank infeed on flank) "–" (for alternating flank infeed)
NSP	real	Starting point offset for the first thread (enter without sign)
NRC	int	Number of rough cuts (enter without sign)
NID	int	Number of noncuts (enter without sign)
VARI	int	Definition of the machining type for the thread Value range: 1 ... 4
NUMT	int	Number of threads (enter without sign)
_VRT	real	Variable retraction distance based on initial diameter, incremental (enter without sign)

© Siemens AG 2004 All rights reserved.

Function



With this cycle you can machine cylindrical and tapered outside and inside threads with constant lead in longitudinal or face machining. The thread can be single or multiple. In multiple thread cutting, the threads are cut one after the other.
Infeed is automatic. You can select either constant infeed per cut or constant cross-section of cut.
A right-hand or left-hand thread is determined by the direction of rotation of the spindle programmed before the cycle call.
Feedrate and spindle override both have no effect in thread travel blocks. The spindle override must not be changed during thread machining.

References: /PG/, Programming Guide: Fundamentals
Chapter Thread cutting with constant lead, G33

A speed-controlled spindle with a position measuring system is required to operate this cycle.

Sequence

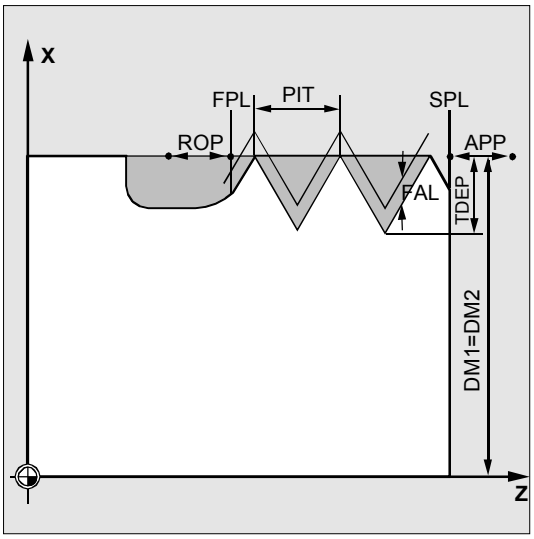
Position reached before the beginning of the cycle:
The starting position is any position from which the programmed thread starting point + arc-in section can be approached without collision.

The cycle implements the following motion sequence:

- Approach to the starting point determined by the cycle at the beginning of the arc-in section for the first thread with G0.
- Infeed for roughing according to the infeed type defined under VARI.
- Thread cutting is repeated according to the number of roughing cuts programmed.
- In the next cut with G33 the final machining allowance is removed.
- This cut is repeated according to the number of programmed noncuts.
- The total motion sequence is repeated for each additional thread.

© Siemens AG 2004 All rights reserved.

Explanation of parameters



PIT and MPIT (value and thread size)
The thread pitch is a paraxial value and entered without a sign. If metric cylindrical threads are being machined it is also possible to define the thread pitch in parameter MPIT as a thread size (M3 to M60). These two parameters should be used as alternatives. If they contain conflicting values, the cycle generates alarm 61001 "Thread pitch wrong"
and is aborted.

DM1 and DM2 (diameter)
This parameter is set to program the thread diameter of the start and end points of the thread.
With an inside thread, this corresponds to the tap hole diameter.

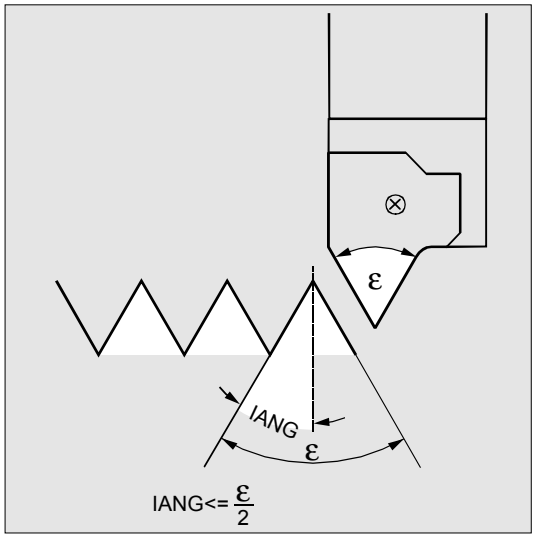
Connection between SPL, FPL, APP and ROP (starting point, end point, arc-in section and arc-out section)
The programmed starting point (SPL) and end point (FPL) are the basis of the thread. However, the starting point used in the cycle is the starting point brought forward by the arc-in section APP and, in the same way, the end point is the programmed end point brought back by the arc-out section ROP. The starting point defined by the cycle always lies
1mm outside the programmed thread diameter in the facing axis. This retraction plane is automatically generated by the control.

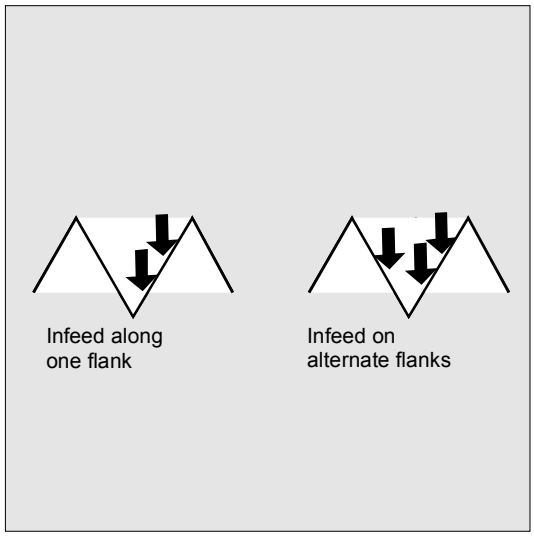
Connection between TDEP, FAL, NRC and NID (thread depth, final machining allowance, number of cuts)
The programmed final machining allowance has an effect parallel to the axis and is subtracted from the preset thread depth TDEP and the remainder is divided into roughing cuts.
The cycle automatically calculates the individual actual infeed depths depending on the parameter VARI.
The thread depth to be machined is divided into infeeds with the same cross-section of cut so that the cutting pressure remains constant for all rough cuts. Infeed is then performed with differing values for the infeed depth.
In a second method, the total thread depth is divided into constant infeed depths. The cross-section of cut

© Siemens AG 2004 All rights reserved.

gets larger from cut to cut. However, if the values for the thread depth are small, this method can create better cutting conditions.
The final machining allowance FAL is removed in one cut after roughing. After this, the noncuts programmed under parameter NID are executed.

IANG (infeed angle)
With parameter IANG you define the infeed angle. If infeed is to be performed at right angles to the cutting direction in the thread, this parameter must be assigned the value zero, i.e. this parameter can also be omitted from the parameter list as it is then automatically assigned the default value zero. If infeed is to be performed along the flank, the absolute value of this parameter must be no more than half the flank angle of the tool.
The sign entered for this parameter defines how this infeed is performed. If a positive value is entered, infeed is always performed on the same flank, if a negative value is entered, infeed is performed alternately on both flanks. The infeed type on both flanks alternately can only be used for cylindrical threads. However, if a negative value is assigned to parameter IANG for a tapered thread, the cycle automatically performs a flank infeed along one flank.

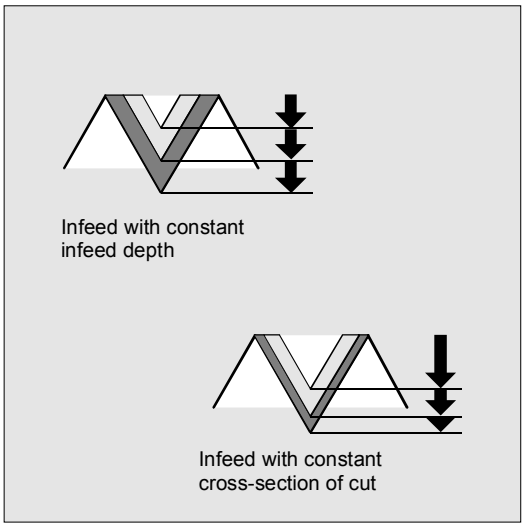




© Siemens AG 2004 All rights reserved.

NSP (starting point offset)
Under this parameter you can program the angular value that defines the point of the first cut for the first thread turn on the circumference on the turned part. This value is a starting point offset. The parameter can be assigned any value between 0.0001 and +359.9999 degrees. If no starting point offset has been entered or the parameter has been omitted from the parameter list, the first thread automatically starts at the zero degrees mark.

VARI (machining type)
With parameter VARI, you define if machining is to be internal or external and with which technology the infeed will be machined during roughing. The parameter VARI can be one of the values between 1 and 4 with the following meaning:



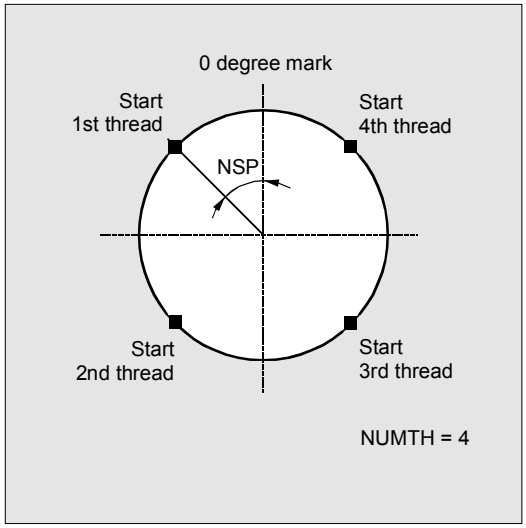
Value	Outside/inside	Const. infeed/const. cross-section of cut
1	External	Constant infeed
2	Internal	Constant infeed
3	External	Constant cross-section of cut
4	Internal	Constant cross-section of cut

If another value is assigned to parameter VARI, the cycle is aborted and alarm 61002 "Machining type incorrectly programmed"
is output.

© Siemens AG 2004 All rights reserved.

NUMT (number of thread starts)

With parameter NUMT you specify the number of thread starts for a multiple thread. If you require a single thread, either assign the value zero to the parameter or omit it from the parameter list.
The thread starts are distributed uniformly around the circumference of the turned part, the first thread is defined in parameter NSP.
If a multiple-start thread with a non-uniform distribution of threads around the circumference is to be machined, the cycle must be called for every thread start and the corresponding starting point offset must be programmed.



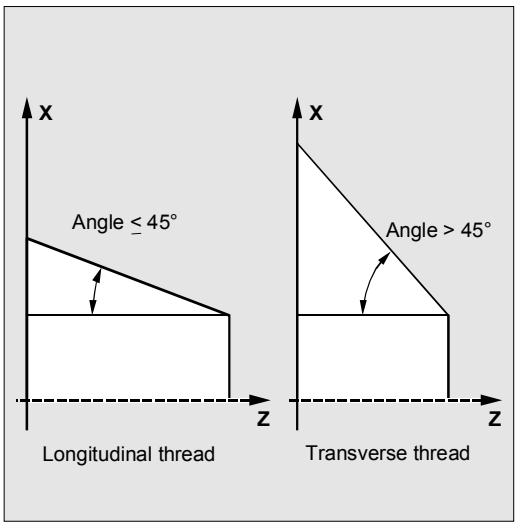
_VRT (variable retraction path)

The retraction path can be programmed on the basis of the initial thread diameter in parameter _VRT.
When _VRT = 0 (parameter not programmed), the retraction
path is 1mm. The retraction path is always measured in the programmed system of units inch or metric.

Other Information

Difference between a longitudinal thread and a face thread

The cycle automatically calculates whether a longitudinal or face thread is to be machined. This depends on the angle of the taper on which the thread is to be machined. If the angle at the taper ≤45 degrees, the longitudinal axis thread is machined, otherwise it will be the face thread.

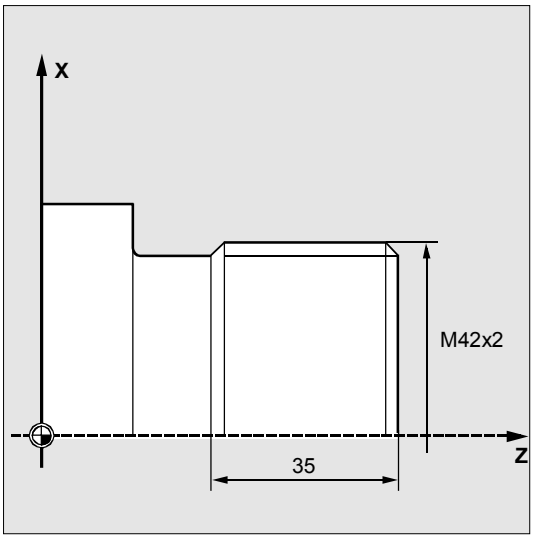


© Siemens AG 2004 All rights reserved.

Programming example

Thread cutting

With this program you can cut a metric outside thread M42x2 with flank infeed. The infeed is performed with a constant cross-section of cut. Five roughing cuts are made to a thread depth of 1.23mm without final machining allowance. After machining has been completed, 2 noncuts are performed.




```
DEF REAL MPIT=42, SPL=0, FPL=-35,          ;Definition of parameters with
DM1=42, DM2=42, APP=10, ROP=3,             ;value assignments
TDEP=1.23, FAL=0, IANG=30, NSP=0
DEF INT NRC=5, NID=2, VARI=3, NUMT=1

N10 G0 G18 G90 Z100 X60                    ;Selection of starting position
N20 G95 D1 T1 S1000 M4                     ;Specification of technology values
N30 CYCLE97 ( , MPIT, SPL, FPL, DM1, ->    ;Cycle call
-> DM2, APP, ROP, TDEP, FAL, IANG, ->
-> NSP, NRC, NID, VARI, NUMT)
N40 G90 G0 X100 Z100                       ;Approach next position
N50 M30                                    ;End of program
```

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

4.8 Thread chaining – CYCLE98



Programming

```
CYCLE98 (PO1, DM1, PO2, DM2, PO3, DM3, PO4, DM4, APP, ROP, TDEP, FAL,
IANG, NSP, NRC, NID, PP1, PP2, PP3, VARI, NUMT, _VRT)
```

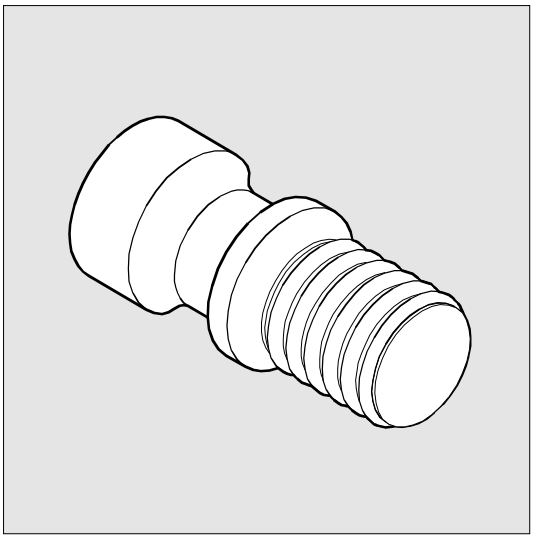


Parameters

PO1	real	Starting point of the thread in the longitudinal axis
DM1	real	Diameter of the thread at the starting point
PO2	real	First intermediate point in the longitudinal axis
DM2	real	Diameter at the first intermediate point
PO3	real	Second intermediate point
DM3	real	Diameter at the second intermediate point
PO4	real	End point of the thread in the longitudinal axis
DM4	real	Diameter at the end point
APP	real	Arc-in section (enter without sign)
ROP	real	Arc-out section (enter without sign)
TDEP	real	Thread depth (enter without sign)
FAL	real	Final machining allowance (enter without sign)
IANG	real	Infeed angle Value range: "+" (for flank infeed on flank) "–" (for alternating flank infeed)
NSP	real	Starting point offset for the first thread (enter without sign)
NRC	int	Number of rough cuts (enter without sign)
NID	int	Number of noncuts (enter without sign)
PP1	real	Thread pitch 1 as value (enter without sign)
PP2	real	Thread pitch 2 as value (enter without sign)
PP3	real	Thread pitch 3 as value (enter without sign)
VARI	int	Definition of the machining type for the thread Value range: 1 ... 4
NUMT	int	Number of threads (enter without sign)
_VRT	real	Variable retraction distance based on initial diameter, incremental (enter without sign)

© Siemens AG 2004 All rights reserved.

Function



With this cycle you can produce several concatenated cylindrical or tapered threads with a constant lead in longitudinal or face machining, all of which can have different thread leads.
The thread can be single or multiple. In multiple thread cutting, the threads are cut one after the other.
Infeed is automatic. You can select either constant infeed per cut or constant cross-section of cut.
A right-hand or left-hand thread is determined by the direction of rotation of the spindle programmed before the cycle call.
Feedrate and spindle override both have no effect in thread travel blocks. The spindle override must not be changed during thread machining.

References: /PG/, Programming Guide: Fundamentals
Chapter Thread cutting with constant lead, G33

Sequence

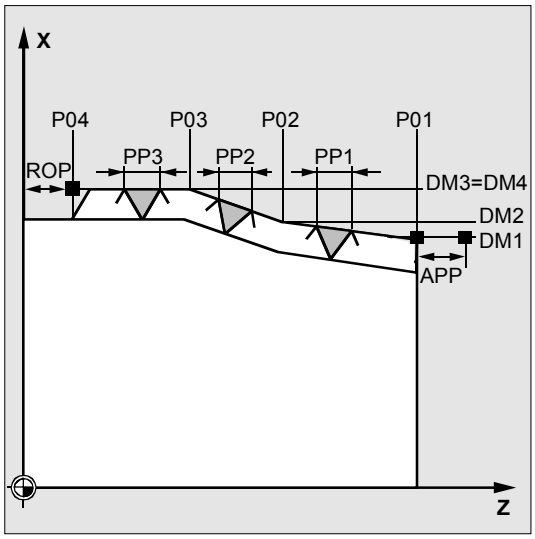
Position reached before the beginning of the cycle:
The starting position is any position from which the programmed thread starting point + arc-in section can be approached without collision.

The cycle implements the following motion sequence:

- Approach to the starting point determined by the cycle at the beginning of the arc-in section for the first thread with G0.
- Infeed to commence roughing according to the infeed type defined under VARI.
- Thread cutting is repeated according to the number of roughing cuts programmed.
- In the next cut with G33 the final machining allowance is cut.
- This cut is repeated according to the number of programmed noncuts.
- The total motion sequence is repeated for each additional thread.

© Siemens AG 2004 All rights reserved.

Explanation of parameters



PO1 and DM1 (starting point and diameter)
With these parameters you define the original starting point of the thread chain. The starting point calculated by the cycle that is approached at the beginning with G0 is the length of the arc-in section in front of the programmed starting point (starting point A).

PO2, DM2 and PO3, DM3 (intermediate point and diameter)
With these parameters you define two intermediate points in the thread.

PO4 and DM4 (endpoint and diameter)
The original end point of the thread is programmed under parameters PO4 and DM4.

With an inside thread, DM1...DM4 corresponds to the tap hole diameter.

Connection between APP and ROP (arc-in, arc-out sections)
The starting point used in the cycle is the starting point brought forward by the arc-in section APP and, in the same way, the end point is the programmed end point brought back by the arc-out section ROP.
The starting point defined by the cycle always lies 1mm outside the programmed thread diameter in the facing axis. This retraction plane is automatically generated by the control.

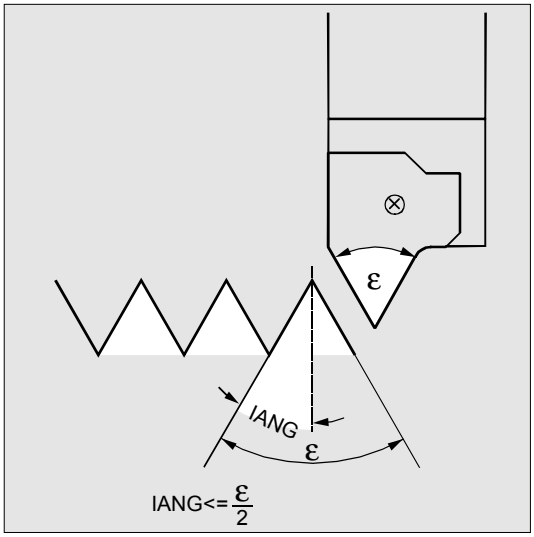
Connection between TDEP, FAL, NRC and NID (thread depth, final machining allowance, number of rough cuts and noncuts)
The programmed final machining allowance is subtracted from the defined thread depth TDEP and the remainder divided into rough cuts. The cycle automatically calculates the individual actual infeed depths depending on the parameter VARI. The thread depth to be machined is divided into infeeds with the same cross-section of cut so that the cutting pressure remains constant for all rough cuts. Infeed is then performed with differing values for the infeed depth.

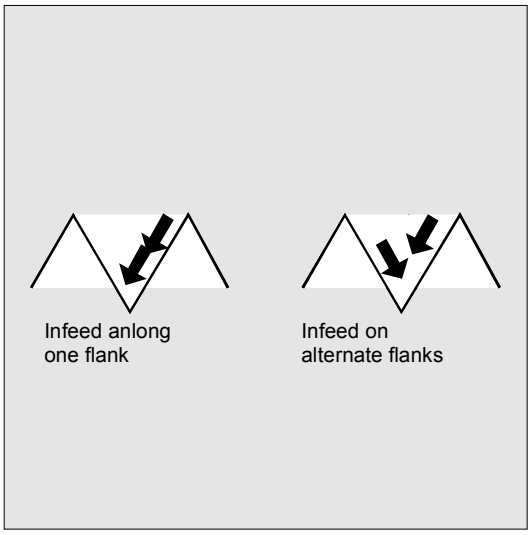
In a second method, the total thread depth is divided into constant infeed depths. The cross-section of cut gets larger from cut to cut. However, if the values for the thread depth are small, this method can create

© Siemens AG 2004 All rights reserved.

better cutting conditions.
The final machining allowance FAL is removed in one cut after roughing. After this, the noncuts programmed under parameter NID are executed.

IANG (infeed angle)
With parameter IANG you define the infeed angle. If infeed is to be performed at right angles to the cutting direction in the thread, this parameter must be assigned the value zero, i.e. this parameter can also be omitted from the parameter list as it is then automatically assigned the default value zero. If infeed is to be performed along the flank, the absolute value of this parameter must be no more than half the flank angle of the tool.
The sign entered for this parameter defines how this infeed is performed. If a positive value is entered, infeed is always performed on the same flank, if a negative value is entered, infeed is performed alternately on both flanks. The infeed type on both flanks alternately can only be used for cylindrical threads. However, if a negative value is assigned to parameter IANG for a tapered thread, the cycle automatically performs a flank infeed along one flank.



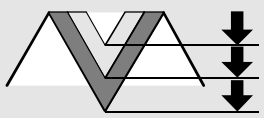


NSP (starting point offset)
Under this parameter you can program the angular value that defines the point of the first cut for the first thread turn on the circumference on the turned part. This value is a starting point offset. The parameter can be assigned any value between 0.0001 and +359.9999 degrees. If no starting point offset has been entered or the parameter has been omitted from the parameter list, the first thread automatically starts at the zero degrees mark.

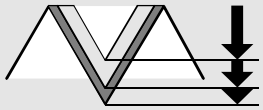
© Siemens AG 2004 All rights reserved.

PP1, PP2 and PP3 (thread pitch)
With these parameters you determine the thread pitch from the three sections of the thread chain. The pitch value must be entered as a paraxial value without a sign.

VARI (machining type)
With parameter VARI, you define if machining is to be internal or external and with which technology the infeed will be machined during roughing. The parameter VARI can be one of the values between 1 and 4 with the following meaning:



Infeed with constant infeed depth

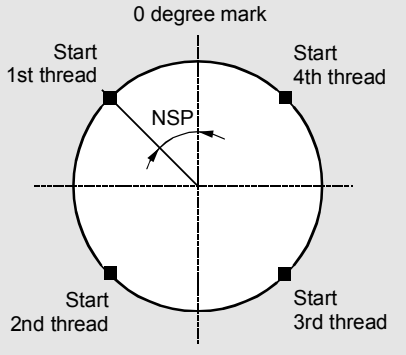


Infeed with constant cross-section of cut

Value	Outside/inside	Const. infeed/const. cross-section of cut
1	External	Constant infeed
2	Internal	Constant infeed
3	External	Constant cross-section of cut
4	Internal	Constant cross-section of cut

If another value is assigned to parameter VARI, the cycle is aborted and the following alarm:
61002 "Machining type incorrectly programmed".

NUMT (number of thread starts)
With parameter NUMT you specify the number of thread starts for a multiple thread. If you require a single thread, either assign the value zero to the parameter or omit it from the parameter list.
The thread starts are distributed uniformly around the circumference of the turned part, the first thread is defined in parameter NSP.
If a multiple-start thread with a non-uniform distribution of threads around the circumference is to be machined, the cycle must be called for every thread start and the corresponding starting point offset must be programmed.

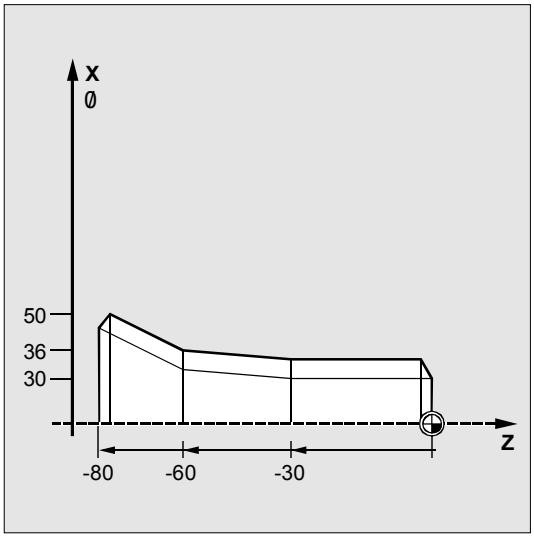


© Siemens AG 2004 All rights reserved.

_VRT (variable retraction path)
The retraction path can be programmed on the basis of the initial thread diameter in parameter _VRT.
When _VRT = 0 (parameter not programmed), the retraction
path is 1mm. The retraction path is always measured in the programmed system of units inch or metric.

Programming example

Thread chain
With this program you can produce a chain of threads starting with a cylindrical thread. Infeed is perpendicular to the thread. Neither a final machining allowance nor a starting point offset have been programmed. Five roughing cuts and one noncut are performed.
The machining type defined is longitudinal, outside, with constant cross-section of cut.




```
N10 G18 G95 T5 D1 S1000 M4                    ;Specification of technology values
N20 G0 X40 Z10                                ;Approach starting position
N30 CYCLE98 (0, 30, -30, 30, -60, ->          ;Cycle call
-> 36, -80, 50, 10, 10, 0.92, , , , ->
-> 5, 1, 1.5, 2, 2, 3, 1)
N40 G0 X55                                    ;Traverse in each axis separately
N50 Z10
N60 X40
N70 M30                                       ;End of program
```

-> Must be programmed in a single block

© Siemens AG 2004 All rights reserved.

4.9 Thread recutting

Thread cutting cycles CYCLE97 and CYCLE98 can be used for thread recutting.

Function

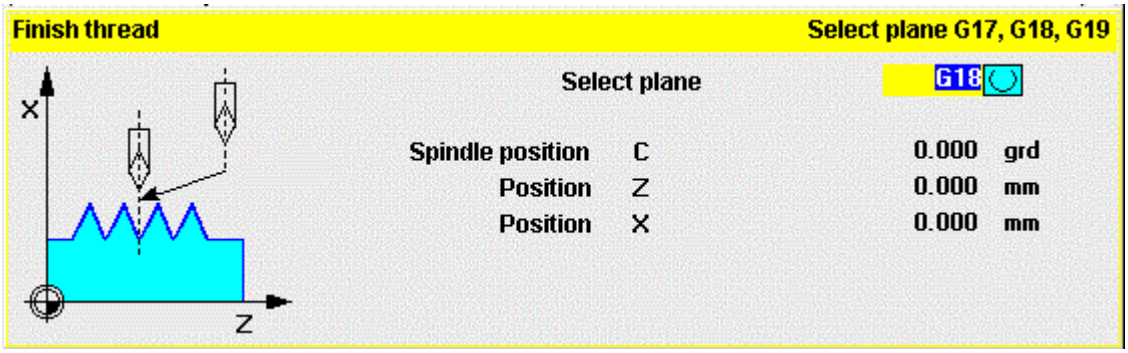
The angular offset of a thread start resulting from tool breakage or remeasurement is taken into account and compensated for by the "Thread recut" function.
This function can be executed in JOG mode in the Machine operating area.

The cycles calculate an additional offset angle for each thread, which is applied in addition to the programmed starting point offset, from the data stored in the thread start during synchronization.

Preconditions
The channel in which the thread recutting program must be executed is already selected; the relevant axes must already be referenced. The channel is in the Reset state, the spindle is stationary.

Sequence

- Select JOG in "Machine" operating area.
- Press soft key "Finish thread"
  ➔ Open screen form for this function.



- Thread into thread start using the threading tool.
- Select soft key "Sync Point" when the cutting tool is positioned exactly in the thread start.
- Press soft key "Cancel" to return to the next-higher soft key menu without activating the function (no data are then stored in the NC).
- Select soft key "OK" to transfer all values to the GUD in the NC.

© Siemens AG 2004 All rights reserved.

- Then retract the tool and move it to its starting position.
- Select "Automatic" and position the program pointer using block search in front of the thread cycle call.
- Start the program with NC start.



Miscellaneous functions

You can delete values stored earlier by selecting another soft key labeled "Delete".

If several spindles are operating in the channel, another box is displayed in the screen form in which you can select a spindle to machine the thread.



Startup

Thread recutting requires operation in the JOG basic display. This is activated with soft key HS8 "Finish thread" in file MA_JOG.COM .

- Start-up for thread recutting for HMI Advanced:
  First open the MA_JOG.COM file and then delete the semicolon preceding the following lines:
  - ;HS8=($80720,,se1)
  - ;PRESS(HS8)
  - ;LM("GENS","drehen2.com")
  - ;END_PRESS

  The file is in the "Standard Cycles" directory. The soft key is then active. The HMI must then be restarted.
- Start-up for thread recutting for HMI Embedded:
  First open the COMMON.COM file and then delete ";" in front of SC108. The file is in the "User Cycles" directory. The HMI must then be restarted.

© Siemens AG 2004 All rights reserved.

4.10 Extended stock removal cycle - CYCLE950

The extended stock removal cycle is an option.
It requires SW 6 in the NCK and HMI Advanced.

Programming

```
CYCLE950 (_NP1, _NP2, _NP3, _NP4, _VARI, _MID, _FALZ, _FALX, _FF1, _FF2,
_FF3, _FF4, _VRT, _ANGB, _SDIS, _NP5, _NP6, _NP7, _NP8, _APZ, _APZA, _APX,
_APXA, _TOL1)
```

Parameters

_NP1	string	Name of the contour subroutine for the finished part contour
_NP2	string	Label / block number start of finished part contour, optional (this can be used to define contour sections)
_NP3	string	Label / block number end of finished part contour, optional (this can be used to define contour sections)
_NP4	string	Name of the stock removal program to be generated
_VARI	integer	Type of machining: (enter without sign) ONES DIGIT: Values: 1...Longitudinal 2...Face 3...Parallel to contour TENS DIGIT: Values: 1...Programmed infeed direction X- 2...Programmed infeed direction X+ 3...Programmed infeed direction Z- 4...Programmed infeed direction Z+ HUNDREDS DIGIT: Values: 1...Roughing 2...Finishing 3...Complete THOUSANDS DIGIT: Values: 1...With rounding 2...Without rounding (liftoff) TEN THOUSANDS DIGIT: Values: 1...Do not machine relief cuts 2...Do not machine relief cuts HUNDRED THOUSANDS DIGIT: Values: 1...Programmed machining direction X- 2...Programmed machining direction X+ 3...Programmed machining direction Z- 4...Programmed machining direction Z+
_MID	real	Infeed depth (enter without sign)
_FALZ	real	Final machining allowance in the longitudinal axis (enter without sign)

© Siemens AG 2004 All rights reserved.

_FALX	real	Final machining allowance in the facing axis (enter without sign)
_FF1	real	Feedrate for longitudinal roughing
_FF2	real	Feedrate for face roughing
_FF3	real	Feedrate for finishing
_FF4	real	Feedrate at contour transition elements (radius, chamfer)
_VRT	real	Liftoff distance for roughing, incremental (enter without sign)
_ANGB	real	Liftoff angle for roughing
_SDIS	real	Safety distance for avoiding obstacles, incremental
_NP5	string	Name of contour program for blank contour
_NP6	string	Label / block number start of blank contour, optional (this can be used to define contour sections)
_NP7	string	Label / block number end of blank contour, optional (this can be used to define contour sections)
_NP8	string	Name of contour program for updated blank contour
_APZ	real	Axial value for defining blank for longitudinal axis
_APZA	int	Absolute or incremental evaluation of parameter _APZ 90=absolute, 91=incremental
_APX	real	Axial value for defining blank for facing axis
_APXA	int	Absolute or incremental evaluation of parameter _APX 90=absolute, 91=incremental
_TOL1	real	Blank tolerance

© Siemens AG 2004 All rights reserved.



Function

With the extended stock removal cycle CYCLE950 you can machine a contour programmed with paraxial or parallel-contour stock removal. Any blank can be defined and is considered during stock removal. The finished part contour must be continuous and may contain any number of relief cut elements. You can specify a blank as a contour or by means of axial values.

Contours can be machined in the longitudinal and facing directions with this cycle. You can freely select a technology (roughing, finishing, complete machining, machining and infeed directions). It is possible to update a blank.

For roughing, the programmed infeed depth is observed precisely; the last two roughing steps are divided equally. Roughing is performed up to the final machining allowance programmed.

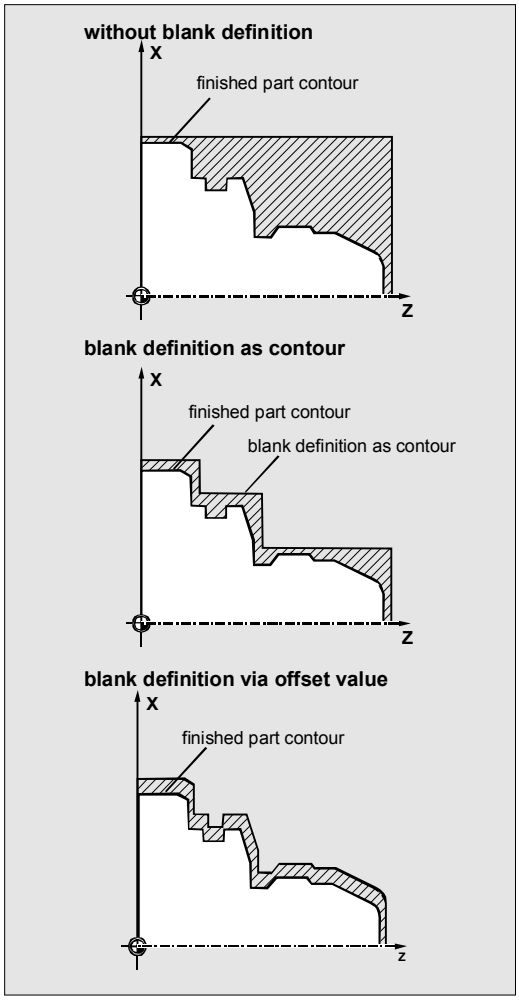
Finishing is performed in the same direction as roughing.

The tool radius compensation is automatically selected and deselected by the cycle.

© Siemens AG 2004 All rights reserved.

New functions compared to CYCLE95:

- You can define a blank either by programming a contour, specifying an allowance on the finished-part contour or entering a blank cylinder (or hollow cylinder in the case of internal machining) from which stock must be removed.
- It is possible to detect residual material that cannot be machined with the current tool. The cycle can generate an updated blank, which is stored as a program in the parts program memory.
- You can specify the contours for stock removal:
  - in a separate program,
  - in the calling main program or
  - as section of any given program.
- During roughing, it is possible to choose between paraxial and contour-parallel machining.
- During roughing, you have the option of machining along the contour so that no corners are left over, or removing stock immediately at the roughing intersection.
- The angle for stock removal at the contour during roughing is programmable.
- Optionally, relief cuts can be machined or skipped during roughing.



Sequence

Position reached before the beginning of the cycle:
The initial position can be any position from which the blank contour can be approached collision-free. The cycle calculates collision-free approach movements to the starting point for machining but does not consider the tool holder data.

Movement for paraxial roughing:

- The starting point for roughing is calculated internally in the cycle and approached with G0.
- The infeed to the next depth, calculated in accordance with the specifications in parameter _MID, is carried out with G0, and paraxial roughing then performed with G1. The feedrate during roughing is calculated internally in the cycle according to the path as the feedrate that results from the values specified for longitudinal and face feed (_FF1 and _FF2).

© Siemens AG 2004 All rights reserved.

- For "Rounding along contour", the previous intersection is approached parallel to the contour.
- When the previous intersection is reached or for machining "Without rounding along contour", the tool is lifted off at the angle programmed in _ANGB and then retracted to the starting point for the next infeed with G0. If the angle is 45 degrees, the programmed liftoff path _VRT is also followed precisely; it is not exceeded for other angles.
- This procedure is repeated until the full depth of the machining section has been reached.

Sequence of motions for roughing in parallel with contour:

- The starting point for roughing and the individual infeed depths are calculated as for paraxial roughing and approached with G0 or G1.
- Roughing is carried out in contour-parallel paths.
- Liftoff and retraction is carried out in the same way as for paraxial roughing.

© Siemens AG 2004 All rights reserved.

Explanation of parameters

_NP1, _NP2, _NP3 (contour programming finished part)
The finished part contour can be programmed optionally in a separate program or in the current main program that calls the routine. The data are transferred to the cycle via parameters _NP1 – Name of the program or _NP2, _NP3 – ID of program section from ... to using block numbers or labels.

So there are three options for contour programming:

- The contour is defined in a separate program – in which case only _NP1 need be programmed;
  (see programming example 1)
- The contour is defined in the calling program in which case only _NP2 and _NP3 have to be programmed;
  (see programming example 2)
- The stock removal contour is part of a program but not part of the program that calls the cycle in which case all three parameters must be programmed.

When the contour is programmed as a program section, the last contour element (block with label or block number end of blank contour) must not contain a radius or chamfer.
Write the program name in _NP1 with path name and program type.
Example:

```
_NP1="/_N_SPF_DIR/_N_PART1_SPF"
```

_NP4 (name of the stock removal program)
The stock removal cycle generates a program for the travel blocks that are required for stock removal between the blank and the finished part. This program is stored in the same directory as the calling program in the parts program memory if no other path is specified when it is generated. If a path is entered, it is stored accordingly in the file system. The program is a main program (type MPF) if no other type is specified.
Parameter _NP4 defines the name of this program.

© Siemens AG 2004 All rights reserved.

_VARI (machining type)

Parameter _VARI defines the type of machining.

Possible settings are:

Unit's place:

1=Longitudinal

2=Face

3=Parallel to the contour

Tens digit:

1=Programmed infeed direction X-

2=Programmed infeed direction X+

3=Programmed infeed direction Z-

4=Programmed infeed direction Z+

Hundreds digit:

1=Roughing

2=Finishing

3=Complete

Thousands digit:

1=With rounding

2=Without rounding (liftoff)

The choice of with or without rounding at the contour determines whether the stock removal occurs immediately at the roughing point or whether machining is continued along the contour to the previous roughing point so that no corners are left.

Ten thousands digit:

1=Machine relief cuts

2=Do not machine relief cuts

Hundred thousands digit:

1=Programmed machining direction X-

2=Programmed machining direction X+

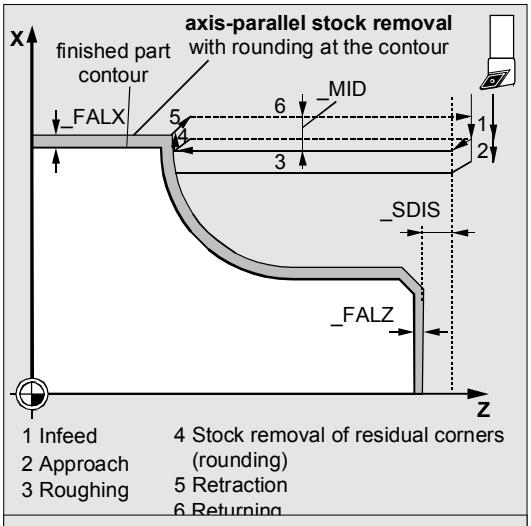
3=Programmed machining direction Z-

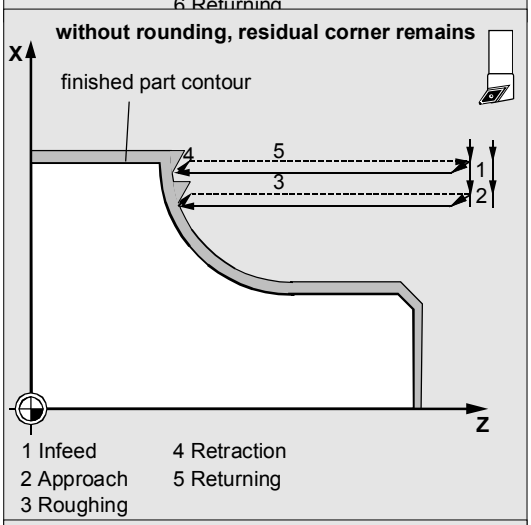
4=Programmed machining direction Z+

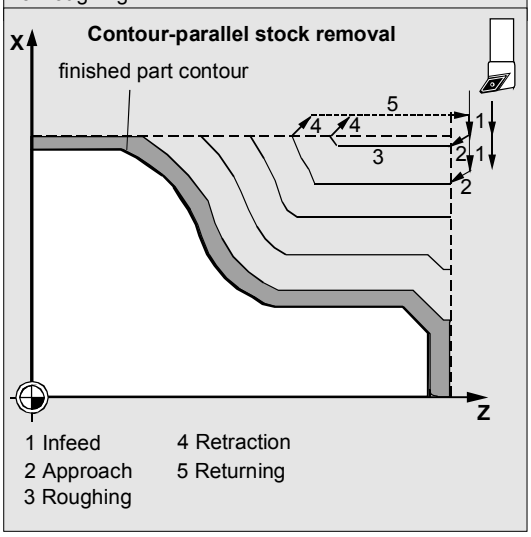
Example:

_VARI=312311 means machining:

longitudinal,
infeed direction X- (i.e. external),
complete;
the workpiece is not rounded along the contour,
relief cuts are machined,
machining direction Z-.







© Siemens AG 2004 All rights reserved.

_MID (infeed depth for roughing)
The infeed depth for roughing is programmed with the parameter _MID. Roughing steps are generated with this infeed until the remaining depth is less than twice the infeed depth. Then two steps are performed each at half of the remaining depth.
_MID is interpreted as a radius or diameter depending on the value of cycle setting data _ZSD[0] if the facing axis is involved in the infeed for roughing.

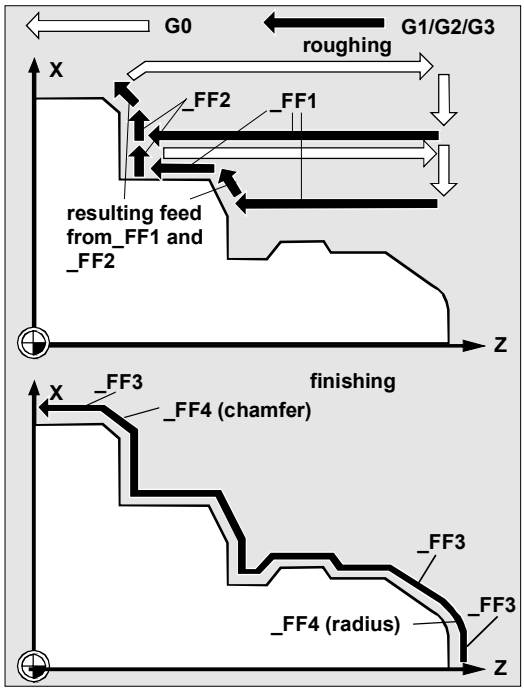
_ZSD[0]=0: _MID is evaluated for radius/diameter programming according to the G group, as a radius with DIAMOF, otherwise as a diameter.
_ZSD[0]=1: _MID is a radius value
_ZSD[0]=2: _MID is a diameter value

When rough cutting parallel to contour, the infeed depth does not act in relation to the specified infeed axis but vertically to the contour. This always results in more cuts than in paraxial rough-cutting with the same value for infeed depth.

_FALZ, _FALX (machining allowance)
The default for a final machining allowance for roughing is provided by the parameters FALZ (for Z axis) and FALX (for X axis). Roughing is always performed to these final machining allowances. If no machining allowances are programmed, stock removal is performed up to the end contour during roughing.

_FF1, _FF2, _FF3 and FF4 (feedrate)
Separate feedrates can be specified for roughing and finishing, as shown in the figure opposite.
Separate feedrates apply for longitudinal (_FF1) and face (_FF2) during roughing. If inclined or circular path sections are traversed when machining the contour, the appropriate feedrate is calculated automatically inside the cycle.
The feedrates programmed at the contour are active during finishing. If none are programmed there, the finishing feedrate in _FF3 and the feedrates at radii and chamfers in _FF4 apply to these contour transition elements.

(see sample program 1 for programming of the parts in the figure below)



© Siemens AG 2004 All rights reserved.

_VRT (liftoff) and _ANGB (lift angle)
The parameter _VRT can be used to program the amount of liftoff during roughing in both axes.
If _VRT=0 (parameter not programmed), liftoff is 1mm.
It is also possible to program the angle at which the axis is retracted from the contour in parameter _ANGB.
If nothing is programmed, the angle is 45°.

_SDIS (safety distance)
Parameter _SDIS determines the amount of clearance for obstructions. This clearance is active for retraction from a relief cut and approach to the next relief cut, for example.
If no value is programmed, the clearance is 1mm.

_NP5, _NP6, _NP7 (contour programming blank)
If a blank is programmed as a contour, it can be programmed as a program name using parameter _NP5 or as a program section with parameters _NP6 and _NP7.
Otherwise, programming is carried out as for finished parts (see _NP1, _NP2, _NP3).

_NP8 (name of contour program for updated blank contour)
Cycle CYCLE950 can detect residual material that cannot be removed with the active tool.
To continue this machining with a different tool, it is possible to generate an updated blank contour automatically. This is stored as a program in the parts program memory. You can specify the program name in parameter _NP8 with or without path details (see sample program 3).
An updated blank contour is always generated when a travel program is generated.

© Siemens AG 2004 All rights reserved.

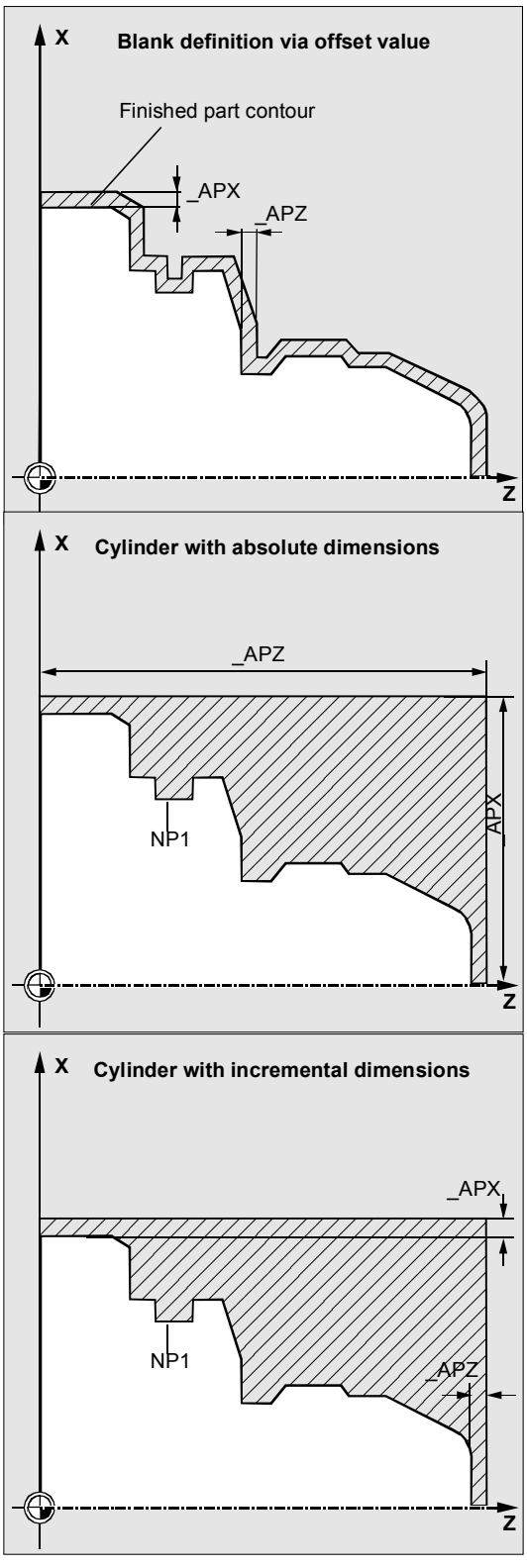
_APZ, _APZA, _APX, _APXA (blank definition)

You can also define a blank by entering the dimensions of a blank cylinder (or hollow cylinder) or as an allowance on the finished-part contour in parameters _APZ and _APX.

You can enter the cylinder dimensions as either absolute or incremental values, although an allowance on the finished-part contour is always interpreted incrementally.

Absolute or incremental values are selected via parameters _APZA and _APXA

(_APZA, _APXA: 90 - absolute 91 - incremental).



© Siemens AG 2004 All rights reserved.

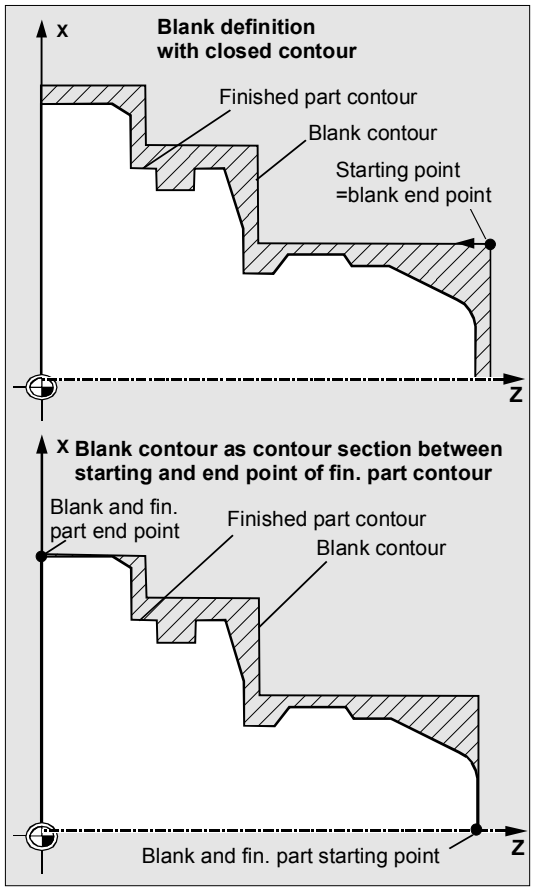
_TOL1 (blank tolerance)
Since a blank does not always correspond exactly to the blank definition when it is cast or forged for example, it makes sense not to travel to the blank contour with G0 for roughing and for the infeed but to activate G1 shortly beforehand to compensate for any tolerances. Parameter _TOL1 defines the distance from the blank at which G1 becomes active.
Traversing is started with G1 at this incremental amount before the blank. If the parameter is not programmed, it has the value 1mm.

Other Information

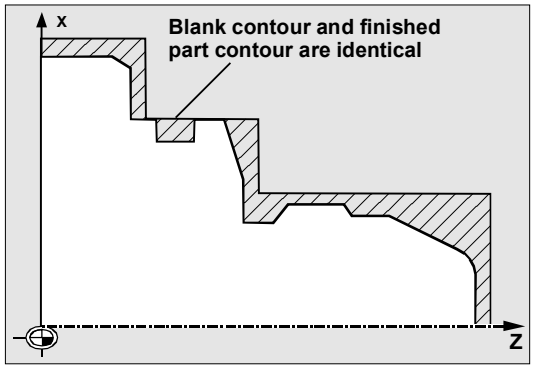
Contour definition
Unlike CYCLE95, one block that contains a link to the current plane is sufficient for contour programming.

For further details of contour definition, see CYCLE95.

Blank contour definition
A blank contour must either be a closed contour (starting point=end point) which encompasses the finished-part contour either partially or fully, or a contour section between the starting and end points of the finished-part contour. The programmed direction is irrelevant.



Blank contours always must be described in a way that they are not partly identical with the finished-part contours, i.e. the machined materials are not combined



© Siemens AG 2004 All rights reserved.

Explanation of the cycle structure

CYCLE950 is used to solve very complex problems during stock removal, which require high processing power in the control. For best timing, the calculation is carried out in the HMI.

The calculation is started in the cycle and a program with traversing blocks for stock removal generated in its result and stored in the file system of the control, where it is called and executed immediately.

This structure means that it is only necessary to perform the calculation the first time a program is executed with CYCLE950 call. When called a second time, the traversing program is available and can be called by the cycle.

Recalculation is performed when:

- A finished contour has been modified;
- A transfer parameter of the cycle has changed;
- A tool with different tool offset data has been activated before the cycle call.

Program storage in the file system

If the contours for CYCLE950 are programmed outside the program that makes the call, the following applies for the search in the file system of the control:

- If the calling program is stored in a workpiece directory, then the programs which define the finished-part or blank contour must also be stored in the same workpiece directory, or at least programmed with path information.
- If the invoked program is located in the "Parts program" directory (MPF.DIR), a search is also made there for the programs is no path is specified.

The cycle creates a program that contains the traversing blocks for stock removal and, optionally, an updated blank contour.

These are either stored in the same directory as the cycle-calling program or in accordance with the specified path.

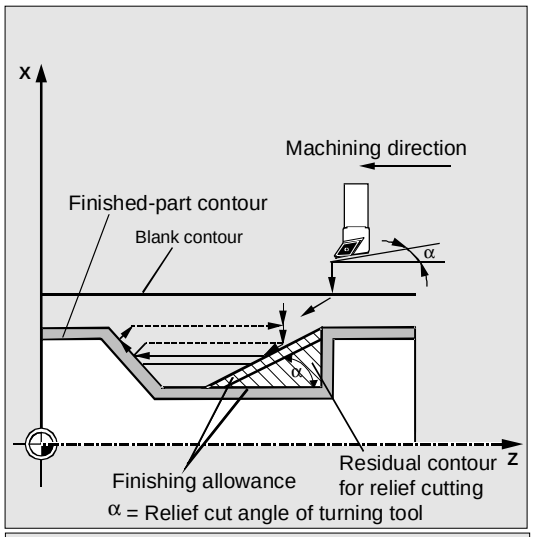
© Siemens AG 2004 All rights reserved.

Note on simulation

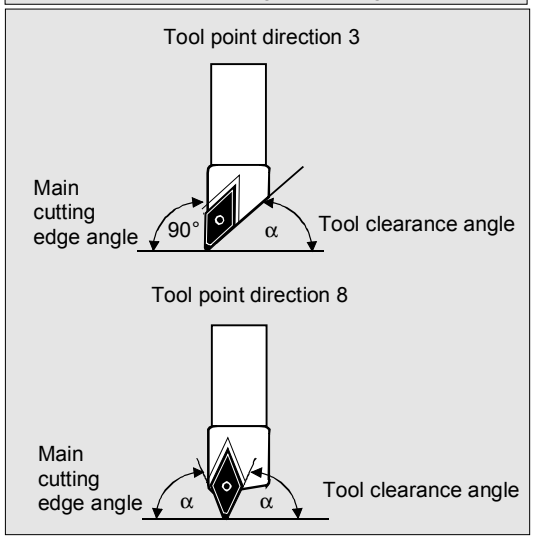
In the simulation of the extended stock removal cycle CYCLE950, the generated programs are saved to the NCU file system. Therefore, only the "NC Active Data" setting is practical since tool offset data are included in program calculation.

Blank updating

The extended stock removal cycle CYCLE950 detects residual material during roughing and is able to generate an updated blank contour outside the machining process, which can be used in a further machining step.



To do this, the cycle internally considers the angle at the tool point.
The relief cut angle of the tool must be entered in the tool offset data (parameter 24).
The cycle defines the main cutting edge angle automatically according to the tool point position.
For tool point positions 1 to 4, the blank update is calculated with a main cutting edge angle of 90°. For tool point positions 5 to 9, the main cutting angle is assumed to be identical to the relief cut angle.
If CYCLE950 is called more than once, each time with blank update, in the same program, different names for the generated blank contours must be assigned; it is not permissible to use the program name (parameter _NP8) more than once.



Extended stock removal cannot be performed in m:n configurations.

© Siemens AG 2004 All rights reserved.

Programming example 1

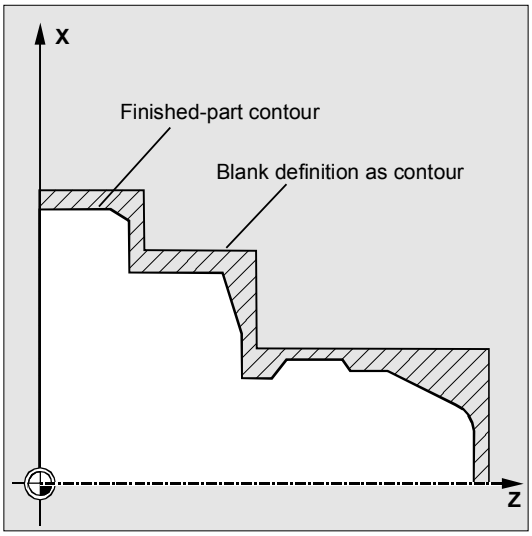
From a preshaped blank, the contour saved in program PART1.MPF is to be machined.

The type of machining for the stock removal process is

- only roughing,
- longitudinal,
- outside,
- with rounding (so that no corners are left over),
- relief cuts are to be machined.

The blank contour is specified in program BLANK1.MPF.

A turning steel with tool point position 3 and a radius of 0.8mm is used.



Machining program:

```
%_N_EXAMPLE_1_MPF
;$PATH=/_N_WKS_DIR/_N_STOCK_REMOVAL_NEW_W
PD
; Example 1: Stock removal with blank
; Sca, 01.04.99
;
; Tool offset data
N10 $TC_DP1[3,1]=500 $TC_DP2[3,1]=3
$TC_DP6[3,1]=0.8 $TC_DP24[3,1]=60
N15 G18 G0 G90 DIAMON
N20 T3 D1
N25 X300
N30 Z150
N35 G96 S500 M3 F2
N45 CYCLE950("Part1",,,"Machine_Part1",
311111,1.25,1,1,0.8,0.7,0.6,0.3,0.5,45,2,
"Blank1",,,,,,,,1)
N45 G0 X300
N50 Z150
N60 M2
```

Finished part contour:

```
%_N_PART1_MPF
;$PATH=/_N_WKS_DIR/_N_STOCK_REMOVAL_NEW_W
PD
; Finished part contour Example 1
;
```


© Siemens AG 2004 All rights reserved.

```
N100 G18 DIAMON F1000
N110 G1 X0 Z90
N120 X20 RND=4
N130 X30 Z80
N140 Z72
N150 X34
N160 Z58
N170 X28 Z55 F300
N180 Z50 F1000
N190 X40
N200 X60 Z46
N210 Z30
N220 X76 CHF=3
N230 Z0
N240 M17
```

Blank contour:

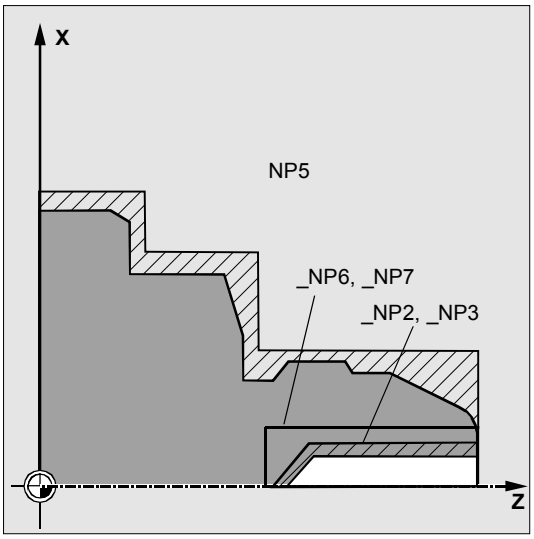
```
%_N_BLANK1_MPF
;$PATH=/_N_WKS_DIR/_N_STOCK_REMOVAL_NEW_W
PD
; Blank contour Example 1
;
N100 G18 DIAMON F1000
N110 G0 X0 Z93
N120 G1 X37
N130 Z55
N140 X66
N150 Z35
N160 X80
N170 Z0
N180 X0
N190 Z93                    ;End point=Starting point
                            ;Blank contour must be closed
N200 M17
```

After machining, the workpiece STOCK_REMOVAL_NEW.WPD contains a new program called MACHINING_PART1.MPF. This program is created during the first program call and contains the traversing motions for machining the contour in accordance with the blank.

© Siemens AG 2004 All rights reserved.

Programming example 2

A simple inside contour is to be machined on the same part as in sample program 1.
A center bore is made first using a diameter-10 drill.
Then, the inside contour is roughed parallel to the contour, since the hole roughly corresponds to the end contour.
This is done by defining a blank contour again for inside machining.
The stock removal contour is located in the same program as the cycle call in the blocks N400 to N420, the blank contour in blocks N430 to N490.



Machining program:

```
%_N_EXAMPLE_2_MPF
;$PATH=/_N_WKS_DIR/_N_STOCK_REMOVAL_NEW_WPD
; Example 2: Stock removal inside parallel
with contour
; Sca, 01.04.99
;
; Tool offset data for turning tool, inside
N100 $TC_DP1[2,1]=500 $TC_DP2[2,1]=6
$TC_DP6[2,1]=0.5 $TC_DP24[2,1]=60
N105 $TC_DP1[1,1]=200 $TC_DP3[1,1]=100
$TC_DP6[1,1]=5
N110 G18 G0 G90 DIAMON
N120 X300
N130 Z150
N140 T1 D1 M6                          ;Change drill with diameter 10
N150 X0                                ;Center drilling in three steps
N160 Z100
N170 F500 S400 M3
N175 G1 Z75
N180 Z76
N190 Z60
N200 Z61
N210 Z45
N220 G0 Z100
N230 X300                              ; Approach tool change point
N240 Z150
N250 T2 D1 M6                          ;Insert turning tool for inside
                                       machining
N260 G96 F0.5 S500 M3
```


© Siemens AG 2004 All rights reserved.

```
N275 CYCLE950("","N400","N420",
"Bearbeite_Teil1_innen",311123,1.25,0,0,
0.8,0.5,0.4,0.3,0.5,45,1,"","N430","N490"
,,,,,,1)
N280 G0 X300
N290 Z150
N300 GOTOF _END                          ;Skip contour definition
N400 G0 X14 Z90                          ;N400 to N420 finished part contour
N410 G1 Z52
N420 X0 Z45
N430 G0 X10 Z90                          ;N430 to N490 blank contour
N440 X16
N450 Z40
N460 X0
N470 Z47
N480 X10 Z59
N490 Z90
N500 _END:M2
```

© Siemens AG 2004 All rights reserved.

Programming example 3

The same part as in sample program 1 should now be machined in two steps.

In the first machining step (N45), roughing is carried out using a tool with tool point position 9 and a large radius with deep infeed depth and no blank specified. The result to be generated is an updated blank with the name blank3.MPF.
The type of machining for this step is:

- only roughing,
- longitudinal,
- outside,
- With rounding,
- relief cuts are not to be machined.

In the second machining step (N75), the residual material on this blank is machined with a different tool and then finished.
The type of machining for this step is:

- complete machining (roughing and finishing)
- longitudinal,
- outside,
- with rounding (so that no corners are left over),
- relief cuts are to be machined.

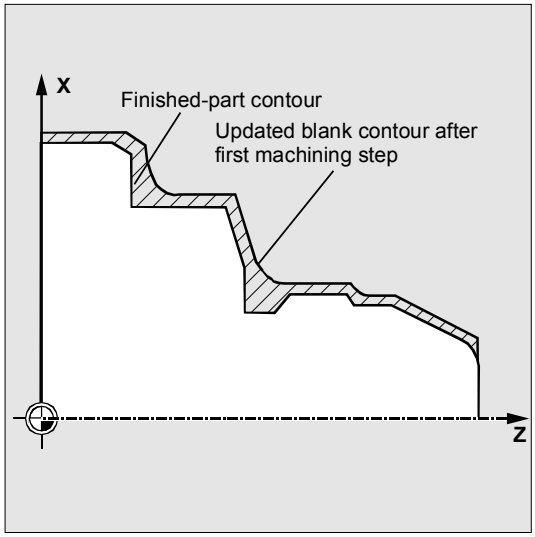
Machining program:

```
%_N_EXAMPLE_3_MPF
;$PATH=/_N_WKS_DIR/_N_STOCK_REMOVAL_NEW_WPD
; Example 3: Stock removal in two steps
with blank updating
; Sca, 09.04.99
;
; Tool offset data
; T3: Roughing steel for rough machining,
tool point position 9, radius 5
N05 $TC_DP1[3,1]=500 $TC_DP2[3,1]=9
$TC_DP6[3,1]=5 $TC_DP24[3,1]=80
; T4: Turning steel for residual material
and finishing
;     Tool point position 3, radius 0.4
```


© Siemens AG 2004 All rights reserved.

```
N10 $TC_DP1[4.1]=500 $TC_DP2[4.1]=3
$TC_DP6[4.1]=0.4 $TC_DP24[4.1]=80
N15 G18 G0 G90 DIAMON
N20 T3 D1                                   ;Tool for roughing
N25 X300
N30 Z150
N35 G96 S500 M3 F2
N45 CYCLE950("Part1",,,"Machine_Part3",
321111,8,1,1,0.8,0.7,0.6,0.5,1,45,6,
"DEFAULT",,,"Blank3",0,91,0,91,1)
N50 G0 X300
N55 Z150
N60 T4 D1                                   ;Tool for roughing residual material and
                                            ;finishing
N65 G96 S500 M3 F2
N75
CYCLE950("Part1",,,"Finish_Part3",311311,
0.5,0.25,0.25,0.8,0.7,0.6,0.5,1,45,6,"Bla
nk3",,,,,,,,1)
N160 M2
```

Finished part contour:
as for sample program 1



© Siemens AG 2004 All rights reserved.

Explanation

Alarm source CYCLE950

Alarm number	Alarm text	Explanation, remedy
61701	"Error in contour description of finished part"	Either none of parameters _NP1, _NP2 or _NP3 is assigned or error in programming of finished-part contour
61702	"Error in contour description of blank"	Either none of parameters _NP5, _NP6 or _NP7 is assigned or error in programming of blank contour
61703	"Internal cycle error while deleting file"	
61704	"Internal cycle error while writing file"	
61705	"Internal cycle error while reading file"	
61706	"Internal cycle error during checksum formation"	
61707	"Internal cycle error during ACTIVATE at MMC"	
61708	"Internal cycle error during READYPROG at MMC"	
61709	"Timeout for contour calculation"	
61720	"Illegal input"	
61721	"Error: unable to determine contour direction"	
61722	"System error"	
61723	"Unable to perform machining"	Use a tool with a larger clearance angle
61724	"No material available"	
61725	"Out of memory, error in contour generation"	
61726	"Internal error: Out of memory _FILECTRL_INTERNAL_ERROR"	
61727	"Internal error: Out of memory _FILECTRL_EXTERNAL_ERROR"	
61728	"Internal error: Out of memory _ALLOC_P_INTERNAL_ERROR"	
61729	"Internal error: Out of memory _ALLOC_P_EXTERNAL_ERROR"	
61730	"Internal error: Invalid Memory"	
61731	"Internal error: Floating-point exception"	
61732	"Internal error: Invalid instruction"	
61733	"Internal error: Floating_Point_Error"	
61734	"Tool point position not compatible with cutting direction"	

© Siemens AG 2004 All rights reserved.

Alarm number	Alarm text	Explanation, remedy
61735	"Finished part lies outside blank contour"	Check definition of blank contour
61736	"Tool insert length < machining depth"	
61737	"Machining_Depth_Of_Cut > Max._Tool_Cutting_Depth"	
61738	"Machining_Cutting _Depth < Min._Tool_Cutting_Depth"	
61739	"Incorrect position of tool for this type of machining"	
61740	"Blank must be a closed contour"	Blank contour must be closed, starting point = end point
61741	"Out of memory"	
61742	"Collision during approach, offset not possible"	

■

© Siemens AG 2004 All rights reserved.

Notes

© Siemens AG 2004 All rights reserved.

Error Messages and Error Handling

5.1 General notes 5-342
5.2 Error handling in cycles 5-342
5.3 Overview of cycle alarms 5-343
5.4 Cycle messages 5-349

© Siemens AG 2004 All rights reserved.

5.1 General notes

If error conditions are detected in the cycles, an alarm is output and execution of the cycle is aborted.
The cycles continue to output messages in the dialog line of the control. These messages do not interrupt execution.



For more information on errors and required responses as well as messages output in the control's dialog line, please refer to the section for the relevant cycle.

5.2 Error handling in cycles

If error conditions are detected in the cycles, an alarm is output and processing is aborted.
Alarms with numbers between 61000 and 62999 are output in the cycles. This range is again subdivided according to alarm responses and acknowledgment criteria.
The text displayed with the alarm number provides an explanation of the cause of the error.

Alarm number	Delete criterion	Alarm responses
61000 ... 61999	NC_RESET	Block preprocessing in the NC is aborted
62000 ... 62999	Backspace key	Program execution is not interrupted; display only

© Siemens AG 2004 All rights reserved.

5.3 Overview of cycle alarms

The alarm numbers are classified as follows:

6	_	X	_	_

- X=0 General cycle alarms
- X=1 Drilling, drilling pattern and cycle alarms
- X=6 Turning cycle alarms

The table below lists the errors that occur in the cycles, when they occur and how to eliminate them.

Alarm number	Alarm text	Source	Explanation, remedy
61000	"No tool offset active"	LONGHOLE SLOT1 SLOT2 POCKET1 to POCKET4 CYCLE71 CYCLE72 CYCLE90 CYCLE93 to CYCLE96	D-correction must be programmed before the cycle call
61001	"Thread lead incorrect"	CYCLE84 CYCLE840 CYCLE96 CYCLE97	Check parameters for thread size and check pitch information (contradict each other)
61002	"Machining type incorrectly defined"	SLOT1 SLOT2 POCKET1 to POCKET4 CYCLE71 CYCLE72 CYCLE76 CYCLE77 CYCLE93 CYCLE95 CYCLE97 CYCLE98	The value assigned to parameter VARI for the machining type is incorrect and must be altered

© Siemens AG 2004 All rights reserved.

Alarm number	Alarm text	Source	Explanation, remedy
61003	"No feedrate programmed in cycle"	CYCLE71 CYCLE72	The parameter for feedrate has been incorrectly set and must be altered.
61005	“3. geometry axis not available"	CYCLE86	With an application on the lathe with no Y-axis in the G18 plane.
61009	"Active tool number = 0"	CYCLE71 CYCLE72	No tool (T) has been programmed before the cycle call.
61010	"Finishing allowance too large"	CYCLE72	The final machining allowance on the base is greater than the total depth and must be reduced.
61011	"Scaling not allowed"	CYCLE71 CYCLE72	A scale factor is currently active that is not permissible for this cycle.
61012	"Different scales in one plane"	CYCLE76 CYCLE77	
61101	"Reference plane incorrectly defined"	CYCLE71 CYCLE72 CYCLE81 to CYCLE90 CYCLE840 SLOT1 SLOT2 POCKET1 to POCKET4 LONGHOLE	Either different values must be entered for the reference plane and the retraction plane if they are relative values or an absolute value must be entered for the depth
61102	"No spindle direction programmed"	CYCLE86 CYCLE87 CYCLE88 CYCLE840 POCKET3 POCKET4	Parameter SDIR (or SDR in CYCLE840) must be programmed
61103	"Number of holes is zero"	HOLES1 HOLES2	No value has been programmed for the number of holes
61104	"Contour violation of the slots/elongated holes"	SLOT1 SLOT2 LONGHOLE	Incorrect parameterization of the milling pattern in the parameters that define the position of the slots/elongated holes in the cycle and their shape
61105	"Cutter radius too large"	SLOT1 SLOT2 POCKET1 to POCKET4 LONGHOLE CYCLE90	The diameter of the milling cutter being used is too large for the figure that is to be machined; either a tool with a smaller radius must be used or the contour must be changed

© Siemens AG 2004 All rights reserved.

Alarm number	Alarm text	Source	Explanation, remedy
61106	"Number or spacing of circle elements"	HOLES2 LONGHOLE SLOT1 SLOT2	Incorrect parameterization of NUM or INDA, the circular elements cannot be arranged in a full circle
61107	"First drilling depth incorrectly defined"	CYCLE83	First drilling depth is inverted in relation to total drilling depth.
61108	"No valid settings for parameters _RAD1 and _DP1"	POCKET3 POCKET4	Parameters _RAD1 and _DP which define the path for depth infeed have been incorrectly set.
61109	"Parameter _CDIR incorrectly defined"	POCKET3 POCKET4	The value of the parameter for milling direction _CDIR has been incorrectly set and must be altered.
61110	"Finishing allowance on base > infeed depth"	POCKET3 POCKET4	The final machining allowance on the base has been set to a higher value than the maximum depth infeed; either reduce final machining allowance or increase depth infeed.
61111	"Infeed width > tool diameter"	CYCLE71 POCKET3 POCKET4	The programmed infeed width is greater than the diameter of the active tool and must be reduced.
61112	"Tool radius negative"	CYCLE72 CYCLE76 CYCLE77 CYCLE90	The radius of the active tool is negative, the setting must be changed to a positive value.
61113	"Parameter _CRAD for corner radius too large"	POCKET3	The parameter for corner radius _CRAD has been set too high and must be reduced.
61114	"Direction of machining G41/G42 incorrectly defined"	CYCLE72	The machining direction of the cutter radius compensation G41/G42 has been incorrectly set.
61115	"Contour approach or return mode (straight line/circle/plane/ space) incorrectly defined"	CYCLE72	The contour approach or return mode has been incorrectly programmed; check parameter _AS1 or AS2.
61116	"Approach or return travel=0"	CYCLE72	The approach or return travel is set to zero and must be increased; check parameter _LP1 or _LP2.
61117	"Active tool radius <= 0"	CYCLE71 POCKET3 POCKET4	The radius of the active tool is negative or zero and must be altered.
61118	"Length or width = 0"	CYCLE71	The length or width of the milling surface is not permissible; check parameters _LENG and _WID.

© Siemens AG 2004 All rights reserved.

Alarm number	Alarm text	Source	Explanation, remedy
61124	"Infeed width is not programmed"	CYCLE71	A value for the infeed width _MIDA must always be programmed for active simulation without a tool.
61125	"Technology selection in parameter _TECHNO incorrectly defined"	CYCLE84 CYCLE840	Check parameter _TECHNO.
61126	"Thread length too short"	CYCLE840	Program lower spindle speed/raise reference plane
61127	"Speed ratio of tapping axis incorrectly defined (machine data)"	CYCLE84 CYCLE840	Check machine data 31050 and 31060 in the appropriate gear stage of the tapping axis
61128	"Plunge angle=0 for insertion with oscillation or helix"	SLOT1	Check parameter _STA2
61180	"No name assigned to swivel data record even though machine data $MN_MM_NUM_TOOL_ CARRIER > 1"	CYCLE800	No name assigned to swivel data record even though several swivel data records exist ($MN_MM_NUM_TOOL_CARRIER>0) or no swivel data record defined ($MN_MM_NUM_TOOL_CARRIER=0)
61181	"NCK software version too old (no TOOLCARRIER functionality)"	CYCLE800	TOOLCARRIER functionality as from NCU 6.3xx
61182	"Name of swivel data record unknown"	CYCLE800	See Swivel cycle start-up CYCLE800 → Kinematics Name (swivel data record)
61183	"Retraction mode GUD7 _TC_FR outside value range 0..2"	CYCLE800	See Swivel cycle start-up CYCLE800 → Retraction; 1st transfer parameter CYCLE800(x,...) is faulty >2
61184	"No solution can be found with current angle inputs"	CYCLE800	
61185	"No or incorrect (min>max) angle ranges declared for rotary axes"	CYCLE800	Check start-up of swivel cycle CYCLE800
61186	"Invalid rotary axis vectors"	CYCLE800	Start-up of swivel cycle CYCLE800: missing or incorrect entry for rotary axis vector V1 or V2
61187	"Block search computation end of block not valid for SWIVEL"	CYCLE800	Select block search with calculation contour

© Siemens AG 2004 All rights reserved.

Alarm number	Alarm text	Source	Explanation, remedy
61188	"No axis name 1st rotary axis declared"		Start-up of swivel cycle CYCLE800: missing entry under rotary axis 1 identifier
61191	"5-axis transformation not set up"	CYCLE832	
61192	"Second 5-axis transformation not set up"	CYCLE832	
61193	"Compressor option not set up"	CYCLE832	
61194	"Spline interpolation option not set up"	CYCLE832	
61200	"Too many elements in machining block"	CYCLE76 CYCLE77	Revise machining block, if necessary deleting elements
61213	"Circle radius too small"	CYCLE77	Enter a larger value for the circle radius.
61215	"Blank dimension incorrectly programmed"	CYCLE76 CYCLE77	Check dimensions of blank spigot. The blank spigot must be larger than the finished spigot.
61601	"Finished part diameter too small"	CYCLE94 CYCLE96	A finished part diameter has been programmed
61602	"Tool width incorrectly defined"	CYCLE93	Parting tool is larger than programmed groove width
61603	"Groove shape incorrectly defined"	CYCLE93	• Radii/chamfers at the groove base do not match the groove width • Face groove on a contour element running parallel to the longitudinal axis is not possible.
61604	"Active tool is in violation of programmed contour"	CYCLE95	Contour violation in relief cut elements as a result of the clearance angle of the tool being used, i.e. use a different tool or check the contour subroutine
61605	"Contour incorrectly programmed"	CYCLE76 CYCLE77 CYCLE95	Illegal relief cut element detected
61606	"Error on contour preparation"	CYCLE95	An error was detected during contour preparation, this alarm is always output with NCK alarm 10930 ... 10934, 15800 or 15810
61607	"Starting point incorrectly programmed"	CYCLE95	The starting point reached before the cycle was called does not lie outside the rectangle described by the contour subroutine
61608	"Wrong tool point direction programmed"	CYCLE94 CYCLE96	A tool point direction between 1 ... 4 that matches the undercut form must be programmed

© Siemens AG 2004 All rights reserved.

Alarm number	Alarm text	Source	Explanation, remedy
61609	"Form incorrectly programmed"	CYCLE94 CYCLE96 LONGHOLE POCKET3 SLOT1	Check parameter for undercut form or shape of slot or pocket.
61610	"No infeed depth programmed"	CYCLE76 CYCLE77 CYCLE96	
61611	"No intersection found"	CYCLE95	No intersection could be calculated with the contour. Check contour programming or change infeed depth
61612	"Thread cannot be recut"	CYCLE97 CYCLE98	
61613	"Undercut position incorrectly defined"	CYCLE94 CYCLE96	Check value in parameter _VARI
61803	"Programmed axis does not exist"	CYCLE83 CYCLE84 CYCLE840	Check parameter _AXN
61807	"Incorrect spindle direction programmed (active)"	CYCLE840	Check parameters SDR and SDAC
62100	"No drilling cycle active"	HOLES1 HOLES2	No modal drilling cycle has been called before the drilling pattern cycle.
62101	"Milling direction incorrect – G3 will be generated"	POCKET3 POCKET4 SLOT2	Check value in parameter CDIR
62105	"Number of columns or rows is zero"	CYCLE800	
62180	"Set rotary axes x.x [deg]"	CYCLE800	Angles to be set for manual rotary axes
62181	"Set rotary axes x.x [deg]"	CYCLE800	Angle to be set for manual rotary axis

© Siemens AG 2004 All rights reserved.

5.4 Cycle messages

The cycles output messages in the dialog line of the control. These messages do not interrupt execution. They provide information about specific cycle behavior and how machining is progressing and are usually displayed for the duration of the machining operation or until the end of the cycle. The following messages can be displayed:

Message text	Source
"Depth: According to value for relative depth"	CYCLE81...CYCLE89, CYCLE840
"Machining elongated hole"	LONGHOLE
"Machining slot"	SLOT1
"Machining circumferential slot"	SLOT2
"Wrong mill direction: G3 will be generated"	SLOT1, SLOT2, POCKET1, POCKET2, CYCLE90
"Changed form of the undercut"	CYCLE94, CYCLE96
"1. Drilling depth: According to value for relative depth"	CYCLE83
"Caution: Final machining allowance ≥ tool diameter!"	POCKET1, POCKET2
"Thread: - longitudinal thread machining"	CYCLE97, CYCLE98
"Thread: - transversal thread machining"	CYCLE97, CYCLE98
"Simulation active, no tool programmed, final contour being traversed"	POCKET1...POCKET4, SLOT1, SLOT2, CYCLE93, CYCLE72
"Simulation active, no tool programmed"	CYCLE71, CYCLE90, CYCLE94, CYCLE96
"Waiting for spindle reversal"	CYCLE840

■

© Siemens AG 2004 All rights reserved.

Notes

© Siemens AG 2004 All rights reserved.

Appendix

A Abbreviations A-352

B Definition of Terms A-361

C References A-381

D Identifiers A-393

E Index I-397

F Commands, identifiers I-399

© Siemens AG 2004 All rights reserved.

A Abbreviations

µC	Micro Controller
AS	Automation System
ASCII	American Standard Code for Information Interchange: American coding standard for the exchange of information
ASIC	Application Specific Integrated Circuit: User switching circuit
ASUB	Asynchronous SUbroutine
AuxF	Auxiliary Function
AV	Job planning
BA	Operating mode
BAG	mode group
BB	Ready
BCD	Binary Coded Decimals: decimals number coded in binary format
BCS	Basic Coordinate System
BIN	Binärdateien (**Bin**ary Files)
BIOS	Basic Input Output System
BOT	BoOT files: boot files for SIMODRIVE 611D
BP	Basic Program
C Bus	Communications bus
CAD	Computer-Aided Design
CAM	Computer-Aided Manufacturing
CNC	Computerized Numerical Control

© Siemens AG 2004 All rights reserved.

COM	Communication
COR	Coordinate Rotation
CP	Communications Processor
CPU	Central Processing Unit: Zentrale Rechnereinheit
CR	Carriage Return
CRC	Cutter Radius Compensation
CRT	Cathode Ray Tube: picture tube
CSB	Central Service Board: PLC module
CSF	Control System Flowchart (programming method for PLC)
CTS	Clear To Send: Signal from serial data interfaces
CUTOM	CUTter radius cOMpensation: Tool radius compensation
DAC	Digital-to-Analog Converter
DB	Data Block in the PLC
DBB	Data Block Byte in the PLC
DBW	Data Block Word in the PLC
DBX	Data Block bit in the PLC
DC	Direct Control: Movement of the rotary axis across the shortest path to the absolute position within one revolution
DCD	Data Carrier Detect
DDE	Dynamic Data Exchange
DIN	Deutsche Industrie Norm (German Industry Standard)
DIO	Data Input/Output: data transfer display
DIR	Directory: Directory
DLL	Dynamic Link Library

© Siemens AG 2004 All rights reserved.

DOE	Data Transmission Equipment
DOS	Disk Operating System
DPM	Dual-Port Memory
DPR	Dual-Port RAM
DRAM	Dynamic Random Access Memory
DRF	Differential Resolver Function: (handwheel)
DRY	Dry Run: Dry run feedrate
DSB	Decoding Single Block: Dekodierungseinzelsatz
DTE	Data Terminal Equipment
DW	Data word
EIA Code	Special punched tape code, number of holes per character always odd
ENC	ENCoder actual-value sensor
EPROM	Erasable Programmable Read Only Memory
ERROR	ERROR from printer
FB	Function Block
FBS	Slimline screen
FC	Function Call: Function block on the PLC
FDB	Product database
FDD	Floppy Disk Drive
FEPROM	Flash-EPROM: read and write memory
FIFO	First In first Out: Memory which works without address specification and whose data are read in the same order in which they were stored.
FIPO	Fine InterPOlator

© Siemens AG 2004 All rights reserved.

FM	Function Module
FPU	Floating Point Unit Gleitpunkteinheit
FRA	Frame block
FRAME	Data block (frame)
FSD	Feed Drive
FST	Feed stop: Feed stop
GUD	Global User Data: Globale Anwenderdaten
HD	Hard Disk Festplatte
HEX	HEXadecimal number
HHU	Handheld Unit
HMS	Hochauflösendes Messsystem
HW	Hardware
I	Input
I/O	Input/Output
I/RF	Power feed/return converter unit on the SIMODRIVE 611(D)
IBN	Startup
IF	Pulse enable of drive module
IK (GD)	Implicit communication (global data)
IKA	Interpolative Compensation: Interpolatory compensation
IM	Interface Module Anschaltungsbaugruppe
IMR	Interface Module Receive: interface module for receiving data
IMS	Interface Module Send: interface module for transmitting data
INC	Increment: Increment

© Siemens AG 2004 All rights reserved.

INI	Initializing Data: Initializing data
IPO	Interpolator
IS	Interface Signal
ISA	Industry Standard Architecture
ISO	International Standardization Organization
ISO code	Special punched tape code, number of holes per character always even
JOG	Jogging: Setup mode
K1 .. K4	Channel 1 to channel 4
$K_{ü}$	Transmission ratio
K_v	Servo gain factor
LAD	LADder logic (programming method for PLC)
LCD	Liquid Crystal Display Flüssigkristallanzeige
LEC	Leadscrew Error Compensation
LED	Light Emitting Diode: Leuchtdiodenanzeige
LF	Line Feed
LR	Position controller
LUD	Local User Data
MB	Megabyte
MC	Measuring Circuit
MCP	Machine control panel
MCS	Machine coordinate system
MD	Machine Data
MDA	Manual Data Automatic: manual input

© Siemens AG 2004 All rights reserved.

MMC	Man-Machine Communication: user interface on numerical control systems for operator control, programming and simulation
MPF	Main Program File: NC part program (main program)
MPI	Multiport Interface
MS	MicroSoft (software manufacturer)
MSD	Main Spindle Drive
NC	Numerical Control
NCK	Numerical Control Kernel: NC kernel with block preparation, traversing range, etc.
NCU	Numerical Control Unit: NCK hardware unit
NRK	Numeric Robotic Kernel: name of NCK operating System
NURBS	Non-Uniform Rational B-Spline
O	Output
OB	Organization Block in the PLC
OC&M	Operator control and monitoring
OEM	Original Equipment Manufacturer
OI	User interface
OP	Operator Panel
OPI	Operator Panel Interface
OPT	Options
Order No.	Machine-readable product designation
OSI	Open Systems Interconnection: standardization for computer communication
P bus	I/O (peripherals) bus
PC	Personal Computer

© Siemens AG 2004 All rights reserved.

PCIN	Name of SW for data exchange with the control
PCMCIA	Personal Computer Memory Card International Association Memory plug-in board normalization
PG	Programming device
PLC	Programmable Logic Control: programmable controller
PLC	Programmable Logic Controller
PMS	Position measuring system
POS	POSitioning
RAM	Random Access Memory: program memory that can be read and written to
REF	REFerence point approach function
REPOS	REPOSition function
RISC	Reduced Instruction Set Computer: type of processor with small instruction set and ability to process instructions at high speed
ROV	Rapid Override: Eingangskorrektur
RPA	R-Parameter Active: memory area in for R-NCK for R parameter numbers
RPY	Roll Pitch Yaw: type of rotation of a coordinate system
RS-232-C	Serial interface (definition of the exchange lines between DTE and DCE)
RTS	Request To Send: RTS, control signal of serialdata interfaces
SBL	Single Block: Single block
SD	Setting Data
SDB	System Data Block
SEA	Setting Data Active: identifier (file type) for setting data
SFB	System Function Block

© Siemens AG 2004 All rights reserved.

SFC	System Function Call
SK	Soft Key
SKP	SKiP: Skip block
SM	Stepper motor
SPF	Sub Program File: Subprogram
SR	Subprogram
SRAM	Static RAM (non-volatile)
SSI	Serial Synchronous Interface Serielle synchrone Schnittstelle
STL	STatement List
SW	Software
SYF	System Files System files
T	Tool T
TC	Tool change
TEA	Testing Data Active: identifier for machine data
TLC	Tool length compensation
TNRC	Tool Nose Radius Compensation
TO	Tool Offset: Tool Compensation
TO	Tool Compensation
TOA	Tool Offset Active identifier (file type) for tool offsets
TRANSMIT	TRANSform Milling Into Turning: Coordinate conversion on turning machine for milling operations
TRC	Tool Radius Compensation
UFR	User Frame: Work offset
WCS	Workpiece coordinate system

© Siemens AG 2004 All rights reserved.

WOP	Workshop-Oriented Programming
WPD	Workpiece Directory: Workpiece directory
ZO	Work offset
ZOA	Zero Offset Active: identifier (file type) for zero offset data

© Siemens AG 2004 All rights reserved.

B Definition of Terms

Important terms are listed in alphabetical order. The symbol "->" precedes terms which are explained under a separate entry in this list

A

Access authorization

The CNC program blocks and data are protected by a 7-level system of access restrictions:

- Three password levels for system manufacturers, machine manufacturers and users and
- Four keyswitch settings which can be evaluated via the PLC.

Alarms

All -> messages and alarms are displayed on the operator panel in plain text with date and time as well as the appropriate symbol for the reset criterion. Alarms and messages are displayed separately.

1. Alarms and messages in the parts program
   Alarms and messages can be displayed directly from the parts program in plain text.
2. Alarms and messages from PLC
   Alarms and messages relating to the machine can be displayed directly from
   the PLC program in plain text. No additional function block packages are required for this purpose.

Approach fixed point

Machine tools can approach fixed points such as a tool change point, loading point, pallet change point etc. in a defined way. The coordinates of these points are stored in the control. Where possible, the control moves these axes in -> rapid traverse.

Approach machine fixed-point

Approach motion towards one of the predefined -> fixed machine points.

Auxiliary functions

Auxiliary functions can be used to pass -> parameters to the -> PLC in -> parts programs, triggering reactions there which are defined by the machine manufacturer.

© Siemens AG 2004 All rights reserved.

B

Backlash compensation Compensation of a mechanical machine backlash, e.g. backlash due to reversal of leadscrews. The backlash compensation can be entered separately for each axis.

Backup battery The backup battery provides non-volatile storage for the -> user program in the -> CPU and ensures that defined data areas and flags, timers and counters are retentive.

Blank Workpiece as it is before the part is machined.

Block "Module" is the term given to any files required for creating and processing programs.

Block A section of a -> part program terminated with a line feed. A distinction is made between -> main blocks and -> subblocks.

Block search The block search function allows selection of any point in the parts program at which machining must start or be continued. The function is provided for the purpose of testing parts programs or continuing machining after an interruption.

Booting Loading the system program after power ON.

C

Channel A channel can execute a -> parts program independently of the other channels. A channel exclusively controls the axes and spindles assigned to it. Part programs runs of various channels can be coordinated by -> synchronization.

Channel structure The channel structure makes it possible to process the -> programs of individual channels simultaneously and asynchronously.

Circular interpolation The -> tool is required to travel in a circle between defined points on the contour at a specified feed while machining the workpiece.

CNC ➔ NC

CNC high-level language The high-level language offers: -> User variables, -> Predefined user variables, -> System variables, -> Indirect programming, -> arithmetic and trigonometric functions, -> Comparison operations and logic operations, -> Program branches and jumps, -> Program coordination (SINUMERIK 840D), -> Macros.

© Siemens AG 2004 All rights reserved.

CNC programming language	The CNC programming language is based on DIN 66025 with high-level language expansions. The -> CNC programming language and -> high-level language expansions support the definition of macros (sequenced statements).
COM	Component of the NC control for the implementation and coordination of communication.
Compensation axis	Axis having a setpoint or actual value modified by the compensation value.
Compensation table	Table containing interpolation points It provides the compensation values of the compensation axis for selected positions on the basic axis.
Compensation value	Difference between the axis position measured by the position sensor and the desired, programmed axis position.
Concurrent positioning axis	Axis that performs an auxiliary movement on a machine tool. (e.g. tool magazine, pallet transport). Positioning axes are axes that do not interpolate with the -> path axes.
Contour	Outline of a -> workpiece
Contour monitoring	The following error is monitored within a definable tolerance band as a measure of contour accuracy. Overloading of the drive, for example, may result in an unacceptably large following error. In such cases, an alarm is output and the axes are stopped.
Coordinate system	See -> machine coordinate system, -> workpiece coordinate system
CPU	Central Processor Unit, -> PLC
Cycle	Protected subroutine for executing a machining process that is repeated on the -> workpiece
Cycle support	The available cycles are listed in menu "Cycle support" in the "Program" operating area. Once the desired machining cycle has been selected, the parameters required for assigning values are displayed in plain text.

© Siemens AG 2004 All rights reserved.

D

Data block

1. A data unit on the -> PLC which can be accessed by -> HIGHSTEP programs.
2. A data unit on the -> NC: Data modules contain data definitions for global user data. These data can be initialized directly when they are defined.

Data transfer program PCIN

PCIN is an auxiliary program for transmitting and receiving CNC user data, e.g. parts programs, tool offsets, etc. via the serial interface. The PCIN program can run under MS-DOS on standard industrial PCs.

Data word

A data unit, two bytes in size, within a -> data block.

Diagnostics

1. Operating area of the control.
2. The control has both a self-diagnosis program as well as test functions for servicing purposes: status, alarm and service displays.

Digital input/output module

Digital modules are signal transducers for binary process signals.

Dimensions specification, metric and inches

Position and lead values can be programmed in inches in the machining program. The control is set to a basic system regardless of the programmable dimensional specification (G70/G71).

DRF

Differential Resolver Function: NC function which generates an incremental zero offset in AUTOMATIC mode in conjunction with an electronic handwheel.

Drift compensation

When the CNC axes are in the constant motion phase, automatic drift compensation is implemented in the analog speed control. (SINUMERIK FM-NC).

E

Editor

The editor makes it possible to create, modify, extend, join and import programs/texts/program blocks.

Electronic handwheel

Electronic handwheels can be used to traverse the selected axes simultaneously in manual mode. The handwheel clicks are analyzed by the increment analyzer.

Exact stop

With a programmed exact stop instruction, the position stated in a block is approached precisely and very slowly, if necessary. In order to reduce the approach time, -> exact stop limits are defined for rapid traverse and feed.

© Siemens AG 2004 All rights reserved.

Exact stop limit
When all path axes reach their exact stop limits, the control responds as if it had reached its destination point precisely. The -> parts program continues execution at the next block.

External zero offset
A zero offset specified by the -> PLC.

F

Fast retraction from contour
When an interrupt is received, it is possible to initiate a motion via the CNC machining program which allows the tool to be retracted quickly from the workpiece contour currently being machined. The retraction angle and the distance retracted can also be parameterized. An interrupt routine can be executed after the rapid retraction. (SINUMERIK FM-NC, 840D).

Feedforward control, dynamic
Contour inaccuracies resulting from following errors can be almost completely eliminated by the dynamic, acceleration-dependent feedforward control function. Feedforward control ensures an excellent degree of machining accuracy even at high tool path velocities. The function can be selected and deselected axis-specifically in the part program.

Feedrate override
The current feedrate setting entered via the control panel or by the PLC is overlaid on the programmed feedrate (0-200 %). The feedrate can also be corrected by a programmable percentage factor (1-200%) in the machining program.

Finished-part contour
Contour of the finished workpiece. See also -> Blank.

Fixed machine point
A point defined uniquely by the machine tool, such as the reference point.

Frame
A frame is an arithmetic rule that transforms one Cartesian coordinate system into another Cartesian coordinate system. A frame contains the components -> zero offset, -> rotation, -> scaling, -> mirroring.

G

General reset
The following memories of the -> CPU are erased by a general reset operation:

- -> Working memory
- Read/write area of the -> load memory
- -> System memory
- -> Backup memory

© Siemens AG 2004 All rights reserved.

Geometry Description of a -> workpiece in the -> workpiece coordinate system.

Geometry axis Geometry axes are for describing a 2- or 3-dimensional range in the workpiece coordinate system.

Global main program/subroutine Each global main program/subroutine may appear only once under its name in the directory. It is not possible to use the same program name in different directories with different contents as a global program.

Ground "Ground" is the term applied to all the electrically inactive, interconnected parts of a piece of equipment which cannot carry any hazardous contact voltage even in the event of a fault.

H

Helical interpolation The helical interpolation function is ideal for machining internal and external threads using form milling cutters and for milling lubrication grooves. The helix comprises two movements:
1. Circular movement in one plane
2. Linear movement perpendicular to this plane.

High-speed digital inputs/outputs As an example, high-speed CNC program routines (interrupt routines) can be started via the digital inputs. High-speed, program-driven switching functions can be initiated via the digital CNC outputs. (SINUMERIK 840D).

HIGHSTEP Combination of the programming features for the -> PLC in the S7-300/400 range.

I

I/O module I/O modules create the link between the CPU and the process. I/O modules are:
- ->Digital input/output modules
- ->Analog input/output modules
- ->Simulator modules

Inch system Measurement system which defines distances in "inches" and fractions of inches.

Increment Travel path length specification based on number of increments. The number of increments can be stored as a -> setting data or selected with keys labeled with 10, 100, 1000, 10 000.

© Siemens AG 2004 All rights reserved.

Incremental dimension A destination for axis traversal is defined by a distance to be covered and a direction referenced to a point already reached. See also -> absolute dimension.

Initialization block Initialization blocks are special -> program blocks. They contain value assignments that are performed before program execution. Initialization blocks are used primarily for initializing predefined data or global user data.

Initialization files An initialization file can be created for each -> workpiece. In it, the various variable value instructions which apply exclusively to one workpiece can be stored.

Interconnecting cables Connecting cables are pre-assembled or user-assembled 2-wire cables with a connector at each end. They are used to connect the -> CPU via the -> multipoint interface (MPI) to a -> programming device or to other CPUs.

Intermediate blocks Movements with selected tool offset (G41/G42) can be interrupted by a limited number of intermediate blocks (blocks without axis motions in the offset plane). When such blocks are used, the tool offset can still be calculated correctly. The permissible number of intermediate blocks read in advance by the control can be set via system parameters.

Interpolator Logical unit of the -> NCK which determines intermediate values for the movements to be traversed on the individual axes on the basis of destination positions specified in the parts program.

Interpolatory compensation Interpolative compensation provides a means of compensating for leadscrew errors (LEC) and measuring-system errors (MSEC) resulting from the production process.

interrupt routine Interrupt routines are special -> subroutines which can be started by events (external signals) in the machining process. The parts program block being processed is aborted and the axis position at the instant of interruption is stored automatically.

Inverse time feedrate On SINUMERIK FM-NC and 840D controls, it is possible to program the time required to traverse the path of a block instead of the feedrate speed for the axis movement (G93).

© Siemens AG 2004 All rights reserved.

J

JOG

Control operating mode (setup operation): The machine can be set up in the JOG mode. Individual axes and spindle can be traversed in jog mode by means of the direction keys. Other functions in Jog mode are -> reference point approach, -> Repos and -> Preset -> (set actual value).

K

Keyswitch

1. **S7-300**: The keyswitch is the mode selector switch on the -> CPU. The keyswitch is operated by means of a removable key.
2. **840D/FM-NC**: The keyswitch on the -> machine control panel has 4 positions that are assigned functions by the operating system of the control. Further, the keyswitch has three different colored keys which can be removed in the specified positions:

$K_{Ü}$

Transmission ratio

K_v

Servo gain factor, a control variable in a control loop.

L

Language

The operator-prompt display texts, system messages and system alarms are available (on diskette) in five system languages:
German, English, French, Italian, and **Spanish**.
The user can select **two** of the listed languages at a time in the control.

Leadscrew error compensation.

Compensation for the mechanical inaccuracies of a leadscrew participating in the feed. The control uses stored deviation values for the compensation.

Limit speed

Minimum/maximum (spindle) speed: The maximum speed of a spindle can be limited by values defined in the machine data, the -> PLC or -> setting data.

Line

A network is the interconnection of several S7-300s and other terminal devices such as a programming device, for example, interlinked by means of -> connecting cables. The networked devices interchange data via the network.

Linear axis

The linear axis is an axis which, in contrast to a rotary axis, describes a straight line.

Linear interpolation

The tool travels along a straight line to the destination point while machining the workpiece.

© Siemens AG 2004 All rights reserved.

Load memory The load memory is the CPU 314 of the -> PLC equivalent of the -> Working memory.

Look Ahead The **Look Ahead** function is a means of optimizing the machining velocity by looking ahead over a parameterizable number of traversing blocks.

Look Ahead for contour violations The control detects and reports the following types of collision:
1. Path is shorter than tool radius.
2. Width of inside corner is less than the tool diameter.

M

Machine Operating area of the control.

Machine axes Physically existent axes on the machine tool.

Machine control panel An operator panel on a machine tool with operating elements such as keys, rotary switches, etc. and simple indicators such as LEDs. It is used to directly influence the machine tool via the PLC.

Machine coordinate system A coordinate system which is related to the axes of the machine tool.

Machine zero A fixed point on the machine tool which can be referenced by all (derived) measuring systems.

Macro techniques Grouping set of instructions under a single identifier. The identifier in the program refers to the collected sequence of instructions.

Main block A block prefixed by ":" containing all the parameters required to start execution of a -> parts program.

Main program -> Parts program identified by a number or name in which other main programs, subroutines or -> cycles may be called.

MDA Control operating mode: Manual Data Automatic: In the MDA mode, individual program blocks or block sequences with no reference to a main program or subroutine can be input and executed immediately afterwards through actuation of the NC start key.

© Siemens AG 2004 All rights reserved.

Measuring circuit

- SINUMERIK FM-NC: The requisite control circuits for axes and spindles are integrated in the control module as standard. A maximum total of 4 axes and spindles can be implemented, with no more than 2 spindles.
- SINUMERIK 840D The signals from the sensors are analyzed in the SIMODRIVE 611D drive modules. The maximum configuration is 8 axes and spindles in total, but including no more than 5 spindles.

Messages

All messages programmed in the parts program and -> alarms detected by the system are displayed on the operator panel in plain text with date and time as well as the appropriate symbol for the reset criterion. Alarms and messages are displayed separately.

Metric system

Standardized system of units: for lengths in millimeters (mm), meters (m), etc.

Mirroring

Mirroring swaps round the signs of the coordinate values of a contour with respect to an axis. It is possible to mirror with respect to more than one axis at a time.

Mounting rail

This rail is used to mount the modules of the S7-300 system.

Multipoint interface

The multipoint interface (MPI) is a 9-pin sub-D port. A parameterizable number of devices can be connected to an MPI for the purpose of communicating with one another:

- Programming devices
- MMI (HMI) systems
- Other automation systems

The "Multipoint Interface MPI" parameter block of the CPU contains the -> parameters which define the properties of the multipoint interface.

N

Name of identifier

In accordance with DIN 66025, identifiers (names) for variables (calculation variables, system variables, user variables), for subroutines, for vocabulary words and words can contain several address letters. These supplements have the same meaning as the words with respect to block format. Identifiers must be unambiguous. It is not permissible to use the same identifier for different objects.

© Siemens AG 2004 All rights reserved.

NC Numerical Control: NC control incorporates all the components of the machine tool control system: -> NCK, -> PLC, -> MMC, -> COM. Note: CNC (computerized numerical control) would be a more appropriate description for the SINUMERIK 840D or FM-NC controls. computerized numerical control.

NCK Numerical Control Kernel: Component of the NC control which executes -> parts programs and essentially coordinates the movements on the machine tool.

Node number The node number is the "contact address" of a -> CPU or the -> programming device or another intelligent I/O module if these devices are exchanging data with one another via a -> network. The node number is assigned to the CPU or the programming device by the S7 tool -> "S7 Configuration".

NRK Numeric Robotic Kernel (operating system of the -> NCK)

NURBS Motion control and path interpolation are implemented internally in the control on the basis of NURBS (Non-Uniform Rational B Splines). A standard procedure is thus available (SINUMERIK 840D) as an internal control function for all modes of interpolation.

O

Oblique-plane machining Drilling and milling operations on workpiece surfaces that do not lie in the coordinate planes of the machine can be performed easily using the function "inclined-surface machining".

OEM The scope for implementing individual solutions (OEM applications) for the SINUMERIK 840D has been provided for machine manufacturers who wish to create their own operator interface or integrate process-oriented functions in the control.

Offset memory Data range in the control in which the tool offset data are stored.

Orientable tool retraction RETTOOL: If machining is interrupted (e.g. when a tool breaks), a program command can be used to retract the tool in a user-specified orientation by a defined distance.

Oriented spindle stop Stops the workpiece spindle with a specified orientation angle, e.g. to perform an additional machining operation at a specific position.

Override Manual or programmable control feature which enables the user to override programmed feedrates or speeds in order to adapt them to a specific workpiece or material.

© Siemens AG 2004 All rights reserved.

P

Parameters

1. The **S7-300** uses two types of parameter:
   - Parameter of a STEP 7 instruction
     A parameter of a STEP 7 instruction is the address of the operand to be processed or a constant.
   - Parameter of a -> parameter block
     A parameter of a parameter block determines the behavior of a module
2. **840D**:
   - Control operating area
   - Calculation parameter. The programmer of the parts program can assign or request the values of the R parameter as required.

Parts program management

The parts program management function can be organized according to -> workpieces. The number of programs and data to be managed determine the size of the user memory. Each file (programs and data) can be given a name consisting of a maximum of 24 alphanumeric characters.

Parts programs

A sequence of instructions to the NC control which combine to produce a specific -> workpiece by performing certain machining operation on a given -> blank Likewise
Performing a certain machining operation on a specific -> blank.

PG

Programming device

PLC

Programmable Logic Control: -> Programmable logic control. Component of the -> NC control: Programmable controller for processing the control logic of the machine tool.

PLC

➔ Programmable Logic Controller

PLC program memory

- SINUMERIK FM-NC: The PLC user program, the user data and the basic PLC program are stored together in the PLC user memory on the CPU 314. The
  S7-CPU314 has a 24 KB user memory for this purpose.
- SINUMERIK 840D The PLC user program, the user data and the basic PLC program are stored together in the PLC user memory. The PLC user memory can be expanded up to 96 KB with memory expansions.

PLC programming

The PLC is programmed with the **STEP 7** software. The STEP 7 programming software is based on the standard **WINDOWS** operating system and incorporates the functionality of STEP5 programming with innovative expansions and developments.

© Siemens AG 2004 All rights reserved.

Polar coordinates
A coordinate system which defines the position of a point on a plane in terms of its distance from the origin and the angle formed by the radius vector with a defined axis.

Polynominal interpolation
Polynomial interpolation provides a means of generating a very wide range of curves, including **straight-line, parabolic and exponential** functions (SINUMERIK 840D).

Power On
Control is switched off and then switched on again.

Preset
The control zero point can be redefined in the machine coordinate system by means of the Preset function. Preset does not cause the axes to move; instead, a new position value is entered for the current axis positions.

Program
1. Operating area of the control.
2. Sequence of instructions to the control.

Program block
Program blocks contain the main and subprograms of the -> Part programs.

Programmable frames
Programmable -> frames can be used to define new coordinate system starting points dynamically while the parts program is running. A distinction is made between absolute definition using a new frame and additive definition with reference to an existing starting point.

Programmable logic controller
Programmable logic controllers (PLC) are electronic controllers whose functions are stored as a program in the control unit. The design and wiring of the unit are not, therefore, dependent on the control functions. Programmable logic controllers have the same structure as a computer, i.e. they consist of a CPU with memory, input/output modules and an internal bus system. The I/Os and programming language are selected according to the requirements of the control technology involved.

Programmable working area limitation
Limitation of the motion space of the tool to a space defined by programmed limitations.

Programming key
Characters and character sequences which have a defined meaning in the programming language
for -> parts programs (see Programming Guide).

Protection zone
Three-dimensional area within a -> working area which the tool tip is not permitted to enter.

© Siemens AG 2004 All rights reserved.

Q

Quadrant error compensation

Contour errors on quadrant transitions caused by frictional fluctuations on guideways can be largely eliminated by means of quadrant error compensation. A circularity test is performed to parameterize the quadrant error compensation function.

R

R parameters

Calculation parameter. The programmer of the -> parts program can assign or request the values of the R parameter as required.

Rapid traverse

The highest speed of an axis. It is used to move the tool from rest to the -> workpiece contour or retract the tool from the contour.

Reference point

A point on the machine tool to which the measurement system of the -> machine axis refers.

Reference Point Approach

If the system of position measuring employed is not based on absolute-value encoders, the control must perform a reference point approach in order to ensure that the measured values supplied by the measuring system are in harmony with the machine coordinate values.

Rententive data

Data areas in data blocks and timers, counters, and markers are retentive if their content is not lost on restart or power-off.

REPOS

1. Reapproach contour, triggered by operator
   REPOS allows the tool to be returned to the interrupt position by means of the direction keys.

2. Programmed contour reapproach
   A selection of approach strategies are available in the form of program commands: Approach point of interruption, approach start of block, approach end of block, approach a point on the path between start of block and interruption.

Rigid tapping

Rigid tapping can be drilled with the help of this function. When the rigid tapping function is used, interpolation of the spindle acting as a rotary axis and the drilling axis ensures that threads are cut exactly to the end of the drilling depth, e.g. tapped blind hole (precondition: spindle is operating in axis mode).

Rotary axis

Rotary axes apply a workpiece or tool rotation to a defined angular position.

© Siemens AG 2004 All rights reserved.

Rotary axis, continuously turning

Depending on the application, the traversing range of a rotary axis can be selected to be limited to less than 360 degrees or to be endlessly turning in both directions. Continuously turning rotary axes are used, for example, for eccentric machining, grinding and winding.

Rotation

Component of a -> frame which defines a rotation of the coordinate system through a specific angle.

Rounding axis

Rounding axes apply a workpiece or tool rotation to an angular position corresponding to an indexing grid. On reaching a grid, the rounding axis is "in position".

S

S7 Configuration

S7 Configuration is a tool for parameterizing modules. S7 Configuration is used to set a variety of
-> parameter blocks of the -> CPU and the I/O modules on the
-> programming device. These parameters are uploaded to the CPU.

S7-300 bus

The S7-300 bus is a serial data bus which supplies modules with the appropriate voltage and via which they exchange data with one another. The connection between the modules is made by means of -> bus connectors.

Safety functions

The control incorporates monitors which are active at all times and which are designed to detect malfunctions in the -> CNC, the programmable controller (-> PLC) and the machine at an early stage in order to minimize the risk of damage to the tool, workpiece or machine. In the event of a fault, the machining operation is interrupted and the drives stopped. The cause of the malfunction is logged and an alarm issued. At the same time, the PLC is notified that a CNC alarm has been triggered.

Scaling

Component of a -> frame which causes axis-specific alterations in the scale.

Serial RS-232 interface

For the purpose of data input and output,

- MMC module MMC100 includes a serial RS232C interface
- MMC modules MMC101 and MMC102 have two RS232 interfaces

available. Machining programs and manufacturer and user data can be imported and exported via these interfaces.

© Siemens AG 2004 All rights reserved.

Services Operating area of the control.

Setting data Data the communicates properties of the machine tool to the NC control in a way defined by the system software.

Simulator module A simulator module is a module
- on which digital input quantities can be simulated via operating elements and
- digital output quantities can be displayed

Soft Key A key whose name appears on an area of the screen. The choice of softkeys displayed is adapted dynamically to the operating situation. The freely assignable function keys (softkeys) are assigned defined functions in the software.

Software limit switches Software limit switches define the limits of the travel range of an axis and prevent the slide contacting the hardware limit switches. Two pairs of values can be assigned per axis and activated separately via the -> PLC.

Spindles The spindle functionality is a two-level construct:
1. Spindles: Speed or position controlled spindle drives analog / digital (SINUMERIK 840D)
2. Auxiliary spindles: Speed-controlled spindle drives function package "aux. spindle", e.g. for power tools.

Spline interpolation Using the spline interpolation function, the control is able to generate a smooth curve from just a small number of specified interpolation points along a setpoint contour.

Standard cycles Standard cycles are provided for machining operations which are frequently repeated:
- Cycles for drilling/milling applications
- Cycles for turning applications (SINUMERIK FM-NC)

The available cycles are listed in menu "Cycle support" in the "Program" operating area. Once the desired machining cycle has been selected, the parameters required for assigning values are displayed in plain text.

© Siemens AG 2004 All rights reserved.

Subblock Block prefixed by "N" containing information for a machining step such as a position parameter.

Subprogram A sequence of instructions of a -> parts program which can be called repetitively with various defining parameters. The subroutine is called from a main program. Every subroutine can be protected against unauthorized read-out and display. -> Cycles are a type of subroutine.

Synchronization Instructions in -> parts programs for coordination of the operations in different -> channels at specific machining points.

Synchronized actions

1. Auxiliary function output
   While a workpiece is being machined, technological functions (-> auxiliary functions) can be output from the CNC program to the PLC. These auxiliary functions control, for example, ancillary equipment on the machine tool such as the sleeve, gripper, chuck, etc.
2. High-speed auxiliary function output
   For time-critical switching functions, the acknowledgement times for the -> auxiliary functions can be minimized and unnecessary stops in the machining process can be avoided.

Synchronized axes Synchronized axes take the same time to traverse as the geometry axes for their path.

System memory The system memory is a memory in the CPU in which the following data are stored:

- Data required by the operating system
- The operands timers, counters, markers

System variables A variable which exists although it has not been programmed by the -> parts program programmer. It is defined by a data type and the variable name preceded by the character **$**. See also -> User-defined variable.

© Siemens AG 2004 All rights reserved.

T

Teach In

Teach In is a means of creating or correcting parts programs. The individual program blocks can be input via the keyboard and executed immediately. Positions approached via the direction keys or handwheel can also be stored. Additional information such as G functions, feedrates or
M functions can be entered in the same block.

Text editor

-> Editor

Tool

A tool is a software tool for entering and changing -> parameters of a parameter block. Tools include:

- -> S7 Configuration
- S7-TOP
- S7-Info

Tool Compensation

By programming a **T function** (5 decades, integer) in the block, you can select the tool. Every T number can be assigned up to nine cutting edges (D addresses). The number of tools to be managed in the control is set at the configuration stage.

Tool edge radius compensation

Contour programming assumes that the tool is pointed. Because this is not actually the case in practice, the curvature radius of the tool used must be communicated to the control which then takes it into account. The curvature center is maintained equidistantly around the contour offset by the radius of curvature.

Tool radius compensation

In order to program a desired -> workpiece contour directly, the control must traverse a path equidistant to the programmed contour, taking into account the radius of the tool used. (G41/G42).

Tool T

A part used on the machine tool for machining. Examples of tools include cutting tools, mills, drills, laser beams, etc.

Transformation

Programming in a Cartesian coordinate system, execution in a non-Cartesian coordinate system (e.g. with machine axes as rotary axes).

Traversing range

The maximum permissible travel range for linear axes is ± 9 decades. The absolute value depends on the selected input and position control resolution and the unit of measurement (inch or metric).

© Siemens AG 2004 All rights reserved.

U

User-defined variable

Users can define variables in the -> parts program or data block (global user data) for their own use. A definition contains a data type specification and the variable name. See also -> System variable.

V

Variable definition

A variable definition includes the specification of a data type and a variable name. The variable names can be used to access the value of the variables.

Velocity control

To be able to achieve an acceptable traverse velocity on very short traverse movements, predictive evaluation over several blocks (-> Look Ahead) can be set.

Vocabulary words

Words with a specific notation which have a defined meaning in the programming language for -> parts programs.

W

Workpiece

Part to be made / machined by the machine tool.

Workpiece contour

Setpoint contour of the -> workpiece to be created/machined.

Workpiece coordinate system

The starting position of the workpiece coordinate system is the -> workpiece origin. In machining operations programmed in the workpiece coordinate system, the dimensions and directions refer to this system.

Workpiece zero

The workpiece origin is the starting point for the -> workpiece coordinate system. It is deifned by distance from the machine zero.

© Siemens AG 2004 All rights reserved.

Y

Zero offset

Specification of a new reference point for a coordinate system through reference to an existing origin and a -> frame settable.

1. settable
   SINUMERIK FM-NC: Four independent zero offsets can be selected for each CNC axis.
   SINUMERIK 840D A configurable number of settable zero offsets are available for each CNC axis. The offsets - which are selected by means of G functions - take effect alternately.
2. External
   In addition to all the offsets which define the position of the workpiece zero, it is possible to superimpose an external zero offset
   - by means of a handwheel (DRF offset) or
   - overlaid by the PLC.
3. Programmable
   It is possible to program zero offsets for all path and positioning axes by means of the TRANS statement.

© Siemens AG 2004 All rights reserved.

C References

General Documentation

/BU/ SINUMERIK & SIMODRIVE, Automation Systems for Machine Tools
Catalog NC 60
Order No.: E86060-K4460-A101-A9
Order No.: E86060-K4460-A101-A9-7600 (English)

/IKPI/ Industrial Communication and Field Devices
Catalog IK PI
Order No.: E86060-K6710-A101-B2
Order No.: E86060-K6710-A101-B2-7600 (English)

/ST7/ SIMATIC
Products for Totally Integrated Automation and Micro Automation
Catalog ST 70
Order No.: E86060-K4670-A111-A8
Order No.: E86060-K4670-A111-A8-7600 (English)

/Z/ MOTION-CONNECT
Connections & System Components for SIMATIC, SINUMERIK, MASTERDRIVES, and SIMOTION
Catalog NC Z
Order No.: E86060-K4490-A001-B1
Order No.: E86060-K4490-A001-B1-7600 (English)

Electronic Documentation

/CD1/ The SINUMERIK System (03.04 Edition)
DOC ON CD
(includes all SINUMERIK 840D/840Di/810D/802- and SIMODRIVE publications)
Order No.: 6FC5298-7CA00-0BG0

© Siemens AG 2004 All rights reserved.

User Documentation

/AUK/	SINUMERIK 840D/810D **AutoTurn** Short Operating Guide Order No.: 6FC5298-4AA30-0BP2	(09.99 Edition)
/AUP/	SINUMERIK 840D/810D **AutoTurn Graphic Programming System** Programming / Setup Order No.: 6FC5298-4AA40-0BP3	(02.02 Edition)
/BA/	SINUMERIK 840D/810D **MMC** Operator's Guide Order No.: 6FC5298-6AA00-0BP0	(10.00 Edition)
/BAD/	SINUMERIK 840D/840Di/810D Operator's Guide: **HMI Advanced** Order No.: 6FC5298-6AF00-0BP3	(03.04 Edition)
/BAH/	SINUMERIK 840D/840Di/810D Operator's Guide **HT 6** Order No.: 6FC5298-0AD60-0BP3	(03.04 Edition)
/BAK/	SINUMERIK 840D/840Di/810D **Short Operating Guide** Order No.: 6FC5298-6AA10-0BP0	(02.01 Edition)
/BAM/	SINUMERIK 810D/840D **ManualTurn** Operator's Guide Order No.: 6FC5298-6AD00-0BP0	(08.02 Edition)
/BAS/	SINUMERIK 840D/840Di/810D Operator's Guide **ShopMill** Order No.: 6FC5298-6AD10-0BP2	(11.03 Edition)
/BAT/	SINUMERIK 840D/810D Operator's Guide **ShopTurn** Order No.: 6FC5298-6AD50-0BP2	(06.03 Edition)
/BEM/	SINUMERIK 840D/810D Operator's Guide: **HMI Embedded** Order No.: 6FC5298-6AC00-0BP3	(03.04 Edition)
/BNM/	SINUMERIK 840D840Di//810D User's Guide **Measuring Cycles** Order No.: 6FC5298-6AA70-0BP3	(03.04 Edition)
/BTDI/	SINUMERIK 840D840Di//810D Motion Control Information System (MCIS) User's Guide **Tool Data Information** Order No.: 6FC5297-6AE01-0BP0	(04.03 Edition)
/CAD/	SINUMERIK 840D/840Di/810D Operator's Guide **CAD Reader** Order No.: (part of the online help)	(03.02 Edition)

© Siemens AG 2004 All rights reserved.

/DA/	SINUMERIK 840D/840Di/810D **Diagnostics Guide** Order No.: 6FC5298-6AA20-0BP4	(03.04 Edition)
/KAM/	SINUMERIK 840D/810D Short Guide **ManualTurn** Order No.: 6FC5298-5AD40-0BP0	(04.01 Edition)
/KAS/	SINUMERIK 840D/810D Short Guide **ShopMill** Order No.: 6FC5298-5AD30-0BP0	(04.01 Edition)
/KAT/	SINUMERIK 840D/810D Short Guide **ShopTurn** Order No.: 6FC5298-6AF20-0BP0	(07.01 Edition)
/PG/	SINUMERIK 840D/840Di/810D Programming Guide **Fundamentals** Order No.: 6FC5298-7AB00-0BP0	(03.04 Edition)
/PGA/	SINUMERIK 840D/840Di/810D Programming Guide **Advanced** Order No.: 6FC5298-7AB10-0BP0	(03.04 Edition)
/PGA1/	SINUMERIK 840D/840Di/810D Lists Manual **System Variables** Order No.: 6FC5298-7AE10-0BP0	(03.04 Edition)
/PGK/	SINUMERIK 840D/840Di/810D Short Guide **Programming** Order No.: 6FC5298-7AB30-0BP0	(03.04 Edition)
/PGM/	SINUMERIK 840D/840Di/810D Programming Guide **ISO Milling** Order No.: 6FC5298-6AC20-0BP2	(11.02 Edition)
/PGT/	SINUMERIK 840D/840Di/810D Programming Guide **ISO Turning** Order No.: 6FC5298-6AC10-0BP2	(11.02 Edition)
/PGZ/	SINUMERIK 840D840Di//810D Programming Guide **Cycles** Order No.: 6FC5298-7AB40-0BP0	(03.04 Edition)
/PI /	PCIN 4.4 Software for Data Transfer to/from **MMC Modules** Order No.: 6FX2060-4AA00-4XB0 (English, German, French) Order from: WK Fürth	
/SYI/	SINUMERIK 840Di **System Overview** Order No.: 6FC5298-6AE40-0BP0	(02.01 Edition)

© Siemens AG 2004 All rights reserved.

Manufacturer/Service Documentation

a) Lists

/LIS/ SINUMERIK 840D/840Di/810D (03.04 Edition)
SIMODRIVE 611D
Lists
Order No.: 6FC5297-7AB70-0BP0

b) Hardware

/ASAL/ SIMODRIVE (10.03 Edition)
Planning Guide General Part for **Asynchronous Motors**
Order No.: 6SN1197-0AC62-0BP0

/APH2/ SIMODRIVE (10.03 Edition)
Planning Guide **Asynchronous Motors 1PH2**
Order No.: 6SN1197-0AC63-0BP0

/APH4/ SIMODRIVE (10.03 Edition)
Planning Guide **Asynchronous Motors 1PH4**
Order No.: 6SN1197-0AC64-0BP0

SIMODRIVE (01.04 Edition)
Planning Guide **Asynchronous Motors 1PH7**
Order No.: 6SN1197-0AC65-0BP0

/APH7S/ SIMODRIVE (01.04 Edition)
Planning Guide **Asynchronous Motors 1PH7**
Order No.: 6SN1197-0AC65-0BP0

/APL6/ MASTERDRIVES VC/MC (01.04 Edition)
Planning Guide **Asynchronous Motors 1PL6**
Order No.: 6SN1197-0AC67-0BP0

/BH/ SINUMERIK 840D840Di//810D (11.02 Edition)
Operator Components Manual
Order No.: 6FC5297-6AA50-0BP3

/BHA/ SIMODRIVE Sensor (03.03 Edition)
Absolute Encoder with Profibus DP User's Guide (HW)
Order No.: 6SN1197-0AB10-0YP2

/EMC/ SINUMERIK, SIROTEC, SIMODRIVE (06.99 Edition)
Planning Guide **EMC Installation Guide**
Order No.: 6FC5297-0AD30-0BP1

The up-to-date declaration of conformity can be viewed on the Internet at
http://www4.ad.siemens.com/
Please enter the ID number 15257461 in the "Search" field (top right) and click on "go".

/GHA/ SINUMERIK/SIMOTION (02.03 Edition)
ADI4 - Analog Drive Interface for 4 Axes
Manual
Order No.: 6FC5297-0BA01-0BP1

© Siemens AG 2004 All rights reserved.

/PFK6/ SIMODRIVE 611, SIMOVERT MASTERDRIVES (05.03 Edition)
Planning Guide **AC Servo Motors 1FK6**
Order No.: 6SN1197-0AD05-0BP0

/PFK7/ SIMODRIVE 611, SIMOVERT MASTERDRIVES (01.03 Edition)
Planning Guide **AC Servo Motors 1FK7**
Order No.: 6SN1197-0AD06-0BP0

PFS6/ SIMOVERT MASTERDRIVES (07.03 Edition)
Planning Guide **AC Servo Motors 1FS6**
Order No.: 6SN1197-0AD08-0BP0

/PFT5/ SIMODRIVE (05.03 Edition)
Planning Guide **AC Servo Motors 1FT5**
Order No.: 6SN1197-0AD01-0BP0

/PFT6/ SIMODRIVE 611, SIMOVERT MASTERDRIVES (01.04 Edition)
Planning Guide **Synchronous Servo Motors 1FT6**
Order No.: 6SN1197-0AD02-0BP0

/PFU/ SINAMICS, SIMOVERT MASTERDRIVES, (09.03 Edition)
MICROMASTER
SIEMOSYN Motors 1FU8
Order No.: 6SN1197-0AC80-0BP0

/PHC/ SINUMERIK 810D (11.02 Edition)
Configuring Manual **(HW)**
Order No.: 6FC5297-6AD10-0BP1

/PHD/ SINUMERIK 840D (11.03 Edition)
Configuring Manual **(HW)**
Order No.: 6FC5297-6AC10-0BP3

/PJAL/ SIMODRIVE 611, SIMOVERT MASTERDRIVES MC (01.04 Edition)
Planning Guide **Synchronous Servo Motors**
General Part for 1FT / 1FK Motors
Order No.: 6SN1197-0AD07-0BP0

/PJFE/ SIMODRIVE (02.03 Edition)
Planning Guide **Built-In Synchronous Motors 1FE1**
Three-Phase AC Motors for Main Spindle Drives
Order No.: 6SN1197-0AC00-0BP4

/PJF1/ SIMODRIVE (12.02 Edition)
Installation Guide **Built-In Synchronous Motors**
1FE1 051.-1FE1 147
Three-Phase AC Motors for Main Spindle Drives
Order No.: 610.43000.02

/PJLM/ SIMODRIVE (06.02 Edition)
Planning Guide **Linear Motors 1FN1, 1FN3**
ALL General Information on the Linear Motor
1FN1 Three-Phase AC Linear Motor 1FN1
1FN3 Three-Phase AC Linear Motor 1FN3
CON Connections
Order No.: 6SN1197-0AB70-0BP3

© Siemens AG 2004 All rights reserved.

/PJM/ SIMODRIVE 611, SIMOVERT MASTERDRIVES (11.00 Edition)
Planning Guide **Servo Motors**
Three-Phase Motors for Feed and Main Spindle Drives
Order No.: 6SN1197-0AA20-0BP4

/PJM2/ SIMODRIVE (07.03 Edition)
Planning Guide **Servo Motors**
Three-Phase Servo Motors for Feed and Main Spindle Drives
Order No.: 6SN1197-0AC20-0BP0

/PJTM/ SIMODRIVE (05.03 Edition)
Planning Guide **Built-In Torque Motors 1FW6**
Order No.: 6SN1197-0AD00-0BP1

/PKTM/ SIMODRIVE, SIMOVERT MASTERDRIVES (03.04 Edition)
Planning Guide **Complete Torque Motors 1FW3**
Order No.: 6SN1197-0AC70-0BP0

/PJU/ SIMODRIVE 611 (02.03 Edition)
Planning Guide **Converters**
Order No.: 6SN1197-0AA00-0BP6

/PMH/ SIMODRIVE Sensor (07.02 Edition)
Planning/Installation Guide
Hollow-Shaft Measuring System SIMAG H
Order No.: 6SN1197-0AB30-0BP1

/PMH2/ SIMODRIVE Sensor (03.04 Edition)
Planning/Installation Guide
Hollow-Shaft Measuring System SIMAG H2
Order No.: 6SN1197-0AB31-0BP1

/PMHS/ SIMODRIVE (12.00 Edition)
Installation Guide **Measuring System for Main Spindle Drives Gear Wheel Encoder SIZAG2**
Order No.: 6SN1197-0AB00-0YP3

/PMS/ SIMODRIVE (03.04 Edition)
Planning Guide
ECO Motor Spindle for Main Spindle Drives
Order No.: 6SN1197-0AD04-0BP1

PPH/ SIMODRIVE (12.01 Edition)
Planning Guide **1PH2 / 1PH4 / 1PH7 Motors**
Three-Phase Asynchronous Motors for Main Spindle Drives
Order No.: 6SN1197-0AC60-0BP0

/PPM/ SIMODRIVE (11.01 Edition)
Planning Guide
Hollow-Shaft Motors for Main Spindle Drives **1PM4 and 1PM6**
Order No.: 6SN1197-0AD03-0BP0

© Siemens AG 2004 All rights reserved.

c) Software

/FB1/ SINUMERIK 840D/840Di/810D/FM-NC (03.04 Edition)
Description of Function **Basic Machine (Part 1)**
(the various manuals are listed below)
Order No.: 6FC5297-7AC20-0BP0

A2	Various Interface Signals
A3	Axis Monitoring, Protection Zones
B1	Continuous Path Mode, Exact Stop and Look Ahead
B2	Acceleration
D1	Diagnostic Tools
D2	Interactive Programming
F1	Travel to Fixed Stop
G2	Velocites, Setpoint/Actual Value Systems, Closed-Loop Control
H2	Output of Auxiliary Functions to PLC
K1	Mode Group, Channels, Program Operation Mode
K2	Axis Types, Coordinate Systems, Frames, Actual-Value System for Workpiece, External Zero Offset
K4	Communication
N2	EMERGENCY STOP
P1	Traverse Axes
P3	Basic PLC Program
R1	Reference Point Approach
S1	Spindles
V1	Feeds
W1	Tool Compensation

/FB2/ SINUMERIK 840D/840Di/810D (03.04 Edition)
Description of Functions **Extended Functions (Part 2)**
including FM-NC: Turning, Stepping Motor
(the various manuals are listed below)
Order No.: 6FC5297-7AC30-0BP0

A4	Digital and Analog NCK I/Os
B3	Several Operator Panels and NCUs
B4	Operation via PG/PC
F3	Remote Diagnostics
H1	Jog with/without Handwheel
K3	Compensations
K5	Mode Groups, Channels, Axis Replacement
L1	FM-NC Local Bus
M1	Kinematic Transformation
M5	Measurements
N3	Software Cams, Position Switching Signals
N4	Punching and Nibbling
P2	Positioning Axes
P5	Oscillation
R2	Rotary Axes
S3	Synchronous Spindles
S5	Synchronized Actions (SW 3 and lower, higher /FBSY/)
S6	Stepper Motor Control
S7	Memory Configuration
T1	Indexing Axes
W3	Tool Change
W4	Grinding

© Siemens AG 2004 All rights reserved.

/FB3/ SINUMERIK 840D/840Di/810D (03.04 Edition)
Descriptionof Functions **Special Functions (Part 3)**
(the various manuals are listed below)
Order No.: 6FC5297-7AC80-0BP0

F2	3-Axis to 5-Axis Transformation
G1	Gantry Axes
G3	Cycle Times
K6	Contour Tunnel Monitoring
M3	Coupled Axes and ESR
S8	Constant Workpiece Speed for Centerless Grinding
S9	Setpoint Switching (S9)
T3	Tangential Control
TE0	Installation and Activation of Compile Cycles
TE1	Clearance Control
TE2	Analog Axis
TE3	Speed/Torque Coupling Master-Slave
TE4	Transformation Package Handling
TE5	Setpoint Exchange
TE6	MCS Coupling
TE7	Retrace Support
TE8	Unclocked Path-Synchronous Switching Signal Output
V2	Preprocessing
W5	3D Tool Radius Compensation

/FBA/ SIMODRIVE 611D/SINUMERIK 840D/810D (03.04 Edition)
Description of Functions **Drive Functions**
(the various sections are listed below)
Order No.: 6SN1197-0AA80-1BP1

DB1	Operational Messages/Alarm Reactions
DD1	Diagnostic Functions
DD2	Speed Control Loop
DE1	Extended Drive Functions
DF1	Enable Commands
DG1	Encoder Parameterization
DL1	Linear Motor MD
DM1	Calculation of Motor/Power Section Parameters and Controller Data
DS1	Current Control Loop
DÜ1	Monitors/Limitations

/FBAN/ SINUMERIK 840D/SIMODRIVE 611 digital (02.00 Edition)
Description of Functions **ANA-Modul**
Order No.: 6SN1197-0AB80-0BP0

/FBD/ SINUMERIK 840D (07.99 Edition)
Description of Functions **Digitizing**
Order No.: 6FC5297-4AC50-0BP0

DI1	Start-Up
DI2	Scanning with Tactile Sensors (scancad scan)
DI3	Scanning with Lasers (scancad laser)
DI4	Milling Program Generation (scancad mill)

/FBDM/ SINUMERIK 840D/840Di/810D (09.03 Edition)
Description of Functions NC Program Management
DNC Machines
Order No.: 6FC5297-1AE81-0BP0

© Siemens AG 2004 All rights reserved.

/FBDN/ SINUMERIK 840D/840Di/810D (03.03 Edition)
Motion Control Information System (MCIS)
Description of Functions **NC Program Management DNC**
Order No.: 6FC5297-1AE80-0BP0
DN1 DNC Plant / DNC Cell
DN2 DNC IFC SINUMERIK, NC Data Transfer via Network

/FBFA/ SINUMERIK 840D/840Di/810D (11.02 Edition)
Description of Functions **ISO Dialects for SINUMERIK**
Order No.: 6FC5297-6AE10-0BP3

/FBFE/ SINUMERIK 840D/810D (04.03 Edition)
Description of Functions **Remote Diagnostics**
Order No.: 6FC5297-0AF00-0BP2
FE1 Remote Diagnosis ReachOut
FE3 Remote Diagnosis pcAnywhere

/FBH/ SINUMERIK 840D/840Di/810D (11.02 Edition)
HMI programming package
Order No.: (included with the software)

Part 1 User's Guide
Part 2 Description of Functions

/FBH1/ SINUMERIK 840D/840Di/810D (03.03 Edition)
HMI programming package
ProTool/Pro Option SINUMERIK
Order No.: (included with the software)

/FBHL/ SINUMERIK 840D/SIMODRIVE 611 digital (10.03 Edition)
Function Description **HLA Module**
Order No.: 6SN1197-0AB60-0BP3

/FBIC/ SINUMERIK 840D/840Di/810D (06.03 Edition)
Motion Control Information System (MCIS)
Description of Functions **TDI Ident Connection**
Order No.: 6FC5297-1AE60-0BP0

/FBMA/ SINUMERIK 840D/810D (08.02 Edition)
Description of Functions **ManualTurn**
Order No.: 6FC5297-6AD50-0BP0

/FBO/ SINUMERIK 840D/810D (09.01 Edition)
Description of Functions Configuring of **OP 030 Operator Interface**
(the various sections are listed below)
Order No.: 6FC5297-6AC40-0BP0
BA Operator's Guide
EU Development Environment (Configuring Package)
PSE Introduction to Configuring of Operator Interface
(IK Screen Kit: Software Update and Configuration

/FBP/ SINUMERIK 840D (03.96 Edition)
Description of Functions **C-PLC Programming**
Order No.: 6FC5297-3AB60-0BP0

© Siemens AG 2004 All rights reserved.

/FBR/ SINUMERIK 840D/840Di/810D (09.01 Edition)
IT Solutions
Description of Functions **Computer Link (SinCOM)**
Order No.: 6FC5297-6AD60-0BP0
HCI Interface for host computer
NPL PLC/NCK interface

/FBSI/ SINUMERIK 840D/SIMODRIVE (11.03 Edition)
Description of Functions SINUMERIK **Safety Integrated**
Order No.: 6FC5297-6AB80-0BP2

/FBSP/ SINUMERIK 840D/840Di/810D (08.03 Edition)
Description of Functions **ShopMill**
Order No.: 6FC5297-6AD80-0BP1

/FBST/ SIMATIC (01.01 Edition)
Description of Functions **FM STEPDRIVE/SIMOSTEP**
Order No.: 6SN1197-0AA70-0YP4

/FBSY/ SINUMERIK 840D/810D (10.02 Edition)
Description of Functions **Synchronized Actions**
Order No.: 6FC5297-6AD40-0BP2

/FBT/ SINUMERIK 840D/810D (01.02 Edition)
Description of Functions **ShopTurn**
Order No.: 6FC5297-6AD70-0BP1

/FBTC/ SINUMERIK 840D/810D (01.02 Edition)
IT Solutions
Description of Functions **Tool Data Communication SinTDC**
Order No.: 6FC5297-5AF30-0BP0

/FBTD/ SINUMERIK 840D/810D (02.01 Edition)
IT Solutions
Description of Functions **Tool Information Systems (SinTDI)**
with Online Help
Order No.: 6FC5297-6AE00-0BP0

/FBTP/ SINUMERIK 840D/840Di/810D (01.03 Edition)
Motion Control Information System (MCIS)
Description of Functions **Preventive Maintenance TPM**
Order No.: Document is included with the software

/FBU/ SIMODRIVE 611 universal/universal E (03.04 Edition)
Description of Functions **Closed-Loop Control Component for Speed Control and Positioning**
Order No.: 6SN1197-0AB20-0BP9

/FBU2/ SIMODRIVE 611 **universal** (04.02 Edition)
Installation Guide
(included with every SIMODRIVE 611)

/FBW/ SINUMERIK 840D/810D (03.04 Edition)
Description of Functions **Tool Management**
Order No.: 6FC5297-6AC60-0BP1

/HBA/ SINUMERIK 840D/840Di/810D (03.02 Edition)
Manual **@Event**
Order No.: 6AU1900-0CL20-0AA0

© Siemens AG 2004 All rights reserved.

/HBI/ SINUMERIK 840Di (09.03 Edition)
Manual **SINUMERIK 840Di**
Order No.: 6FC5297-6AE60-0BP2

/INC/ SINUMERIK 840D840Di//810D (06.03 Edition)
System Overview
Commissioning Tool SINUMERIK SinuCOM NC
Order No.: (part of the Online Help for the Startup Tool)

/PJE/ SINUMERIK 840D/810D (08.01 Edition)
Description of Functions **HMI Embedded Configuring Package**
Software Update, Configuration, Installation
Order No.: 6FC5297-6EA10-0BP0
(the document PS Configuring Syntax is supplied with the software and available as a pdf file)

/POS1/ SIMODRIVE **POSMO A** (08.03 Edition)
User Guide
Decentralized Positioning Motor on PROFIBUS DP
Order No.: 6SN2197-0AA00-0BP6

/POS2/ SIMODRIVE **POSMO A** (05.03 Edition)
Installation Guide (included with every POSMO A)

/POS3/ SIMODRIVE POSMO SI/CD/CA (03.04 Edition)
User Guide **Distributed Servo Drive Systems**
Order No.: 6SN2197-0AA20-0BP6

/POS4/ SIMODRIVE **POSMO SI** (08.03 Edition)
Installation Guide (included with every POSMO SI)

/POS5/ SIMODRIVE **POSMO CD/CA** (08.03 Edition)
Installation Guide (included with every POSMO CD/CA)

/S7H/ SIMATIC S7-300 (2002 Edition)
Installation Manual **Technological Functions**
Order No.: 6ES7398-8AA03-8BA0
- Reference Manual: CPU Data (HW Description)
- Reference Manual: Module Data

/S7HT/ SIMATIC S7-300 (03.97 Edition)
Manual **STEP 7, Fundamentals, V. 3.1**
Order No.: 6ES7810-4CA02-8BA0

/S7HR/ SIMATIC S7-300 (03.97 Edition)
Manual **STEP 7, Reference Manuals, V. 3.1**
Order No.: 6ES7810-4CA02-8AR0

/S7S/ SIMATIC S7-300 (04.02 Edition)
FM 353 for Stepper Drive Positioning Module
Order together with configuring package

/S7L/ SIMATIC S7-300 (04.02 Edition)
FM 354 for Servo Drive Positioning Module
Order together with configuring package

/S7M/ SIMATIC S7-300 (01.03 Edition)
FM 357.2 Multimodule for Servo and Stepper Drives
Order together with configuring package

© Siemens AG 2004 All rights reserved.

/SP/ SIMODRIVE 611-A/611-D
SimoPro 3.1
Program for Configuring Machine-Tool Drives
Order No.: 6SC6111-6PC00-0BA❑
Order from: WK Fürth

d) Start-up

/BS/ SIMODRIVE 611 analog (10.00 Edition)
Description **Start-Up Software for**
Main Spindle and Asynchronous Motor Modules Version 3.20
Order No.: 6SN1197-0AA30-0BP1

/IAA/ SIMODRIVE 611 analog (10.00 Edition)
Installation and Start-Up Guide
Order No.: 6SN1197-0AA60-0BP6

/IAC/ SINUMERIK 810D (11.02 Edition)
Installation and Start-Up Guide
(including description of SIMODRIVE 611D start-up software)
Order No.: 6FC5297-6AD20-0BP1

/IAD/ SINUMERIK 840D/SIMODRIVE 611D (03.04 Edition)
I**nstallation and Start-Up Guide**
(including description of SIMODRIVE 611D start-up software)
Order No.: 6FC5297-7AB10-0BP0

/IAM/ SINUMERIK 840D/840Di/810D (03.04 Edition)
HMI Installation and Start-Up Guide
Order No.: 6FC5297-6AE20-0BP3
AE1 Updates/Options
BE1 Expanding the Operator Interface
HE1 Online Help
IM2 Start-Up HMI Embedded
IM4 Start-Up HMI Advanced
TX1 Setting Foreign Language Texts

© Siemens AG 2004 All rights reserved.

D Identifiers

List of input/output variables for cycles

Name	*Meaning in English*	*Meaning in German*
AD	Allowance depth	Basic pocket depth from reference plane
AFSL	Angle for slot length	Winkel für die Nutlänge
ANG1, ANG2	Flank angle	Flankenwinkel
ANGB	Liftoff angle for roughing	Liftoff angle for roughing
AP1	Unfinished dimension in plane	Rohmaß Taschenlänge/Taschenradius
AP2	Unfinished dimension in plane	Basic size pocket width
APP	Approach path	Einlaufweg
APX	Axial value for defining blank for facing axis	Axial value for defining blank for facing axis
APXA	Absolute or incremental evaluation of parameter APX	Bewertung des Parameters _APX absolut oder inkrementell
APZ	Axial value for defining blank for longitudinal axis	Axial value for defining blank for longitudinal axis
APZA	Absolute or incremental evaluation of parameter APZ	Bewertung des Parameters _APZ absolut oder inkrementell
AS1, AS2	Direction of approach/approach path	Spezifikation der Anfahrrichtung/-bahn
AXN	Tool axis	Tool axis
BNAME	Name for program of drill positions	Name for program of drill positions
CDIR	Circle direction	Drehrichtung, Fräsrichtung
CPA	Center point, abscissa	Center point of hole circle, abscissa (absolute)
CPO	Center point,ordinate	Center point of hole circle, ordinate (absolute)
CRAD	Corner radius	Corner radius
DAM	Degression value, Path for roughing interrupt	Degressionsfaktor / Weglänge
DBH	Distance between holes	Abstand zwischen den Bohrungen
DIAG	Groove depth	Einstichtiefe
DIATH	Diameter of thread	Nominal diameter, outside diameter of thread
DIS1	Distance	programmierbarer Vorhalteabstand
DIS1	Distance between columns	Abstand der Spalten
DIS2	Number of lines, Distance between rows	Abstand der Zeilen
DM1... DM4	Diameter	Diameter of the thread at the starting point
DP	Depth	Depth (absolute)
DP1	First depth	Insertion depth
DPR	Depth, relative	Tiefe relativ zur Referenzebene

© Siemens AG 2004 All rights reserved.

DT	Dwell time	Dwell time for chip breaking during roughing
DTB	Dwell time at bottom	Verweilzeit auf Endbohrtiefe/am Einstichgrund
DTD	Dwell time at depth	Dwell time at final drilling depth
DTS	Dwell time at starting point	Verweilzeit am Anfangspunkt
ENC	Tapping with/without encoder	Gewindebohren mit/ohne Geber
FAL	Finish allowance	konturgerechtes Schlichtaufmaß am Nutrand/Taschenrand
FAL1	Finish allowance on groove base	Final machining allowance on the base of groove
FAL2	Finish allowance on flanks	Final machining allowance on the flanks
FALD	Finish allowance depth	Schlichtaufmaß am Grund
FALZ	Finish allowance, z axis	Schlichtaufmaß in der Längsachse
FALX	Finish allowance, x axis	Schlichtaufmaß in der Planachse
FDEP	First depth	First drilling depth (absolute)
FDIS	First distance	Abstand der ersten Bohrung vom Bezugspunkt
FDP1	Overrun path in direction to plane	Überlaufweg in Richtung der Ebenenzustellung
FDPR	First depth, relative	erste Bohrtiefe relativ zur Referenzebene
FF1	Feedrate for roughing	Vorschub für Schruppen
FF2	Feedrate for insertion	Vorschub zum Eintauchen
FF3	Feedrate for finishing	Feedrate for finishing
FF4	Feedrate for contour transition elements	Vorschub an Konturübergangselementen
FFCP (SW 6.3 and higher)	Feedrate for circular positioning	Vorschub für Zwischenpositionierung auf Kreisbahn
FFD	Feedrate for depth	Feedrate for depth infeed
FFP1	Feedrate surface	Feedrate for surface machining
FFP2	Feedrate for finishing	Feedrate for finishing
FFR	Feedrate	Feed
FORM	Definition of form	Definition of the form
FPL	Final point along longitudinal axis	Endpunkt in der Längsachse
FRF	Feedrate factor	Vorschubfaktor
IANG	Infeed angle	Infeed angle
INDA	Incremental angle	Indexing angle
IDEP	Infeed depth	Zustelltiefe
KNAME	Name of the contour subroutine	Name of the contour subroutine
LSANF	Block number/label identifying start of contour definition	Block number/label identifying start of contour definition
LSEND	Block number/label identifying end of contour definition	Block number/label identifying end of contour definition
LP1	Length of approach travel, radius	Länge des Anfahrwegs, Radius

© Siemens AG 2004 All rights reserved.

LP2	Length of return travel, radius	Länge des Abfahrwegs, Radius
KDIAM	Internal diameter of thread	Core diameter, inside diameter of thread
LENG	Elongated hole length, pocket length	Langlochlänge, Taschenlänge
MDEP	Minimum depth	Minimum drilling depth
MID	Maximum infeed depth	maximale Zustelltiefe für eine Zustellung
MIDA	Maximum infeed width	maximale Zustellbreite
MIDF	Maximum infeed depth for finishing	Maximum infeed depth for finishing
MPIT	Thread lead as thread size	Thread pitch as thread size
NID	Number of noncuts	Anzahl der Leerschnitte
NP1 ... NP8	Name/Label ...	Name des Konturunterprogramms der Fertigteilkontur/Label
NPP	Name of parts program	Name of the contour subroutine
NRC	Number of roughing cuts	Anzahl der Schruppschnitte
NSP	Start point offset first thread	Startpunktversatz für den ersten Gewindegang
NUM	Number of holes	Number of drill holes
NUM1	Number of columns	Number of columns
NUM2	Number of lines	Number of rows
NUMT	Number of threads	Anzahl der Gewindegänge
PA	Reference point, abscissa	Bezugspunkt Tasche
PO	Reference point, ordinate	Bezugspunkt Tasche
PO1 ... PO4	Point in longitudinal axis	Anfangspunkt/Zwischenpunkt/Endpunkt des Gewindes in der Längsachse
PIT	Thread lead	Gewindesteigung als Wert
PNAME	Name for pocket milling machining program	Name for pocket milling machining program
POSS	Position for oriented spindle stop	spindle position
PP1 ... PP3	Thread pitch 1...3 as value	Gewindesteigung 1...3 als Wert
PRAD	Pocket radius	Pocket radius
RAD	Radius	Radius des Kreises
RAD1	Radius	Radius der Helixbahn beim Eintauchen
RCO1, RCO2	Radius/chamfer outside	Radius/Fase, außen
RCI1, RCI2	Radius/chamfer inside	Radius/Fase, innen
RFF	Retract feed	Retraction feedrate
RFP	Reference plane	Reference plane (absolute)
ROP	Run out path	Auslaufweg
RPA	Retract path, abscissa	Rückzugsweg in der Abszisse
RPAP	Retract path, applicate	Rückzugsweg in der Applikate
RPO	Retract path, ordinate	Rückzugsweg in der Ordinate
RL	Bypass contour	Umfahren der Kontur mittig
RTP	Retract plane	Retraction plane (absolute)
SDAC	Spindle direction after cyle	Drehrichtung nach Zyklusende
SDIR	Spindle direction	Spindle direction
SDIS	Safety distance	Safety distance
SDR	Spindle direction for retraction	Drehrichtung für Rückzug

© Siemens AG 2004 All rights reserved.

SPCA	Reference point, abscissa	Abscissa of a reference point on the straight line (absolute)
SPCO	Reference point, ordinate	Ordinate of this reference point (absolute)
SPD	Starting point in the facing axis	Anfangspunkt in der Planachse
SPL	Starting point along longitudinal axis	Starting point in the longitudinal axis
SSF	Speed for finishing	Speed for finishing
SST	Speed for tapping	Speed for tapping
SST1	Speed for retraction	Speed for retraction
STA, STA1	Angle	Angle
STA2	Insertion angle	Maximum insertion angle for oscillation movement
TDEP	Thread depth	Gewindetiefe
TN	Name of stock removal tool	Name of stock removal tool
TOL1	Blank tolerance	Blank tolerance
TYPTH	Type of thread	Gewindetyp
VARI	Working	Machining type
VRT	Variable return path	variabler Rückzugsbetrag / Rückzugsweg
WID	(Pocket) width	Taschenbreite
WIDG	Groove width	Einstichbreite

■

© Siemens AG 2004 All rights reserved.

E Index

A
Absolute drilling depth 2-52, 3-118, 3-123, 3-137, 3-159, 3-197
Auxiliary cycle routines 1-20
Axis assignment 1-21
B
Behavior when quantity parameter is zero 2-94
Blank 4-322
Blank updating 4-331
Blueprint programming 1-30
Boring 2-49
Boring 1 2-78
Boring 2 2-81
Boring 3 2-85
Boring 4 2-87
Boring 5 2-89
Boring cycle 2-49
C
Call 1-21, 2-50
Call conditions 1-21
Centering 2-51
Circumferential slot - SLOT2 3-129
Contour 1-line 1-30
Contour 2-line 1-30
Contour 3-line 1-30
Contour definition 4-294, 4-329
Contour monitoring 4-272, 4-297
Contour programming 4-324
CONTPRON 4-295
cycle alarms 5-343
Cycle call 1-23
Cycle setting data, milling 3-108
Cycle setting data, Turning 4-272
Cycle simulation 1-26
Cycle support for user cycles 1-35
Cycle support in the program editor 1-27
CYCLE60 (SW 6.4 and later) 3-259
CYCLE71 3-157
CYCLE72 3-163
CYCLE73 3-182, 3-188
CYCLE74 3-182, 3-183
CYCLE75 3-182, 3-185
CYCLE76 3-173
CYCLE77 3-178
CYCLE800 3-210
CYCLE801 2-102
CYCLE81 2-51
CYCLE82 2-54
CYCLE83 2-56
CYCLE832 (SW 6.3 and later) 3-246
CYCLE84 2-63
CYCLE840 2-70
CYCLE85 2-78
CYCLE86 2-81
CYCLE87 2-85
CYCLE88 2-87
CYCLE89 2-89
CYCLE90 3-109
CYCLE93 4-274
CYCLE94 4-283
CYCLE95 4-287
CYCLE950 4-319
CYCLE96 4-300
CYCLE97 4-304
CYCLE98 4-311
D
Deep-hole drilling 2-56
Deep-hole drilling with chip breaking 2-58
Deep-hole drilling with swarf removal 2-57
Definition files for cycles 1-41
Dot matrix 2-102
Drill pattern cycles 1-19, 2-94
Drilling 2-51
Drilling cycles 1-19, 2-48
Drilling pattern cycles without drilling cycle call 2-94
Drilling, counterboring 2-54
E
Elongated holes on a circle - LONGHOLE 3-116
Engraving cycle – CYCLE60 (SW 6.4 and later) 3-259
Extended stock removal cycle - CYCLE950 4-319
F
Face thread 4-309
FGROUP 3-109
Form of delivery for cycles in HMI Advanced 1-43
Free contour programming 1-30
G
Geometrical parameters 2-49
Grooving cycle - CYCLE93 4-274

© Siemens AG 2004 All rights reserved.

H
High speed settings – CYCLE832 (SW 6.3 and later) 3-246
Hole circle 2-99
HOLES1 2-95
HOLES2 2-99
I
Inside threads 3-111
L
Level definition 1-21
LONGHOLE 3-116
Longitudinal thread 4-309
M
Machine data 1-40, 1-44
Machining parameters 2-49
Machining plane 1-21
MCALL 2-91
Messages 1-22, 5-349
Milling circular pockets - POCKET2 3-139
Milling circular pockets - POCKET4 3-152
Milling circular spigots - CYCLE77 3-178
Milling Cycles 1-19
Milling rectangular pockets - POCKET1 3-135
Milling rectangular pockets - POCKET3 3-143
Milling rectangular spigots - CYCLE76 3-173
Modal call 2-91
O
Outside threads 3-110
Overview of cycle alarms 5-343
Overview of cycles 1-18
P
Parallel-contour 4-321
Parameter list 1-23
Path milling 3-163
Planfräsen 3-157
Plausibility checks 2-94
Pocket milling with islands 3-182
Pocket milling with islands - CYCLE73 3-188
POCKET1 3-135
POCKET2 3-139
POCKET3 3-143
POCKET4 3-152
POCKET450 3-152
R
Reference plane 2-52, 3-197
Relative drilling depth 2-52, 3-118, 3-137, 3-159, 3-197
Relative drilling depth 3-123
residual material 4-322
Retraction plane 2-52, 3-197
Return conditions 1-21
Rigid tapping 2-63
row of holes 2-95
S
Safety distance 2-52, 3-197
SETMS 3-108
Simulation without tool 1-26
SLOT1 3-121
SLOT2 3-129
Slots on a circle - SLOT1 3-121
Spindle handling 4-271
SPOS 2-65, 2-66
Starting point 4-297
Stock removal cycle - CYCLE95 4-287
Swiveling – CYCLE800 3-210
T
Tapping with compensating chuck 2-70
Tapping with compensating chuck with encoder 2-72
Tapping with compensating chuck without encoder 2-71
Thread chaining - CYCLE98 4-311
Thread cutting - CYCLE97 4-304
Thread milling 3-109
Thread recutting 4-317
Thread undercut - CYCLE96 4-300
Tool clearance angle 4-272
Transfer island contour - CYCLE75 3-185
Transfer pocket edge contour - CYCLE74 3-183
Turning cycles 1-20
U
Undercutting cycle - CYCLE94 4-283
Upgrading cycles in HMI Advanced 1-43

© Siemens AG 2004 All rights reserved.

F Commands, identifiers

C
CYCLE60 (SW 6.4 and later) 3-259
CYCLE71 3-157
CYCLE72 3-163
CYCLE73 3-188
CYCLE74 3-183
CYCLE75 3-185
CYCLE76 3-173
CYCLE77 3-178
CYCLE800 3-210
CYCLE801 2-102
CYCLE81 2-51
CYCLE82 2-54
CYCLE83 2-56
CYCLE832 (SW 6.3 and later) 3-246
CYCLE84 2-63
CYCLE840 2-70
CYCLE85 2-78
CYCLE86 2-81
CYCLE87 2-85
CYCLE88 2-87
CYCLE89 2-89
CYCLE90 3-109
CYCLE93 4-274
CYCLE94 4-283
CYCLE95 4-287
CYCLE950 4-319
CYCLE96 4-300
CYCLE97 4-304
CYCLE98 4-311
H
HOLES1 2-95
HOLES2 2-99
L
LONGHOLE 3-116
P
POCKET1 3-135
POCKET2 3-139
POCKET3 3-143
S
SLOT1 3-121
SLOT2 3-129

© Siemens AG 2004 All rights reserved.

Notes

© Siemens AG 2004 All rights reserved.

To SIEMENS AG A&D MC BMS P.O. Box 3180 D-91050 Erlangen (Tel. ++49-180-5050-222 [Hotline] Fax +49 (0) 9131 98 -2176 [documentation] Email: motioncontrol.docu@erlf.siemens.de	Suggestions Corrections For Publication/Manual: SINUMERIK 840D/840Di/810D Cycles User Documentation
From Name Company/Dept. Address: Zip code: Town: Phone: / Fax: /	Programming Guide Order No.: 6FC5298-7AB40-0BP0 Edition: 03.04 Should you come across any printing errors when reading this publication, please notify us on this sheet. Suggestions for improvement are also welcome.

Suggestions and/or corrections

Overview of SINUMERIK 840D/840Di/810D Documentation (03.2004)

General Documentation

- *SINUMERIK* 840D/810D — Brochure
- *SINUMERIK* 840D/840Di/810D/ — Catalog Ordering Info. NC 60 ***)**
- *SINUMERIK SIROTEC SIMODRIVE* Accessories — Catalog Accessories NC-Z

User Documentation

- *SINUMERIK* 840D/810D/FM-NC — **AutoTurn**
  - Short Guide
  - Programming/ Setup
- *SINUMERIK* 840D/840Di/810D — Operator's Guide **– HT 6**
- *SINUMERIK* 840D/840Di/810D — **Diagnostics Guide *)**
- *SINUMERIK* 840D/840Di/810D — **Operator's Guide *)**
  - Short Guide
  - HMI Embedded
  - HMI Advanced

User Documentation

- *SINUMERIK* 840D/840Di/810D — **Program. Guide**
  - Short Guide
  - Fundamentals ***)**
  - Advanced ***)**
  - Cycles
  - Measuring Cycles
  - ISO Turning/Milling
- *SINUMERIK* 840D/810D — Operator's Guide
  - **ManualTurn**
  - Short Guide ManualTurn
  - **ShopMill**
  - Short Guide ShopMill
  - **ShopTurn**
  - Short Guide ShopTurn
- *SINUMERIK* 840Di — **System Overview**

Manufacturer/Service Documentation

- *SINUMERIK* — **Configuring** (HW) ***)**
  - 810D
  - 840D
- *SINUMERIK* 840D/840Di/810D — **Operator Components** (HW) ***)**
- *SINUMERIK* 840D/810D — Description of Functions
  - **ManualTurn**
  - **ShopMill**
  - **ShopTurn**
- *SINUMERIK* 840D/840Di/810D — Description of Functions **Synchronized Actions**

Manufacturer/Service Documentation

- *SINUMERIK SIMODRIVE* 611D 840D/810D — Description of Functions **Drive Functions *)**
- *SINUMERIK* 840D/840Di/810D — **Description of Functions**
  - Basic Machine ***)**
  - Extended Functions
  - Special Functions
- *SINUMERIK* 840D/840Di/810D — Description of Functions **Tool Management**
- *SINUMERIK* 840D/810D — **Configuring Kit** HMI Embedded
- *SINUMERIK* 840D/810D — Description of Functions Operator Interface **OP 030**
- *SINUMERIK* 840D/810D — **MCIS**
  - Computer Link
  - Tool Data Information System
  - NC Data Management
  - NC Data Transfer
  - Tool Data Communication

Manufacturer/Service Documentation

- *SINUMERIK SIMODRIVE* — Description of Functions SINUMERIK **Safety Integrated**
- *SINUMERIK* 840D — Description of Functions **Digitizing**
- *SINUMERIK SIMODRIVE* — **Installation & Start-Up Guide *)**
  - 810D
  - 840D/611D
  - HMI
- *SINUMERIK SIMODRIVE* 840D/840Di 810D 611D
  - **Lists *)**
  - **List Syst. Var.**
- *SINUMERIK SIMODRIVE* 840D 611D — Description of Functions **Linear Motor**
- *SINUMERIK SIMODRIVE* 840D 611D — Description of Functions
  - **Hydraulics Module**
  - **Analog Module**
- *MOTION CONTROL SYSTEMS* — **EMC** Guidelines

Manufacturer/Service Documentation

- *SINUMERIK* 840D/840Di/810D — Description of Functions **ISO Dialects** for SINUMERIK
- *SINUMERIK* 840Di — **Manual** (HW + Installation and Start-Up)
- *SINUMERIK* 840D/810D — Description of Functions **Remote Diagnosis**
- *SINUMERIK* 840D/840Di/810D — **Manual @ Event**

Electronic Documentation

- *SINUMERIK SIMODRIVE* 840D/840Di/810D 611, Motors — **DOC ON CD *)** The SINUMERIK System

---

***)** These documents are a minimum requirement

Siemens AG
Automation & Drives
Motion Control Systems
P. O. Box 3180, D – 91050 Erlangen
Germany

www.siemens.com

© Siemens AG, 2004
Subject to change without prior notice
Order No. 6FC5298-7AB40-0BP0

Printed in Germany